TWO ABELIAN STATEMENTS EQUIVALENT TO THE HODGE CONJECTURE

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ABSTRACT. The Hodge conjecture is equivalent to two statements on abelian varieties, and the algebraic locus of a flat Hodge class is the whole base or a meagre set. For every real quadratic field we construct K3 surfaces with that real multiplication on all of whose powers the conjecture holds, and we prove the link classes of Mumford fourfolds algebraic on a dense set of special points.

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1. INTRODUCTION

The Hodge conjecture asks that every rational cohomology class of type (p,p) on a smooth complex projective variety be a rational combination of classes of algebraic subvarieties. Hodge raised it in 1950 [43]; Lefschetz had proved it for divisors [52]. It is known in a scattered family of cases and open in general. This paper does three things. It proves that the conjecture is equivalent to two statements about abelian varieties. It proves that the locus on which either statement is known has the same shape in every family, dense and either everything or meagre, so that each statement asks that algebraicity propagate from a dense set of special points to the whole family. And it goes to the first class each statement has to reach, proves what can be proved there, and names the step at which it stops. The conjecture is not proved here. The new theorems are the reduction and the dichotomy; real multiplication made algebraic by $\mathbb{Q}(\sqrt{6})$ on K3 surfaces, by the cyclic cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ on the surfaces of Mumford type, and by every real quadratic field in transcendental rank six; the Hodge conjecture on every power of the Fricke K3 surfaces of levels 5 and 7; the link classes

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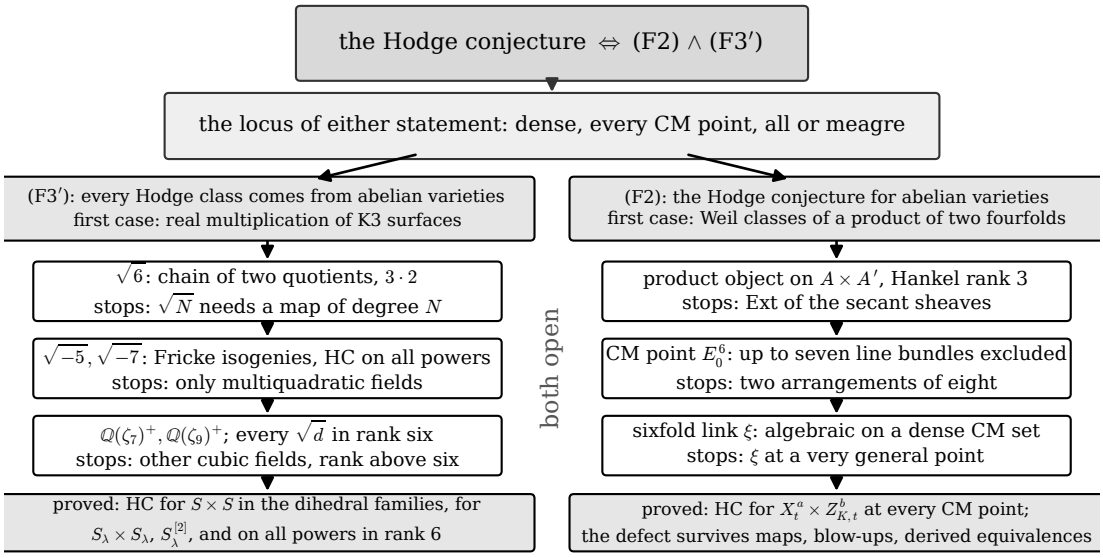


FIGURE 1. The plan of the paper. The conjecture is equivalent to (F2) and (F3′) (Theorem 1.1, Proposition 3.9); the locus of either statement is dense, contains every CM point and is all or meagre (Section 4). For (F3′) the first class is the real multiplication of K3 surfaces: the chain of quotients for $\sqrt{6}$ (Theorem 5.5), the Fricke isogenies (Theorem 5.9), the cyclic cubic fields on the surfaces of Mumford type (Theorem 5.14), every real quadratic field in rank six (Theorem 5.23), and the steps at which each stops (Propositions 5.11 and 5.15). For (F2) the first class is the Weil classes of a product of two fourfolds: the product object (Theorem 6.3), the point of complex multiplication (Lemma 6.8 to Remark 6.28) and the sixfold link (Theorems 6.47 and 6.51); the defect survives the operations of Proposition 7.2.

TABLE 1. The main results. "Corollary" marks a statement whose decisive input is a published theorem, named in the text.

Statement	Where	Status
HC $\iff$ (F2) + (F3′), (F2) $\iff$ **AB**	Thm. 1.1	proved
algebraic locus is all or meagre; dense for Weil classes	Thm. 1.2	proved
real multiplication by $\sqrt{6}$ through chains of quotients	Thm. 1.3	proved
Hodge conjecture on all powers of the Fricke families, $N = 5, 7$	Thm. 5.9	proved
real multiplication by $\mathbb{Q}(\zeta_7)^+$, $\mathbb{Q}(\zeta_9)^+$	Thm. 5.14	proved
every $\mathbb{Q}(\sqrt{d})$ in rank six, all powers	Thm. 5.23	corollary [32]
link classes on the dense set Σ_K	Thm. 6.51	corollary [57]
E_0^6: at most seven line bundles; eight outside two arrangements	Thm. 6.25, Rem. 6.28	proved; enumeration
(F2) and (F3′)	Sect. 7	open

of Mumford fourfolds on a dense set of points of complex multiplication; and, at the point E_0^6, the exclusion of every twisted complex of line bundles and distinct threefolds with at most seven line bundles, and of all but two arrangements of eight. Figure 1 draws the plan, and Table 1 lists the main results.

1.1. The two statements

Write **HC** for the Hodge conjecture for all smooth complex projective varieties and **AB** for its restriction to abelian varieties. For a smooth projective variety X and an integer p let $\mathrm{Hdg}^p(X)$ be the

Hodge classes of degree $2p$, and let $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$ be the span of the classes $\Gamma_*\gamma$, where B runs over the complex abelian varieties (the point included), γ over the Hodge classes on B, and Γ over the algebraic correspondences from B to X. The two statements are

(F2) every Hodge class on every complex abelian variety is algebraic, in the form in which it first appears in the closure computation of Appendix A.6: the Hodge classes of the smallest known instance, the exceptional classes on the square of a Mumford fourfold, are algebraic together with everything that can be built from them;

(F3′) $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ for every X and every p: every Hodge class is algebraic modulo the images, under algebraic correspondences, of Hodge classes of abelian varieties.

Theorem 1.1 (Reduction)**. HC** *holds if and only if* (F2) *and* (F3′) *both hold;* (F2) *is equivalent to* **AB**, *and* (F3′) *holds on every abelian variety, on every variety dominated by a product of curves or by an abelian variety, and on every variety in degrees* $2p$ *with* $p \in \{0, 1, \dim X - 1, \dim X\}$.

The content of the theorem is in the second sentence: (F2) is not a special case of the conjecture for abelian varieties but the whole of it, carried by its hardest classes, and (F3′) is exactly what the conjecture asks of the part of the cohomology of X that no abelian variety reaches. Section 3 proves it and places the two statements in the closure graph of the known reductions, with the Lefschetz standard conjecture, André's motivated classes and the variational Hodge conjecture as the other nodes (Fig. 2).

1.2. **The shape of the known locus**

For a smooth projective family $\pi\colon \mathscr{X} \to S$ and a flat Hodge class ξ let $\Sigma(\xi) \subseteq S$ be the set of points at which ξ is algebraic.

Theorem 1.2 (Dichotomy and propagation)**.**

(i) $\Sigma(\xi)$ *is a countable union of Zariski closed subsets of* S*, and either* $\Sigma(\xi) = S$ *or* $\Sigma(\xi)$ *is meagre.*

(ii) (F2) *holds if and only if, for every polarised abelian scheme and every flat Hodge class* ξ*, algebraicity of* ξ *at one point of complex multiplication forces* $\Sigma(\xi) = S$*.*

(iii) *For the Weil classes of every CM field, every* $n \geq 2$ *and every discriminant,* $\Sigma(\xi)$ *contains an explicit point of complex multiplication and is dense.*

(iv) (F3′) *holds for* X *in degree* $2p$ *if and only if a Hodge complement of the abelian part of* $H^{2p}(X, \mathbb{Q})$ *carries no nonzero Hodge class.*

So both statements have the same form. There is a dense set of points, the points of complex multiplication, at which the class is known to be algebraic by explicit cycles; the locus of algebraicity is a countable union of closed strata; and the conjecture is the statement that this union is not meagre. Figure 3 draws it. Section 4 proves the theorem, with the results on Weil type of Appendix A.2.

1.3. **Constructions at the first class**

The rest of the paper is constructive. At each statement we go to the first class that is not known and build the object that would make it algebraic.

At (F3′) the first class lives on powers of a K3 surface S with real multiplication: the transcendental lattice $T(S)_{\mathbb{Q}}$ is a vector space over a totally real field $E = \mathrm{End}_{\mathrm{Hdg}} T(S)_{\mathbb{Q}}$, and the elements of E are Hodge classes on $S \times S$. Section 5 proves:

Theorem 1.3 (Chains of symplectic quotients)**.** *Let* S *be a projective K3 surface and* $e \in E$ *with* $e^2 = n \in \mathbb{Q}_{>0}$*. If there is a chain of projective K3 surfaces* $S = S_0, S_1, \dots, S_r$ *joined by dominant rational maps or rational Hodge isometries whose degrees multiply to* n *modulo squares, then* e *is algebraic on* $S \times S$*. In particular, for every projective K3 surface* S *with* $\mathbb{Q}(\sqrt{6}) \subseteq E$ *whose transcendental space embeds in the invariant lattice of a symplectic automorphism of order three, the real multiplication by* $\sqrt{6}$ *is algebraic, the Hodge conjecture holds for* $S \times S$ *and* $S^{[2]}$*, and such surfaces form two-dimensional families whose very general member has* $E = \mathbb{Q}(\sqrt{6})$ *exactly.*

Varesco [83] proved the case of a single quotient by a symplectic automorphism of prime order, which gives $\sqrt{2}$ and $\sqrt{3}$; van Geemen and Schütt [88] reach $\sqrt{2},\sqrt{3},\sqrt{5},\sqrt{7}$ through rational self-maps and the dihedral families. The chain adds $\sqrt{6}$, the first composite value, with the lattice condition made explicit.

A second construction starts from the universal elliptic curve over the modular curves $X_0(5)$ and $X_0(7)$: the Fricke involution lifts to an isogeny of degree 5 or 7 between the elliptic K3 surface over the base and the one over its image, and on the quadratic twists of the two surfaces by the same pair of parameters it becomes a rational map of that degree between two K3 surfaces of the same family. Lemma 5.8 determines the sign of its square: on the twists at a pair exchanged by the involution the map acts on the transcendental lattice with square $-N$, a complex multiplication by $\mathbb{Q}(\sqrt{-5})$ or $\mathbb{Q}(\sqrt{-7})$, while at the fixed points of the involution, the surfaces of [88], its square is N. Theorem 5.9 shows that every member of the two families has that complex multiplication, algebraic because it is the pullback of the isogeny, and that the Hodge conjecture and (F3′) hold for every power and every Hilbert scheme of points of every member. Proposition 5.11 shows why the method stops: every correspondence component of geometric genus at most one acts on the transcendental lattice as a similarity, and these span only the multiquadratic subfield of E.

The third construction reaches every real quadratic field at once. By a theorem of Floccari [32], a K3 surface whose rational transcendental space embeds isometrically in $U^3_{\mathbb{Q}} \oplus \langle -1 \rangle$ satisfies the Hodge conjecture on all its powers. Section 5.8 constructs, for every squarefree $d > 1$, an explicit quadratic space of signature $(2,4)$ with a self-adjoint square root of d that embeds there, and so obtains:

Theorem 1.4 (Every real quadratic field in rank six)**.** *For every squarefree integer $d > 1$ there is a one-dimensional family of projective K3 surfaces, of Picard number 16 at the very general member, whose endomorphism field is exactly $\mathbb{Q}(\sqrt{d})$ there, and on every power and every Hilbert scheme of points of every member the Hodge conjecture holds.*

For the cubic field of a Mumford fourfold the multiquadratic subfield is $\mathbb{Q}$, so neither quotient chains nor Fricke maps see it. For the two cyclic cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ the quaternion algebra over F ramified at exactly two real places is the even Clifford algebra of a two-dimensional dihedral family of K3 surfaces of Picard number ten, and Theorem 5.14 makes the real multiplication by F algebraic there, through Buskin's theorem; Proposition 5.15 says exactly what the mechanism cannot reach.

At (F2) the first class is a Weil class of a very general abelian $2n$-fold of Weil type with $n \geq 3$, or an exceptional class of a Mumford fourfold. For the Weil classes the tool is the polarised semiregularity criterion (Theorem 6.1, proved in Appendix A.3): a perfect complex $\mathscr{E}$ whose Chern character is a nonzero multiple of a Weil class plus a polynomial in the polarisation, with $\dim \mathrm{Ext}^2(\mathscr{E},\mathscr{E})$ equal to the lower bound $r(\gamma) = (4+\rho)n^2 - 2n$, makes the Weil classes algebraic on every member with full Hodge group. Section 6 proves:

Theorem 1.5 (The product object)**.** *Let A, A' be very general polarised abelian varieties of Weil type for the same imaginary quadratic field, of dimensions $2m$, $2m'$, with algebraic Weil classes, and let $n = m + m' \geq 3$. There is a perfect complex $\mathscr{E}$ on $A \times A'$, a direct sum with shifts of exterior products of secant sheaves and powers of the polarisations, whose Chern character has the polarised shape with Hankel rank $\rho = 3$, so that $r(\gamma) = 7n^2 - 2n$, and whose first-order Hodge locus is the whole tangent space of the polarised Weil family. Every such direct sum has $\dim \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) > r(\gamma)$, and its semiregularity map is not injective. The criterion can therefore be met at a product point only by a convolution of these pieces with nonzero differentials.*

At the point E_0^6 of complex multiplication the Ext groups are computable (Lemma 6.8), and Section 6 excludes, for every polarisation, every convolution of distinct threefolds and at most five line bundles (Theorem 6.25); a sharper Maurer–Cartan rule (Lemma 6.27) and an enumeration extend this to seven line bundles and to all but two arrangements of eight (Remark 6.28), and Section 6.8 reads the support of an object meeting the criterion off its localised obstruction. For the exceptional classes of a Mumford fourfold X we replace the K3 surface that carries their transcendental part by an abelian sixfold Z_K over the same Shimura curve, so that the whole problem stays inside abelian varieties.

Theorem 1.6 (The sixfold link)**.** *Let X be a Mumford fourfold with cubic field F whose Hodge group is the group of Mumford's construction, as it is outside a countable subset of its Shimura curve C, let $K = F(\sqrt{-\delta})$ be a CM extension inside the quaternion algebra, and let Z_K be the abelian sixfold of Shimura type attached to K over C. The Hodge classes in $H^1(X)\otimes H^1(X)\otimes H^2(Z_K)$ that identify the two copies of the nine-dimensional representation T form a line over F, spanned by a class ξ. If ξ is algebraic and K contains no imaginary quadratic field, the exceptional classes of $X\times X$ are algebraic, and so is every Hodge class on every power of X. If $K = F\cdot k$ with k imaginary quadratic, then ξ_t is algebraic at every point t of a countable dense subset of C, and so is every Hodge class on every product of powers of X_t and $Z_{K,t}$.*

The last sentence is Theorem 6.51: at those points X_t is isogenous to $E\times A$ and $Z_{K,t}$ to $E^3\times A$, with E a CM elliptic curve and A a CM threefold, the Hodge ring of every $E^a\times A^b$ is generated by divisors and pull-backs of the Weil plane of the fourfold $A\times E$, and that fourfold carries a polarisation of discriminant one, so that Markman's theorem [57, Thm. 1.5] applies. By the dichotomy the algebraic locus of ξ is then either countable and dense or all of C; which one holds is open.

1.4. What is proved and what is not

Table 7 in Section 7 lists every statement of the paper with its status. The reduction, the dichotomy, the theorems on real multiplication and on the Fricke families, the link at the points of complex multiplication and the exclusions at E_0^6 are proved. (F2) and (F3′) are open, and with them the conjecture. The value of the constructions lies in the precision of the question each one leaves.

Appendix A proves the results on Weil classes, on the polarised criterion, on convolutions of line bundles, on the closure graph and on Mumford fourfolds that the body uses, so that the paper depends only on published work. Appendix C describes the machine checks: a Lean 4 file that verifies the closure graph and the finite combinatorics of Section 6, and programs in Python and Julia that recompute every finite quantity used in the proofs. No proved statement of the paper rests on a computation; the statements that do are marked as such.

2. Conventions

Varieties are smooth, projective and connected over $\mathbb{C}$ unless said otherwise. Cohomology is singular cohomology with rational coefficients, $H^k(X) = H^k(X,\mathbb{Q})$, with its Hodge structure. A Hodge class of degree $2p$ is an element of $H^{2p}(X)\cap H^{p,p}$; their space is $\mathrm{Hdg}^p(X)$. A class is algebraic if it is a rational combination of fundamental classes of subvarieties of codimension p; by Fulton [33], intersection products, pull-backs and push-forwards along morphisms preserve algebraic classes. An algebraic correspondence from Y to X is an algebraic class Γ on $Y\times X$, acting by $\Gamma_*\gamma = \mathrm{pr}_{X*}([\Gamma]\cup \mathrm{pr}_Y^*\gamma)$. The Hodge group $\mathrm{Hg}(V)$ of a rational Hodge structure V is the smallest algebraic subgroup of $\mathrm{GL}(V)$ defined over $\mathbb{Q}$ whose real points contain the circle $h(U(1))$; for a polarisable V it is reductive, the Hodge classes in every tensor construction on V are its invariants, and the Hodge group of a variety is that of its total cohomology [28]. The Hodge conjecture for X is the statement that every Hodge class on X is algebraic; **HC** is the conjecture for all varieties and **AB** for all abelian varieties.

For an abelian variety A of dimension g with $H^1(A) = V$, the cohomology ring is $\bigwedge^\bullet V$. An abelian $2n$-fold A is of Weil type for the imaginary quadratic field K if K acts on A up to isogeny with V a K-vector space of dimension $2n$ on which the two complex embeddings have equal multiplicity n in $H^{1,0}$. The Weil classes are the elements of $\bigwedge_K^{2n} V\subseteq\bigwedge^{2n}V$, a two-dimensional rational space of Hodge classes of degree $2n$ [92]; $K^\times$ acts on it by $\tau\mapsto\tau^{2n}$, and η^k, for η the class of a K-hermitian polarisation, carries $\tau\mapsto N_{K/\mathbb{Q}}(\tau)^k$. For a general CM field we use the setting of Appendix A (Setting A.1): a family of Weil type for a CM field F of degree $2m$ with $n\geq 2$ and discriminant δ has a connected polarised period domain $\mathscr{D}_F$ of dimension mn^2 (Proposition A.4), and its very general member has Hodge group $\mathrm{SU}(V,H)$ for the F-hermitian form H of the polarisation.

A Mumford fourfold is a simple abelian fourfold X whose Hodge group is the $\mathbb{Q}$-form of SL_2^3 given by the norm-one group D^1 of a quaternion algebra D over a totally real cubic field F, split at one real place and definite at the other two [63]. Its own Hodge ring is generated by the polarisation, and $X\times X$ carries exceptional Hodge classes of degree four that are not products of divisor classes and are not

Weil classes of any quotient [65] (Proposition A.51). We write $T = D^0 \subseteq D$ for the trace-zero part, a nine-dimensional F-line on which $\mathrm{Hg}(X)$ acts by conjugation; $T(-1)$ is a Hodge structure of K3 type with Hodge numbers $(1,7,1)$, and the exceptional classes of $X \times X$ are the real multiplication by F on the copy of $T(-1)$ inside $H^1(X) \otimes H^1(X)$.

For a K3 surface S the transcendental lattice $T(S)$ is the orthogonal complement of $\mathrm{NS}(S)$ in $H^2(S,\mathbb{Z})$; $T(S)_{\mathbb{Q}}$ is an irreducible Hodge structure, and $E = \mathrm{End}_{\mathrm{Hdg}} T(S)_{\mathbb{Q}}$ is a totally real field or a CM field, acting by self-adjoint endomorphisms in the first case [93]. The K3 lattice is $\Lambda = U^3 \oplus E_8(-1)^2$, and $\Lambda_{\mathbb{Q}} = \Lambda \otimes \mathbb{Q}$. For a rational quadratic space W we use the discriminant $d(W) \in \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ and the Hasse invariants $\varepsilon_p(W) = \prod_{i<j}(a_i,a_j)_p$ of a diagonalisation $\langle a_1,\dots,a_k\rangle$, with $(\ ,\ )_p$ the Hilbert symbol; the classification of rational quadratic forms by rank, signature, discriminant and Hasse invariants, and the existence theorem for prescribed local invariants, are Serre's [79, Chap. IV].

Perfect complexes on A are objects of the bounded derived category of coherent sheaves; $\mathrm{ch}(\mathscr{E}) \in H^{\mathrm{ev}}(A)$ is the Chern character, additive on triangles, multiplicative on exterior products, with $\mathrm{ch}(\mathscr{E}[1]) = -\mathrm{ch}(\mathscr{E})$. The semiregularity map $\sigma_{\mathscr{E}} \colon \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) \to \bigoplus_q H^{q+2}(A,\Omega^q)$ is that of Buchweitz and Flenner [10], after Bloch [17]; its injectivity implies that the Hodge locus of $\mathrm{ch}(\mathscr{E})$ coincides, near the given point, with the locus where $\mathscr{E}$ deforms [10, 73].

Throughout, an assertion attributed to a reference is cited and not reproved; everything else is proved in the text. The phrase "very general" means outside a countable union of proper closed analytic subsets of the parameter space.

3. The reduction to abelian varieties

3.1. The statement (F3′) and the class $\mathscr{A}$

Definition 3.1. For a variety X and $p \geq 0$ let $\mathrm{Ab}^p(X) \subseteq H^{2p}(X)$ be the span of the classes $\Gamma_*\gamma$ of degree $2p$, where B runs over the complex abelian varieties, the point included, γ over $\mathrm{Hdg}^*(B)$, and Γ over the algebraic cycles on $B \times X$ with rational coefficients. The statement (F3′) is: $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ for every X and every p.

Each $\Gamma_*\gamma$ is a Hodge class, since cup product with an algebraic class and push-forward along a projection are morphisms of Hodge structures of the right bidegree; so $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$. Taking B a point shows that $\mathrm{Ab}^p(X)$ contains every algebraic class, and taking X abelian, $B = X$ and Γ the diagonal shows that (F3′) holds on every abelian variety.

Proposition 3.2. **HC** *holds if and only if* **AB** *and* (F3′) *both hold.*

Proof. If **HC** holds then **AB** holds, and every Hodge class is algebraic, hence in $\mathrm{Ab}^p(X)$. Conversely let $\gamma \in \mathrm{Hdg}^p(X)$. By (F3′), $\gamma = \sum_j \Gamma_{j*}\gamma_j$ with γ_j a Hodge class on an abelian variety B_j, algebraic by **AB**; then $\mathrm{pr}_{B_j}^*\gamma_j \cup [\Gamma_j]$ is algebraic on $B_j \times X$ and so is its push-forward to X. □

Proposition 3.3. *Let $\mathscr{A}$ be the class of varieties X such that $H^*(X)$ is spanned by classes $\Gamma_* y$ with B abelian, Γ an algebraic cycle on $B \times X$ and $y \in H^*(B)$.*

(i) *Every $X \in \mathscr{A}$ satisfies* (F3′).

(ii) *$\mathscr{A}$ contains the abelian varieties and the curves, and is closed under products and under images of surjective morphisms. It contains every product of curves, every symmetric product of a curve, and every variety dominated by a product of curves or by an abelian variety.*

(iii) *Every X of dimension d satisfies* (F3′) *in the degrees $2p$ with $p \in \{0,1,d-1,d\}$. In particular every variety of dimension at most three satisfies it in every degree.*

Proof. (i) Let $\gamma \in \mathrm{Hdg}^p(X)$ and write $\gamma = \sum_j \Gamma_{j*} y_j$ with $y_j \in H^*(B_j)$. After replacing y_j by its component in the degree k_j that Γ_{j*} sends to $H^{2p}(X)$, each Γ_{j*} is a morphism of rational Hodge structures $H^{k_j}(B_j)(m_j) \to H^{2p}(X)$ for a suitable Tate twist, and the sum is a morphism

$$\Phi \colon W = \bigoplus_j H^{k_j}(B_j)(m_j) \longrightarrow H^{2p}(X), \qquad \Phi\big((y_j)_j\big) = \gamma.$$

A correspondence shifts degrees by an even number, so every k_j is even and W is pure of weight $2p$. Polarisable rational Hodge structures form a semisimple abelian category, so $W = \ker\Phi \oplus W'$

with $\Phi|_{W'}\colon W' \to \operatorname{im}\Phi$ an isomorphism of Hodge structures. The Hodge class γ of $\operatorname{im}\Phi$ is therefore $\Phi(y')$ for a Hodge class $y' = (y'_j)_j$ of $W' \subseteq W$, and each component y'_j is a Hodge class on B_j, since the projections of W onto its summands are morphisms of Hodge structures. Hence $\gamma = \sum_j \Gamma_{j*} y'_j \in \mathrm{Ab}^p(X)$.

(ii) The point, abelian varieties and curves are in $\mathscr{A}$: for a curve C, H^0 and H^2 come from the point and $H^1(C) = H^1(J(C))$ by the Abel–Jacobi map. If $X, Y \in \mathscr{A}$ then $H^*(X \times Y) = H^*(X) \otimes H^*(Y)$ is spanned by exterior products of classes $\Gamma_* y \boxtimes \Gamma'_* y' = (\Gamma \times \Gamma')_*(y \boxtimes y')$ with $B \times B'$ abelian. If $f\colon X \to X'$ is surjective, then $f_* f^* = \deg(f)$ on $H^*(X')$ after cutting X by hyperplanes to the dimension of X' if needed, so every class of X' is f_* of a class of X, and $f_* \Gamma_* y = (\Gamma_f \circ \Gamma)_* y$ with Γ_f the graph of f. The listed varieties follow: a symmetric product is the image of a product of curves, and a variety dominated by a product of curves or an abelian variety is such an image.

(iii) Degrees 0 and $2d$ are spanned by the class of X and the class of a point. In degree 2 every Hodge class is a divisor class by Lefschetz [52]. In degree $2d-2$, let θ be a hyperplane class; the hard Lefschetz theorem makes $\theta^{d-2}\cup\colon H^2(X) \to H^{2d-2}(X)$ an isomorphism of Hodge structures, so every Hodge class of degree $2d-2$ is $\theta^{d-2} \cup \alpha$ with α a Hodge class of degree 2, algebraic by Lefschetz; a product of algebraic classes is algebraic. □

Part (iii) covers every variety of dimension at most three whatever its Hodge numbers, the Calabi–Yau threefolds included, for which the Hodge numbers of the toric hypersurface families are tabulated in [13]; the first open degree is the middle degree of a fourfold.

Proposition 3.4 (Fourfolds with few zero-cycles)**.** *Let X be a smooth projective fourfold whose group of zero-cycles is supported on a threefold: $\mathrm{CH}_0(Z)_{\mathbb{Q}} \to \mathrm{CH}_0(X)_{\mathbb{Q}}$ is surjective for some closed $Z \subset X$ of dimension at most three. Every uniruled fourfold is such: a general point lies on a rational curve, which meets an ample divisor D, so the point is rationally equivalent to a point of D, and the points of a dense open set generate $\mathrm{CH}_0(X)$. Then the Hodge conjecture holds for X in degree four, and X satisfies* (F3′) *in every degree.*

Proof. Bloch and Srinivas prove, by the decomposition of the diagonal that the hypothesis provides, that a fourfold whose zero-cycles are supported on a threefold satisfies the Hodge conjecture for cycles of codimension two [19, Thm. 1]. So $\mathrm{Hdg}^2(X) = \mathrm{Alg}^2(X) \subseteq \mathrm{Ab}^2(X)$, the class of a cycle Z being $\Gamma_* 1$ for the point and $\Gamma = Z$; the other degrees are Proposition 3.3(iii). □

For uniruled fourfolds the conclusion in degree four is the theorem of Conte and Murre [23]; the proposition isolates the hypothesis on zero-cycles behind it.

3.2. **The statement (F2)**

For an abelian variety Y let $R(Y) \subseteq H^*(Y)$ be the $\mathbb{Q}$-subalgebra generated by the divisor classes of Y and by the pull-backs f^*w of Weil classes w of abelian varieties B of Weil type along homomorphisms $f\colon Y \to B$. Every element of $R(Y)$ is a Hodge class, and every Hodge class on an abelian variety of dimension at most five lies in $R(Y)$ [66].

Definition 3.5. The statement (F2) is: for every complex abelian variety Y, every Hodge class on Y that does not lie in $R(Y)$ is algebraic.

The classes in $R(Y)$ are the ones the known theorems reach when they reach anything: divisor classes by Lefschetz, Weil classes by Markman on fourfolds of Weil type of discriminant one [57], and on all abelian fourfolds of Weil type and on abelian sixfolds of split Weil type by Markman's secant-sheaf construction [59, 60], and by the propagation results of Appendix A (Theorem A.7) wherever they propagate. So (F2) is the conjecture for abelian varieties with the known classes struck out. The next proposition says that striking them out removes nothing.

Setting 3.6. Let X be a Mumford fourfold, $V = H^1(X)$ with its polarisation ψ, $G = \mathrm{Hg}(X) \subseteq \mathrm{Sp}(V, \psi)$, and $e_0 \in \mathrm{Hdg}^2(X \times X)$ an exceptional class, a Hodge class of degree four that is not a polynomial in divisor classes. Such classes exist [65];[1] X is simple with $\mathrm{End}^0(X) = \mathbb{Q}$ [63].

[1] Mumford constructed [63] abelian fourfolds X whose Hodge group is a $\mathbb{Q}$-form of SL_2^3 acting on $H^1(X)$, of dimension eight, through the tensor product of the three two-dimensional representations; the form comes from a quaternion algebra

Lemma 3.7. *Let Y be an abelian variety. Let $Y_X \subseteq Y$ be the largest abelian subvariety isogenous to a power of X and Y' the identity component of the intersection of the kernels of the homomorphisms $Y \to X$. Then $Y_X \times Y' \to Y$ is an isogeny,* $\mathrm{Hom}(X,Y') = 0$*, and*

$$H^1(Y) = \big(V \otimes Q\big) \oplus H^1(Y'), \qquad Q = \mathrm{Hom}^0(Y_X, X),$$

the first summand embedded by $v \otimes f \mapsto f^*v$. Let $\mathrm{Sp}(V,\psi)$ act on $H^*(Y) = \bigwedge^* H^1(Y)$ through the factor V. Then every class of $R(Y)$ is invariant under $\mathrm{Sp}(V,\psi)$, and for $Y = X^m$ the invariants of $\mathrm{Sp}(V,\psi)$ in $H^*(X^m)$ are polynomials in divisor classes.

Proof. Poincaré reducibility gives $Y \sim Y_X \times Y'$ with $Y_X \sim X^k$ and no factor X in Y'; since X is simple, $\mathrm{Hom}(X,Y') = \mathrm{Hom}(Y',X) = 0$, and since $\mathrm{End}^0(X) = \mathbb{Q}$, Q has dimension k and the displayed map is an isomorphism. A homomorphism $h\colon Y \to Z$ of abelian varieties decomposes up to isogeny as $h_X \times h'$, and h^* acts on H^1 as $(1 \otimes (f \mapsto f \circ h_X)) \oplus h'^*$, which commutes with the action of $\mathrm{GL}(V)$ on the factor V. So pull-backs along homomorphisms, cup products and exterior products are $\mathrm{GL}(V)$-equivariant, and the classes invariant under $\mathrm{Sp}(V,\psi)$ form a subalgebra stable under pull-back. It remains to check the generators of $R(Y)$.

A divisor class on Y has Künneth components in $\bigwedge^2(V \otimes Q)$, $(V \otimes Q) \otimes H^1(Y')$ and $\bigwedge^2 H^1(Y')$, each a Hodge class. The middle one is a homomorphism from Y' to the dual of Y_X, which is zero. The first is a Hodge class on Y_X, hence a G-invariant of $\bigwedge^2(V \otimes Q) = S^2V \otimes \bigwedge^2 Q \oplus \bigwedge^2 V \otimes S^2Q$. Since G contains the Hodge group of a simple abelian fourfold whose only Hodge classes of degree two are the divisor classes, $(S^2V)^G = 0$ and $(\bigwedge^2 V)^G = \mathbb{Q}\psi$; so the component lies in $\psi \otimes S^2Q$, which $\mathrm{Sp}(V,\psi)$ fixes. The third component is fixed because $\mathrm{GL}(V)$ acts trivially on $H^1(Y')$.

For a Weil class, let B be of Weil type for K and $f\colon Y \to B$. The field K acts on $B \sim B_X \times B'$ through $\mathrm{End}^0(B_X) \times \mathrm{End}^0(B')$, because $\mathrm{Hom}(B_X, B') = 0$; so it acts on $Q_B = \mathrm{Hom}^0(B_X, X)$ and on $H^1(B')$, and the eigenspace of $H^1(B, \bar{\mathbb{Q}})$ for an embedding σ of K is $V \otimes Q_{B,\sigma} \oplus H^1(B', \bar{\mathbb{Q}})_\sigma$. A Weil class lies in the sum over σ of the top exterior powers of these eigenspaces, and $\bigwedge^{\mathrm{top}}(V \otimes Q_{B,\sigma} \oplus W_\sigma) = (\det V)^{\otimes \dim Q_{B,\sigma}} \otimes (\det Q_{B,\sigma})^{\otimes 8} \otimes \det W_\sigma$, on which $\mathrm{SL}(V) \supseteq \mathrm{Sp}(V,\psi)$ acts trivially. Pulling back along f preserves invariance.

For $Y = X^m$, $H^*(X^m) = \bigwedge^*(V \otimes \mathbb{Q}^m)$, and the invariants of $\mathrm{Sp}(V,\psi)$ in the tensor algebra of V are spanned by complete contractions with ψ by the first fundamental theorem for the symplectic group [40, Chap. 5]; a contraction of two factors of X^m is the pull-back of the class $\psi \in H^2(X \times X)$ along the projection to those factors, a divisor class, and a contraction within one factor is $\psi \in H^2(X)$. So the invariants are polynomials in divisor classes. □

Theorem 3.8. (F2) *implies* **AB***, and is therefore equivalent to it.*

Proof. **AB** contains (F2). Assume (F2), let A be an abelian variety and $0 \neq \beta \in \mathrm{Hdg}^p(A)$. If $\beta \notin R(A)$ it is algebraic by (F2), so suppose $\beta \in R(A)$.

The line $\mathbb{Q}e_0$ is not stable under $\mathrm{Sp}(V,\psi)$: a stable line would be a one-dimensional representation of the connected semisimple group $\mathrm{Sp}(V,\psi)$, hence trivial, and e_0, which is not a polynomial in divisor classes, would be invariant, against Lemma 3.7. The stabiliser of the line is therefore a proper Zariski closed subgroup, and since $\mathrm{Sp}(V,\psi)(\mathbb{Q})$ is Zariski dense in $\mathrm{Sp}(V,\psi)$ [18, Cor. 18.3] there is $g \in \mathrm{Sp}(V,\psi)(\mathbb{Q})$ with $ge_0 \notin \mathbb{Q}e_0$.

Put $B = X \times X$, $Y = A \times B$, and let b be an ample class on B. The decomposition of Lemma 3.7 for Y is the sum of those for A and B, so $\mathrm{Sp}(V,\psi)$ acts on $H^*(Y) = H^*(A) \otimes H^*(B)$ by $g(x \otimes y) = gx \otimes gy$, and by the lemma it fixes $R(Y)$ pointwise. If $\beta \otimes e_0 = \mathrm{pr}_A^*\beta \cup \mathrm{pr}_B^* e_0$ were in $R(Y)$, then $g\beta \otimes ge_0 = \beta \otimes e_0$ with $\beta \neq 0$, which forces $ge_0 \in \mathbb{Q}e_0$; so $\beta \otimes e_0 \notin R(Y)$. On the other hand $\beta \otimes b^2 \in R(Y)$, because

over a totally real cubic field F that is split at one real place and ramified at the other two, and the corestriction to $\mathbb{Q}$ of that algebra is the matrix algebra, which is what makes the representation rational. Such an X is simple, has $\mathrm{End}^0(X) = \mathbb{Q}$, is not of CM type, and its Hodge group is smaller than the full symplectic group although no endomorphism accounts for the loss; its Shimura curve is compact. The exceptional classes of [65] live in degree four on $X \times X$ (equivalently in End of H^1 as a Hodge structure of weight zero) and span the invariants of the Hodge group that are not generated by the polarisation, and no one has proved them algebraic. The fourfold enters this paper because it is the smallest variety on which (F2) has an unsettled instance that no CM point reaches directly.

TABLE 2. The rule set. A rule with two premises needs both. The closure of a set of statements under these rules, and the list of minimal sets whose closure contains **HC**, are computed by hand in the text and checked in Lean (Appendix C).

premises	conclusion	proof
HC	(F3), (F2), (F3′), **AB**, (M)	definitions; [6]
(F3)	**HC**	Prop. A.49
(F2)	**AB**	Theorem 3.8
AB	(F2)	definition
AB, (F3′)	**HC**	Proposition 3.2
(L), (M)	**HC**	[6]; Thm. A.50
(L)	**AB**	Thm. A.50
(V)	**AB**	Thm. A.50

the pull-back of a generator of $R(A)$ or $R(B)$ along a projection is a generator of $R(Y)$ and $R(Y)$ is an algebra. Hence also

$$\beta \otimes (e_0 + b^2) = \beta \otimes e_0 + \beta \otimes b^2 \notin R(Y).$$

The classes $\beta \otimes e_0$ and $\beta \otimes (e_0 + b^2)$ are Hodge classes of degree $2p+4$ on Y outside $R(Y)$, so (F2) makes both algebraic, and their difference $\beta \otimes b^2$ is algebraic. Then

$$\mathrm{pr}_{A*}\big((\beta \otimes b^2) \cup \mathrm{pr}_B^* b^6\big) = \Big(\int_B b^8\Big)\beta,$$

a push-forward of a product of algebraic classes, is algebraic, and $\int_B b^8 > 0$ because b is ample on the eightfold B. So β is algebraic. □

Theorem 1.1 is Propositions 3.2, 3.3 and Theorem 3.8 together.

3.3. **The closure graph**

Appendix A.6 records the implications between the conjecture and the statements that have been used to approach it: the Lefschetz standard conjecture (L) for all varieties, the conjecture (M) that every Hodge class is motivated in André's sense [6], the variational Hodge conjecture (V), the statement (F3) that the conjecture holds for all varieties that are not abelian, and the two statements of this paper.[2] The rules are listed in Table 2, each with the place where it is proved; the first two rows are Propositions 3.2 and Theorem 3.8.

Proposition 3.9. *Under the rules of Table 2 the closure of a set of statements contains* **HC** *if and only if the set contains one of*

$$\{\mathrm{F3}\},\ \{\mathrm{L},\mathrm{M}\},\ \{\mathrm{L},\mathrm{F3}'\},\ \{\mathrm{V},\mathrm{F3}'\},\ \{\mathrm{F2},\mathrm{F3}'\},\ \{\mathbf{AB},\mathrm{F3}'\}, \tag{3.1}$$

and these six sets are the minimal ones with this property.

Proof. Each listed set derives **HC**: $\{\mathrm{F3}\}$ by the second row; $\{\mathrm{L},\mathrm{M}\}$ by the sixth; the other four through **AB**, which (F2), (L) and (V) each give, and the fifth row. For the converse, call a set closed if it is stable under the rules. A closed set without **HC** contains no (F3), does not contain (L) and (M) together, and, if it contains (F3′), contains none of **AB**, (F2), (L), (V), since each of these yields **AB** and then **HC**. The maximal closed sets without **HC** are therefore

$$M_1 = \{\mathrm{F2}, \mathbf{AB}, \mathrm{L}, \mathrm{V}\},$$

[2]For a smooth projective variety X of dimension d with hyperplane class h, (L) asks that the inverse of the hard Lefschetz isomorphism $h^{d-2p}\colon H^{2p}(X,\mathbb{Q}) \to H^{2d-2p}(X,\mathbb{Q})$ be induced by an algebraic cycle on $X \times X$ for every p; it is known for abelian varieties, surfaces and the varieties whose cohomology is spanned by algebraic classes, and open for a general threefold. A Hodge class is motivated in André's sense when it is obtained from classes of algebraic cycles by cup products, pull-backs, push-forwards and the inverse Lefschetz operators, on X and on the products $X \times Y$; (M) asks that every Hodge class be of this kind, which is weaker than the conjecture by exactly the use of the inverse Lefschetz operator. (V) asks that a flat Hodge class in a smooth projective family be algebraic on every fibre once it is algebraic on one; Grothendieck observed that (V) follows from the conjecture, and the converse is open. (F3) concerns the varieties that are not abelian and makes no statement about abelian varieties at all; it is the complement, inside the conjecture, of (F2), and the pair (F2), (F3) is the coarsest division of the conjecture that this paper refines.

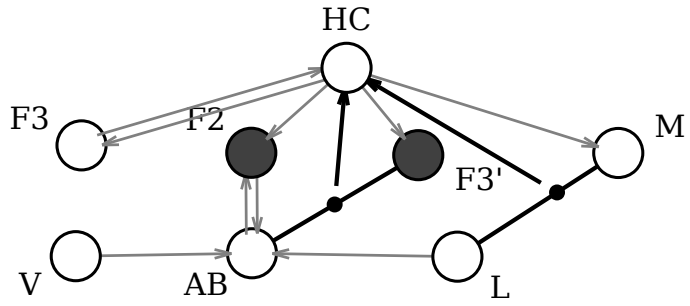


FIGURE 2. The closure graph of Table 2. Grey arrows are rules with one premise; a black fork joins the two premises of a rule that needs both. The filled nodes are (F2) and (F3′).

$$M_2 = \{\mathrm{F2}, \mathbf{AB}, \mathrm{V}, \mathrm{M}\},$$
$$M_3 = \{\mathrm{F3}', \mathrm{M}\},$$

each of which is closed by inspection of the table. The closure of a set Σ contains **HC** exactly when Σ lies in none of the M_i. If Σ contains (F3) this holds. Otherwise Σ needs an element outside M_1, that is (F3′) or (M); one outside M_2, that is (F3′) or (L); and one outside M_3, that is one of (F2), **AB**, (L), (V). The minimal choices are (F3′) with one of (F2), **AB**, (L), (V), and (M) with (L). This gives the six sets, and each is minimal because removing any element lands in some M_i. The same computation is carried out mechanically from the rule list in the Lean file of Appendix C, which returns these six sets and no others. □

Figure 2 draws the graph. The two statements of this paper are the filled nodes; every route to **HC** other than (F3), which is **HC** itself, and (L)+(M) passes through (F3′).

4. THE ALGEBRAIC LOCUS IN A FAMILY

Definition 4.1. Let $\pi\colon \mathscr{X} \to S$ be a smooth projective morphism to a smooth connected complex quasi-projective variety, and let ξ be a global section of $R^{2p}\pi_*\mathbb{Q}$ that is a Hodge class at one point. The algebraic locus of ξ is the set $\Sigma(\xi)$ of $s \in S$ at which $\xi_s \in H^{2p}(\mathscr{X}_s)$ is algebraic.

By the theorem of the fixed part [26, 4], the flat sections of a polarisable variation form a sub-Hodge structure, constant along S; so ξ_s is a Hodge class at every point once it is one somewhere.[3]

Lemma 4.2 (Dichotomy)**.**

[3]The theorem says that for a polarisable variation of Hodge structure $\mathscr{V}$ over a smooth connected quasi-projective base S, the space $H^0(S, \mathscr{V})$ of global flat sections, which is the subspace of a fibre $\mathscr{V}_s$ fixed by the monodromy group, is a sub-Hodge structure of $\mathscr{V}_s$, and the Hodge structure it carries does not depend on s. Deligne proved it for the variations $R^k\pi_*\mathbb{Q}$ of a smooth projective morphism, where it follows from the degeneration of the Leray spectral sequence and the weight filtration of a smooth compactification; André proved it for every polarisable variation by showing that the connected monodromy group is a normal subgroup of the derived group of the generic Mumford–Tate group. In the situation of the text the section ξ is flat, so its value at one point is fixed by the monodromy; if that value has type (p,p) at one point, the sub-Hodge structure of flat sections contains it as a Hodge class, and since that Hodge structure is the same at every point, ξ_s has type (p,p) everywhere.

(i) $\Sigma(\xi)$ *is a countable union of Zariski closed subsets of* S.

(ii) *Either* $\Sigma(\xi) = S$ *or* $\Sigma(\xi)$ *is meagre in* S^{an}.

(iii) *The following are equivalent:* $\Sigma(\xi) = S$; $\Sigma(\xi)$ *has an interior point;* $\Sigma(\xi)$ *is not meagre;* ξ_s *is algebraic for the very general* s. *If* $\Sigma(\xi)$ *is dense they are equivalent to* $\Sigma(\xi)$ *being closed.*

Proof. (i) Fix a relatively ample line bundle on $\mathscr{X}$. For a Hilbert polynomial P the relative Hilbert scheme $\mathrm{Hilb}^P(\mathscr{X}/S)$ is projective over S [38], and the class of the fibre of the universal family in $H^{2p}(\mathscr{X}_s)$ is locally constant on each connected component of it, since a flat family of subschemes over a connected base has locally constant cycle class. A rational combination $\sum_i a_i[Z_i]$ equal to ξ_s is a point of a fibre product $\prod_S \mathrm{Hilb}^{P_i}(\mathscr{X}/S)$, projective over S and with countably many connected components; on each component the condition $\sum a_i[Z_i] = \xi$ is either empty or holds throughout, because both sides are flat sections. So $\Sigma(\xi)$ is the union, over the countably many tuples (P_i, a_i) and components, of images of proper morphisms to S, each Zariski closed.

(ii) A countable union of proper Zariski closed subsets of an irreducible S has empty interior in S^{an}, and each is closed, so the union is meagre; if one of the closed sets is S itself, $\Sigma(\xi) = S$.

(iii) By the Baire category theorem[4] S^{an} is not meagre in itself, so $\Sigma(\xi) = S$ excludes meagreness, and a set with an interior point is not meagre; by (ii) not meagre forces $\Sigma(\xi) = S$. The complement of a countable union of proper closed subvarieties is the very general locus, so algebraicity at the very general point means that one of the closed sets is all of S. A dense set that is also closed is S. □

Lemma 4.3 (Points of CM type are dense)**.** *Let* G *be a connected reductive group over* $\mathbb{Q}$, $u_0\colon U(1) \to G_{\mathbb{R}}$ *a homomorphism and* $\mathscr{D}$ *the orbit of* u_0 *under conjugation by* $G(\mathbb{R})^+$. *The points* $u \in \mathscr{D}$ *for which* $u(U(1))$ *lies in the real points of a maximal torus of* G *defined over* $\mathbb{Q}$ *are dense in* $\mathscr{D}$.

Proof. Let $u_1 \in \mathscr{D}$ and W a neighbourhood of u_1. A maximal torus T_1 of $G_{\mathbb{R}}$ contains $u_1(U(1))$, since the centraliser of that compact torus is reductive and connected. Pick a regular element Y_1 in the Lie algebra $\mathfrak{t}_1$ of T_1. The map $(g, Y) \mapsto \mathrm{Ad}(g)Y$ from $G(\mathbb{R})^+ \times \mathfrak{t}_1$ to $\mathfrak{g}_{\mathbb{R}}$ is submersive at $(1, Y_1)$ because $\mathfrak{g}_{\mathbb{R}} = \mathfrak{t}_1 \oplus [\mathfrak{g}_{\mathbb{R}}, Y_1]$, so the image of a small neighbourhood $N_1 \times N_2$, with $N_1 u_1 N_1^{-1} \subseteq W$ and N_2 made of regular elements, contains an open set of $\mathfrak{g}_{\mathbb{R}}$, hence a rational point $Y = \mathrm{Ad}(g)Y'$. The identity component T of the centraliser of the rational regular semisimple Y is a maximal torus over $\mathbb{Q}$ with $T_{\mathbb{R}} = gT_1g^{-1}$, and $u = gu_1g^{-1} \in W$ has $u(U(1)) \subseteq T_{\mathbb{R}}$. □

When $G = \mathrm{Hg}(A)$ and u is the Hodge cocharacter of an abelian variety A, the condition of the lemma says that $\mathrm{Hg}(A)$ lies in a $\mathbb{Q}$-torus, and then A is of CM type: a maximal $\mathbb{Q}$-torus of $\mathrm{GL}(H^1(A))$ containing it is the unit group of a commutative semisimple algebra L of dimension $2\dim A$ commuting with $\mathrm{Hg}(A)$, and the commutant of $\mathrm{Hg}(A)$ is $\mathrm{End}^0(A)$.

Proposition 4.4 (A family with a CM member through every Hodge class)**.** *Let* A_0 *be an abelian variety and* γ *a Hodge class on it. There are a polarised abelian scheme* $\pi\colon \mathscr{A} \to S$ *over a smooth connected quasi-projective base, a flat section* ξ *of* $R^{2p}\pi_*\mathbb{Q}$, *a point* s_0 *with* $\mathscr{A}_{s_0} \cong A_0$ *and* $\xi_{s_0} = \gamma$, *and a point* c *with* $\mathscr{A}_c$ *of CM type; the points of CM type are dense in* S.

Proof. Let $G = \mathrm{Hg}(A_0)$, let u_0 be the Hodge cocharacter of A_0 and $\mathscr{D}$ its $G(\mathbb{R})^+$-orbit. Every $u \in \mathscr{D}$ defines a polarisable Hodge structure of weight one on $H^1(A_0)$ with Hodge group contained in G, so an abelian variety A_u up to isogeny; fixing the lattice $H^1(A_0, \mathbb{Z})$, the polarisation of A_0 and a level structure of level $N \geq 3$, the quotient $S = \Gamma\backslash\mathscr{D}$ by a neat congruence subgroup $\Gamma \subseteq G(\mathbb{Q})$ is a smooth quasi-projective variety carrying a polarised abelian scheme with these fibres [63, 26]. The class γ is invariant under G, so it is a flat section ξ of $R^{2p}\pi_*\mathbb{Q}$, and $\xi_{s_0} = \gamma$ at the image s_0 of u_0. By Lemma 4.3 the points of CM type are dense in $\mathscr{D}$, hence in S. □

[4] A subset of a topological space is meagre when it is a countable union of nowhere dense sets, that is, of sets whose closure has empty interior. The Baire category theorem states that a complete metric space, or a locally compact Hausdorff space, is not meagre in itself; a connected complex manifold such as S^{an} is locally compact, so the theorem applies to it. A proper Zariski closed subset of an irreducible variety has empty interior in the analytic topology, because a holomorphic function vanishing on an open set vanishes identically on a connected manifold; so a countable union of proper Zariski closed subsets is meagre, and the alternative in (ii) is exactly whether one of the countably many closed strata of (i) is the whole of S.

Theorem 4.5 ((F2) is propagation from CM points)**.** *The following are equivalent.*

(i) (F2).

(ii) **AB**.

(iii) *For every polarised abelian scheme $\pi\colon \mathscr{A} \to S$ and every flat Hodge class ξ, algebraicity of ξ_s at one point s forces $\Sigma(\xi) = S$.*

(iv) *The same for the points s of CM type.*

(v) *For every π, ξ with ξ_c algebraic at a point c of CM type, $\Sigma(\xi)$ has an interior point.*

Proof. (i) and (ii) are Theorem 3.8. If (ii) holds, ξ_t is a Hodge class at every t by the theorem of the fixed part and algebraic by (ii), so (iii) holds; (iv) is a case of (iii), and (iv) and (v) are equivalent by Lemma 4.2(iii).

Assume (iv). Let F be a CM field and $n \geq 2$. The family of abelian varieties of split Weil type for F over its Shimura variety is a polarised abelian scheme on which the Weil classes are flat sections, and by Theorem A.6 they are algebraic at a member isogenous to a product of abelian varieties with complex multiplication by F, a member of CM type; so by (iv) they are algebraic on every member, and by pull-back along an F-linear isogeny on every abelian variety of split Weil type for F. By André's theorem [5], every Hodge class on an abelian variety of CM type is a sum of pull-backs, along homomorphisms, of Weil classes of abelian varieties of split Weil type; so **AB** holds for every abelian variety of CM type. Now let A_0 be any abelian variety and γ a Hodge class on it, and take π, ξ, s_0, c from Proposition 4.4. Then ξ_c is algebraic, (iv) gives $\Sigma(\xi) = S$, and $\gamma = \xi_{s_0}$ is algebraic. This is (ii). □

Proposition 4.6 (The Weil classes)**.** *For every CM field F, every $n \geq 2$ and every discriminant, the algebraic locus of the Weil classes in the polarised period domain $\mathscr{D}_F$ contains an explicit point of CM type and is dense, and it is either $\mathscr{D}_F$ or meagre.*

Proof. This is Theorem A.8, items (i) to (iii). The density is the Hecke orbit of the base point: the group $G_F(\mathbb{Q})$ of the unitary group of the hermitian form acts on $\mathscr{D}_F$ through F-linear isogenies that preserve the polarisation up to a scalar, hence preserve the algebraicity of the Weil classes, and its closure contains $G_F(\mathbb{R})^+$, which acts transitively. The dichotomy is Lemma 4.2. □

Proposition 4.7 ((F3′) and the abelian part)**.** *Let $H^{2p}_{\mathrm{ab}}(X) \subseteq H^{2p}(X)$ be the sum of the images of all maps $\Gamma_*\colon H^k(B) \to H^{2p}(X)$ with B abelian and Γ an algebraic cycle on $B \times X$. Then $H^{2p}_{\mathrm{ab}}(X)$ is a sub-Hodge structure, $\mathrm{Ab}^p(X) = \mathrm{Hdg}^p(X) \cap H^{2p}_{\mathrm{ab}}(X)$, and (F3′) holds for X in degree $2p$ if and only if a Hodge complement $T^{2p}_{\mathrm{ab}}(X)$ of $H^{2p}_{\mathrm{ab}}(X)$ carries no nonzero Hodge class.*

Proof. Each Γ_* is a morphism of Hodge structures, so its image and the sum of finitely many images, which is all of $H^{2p}_{\mathrm{ab}}(X)$ by dimension, are sub-Hodge structures. A Hodge class in $H^{2p}_{\mathrm{ab}}(X)$ is $\sum \Gamma_{j*} y_j$, and the argument of Proposition 3.3(i) replaces each y_j by a Hodge class; so $\mathrm{Ab}^p(X) = \mathrm{Hdg}^p(X) \cap H^{2p}_{\mathrm{ab}}(X)$. A Hodge complement exists because the Hodge group is reductive, and the Hodge classes of $H^{2p}(X)$ are the direct sum of those of the two summands, which gives the last assertion. □

Theorem 1.2 is Lemma 4.2, Theorem 4.5, Proposition 4.6 and Proposition 4.7. Figure 3 draws the locus of Proposition 4.6.

Both statements are now of one shape. There is a dense set of points at which the class is algebraic by an explicit cycle; the locus is a countable union of closed strata; the statement is that this union is not meagre, that is, that algebraicity propagates from the dense set. The rest of the paper goes to the first point where the propagation is not known and builds the object that would carry it.

5. Constructions at (F3′): real multiplication on K3 surfaces

5.1. Where the first class is

Let S be a projective K3 surface, $T = T(S)_{\mathbb{Q}}$ and $E = \mathrm{End}_{\mathrm{Hdg}} T$. The Hodge classes of $S \times S$ are $H^0 \otimes H^4$, $H^4 \otimes H^0$ and $\mathrm{End}_{\mathrm{Hdg}} H^2(S) = \mathrm{End}(\mathrm{NS}_{\mathbb{Q}}) \oplus E$, since T is irreducible with no class of type $(1,1)$ and $\mathrm{Hom}_{\mathrm{Hdg}}(T, \mathrm{NS}_{\mathbb{Q}}) = 0$. The first two pieces and $\mathrm{End}(\mathrm{NS}_{\mathbb{Q}})$ are spanned by products of

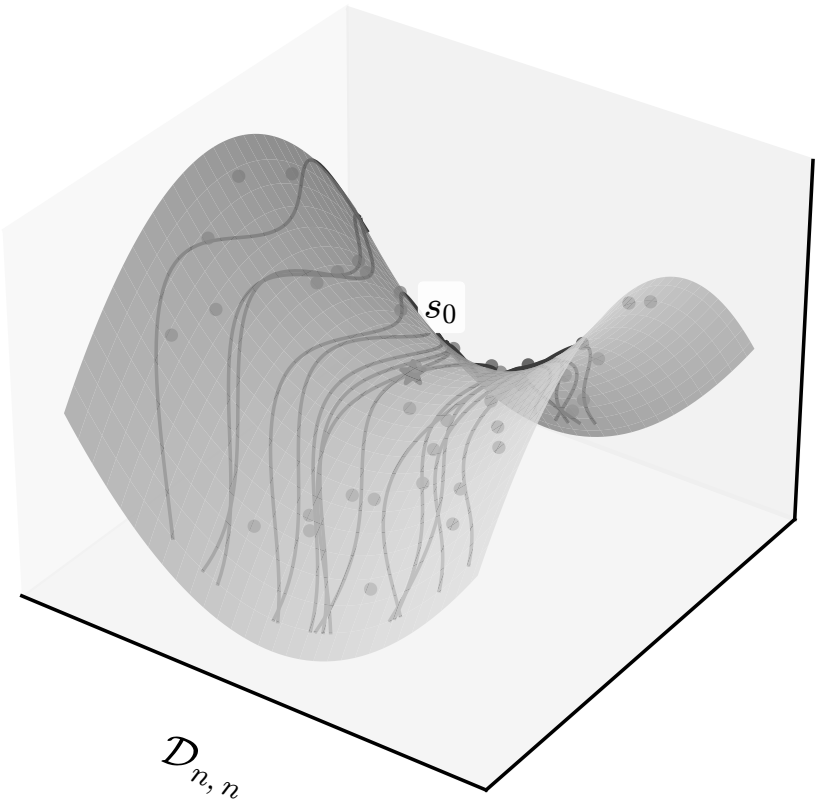


FIGURE 3. The period domain $\mathscr{D}_{n,n}$ of a Weil family as a surface. The star is the CM base point, the dots and curves are strata of the algebraic locus, dense by the Hecke orbit. The locus is a countable union of closed strata; it is either the whole surface or meagre.

divisor classes. So the Hodge conjecture for $S \times S$ is the algebraicity of E, and by Proposition 4.7 the same classes decide (F3′) for $S \times S$ once T is not reached from abelian varieties. When E is a CM field it is spanned by its elements of norm one, which are Hodge isometries of T and algebraic by Buskin's theorem [21]; see [83, Rem. 2.16]. When E is totally real, the Rosati involution is the identity on it [93], every $e \in E$ is self-adjoint for the intersection form q, and $q(ex,ey) = q(e^2x,y)$. The first class is a real multiplication.

5.2. **Chains of correspondences**

Lemma 5.1. *Let $f\colon S \dashrightarrow S'$ be a dominant rational map of K3 surfaces of degree d and Γ_f the closure of its graph. Then $f_* = \Gamma_{f*}$ and $f^* = \Gamma^t_{f*}$ are morphisms of Hodge structures, f^* maps $T(S')_{\mathbb{Q}}$ isomorphically onto $T(S)_{\mathbb{Q}}$ with $q_S(f^*a, f^*b) = d\, q_{S'}(a,b)$, and $f_* = d\,(f^*)^{-1}$ on $T(S)_{\mathbb{Q}}$.*

Proof. The graph is an algebraic cycle, so both maps are induced by algebraic classes and are morphisms of Hodge structures. Pull-back of a nowhere vanishing 2-form is a nonzero 2-form, so $f^*(T(S')_{\mathbb{Q}})$ is a sub-Hodge structure of $H^2(S)$ that meets $H^{2,0}(S)$; the smallest sub-Hodge structure of $H^2(S)$ containing $H^{2,0}$ is $T(S)_{\mathbb{Q}}$, so $f^*T(S')_{\mathbb{Q}} \supseteq T(S)_{\mathbb{Q}}$, and since $T(S')_{\mathbb{Q}}$ is irreducible and f^* is

injective on it ($f_*f^* = d$), equality of dimensions gives $f^*T(S')_{\mathbb{Q}} = T(S)_{\mathbb{Q}}$. The projection formula gives $q_S(f^*a, f^*b) = q_{S'}(f_*f^*a, b) = d\,q_{S'}(a,b)$ and $f_*f^* = d$. □

Lemma 5.2. *A Hodge isometry $\varphi\colon T(S)_{\mathbb{Q}} \to T(S')_{\mathbb{Q}}$ is induced by an algebraic class on $S \times S'$.*

Proof. The spaces $\mathrm{NS}(S)_{\mathbb{Q}}$ and $\mathrm{NS}(S')_{\mathbb{Q}}$ are the orthogonal complements of $T(S)_{\mathbb{Q}}$ and $T(S')_{\mathbb{Q}}$ in the two copies of $\Lambda_{\mathbb{Q}}$, isometric to each other through φ; by Witt's cancellation theorem the complements are isometric, so φ extends to a Hodge isometry $H^2(S) \to H^2(S')$, which is algebraic by Buskin's theorem [21] (see [46, Thm. 0.1]). The projector of $H^2(S)$ onto $T(S)_{\mathbb{Q}}$ along $\mathrm{NS}(S)_{\mathbb{Q}}$ is algebraic, being 1 minus the projector onto NS, which is a sum $\sum_{i,j} c_{ij} D_i \otimes D_j$ over a basis of divisor classes with (c_{ij}) the inverse Gram matrix; composing gives φ on T and zero on NS. □

Theorem 5.3 (Chains)**.** *Let S be a projective K3 surface with E totally real and $e \in E$ with $e^2 = n \in \mathbb{Q}_{>0}$. Suppose there are projective K3 surfaces $S = S_0, S_1, \dots, S_r$ and, for each i, either a dominant rational map $S_{i-1} \dashrightarrow S_i$ of degree n_i, a dominant rational map $S_i \dashrightarrow S_{i-1}$ of degree n_i, or a Hodge isometry $T(S_{i-1})_{\mathbb{Q}} \cong T(S_i)_{\mathbb{Q}}$ (and then $n_i = 1$), such that*

$$\prod_i n_i^{\pm 1} \equiv n \pmod{\mathbb{Q}^{\times 2}},$$

the sign being $+$ for maps leaving S_{i-1} and $-$ for maps arriving at it. Then e is algebraic on $S \times S$.

Proof. Composing along the chain the maps f_* for maps leaving S_{i-1}, f^* for maps arriving, and the classes of Lemma 5.2 for the isometries, gives an algebraic isomorphism $\phi\colon T(S_r)_{\mathbb{Q}} \to T(S)_{\mathbb{Q}}$ of Hodge structures with $q_S(\phi a, \phi b) = \mu\, q_{S_r}(a,b)$, where $\mu = \prod n_i^{\pm 1}$ by Lemma 5.1 (a map leaving contributes $f_* = n_i (f^*)^{-1}$, of multiplier n_i; a map arriving contributes f^*, of multiplier n_i in the other direction, that is n_i^{-1} for the composite in the direction of ϕ). Write $\mu = nc^2$ with $c \in \mathbb{Q}^\times$. Since e is self-adjoint with $e^2 = n$, it is a Hodge similarity of $T(S)_{\mathbb{Q}}$ of multiplier n. So $\psi = (c^{-1}\phi)^{-1} \circ e\colon T(S)_{\mathbb{Q}} \to T(S_r)_{\mathbb{Q}}$ is a Hodge isometry, algebraic by Lemma 5.2, and $e = (c^{-1}\phi) \circ \psi$ is a composite of algebraic correspondences, hence algebraic on $S \times S$. □

The case $r = 1$ with $S \dashrightarrow S_1$ the quotient by a symplectic automorphism of prime order p is Varesco's theorem [83, Thm. 2.1], which gives $e = \sqrt{p}$ and is used for $p = 2, 3$. What a chain adds is a composite multiplier.

5.3. **Real multiplication by $\mathbb{Q}(\sqrt{6})$**

A symplectic automorphism of finite order on a K3 surface acts trivially on $T(S)$, its quotient has a K3 resolution Y [68], and $T(Y)_{\mathbb{Q}} \cong T(S)_{\mathbb{Q}}(1/\mathrm{ord})$ by Lemma 5.1. The invariant lattice of a symplectic involution is $U^3 \oplus E_8(-2)$, and a K3 surface has a symplectic involution exactly when $E_8(-2)$ embeds primitively in its Néron–Severi lattice [62, 87]. The invariant lattice of a symplectic automorphism of order three is $U \oplus U(3)^2 \oplus A_2^2$, and such an automorphism exists exactly when the Coxeter–Todd lattice $K_{12}(-2)$ embeds primitively in the Néron–Severi lattice [39, Thm. 4.1]. Put

$$V_2 = \big(U^3 \oplus E_8(-2)\big)_{\mathbb{Q}}, \qquad V_3 = \big(U \oplus U(3)^2 \oplus A_2^2\big)_{\mathbb{Q}},$$

of ranks 14 and 10 and signatures $(3,11)$ and $(3,7)$; over $\mathbb{Q}$, $U(3) \cong U$ and $E_8(-2) \cong \langle -1\rangle^8 \cong \langle -2\rangle^8$. Varesco [83, Props. 2.5 and 2.11] shows that a projective K3 surface whose transcendental space embeds isometrically in V_2 (resp. V_3) is Hodge isometric to one with a symplectic involution (resp. a symplectic automorphism of order three).

Lemma 5.4 (Embedding criterion)**.** *Let W and V be rational quadratic spaces with $\mathrm{rk}\,V - \mathrm{rk}\,W \geq 3$. Then W embeds isometrically in V if and only if the signature of V dominates that of W in both signs. In that case a complement W' exists with $V \cong W \oplus W'$.*

Proof. Necessity is clear. For sufficiency put $k = \mathrm{rk}\,V - \mathrm{rk}\,W \geq 3$ and let (s_+, s_-) be the difference of signatures, $s_+ + s_- = k$. A quadratic space W' of rank k and signature (s_+, s_-) with prescribed discriminant $d' = d(V)d(W)$ and prescribed Hasse invariants ε'_p exists provided the ε'_p equal 1 for almost all p, $\prod_{p \leq \infty} \varepsilon'_p = 1$, and ε'_∞, d' match the signature [79, Chap. IV, Prop. 7], with no further condition when $k \geq 3$. Take $\varepsilon'_p = \varepsilon_p(V)\varepsilon_p(W)(d(W), d')_p$, the value forced by the formula $\varepsilon_p(W \oplus$

$W') = \varepsilon_p(W)\varepsilon_p(W')(d(W),d(W'))_p$. The product formula for these holds because it holds for V, for W and for the Hilbert symbol. Then $W \oplus W'$ and V have the same rank, signature, discriminant and Hasse invariants at every place, so they are isometric by the Hasse–Minkowski theorem [79, Chap. IV, Thm. 9], and W embeds in V. □

Theorem 5.5 (Two-dimensional families with algebraic real multiplication by $\mathbb{Q}(\sqrt{6})$)**.** *Let S be a projective K3 surface with $\mathbb{Q}(\sqrt{6}) \subseteq E$ whose transcendental space $T = T(S)_{\mathbb{Q}}$ embeds isometrically in V_3.*

(i) *$\sqrt{6} \in E$ is algebraic on $S \times S$.*

(ii) *The Hodge conjecture, and* (F3′)*, hold for $S \times S$ and for the Hilbert square $S^{[2]}$.*

(iii) *Such surfaces exist in two-dimensional families, whose very general member has* $\operatorname{rk} T = 8$ *and $E = \mathbb{Q}(\sqrt{6})$ exactly. One such family is the one whose transcendental space is E^4 with the trace form of $q_E = \langle -2,-2,-2+\sqrt{6},-1+\sqrt{6}\rangle$ over $E = \mathbb{Q}(\sqrt{6})$.*

Proof. (i) By [83, Prop. 2.11] S is Hodge isometric to a K3 surface S' with a symplectic automorphism σ of order three, and by Lemma 5.2 the isometry is algebraic, so we may replace S by S'. Let Y be the minimal resolution of S'/σ and $f_1 \colon S' \dashrightarrow Y$ the quotient map, of degree three, so that $T(Y)_{\mathbb{Q}} \cong T(1/3)$, of rank $\operatorname{rk} T \le 8$ and signature $(2, \operatorname{rk} T - 2)$: the rank is at most 8 because T embeds in V_3 of rank 10 and signature $(3,7)$ with a complement that must contain the remaining positive direction, so the complement has rank at least two. Now $T(Y)_{\mathbb{Q}}$ has rank at most 8 and signature dominated by $(3,11)$, and $14-8 \ge 3$, so by Lemma 5.4 it embeds in V_2. By [83, Prop. 2.5] Y is Hodge isometric to a K3 surface Y' with a symplectic involution ι; let Z be the resolution of Y'/ι and $f_2 \colon Y' \dashrightarrow Z$ the quotient, of degree two. The chain $S' \dashrightarrow Y \cong Y' \dashrightarrow Z$ has multiplier $3 \cdot 1 \cdot 2 = 6$, and Theorem 5.3 applies to $e = \sqrt{6}$ (Fig. 4).

(ii) By the first paragraph of this section it remains to make E algebraic. E contains $\mathbb{Q}(\sqrt{6})$, and either $E = \mathbb{Q}(\sqrt{6})$ or E is a CM field: a totally real E has $\dim_E T \ge 3$, because the complex line $H^{2,0}$ lies in one eigenspace $T_\sigma \otimes \mathbb{C}$ and $\overline{H^{2,0}}$ in the same one, so that $T_\sigma \otimes \mathbb{R}$ has signature $(2,*)$ and $\dim_E T \ge 3$ follows from q being negative definite on the other eigenspaces together with $T \otimes \mathbb{R}$ having exactly two positive directions; and $[E:\mathbb{Q}] \cdot \dim_E T = \operatorname{rk} T \le 8$ then forces $[E:\mathbb{Q}] \le 2$. If $E = \mathbb{Q}(\sqrt{6})$, (i) suffices. If E is CM, it is spanned by its norm-one elements, Hodge isometries, algebraic by Buskin's theorem. So every Hodge class on $S \times S$ is algebraic, and (F3′) follows. For $S^{[2]}$ the Chow motive is a direct summand of twists of $h(S \times S)$, $h(S)$ and $\mathbf{1}$ [25], so every Hodge class on $S^{[2]}$ is the image under an algebraic correspondence of a Hodge class on $S \times S$ or S, and both statements follow.

(iii) Let $E = \mathbb{Q}(\sqrt{6})$ with real embeddings σ_1, σ_2 ($\sqrt{6} \mapsto \pm\sqrt{6}$), and let $q_E = \langle a_1,a_2,a_3,a_4\rangle$ with a_1,a_2 negative at both places and a_3,a_4 positive at σ_1, negative at σ_2. The $\mathbb{Q}$-space $T = E^4$ with the form $\operatorname{Tr}_{E/\mathbb{Q}} \circ q_E$ has rank 8 and signature $(2,2)+(0,4) = (2,6)$, and multiplication by $\sqrt{6}$ is self-adjoint of square 6. For $q_E = \langle -2,-2,-2+\sqrt{6},-1+\sqrt{6}\rangle$ one computes $d(T) = 10$, $\varepsilon_2 = 1$, $\varepsilon_3 = -1$, $\varepsilon_5 = 1$ and $\varepsilon_p = 1$ otherwise, while V_3 has $d = -1$, $\varepsilon_2 = 1$, $\varepsilon_3 = -1$, $\varepsilon_5 = 1$; and $T \oplus \langle 1,-10\rangle$ has the invariants of V_3 at every place, so $V_3 \cong T \oplus \langle 1,-10\rangle$ by Hasse–Minkowski, with a complement containing a positive vector (Table 3; the Hilbert symbols are recomputed in the programs of Appendix C). Embed V_3 in $\Lambda_{\mathbb{Q}}$ as the invariant lattice of an order-three symplectic action, with orthogonal complement $K_{12}(-2)$, and let $N \subseteq \Lambda$ be the primitive sublattice with $N_{\mathbb{Q}} = T^{\perp}$; it contains $K_{12}(-2)$ primitively and a positive class. The Hodge structures of K3 type on T for which $\sqrt{6}$ is a Hodge endomorphism are the periods $\omega \in \mathbb{P}(T_{\sigma_1} \otimes \mathbb{C})$ with $q(\omega) = 0$, $q(\omega,\bar\omega) > 0$, where T_{σ_1} is the σ_1-eigenspace, of signature $(2,2)$; this is a domain of dimension two [83, Prop. 2.3]. For each such ω the surjectivity of the period map gives a K3 surface S_ω with $T(S_\omega)_{\mathbb{Q}} \subseteq T$, projective because N contains a positive class, carrying an order-three symplectic automorphism because $K_{12}(-2) \subseteq N$ primitively. For ω outside a countable union of proper closed analytic subsets, $T(S_\omega)_{\mathbb{Q}} = T$ and $E = \mathbb{Q}(\sqrt{6})$, by the argument of Theorem A.60, which needs only $\dim_E T \ge 3$. □

In Varesco's families E is $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{3})$ [83, Thms. 2.9, 2.15]; van Geemen and Schütt obtain $\sqrt{2}$, $\sqrt{3}$, $\sqrt{7}$ from rational self-maps of elliptic K3 surfaces [88, §§6–7]. The lattice condition of

TABLE 3. Rational invariants used in Theorem 5.5(iii). Hasse invariants are $+1$ at every prime not shown.

space	rank	signature	d	ε_2	ε_3	ε_5
$T = \mathrm{Tr}_{E/\mathbb{Q}} q_E$	8	$(2,6)$	10	$+1$	-1	$+1$
$\langle 1,-10\rangle$	2	$(1,1)$	-10	$+1$	$+1$	$+1$
$T \oplus \langle 1,-10\rangle$	10	$(3,7)$	-1	$+1$	-1	$+1$
V_3	10	$(3,7)$	-1	$+1$	-1	$+1$
V_2	14	$(3,11)$	-1	-1	$+1$	$+1$

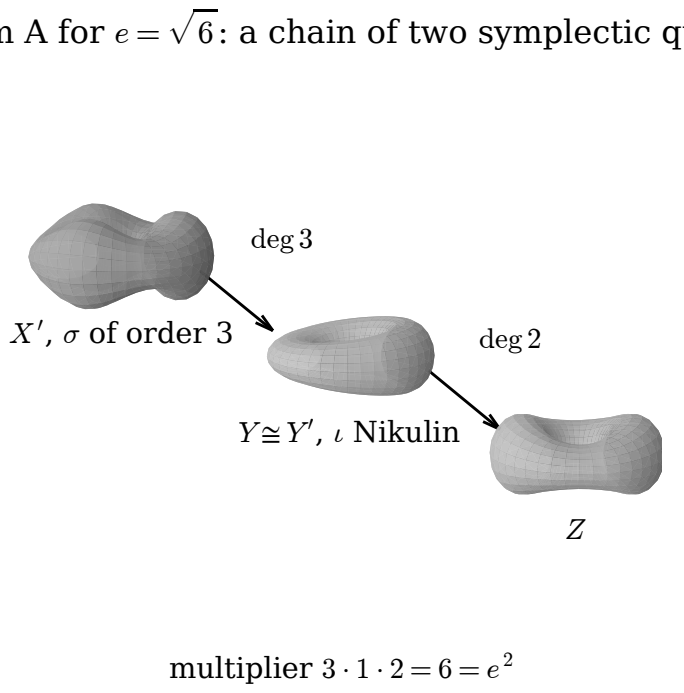


FIGURE 4. The chain of Theorem 5.5: the quotient of S' by a symplectic automorphism of order three, a Hodge isometry (algebraic by Buskin), and the quotient of Y' by a Nikulin involution. The multipliers compose to 6.

Theorem 5.5 is a genuine restriction: among the 165 sign-admissible diagonal forms q_E with small coefficients, 98 embed in V_3 and 67 do not (Appendix C).

5.4. **Isogenies of elliptic surfaces**

Varesco's quotients and the chains of Theorem 5.3 produce the square roots of the orders of symplectic automorphisms, 2 and 3, and their products. A different source of similarities is a rational self-map of a K3 surface, and the self-maps that exist in families come from isogenies of elliptic fibrations: van Geemen and Schütt obtain $\sqrt{2}$ and $\sqrt{3}$ from isogenies of degree 2 and 3 in this way, and for degrees 5 and 7 they give one surface each [88, §§6–7]. We show that the degrees 5 and 7 come in one-dimensional families, that in the families the isogeny is a complex multiplication by $\mathbb{Q}(\sqrt{-5})$ or $\mathbb{Q}(\sqrt{-7})$ and not a real one, the sign of its square being decided by how the twisting function behaves under the Fricke involution, and that the families satisfy the Hodge conjecture on all their powers.

Let $N \in \{5,7\}$ and write N^k for 125 when $N = 5$ and 49 when $N = 7$. The modular curve $X_0(N)$ is rational, with Fricke's coordinate t in which the involution $(E,C) \mapsto (E/C, E[N]/C)$ is $w(t) = N^k/t$

[88, §7.6].[5] The universal curve with a cyclic subgroup C_t of order N over $\mathbb{P}^1_t$ is

$$\mathscr{E}_t\colon\ y^2 = x^3 - 27c_4(t)\,x - 54c_6(t), \tag{5.1}$$

with, for $N=7$,

$$c_4 = (t^2+13t+49)(t^2+5t+1),$$
$$c_6 = (t^2+13t+49)(t^4+14t^3+63t^2+70t-7),$$

and for $N=5$,

$$c_4 = (t^2+10t+5)(t^2+22t+125),$$
$$c_6 = (t^2+4t-1)(t^2+22t+125)^2.$$

The discriminant is $1728\,t\,(t^2+13t+49)^2$ for $N=7$ and $1728\,t\,(t^2+22t+125)^3$ for $N=5$, of degree 5 and 7. So the relatively minimal model R_N of (5.1) over $\mathbb{P}^1_t$ is a rational elliptic surface with fibres of type I_1 at $t=0$, I_N at $t=\infty$, and at the two roots e_1, e_2 of the quadratic factor, where c_4 vanishes, of type II for $N=7$ and III for $N=5$; the Euler numbers add up to $1+N+2\cdot 2=12$ and $1+N+2\cdot 3=12$. The roots satisfy $e_1e_2 = N^k$, so w exchanges them and exchanges 0 with ∞.

Lemma 5.6 (The Fricke isogeny over the function field)**.** *$\mathscr{E}_t/C_t \cong \mathscr{E}_{w(t)}$ over $\mathbb{C}(t)$.*

Proof. The two curves have the same j-invariant, because w is the Fricke involution of $X_0(N)$. Two elliptic curves over $\mathbb{C}(t)$ with the same non-constant j-invariant differ by a quadratic twist $\chi \in \mathbb{C}(t)^\times/\mathbb{C}(t)^{\times 2}$, and a twist ramified at a point of $\mathbb{P}^1$ changes the Kodaira type of the fibre there, exchanging I_n with I_n^*, II with IV* and III with III*. So it suffices to show that $\mathscr{E}_t/C_t$ has the same fibre types as $\mathscr{E}_{w(t)}$, which are I_N at 0, I_1 at ∞ and II or III at e_1, e_2: then χ is unramified on $\mathbb{P}^1$, hence constant, hence trivial over $\mathbb{C}$.

At $t=0$ the fibre of $\mathscr{E}$ is I_1, the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{C}((t))$, whose only cyclic subgroup of order N rational over $\mathbb{C}((t))$ is μ_N; the quotient $\mathbb{G}_m/q^{N\mathbb{Z}}$ is I_N. At e_i the local monodromy $M \in \mathrm{SL}_2(\mathbb{Z})$ of $\mathscr{E}$ has order 6 or 4 and irreducible characteristic polynomial; the isogeny is given on the period lattices by a matrix $A \in M_2(\mathbb{Z})$ with $\det A = N$, and the monodromy of the quotient is $M' = AMA^{-1}$. The $\mathrm{SL}_2(\mathbb{Z})$-classes with this characteristic polynomial are those of M and M^{-1}, which are the types II and II*, or III and III*. If $M' = gM^{-1}g^{-1}$ with $g \in \mathrm{SL}_2(\mathbb{Z})$, then $B = g^{-1}A$ conjugates M to M^{-1}. All rational matrices doing so form the coset $B_0\,\mathbb{Q}(M)^\times$, where $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ for $M = \left(\begin{smallmatrix} 0 & -1 \\ 1 & \mathrm{tr}M \end{smallmatrix}\right)$ has determinant -1 and the elements of the imaginary quadratic field $\mathbb{Q}(M)$ have positive norm; so $\det B < 0$, against $\det B = N$. Hence the quotient has the type of M at e_1, e_2. At ∞ the j-invariant of the quotient has a simple pole, so the fibre is I_1 or I_1^*, and the Euler numbers of the other fibres add up to 11; since the total is 12 the fibre is I_1. □

For $p \neq q$ in $\mathbb{P}^1_t$ let $u_{p,q} = (t-p)(t-q)$, and let $X_{p,q}$ be the elliptic K3 surface whose generic fibre is the quadratic twist $\mathscr{E}_t \otimes u_{p,q}$: the twist multiplies c_4 by u^2 and c_6 by u^3, giving Weierstrass coefficients of degrees 8 and 12, and adds two fibres of type I_0^* at p and q when these are not among $0, \infty, e_1, e_2$, by Tate's algorithm; the same bookkeeping of Weierstrass coefficients, fibre types and their resolutions in higher dimension is carried out in [14]. For a double cover $\psi\colon \mathbb{P}^1_s \to \mathbb{P}^1_t$ let X_ψ be the relatively minimal model of $\psi^*\mathscr{E}$.

Lemma 5.7.

(i) *$X_{p,q}$ has Picard number at least 16: its Néron–Severi lattice contains $U \oplus A_6 \oplus D_4^2$ for $N=7$ and $U \oplus A_4 \oplus A_1^2 \oplus D_4^2$ for $N=5$. The same bound holds for X_ψ, whose fibres $2\mathrm{I}_1 + 2\mathrm{I}_N$, together*

[5] The curve $X_0(N)$ parametrises elliptic curves with a cyclic subgroup of order N; it has genus zero exactly for $N \le 10$ and $N \in \{12, 13, 16, 18, 25\}$, and for $N=5$ and $N=7$ a coordinate is the Hauptmodul $t = \big(\eta(\tau)/\eta(N\tau)\big)^k$ with $k = 12/(N-1)$, so $k=3$ for $N=5$ and $k=2$ for $N=7$; this is why N^k is 125 in one case and 49 in the other. The Fricke involution sends τ to $-1/N\tau$, which exchanges the two cusps 0 and ∞ of $X_0(N)$ and, on the moduli interpretation, replaces (E, C) by the quotient E/C with the image of $E[N]$ as its distinguished subgroup. Applying it twice gives $E/E[N] \cong E$, so it is an involution, and in the coordinate t it reads $t \mapsto N^k/t$ because $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ turns the quotient of eta functions into its reciprocal up to the constant $N^{k/2}$, squared to N^k by the exponent.

with four fibres of type III *when* $N = 5$*, and the pulled-back Mordell–Weil group of* R_N*, of rank* 2*, give rank at least* 16 *by the Shioda–Tate formula.*

(ii) *The fibrewise isogeny by* C_t *is a dominant rational map of degree* N

$$f\colon X_{p,q} \dashrightarrow X_{w(p),w(q)}, \qquad f\colon X_\psi \dashrightarrow X_{w\circ\psi}.$$

(iii) *The deck involution of* ψ *times* $[-1]$ *on the fibres is a symplectic involution* ι *of* X_ψ*, and the minimal resolution of* X_ψ/ι *is* $X_{p,q}$ *with* $\{p,q\}$ *the branch locus of* ψ*. Hence* $T(X_\psi)_{\mathbb{Q}} \cong T(X_{p,q})_{\mathbb{Q}}(1/2)$ *as in Lemma 5.1, and the two surfaces have the same Picard number.*

Proof. (i) Shioda–Tate with the fibres listed, and for X_ψ the Mordell–Weil rank $8-6=2$ of R_N. (ii) By Lemma 5.6 the quotient of the generic fibre is $\mathscr{E}_{w(t)} \otimes u_{p,q}(t)$, and

$$u_{p,q}(w(t)) = \Big(\frac{N^k}{t} - p\Big)\Big(\frac{N^k}{t} - q\Big) = \frac{pq}{t^2}\, u_{w(p),w(q)}(t), \tag{5.2}$$

a constant times a square times $u_{w(p),w(q)}$. So $\mathscr{E}_{w(t)} \otimes u_{p,q}(t) \cong \mathscr{E}_{w(t)} \otimes u_{w(p),w(q)}(w(t))$, which is the pull-back along the automorphism w of the base of the generic fibre of $X_{w(p),w(q)}$; pulling back along an automorphism of the base gives an isomorphic surface. For X_ψ, $(\psi^*\mathscr{E})/C = \psi^*(\mathscr{E}/C) \cong \psi^* w^* \mathscr{E}$. (iii) The quadratic twist by the function defining the double cover is the quotient of the base change by the deck involution composed with $[-1]$ on the fibres; the involution is symplectic with eight fixed points, the 2-torsion of the two fibres over the ramification points, and the quotient's resolution is the twist. □

Lemma 5.8 (The sign of the Fricke isogeny)**.** *Let* $\varphi_t\colon \mathscr{E}_t \to \mathscr{E}_{w(t)}$ *be an isogeny over* $\mathbb{C}(t)$ *with kernel* C_t*, which exists by Lemma 5.6 and is unique up to sign, and let* $\varphi_t^*(dx/y) = \lambda(t)\,dx/y$ *in the models* (5.1)*.*

(i) $\varphi_{w(t)} \circ \varphi_t = [-N]$ *on* $\mathscr{E}_t$*, and* $\lambda(t)\lambda(w(t)) = -N$*.*

(ii) *Let* $u \in \mathbb{C}(t)$ *satisfy* $u(w(t)) = c(t)^2 u(t)$ *with* $c \in \mathbb{C}(t)$*, let* X *be the elliptic K3 surface with generic fibre* $\mathscr{E}_t \otimes u$*, and let* $f\colon X \dashrightarrow X$ *be the fibrewise isogeny by* C_t *followed by the isomorphism* $(x,y) \mapsto (c^2x, c^3y)$ *from* $\mathscr{E}_{w(t)} \otimes u(t)$ *to* $\mathscr{E}_{w(t)} \otimes u(w(t))$*, the fibre of* X *over* $w(t)$*. Then* f^* *acts on* $H^{2,0}(X)$ *by a constant* Λ *with*

$$\Lambda^2 = -\frac{N}{c(t)\,c(w(t))}.$$

For $u = (t-p)(t-N^k/p)$ *one has* $c(t)c(w(t)) = 1$ *and* $\Lambda^2 = -N$*; for* $u = t^2 - N^k$ *one has* $c(t)c(w(t)) = -1$ *and* $\Lambda^2 = N$*.*

Proof. (i) The composite is an endomorphism of $\mathscr{E}_t$ over $\mathbb{C}(t)$ of degree N^2. Since j is not constant, $\mathrm{End}_{\mathbb{C}(t)}(\mathscr{E}_t) = \mathbb{Z}$, so the composite is $[N]$ or $[-N]$, and changing φ to $-\varphi$ does not change it. We read off the sign on the upper half plane $\mathfrak{H}$. Let $\tau \mapsto t(\tau)$ be the quotient map $\mathfrak{H} \to X_0(N)$. By the moduli interpretation of $X_0(N)$ the pair $(\mathscr{E}_{t(\tau)}, C_{t(\tau)})$ is $(\mathbb{C}/(\mathbb{Z}\tau+\mathbb{Z}), \langle 1/N\rangle)$, analytically in τ, and the Fricke involution lifts to $\tau \mapsto w(\tau) = -1/N\tau$. The map $f_\tau(z) = z/\tau$ carries $\mathbb{Z}\tau+\mathbb{Z}$ into $\mathbb{Z}w(\tau)+\mathbb{Z} = \mathbb{Z}/N\tau+\mathbb{Z}$, so it is an isogeny $\mathbb{C}/(\mathbb{Z}\tau+\mathbb{Z}) \to \mathbb{C}/(\mathbb{Z}w(\tau)+\mathbb{Z})$, and its kernel $\tau(\mathbb{Z}w(\tau)+\mathbb{Z})/(\mathbb{Z}\tau+\mathbb{Z}) = (\mathbb{Z}/N+\mathbb{Z}\tau)/(\mathbb{Z}\tau+\mathbb{Z})$ is $\langle 1/N\rangle$. Its composite with the map of the same form at $w(\tau)$ is $f_{w(\tau)}(f_\tau(z)) = z/\tau w(\tau) = -Nz$. On the set $\mathfrak{H}'$ of τ with $j(\tau) \notin \{0, 1728\}$ the automorphisms of $\mathscr{E}_\tau$ are ± 1, so $\varphi_{t(\tau)} = \varepsilon(\tau) f_\tau$ with $\varepsilon(\tau) \in \{\pm 1\}$; ε is continuous because φ is, hence constant on $\mathfrak{H}'$, which is $\mathfrak{H}$ minus a discrete set and connected. So $\varphi_{w(t)} \circ \varphi_t = \varepsilon^2 f_{w(\tau)} \circ f_\tau = [-N]$ on $\mathfrak{H}'$, hence identically. Pulling back dx/y along the composite gives $\lambda(t)\lambda(w(t))\,dx/y$ on one side and $-N\,dx/y$ on the other.

(ii) Over $\mathbb{C}(t)(\sqrt{u})$ the twist by u is the same change of coordinates $(x,y) \mapsto (ux, u^{3/2}y)$ on $\mathscr{E}_t$ and on $\mathscr{E}_{w(t)}$, which multiplies dx/y by $u^{-1/2}$ on both, so the twisted isogeny pulls dx/y back to $\lambda(t)\,dx/y$. The isomorphism $(x,y) \mapsto (c^2x, c^3y)$ pulls dx/y back to $c(t)^{-1}dx/y$, and the holomorphic two-form of X, which is $dx/y \wedge dt$ on the Weierstrass model, restricts to the fibre over $w(t)$ as $dx/y \wedge d(w(t)) = w'(t)\,dx/y \wedge dt$. Hence $f^*(dx/y \wedge dt) = \Lambda(t)\,dx/y \wedge dt$ with $\Lambda(t) = w'(t)\lambda(t)/c(t)$. The pull-back of a holomorphic two-form along a dominant rational map of K3 surfaces is holomorphic off the finitely many points of indeterminacy, hence everywhere, so f^* preserves the line

$H^{2,0}(X)$ and Λ is a constant. Then $\Lambda^2=\Lambda(t)\Lambda(w(t))=w'(t)w'(w(t))\,\lambda(t)\lambda(w(t))/c(t)c(w(t))$, where $w'(t)w'(w(t))=(w\circ w)'(t)=1$ and $\lambda(t)\lambda(w(t))=-N$ by (i). For $u=(t-p)(t-N^k/p)$ equation (5.2) with $q=N^k/p$ gives $u(w(t))=(N^k/t^2)\,u(t)$, so $c(t)=N^{k/2}/t$ for a fixed square root $N^{k/2}$, and $c(t)c(w(t))=N^k/t\,w(t)=1$. For $u=t^2-N^k$, $u(w(t))=N^{2k}/t^2-N^k=-(N^k/t^2)\,u(t)$, so $c(t)=iN^{k/2}/t$ and $c(t)c(w(t))=-1$. □

The archive [8] computes λ itself: the kernel polynomial of φ_t is the factor of degree $(N-1)/2$ of the N-th division polynomial over $\mathbb{Z}[t,x]$, Vélu's formulas give the quotient curve, and the isomorphism to (5.1) at $w(t)$ has $\lambda(t)^2=-t^2/25$ for $N=5$ and $-t^2/7$ for $N=7$, so that $\lambda(t)^2\lambda(w(t))^2=N^2$ and $\Lambda^2=-N$ on the pair twists directly; the program `julia/fricke_sign.jl` certifies the kernel polynomial (it divides the division polynomial, and it and its cofactor are irreducible modulo a prime) and carries out the rest, and the Lean file checks the polynomial identity it reduces to.

Theorem 5.9 (Fricke families)**.** *Let $N\in\{5,7\}$ and $K=\mathbb{Q}(\sqrt{-N})$. Let $\mathfrak{A}_N$ be the family of surfaces $X_p=X_{p,N^k/p}$ for $p\notin\{0,\infty,e_1,e_2,\pm\sqrt{N^k}\}$, and $\mathfrak{B}_N$ the family of surfaces X_ψ with ψ branched over a pair $\{b,N^k/b\}$.*

(i) *Every member X has a dominant rational self-map f of degree N with $f^*\in E=\mathrm{End}_{\mathrm{Hdg}}T(X)_{\mathbb{Q}}$, $f^*f_*=N$, $(f^*)^2=-N$ and $f_*=-f^*$. So $K\subseteq E$, with $\sqrt{-N}=\pm f^*$ algebraic on $X\times X$, and E is a CM field.*

(ii) *The families are not isotrivial, and every member has $\mathrm{rk}\,T(X)\in\{2,4,6\}$. At all but countably many members $E=K$ and the Hodge group of $T(X)$ is the unitary group of the K-hermitian form h on $T(X)$ with $\mathrm{Tr}_{K/\mathbb{Q}}h=q_X$; at the others $T(X)$ is of CM type.*

(iii) *For every member X and every $n\geq 1$ the Hodge conjecture holds for X^n and for the Hilbert scheme $X^{[n]}$, and* (F3′) *holds there.*

Van Geemen and Schütt state that the Fricke parametrisations give one-dimensional families with complex multiplication by $\mathbb{Q}(\sqrt{-5})$ and $\mathbb{Q}(\sqrt{-7})$, and give the details for the two surfaces at the fixed points of w only [88, §7.6]; (i) and (ii) make their statement explicit, and (iii) is what the complex multiplication gives.

Proof. (i) w exchanges p and N^k/p, so Lemma 5.7(ii) gives $f\colon X_p\dashrightarrow X_p$, the map of Lemma 5.8(ii) for $u=(t-p)(t-N^k/p)$; for $\mathfrak{B}_N$, $w\circ\psi$ has the same branch locus as ψ, so $w\circ\psi=\psi\circ\sigma$ for a Möbius transformation σ and $X_{w\circ\psi}\cong X_\psi$. The correspondence of a dominant rational map preserves NS and its transpose does too, so it preserves $T=\mathrm{NS}^\perp$ and acts there as a morphism of Hodge structures, nonzero since $f^*\omega=\Lambda\omega$ with $\Lambda\neq 0$; on a resolution S of the graph with $p_1\colon S\to X$ birational and p_2 of degree N, both p_1^*T and p_2^*T are irreducible sub-Hodge structures of $H^2(S)$ containing its unique $(2,0)$-line, so they coincide, and $f_*f^*=p_{2*}p_1^*p_{1*}p_2^*=N$ on T. The field E acts faithfully on the line $H^{2,0}(X)$, and f^* acts there by Λ with $\Lambda^2=-N$ by Lemma 5.8(ii); so $(f^*)^2=-N$ in E, and $f_*=N(f^*)^{-1}=-f^*$. For X_ψ the generic fibre is $\mathscr{E}_{\psi(s)}$ over $\mathbb{C}(s)$, f is the fibrewise isogeny over σ, and σ is an involution: in the coordinate s with $s^2=(t-b)/(t-N^k/b)$ one computes $(w(t)-b)/(w(t)-N^k/b)=b^2/N^ks^2$, so $\sigma(s)=\kappa/s$ with $\kappa^2=b^2/N^k$; the argument of Lemma 5.8(ii) with $c=1$ and σ in place of w gives $\Lambda^2=\lambda(t)\lambda(w(t))\,\sigma'(s)\sigma'(\sigma(s))=-N$. The adjoint of f^* for q_X is $f_*=-f^*$, so the involution of E induced by q_X is not the identity; by Zarhin's theorem E is totally real or a CM field [93, Thm. 1.5.1], so it is a CM field, and $K=\mathbb{Q}+\mathbb{Q}f^*\subseteq E$. The class of the graph of f is algebraic, its action on $H^2(X)$ is f^* on T and an endomorphism of $\mathrm{NS}(X)_{\mathbb{Q}}$, which is a sum of algebraic classes of $\mathrm{NS}\otimes\mathrm{NS}$; so f^*, extended by zero on NS, is an algebraic class on $X\times X$.

(ii) The elliptic fibration of X_p has labelled singular fibres I_1 at 0, I_N at ∞, II or III at e_1,e_2 and I_0^* at p and N^k/p. An isomorphism of elliptic fibrations induces a Möbius transformation preserving this configuration; the transformations preserving $\{0,\infty\}$ and $\{e_1,e_2\}$ are $t\mapsto\lambda t$ and $t\mapsto\lambda/t$ with $\lambda e_1\in\{e_1,e_2\}$, and since $e_1+e_2\neq 0$ and $e_1^2\neq e_2^2$ they are the identity and w, which preserve $\{p,N^k/p\}$. So X_p and $X_{p'}$ have isomorphic fibrations only when $p'\in\{p,N^k/p\}$. A K3 surface has finitely many elliptic fibrations up to automorphisms [81], so the map $p\mapsto[X_p]$ to the moduli space has finite fibres and the family is not isotrivial; by local Torelli its period map has uncountable image. By (i) $T(X)_{\mathbb{Q}}$ is a K-vector space, so its rank is even, and it is at most 6 by Lemma 5.7(i). It is also an E-vector

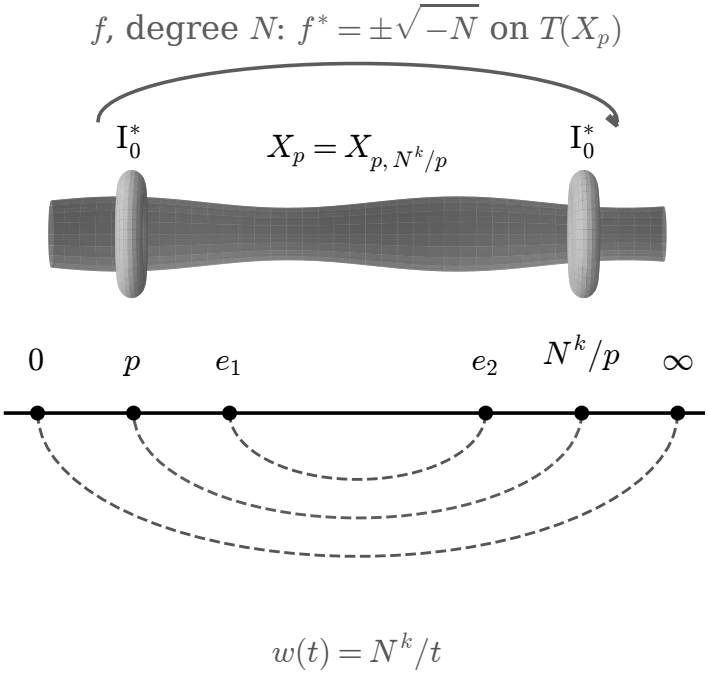


FIGURE 5. The family $\mathfrak{A}_N$ of Theorem 5.9. Over the Fricke coordinate t of $X_0(N)$ the involution $w(t) = N^k/t$ exchanges the cusps $0, \infty$ and the elliptic points e_1, e_2; the twist X_p at a pair $\{p, N^k/p\}$ exchanged by w is carried to itself by the fibrewise isogeny f of degree N, and f^* is the complex multiplication by $\sqrt{-N}$ on the transcendental lattice.

space, so $[E:\mathbb{Q}]$ divides $\operatorname{rk} T(X)$; if $E \neq K$ then $[E:\mathbb{Q}] \in \{4,6\}$, $\dim_E T(X) = 1$ and $T(X)$ is of CM type. A Hodge structure of CM type on a sublattice of $M^\perp$ is determined by its endomorphism field, a subfield of the endomorphism algebra of the sublattice, and by an eigenline of that field, so the periods of such members form a countable subset of the period domain of $M^\perp$; the period map of the family is a non-constant holomorphic map from a connected curve, its fibres over a countable set are countable, so all but countably many members have $E = K$. There the Hodge group of $T(X)$ is the unitary group of the K-hermitian form h with $\operatorname{Tr}_{K/\mathbb{Q}} h = q_X$ by Zarhin's theorem for CM fields [93, Thm. 2.3.1]. The same holds for $\mathfrak{B}_N$ by Lemma 5.7(iii).

(iii) By (i) E is a CM field containing K. If $E = K$ then $E = \mathbb{Q} + \mathbb{Q}f^*$ is algebraic. Otherwise every element of E of norm one to its maximal real subfield is a Hodge isometry of T, extended by the identity on NS to a Hodge isometry of $H^2(X,\mathbb{Q})$, algebraic by Buskin's theorem [21]. The algebraic classes form a $\mathbb{Q}$-subalgebra $E' \subseteq E$ containing these elements; for $x \in E$ with $\bar{x} \neq x$ and $t_1 \neq t_2$ in $\mathbb{Q}$ it contains $(t_i + x)/(t_i + \bar{x})$, hence their quotient $1 + (t_2 - t_1)/(t_1 + \bar{x})$, hence $\bar{x}$; so $E' = E$. In both cases $\operatorname{End}_{\mathrm{Hdg}} H^2(X) = \operatorname{End}(\mathrm{NS}) \oplus E$ is algebraic, and $\operatorname{Hdg}^2(X \times X) = \operatorname{End}_{\mathrm{Hdg}} H^2(X) \oplus H^0 \otimes H^4 \oplus H^4 \otimes H^0$ since $H^1 = H^3 = 0$, so the Hodge conjecture holds for $X \times X$. For a K3 surface whose endomorphism field is a CM field the Hodge classes in the tensor algebra of T are generated by those of $T \otimes T$, and the Hodge conjecture holds for every power [84, Thm. 2.2, Cor. 2.3], [75, Prop. 5.2, Thm. 5.4]. It passes to the Hilbert schemes $X^{[n]}$, whose motives are direct summands of motives of powers of X [25], and (F3′) on a variety follows from the Hodge conjecture on it. □

The surface of [88, §7.6] with real multiplication by $\sqrt{7}$ is the twist at $\{7,-7\}$, the two fixed points of w, and the surface of [88, Prop. 7.7] is the twist at the fixed points $\pm 5\sqrt{5}$ for $N = 5$, which has Picard number 18 and complex multiplication as well: there $u = t^2 - N^k$, $c(t)c(w(t)) = -1$ and $\Lambda^2 = N$ by

Lemma 5.8(ii), so f^* is the real multiplication by $\sqrt{N}$. The families $\mathfrak{A}_N$ are the twists at the pairs exchanged by w, where $c(t)c(w(t))=1$ and $\Lambda^2=-N$; the two signs are the two values of the constant by which the twisting function changes under w once the square factor N^k/t^2 is removed. The chains of Theorem 5.3 apply to every member, and a member whose field E contained $\sqrt{2}$, $\sqrt{3}$ or $\sqrt{6}$ would have that element algebraic as well; by Theorem 5.9(ii) such a member is of CM type, and we know none.

Remark 5.10 (The Picard number)**.** Theorem 5.9 does not say which of the three ranks the very general member has. The archive decides it by counting points. The surface X_1, the twist at $\{1,N^k\}$, has a Weierstrass model over $\mathbb{Z}$ with good reduction at $\ell=11$ for $N=7$ and at $\ell=7$ for $N=5$, primes split in K; the program `julia/fricke_picard.jl` counts its points over $\mathbb{F}_{\ell^i}$, $i\le 3$, fibre by fibre, subtracts the trace of Frobenius on the trivial lattice, and completes the characteristic polynomial of Frobenius on the rank-six complement of the trivial lattice by the functional equation, whose sign is fixed by the nonzero middle coefficient. The polynomials are $x^6-726x^3+11^6$ for $N=7$ and $x^6+6x^5+56x^4+392x^3+2744x^2+14406x+7^6$ for $N=5$; in the variable x/ℓ neither has a root of unity among its roots, each is irreducible over $\mathbb{Q}$ and splits over K into a cubic and its conjugate, as the action of f^* requires. By the Tate conjecture for K3 surfaces over finite fields, used in the same way in [88, §7], $\rho(X_1)=16$, so the very general member of $\mathfrak{A}_N$ and of $\mathfrak{B}_N$ has $\rho=16$, $\operatorname{rk}T=6$, $E=K$ with $\dim_K T=3$, and its Hodge group is the unitary group of a hermitian form of signature $(1,2)$ on K^3. At the primes inert in K, $\ell=5$ for $N=7$ and $\ell=11$ for $N=5$, the odd traces vanish, as they must when Frobenius exchanges the two eigenspaces of f^*. No statement of this section rests on this computation.

5.5. **The reach of similarities**

Every correspondence used so far acts on transcendental spaces as a Hodge similarity. The next proposition says that this is not an accident of the examples but a property of every correspondence whose components are small, and it measures what a correspondence must look like to act otherwise.

Proposition 5.11 (Structure of the transcendental action)**.** *Let X, X' be projective K3 surfaces, $\Gamma\in \mathrm{CH}^2(X\times X')_{\mathbb{Q}}$, and $\gamma_T\colon T(X)_{\mathbb{Q}}\to T(X')_{\mathbb{Q}}$ the map induced on transcendental spaces by composing the action of Γ with the orthogonal projections.*

(i) *γ_T is zero or an isomorphism of Hodge structures.*

(ii) *If $\gamma_T\neq 0$ then $a=\gamma_T^{\dagger}\gamma_T$ lies in the maximal totally real subfield of $E=\mathrm{End}_{\mathrm{Hdg}}T(X)_{\mathbb{Q}}$ and is totally positive, and γ_T is a similarity exactly when $a\in\mathbb{Q}$.*

(iii) *Let Γ be irreducible, S a resolution of its normalisation with projections p_1,p_2 of degrees d_1,d_2 to X, X'. If $p_1^*T(X)_{\mathbb{Q}}=p_2^*T(X')_{\mathbb{Q}}$ inside $H^2(S,\mathbb{Q})$, then γ_T is a similarity of multiplier d_1d_2. This holds whenever $p_g(S)\le 1$.*

(iv) *If Γ is irreducible and γ_T is not a similarity, then $p_1^*T(X)_{\mathbb{Q}}\cap p_2^*T(X')_{\mathbb{Q}}=0$, so $p_g(S)\ge 2$ and the transcendental rank of S is at least twice that of X.*

(v) *The $\mathbb{Q}$-algebra generated by the actions on $T(X)_{\mathbb{Q}}$ of all correspondences from X to itself whose irreducible components either fail to dominate a factor or have $p_g\le 1$ is the multiquadratic subfield $E_2\subseteq E$ generated by the square roots of the products d_1d_2 that occur. When $[E:\mathbb{Q}]$ is odd it is $\mathbb{Q}$.*

Proof. (i) $T(X)_{\mathbb{Q}}$ is an irreducible Hodge structure [93], and the image of γ_T is a sub-Hodge structure of $H^2(X')$; if nonzero it contains the $(2,0)$-part, hence $T(X')_{\mathbb{Q}}$, and γ_T is injective by irreducibility. (ii) a is a Hodge endomorphism of $T(X)$, so $a\in E$, and it is self-adjoint, so it lies in the fixed field of the adjoint involution, the maximal totally real subfield. The form $q'(\gamma x,\gamma y)=q(x,ay)$ is a polarisation of the Hodge structure $T(X)$ up to sign on each real eigenspace of E, and comparison with q gives $\sigma(a)>0$ at every real place. A similarity of multiplier μ has $a=\mu$. (iii) Both p_i^*T are irreducible sub-Hodge structures of $H^2(S)$ with a $(2,0)$-part; if $p_g(S)\le 1$ they share the unique $(2,0)$-line, so they meet and coincide. When they coincide, $\gamma_T=p_{2*}p_1^*=d_2(p_2^*)^{-1}p_1^*$ on T, so $p_2^*\gamma x=d_2\,p_1^*x$; since $q_S(p_1^*x,p_1^*y)=d_1q(x,y)$ and $q_S(p_2^*x',p_2^*y')=d_2q'(x',y')$, this gives $d_2\,q'(\gamma x,\gamma y)=q_S(p_2^*\gamma x,p_2^*\gamma y)=d_2^2\,q_S(p_1^*x,p_1^*y)=d_1d_2^2\,q(x,y)$, that is $q'(\gamma x,\gamma y)=d_1d_2\,q(x,y)$. (iv) If the two subspaces met they would

coincide by irreducibility and (iii) would apply; so they are independent, their $(2,0)$-lines are distinct, and their sum has rank $2\operatorname{rk}T(X)$ inside the transcendental part of $H^2(S)$. (v) A component that does not dominate a factor acts through NS or through H^0 and H^4, hence by zero on T. The others act by similarities by (iii), a composite of similarities returning to X is an element $\Phi \in E$ with $\Phi^\dagger\Phi \in \mathbb{Q}$, and for totally real E this is $\Phi^2 \in \mathbb{Q}$; such elements generate E_2. For a CM field E the elements with $\Phi^\dagger\Phi \in \mathbb{Q}$ are the $\mathbb{Q}^\times$-multiples of the norm-one elements, which generate E, so the statement is about the totally real case, which is the one that matters for (F3′). An odd-degree field has $E_2 = \mathbb{Q}$. □

The correspondences of Lemmas 5.1 and 5.2, the Fourier–Mukai kernels between a K3 surface and its moduli spaces of sheaves, the rational Hodge isometries of hyper-Kähler varieties of $\mathrm{K3}^{[n]}$ type [58], the Fricke isogenies of Theorem 5.9 and all their composites fall under (iii) or (v). A correspondence that realises an element of a cubic field must therefore have a component whose resolution has $p_g \geq 2$ and transcendental rank at least twice that of the surface. Such correspondences exist, and the only ones we know come from finite groups.

Proposition 5.12 (Hecke correspondences)**.** *Let Z be a smooth projective surface with an action of a finite group G, $H \leq G$ a subgroup, and X a smooth projective surface with $p_g(X) = 1$ birational to Z/H; write $\pi\colon Z \dashrightarrow X$.*

(i) *For $g \in G$ the correspondence $\Gamma_g = (\pi \times \pi)_*[\text{graph of } g]$ acts on $H^2(X,\mathbb{Q})$ as the Hecke operator of the double coset HgH, and these operators span a $\mathbb{Q}$-algebra isomorphic to a quotient of $\mathrm{End}_G\mathbb{Q}[G/H]$ acting by Hodge endomorphisms.*

(ii) *Exactly one simple factor of this algebra acts nontrivially on $H^{2,0}(X)$; it is a field, it preserves $T(X)_\mathbb{Q}$, and it embeds into E by algebraic correspondences. If $H^{2,0}(Z)^G = 0$ this factor is not the one on which G acts trivially.*

(iii) *Every field obtained in this way is abelian over $\mathbb{Q}$. In particular a totally real cubic field F whose Galois closure has group S_3 is realised by no correspondence built from a finite group acting on a cover of X.*

Proof. (i) Let $\tilde{Z} \to Z$ be an equivariant resolution and $\pi\colon \tilde{Z} \to X$ the quotient by H followed by the resolution; then $\pi_*\pi^* = |H|$ and $\pi_* h^* = \pi_*$ for $h \in H$, so Γ_g acts as $\pi_* g^* \pi^* = (\pi^*)^{-1} \circ \sum_{h \in HgH/H} h^* \circ \pi^*$ on $H^2(X,\mathbb{Q}) = H^2(\tilde{Z},\mathbb{Q})^H$, the Hecke operator of HgH. The Hecke operators of (G,H) span $\mathrm{End}_G\mathbb{Q}[G/H]$, a semisimple algebra, and the action on $H^2(\tilde{Z},\mathbb{Q})^H$ factors through it. (ii) The idempotents of the simple factors are rational combinations of Hecke operators, so they are Hodge endomorphisms of $H^2(X)$; they sum to 1 and act on the line $H^{2,0}(X)$ by 0 or 1, so exactly one of them, e_V, acts as 1. A simple factor with nonzero image in the commutative ring $\mathrm{End}_{\mathrm{Hdg}}H^{2,0}(X) = \mathbb{C}$ through a ring homomorphism to a field is commutative, hence a field $\mathscr{H}_V$. For $f \in \mathscr{H}_V$, $f(T(X)_\mathbb{Q})$ is a sub-Hodge structure of the transcendental part of $H^2(X)$; if $f\omega \neq 0$ it contains ω, hence $T(X)_\mathbb{Q}$, and if $f\omega = 0$ it has no $(2,0)$-part and lies in $\mathrm{NS} \cap T = 0$. So $\mathscr{H}_V \to \mathrm{End}_{\mathrm{Hdg}}T(X)_\mathbb{Q} = E$ is a homomorphism from a field, injective. The factor on which G acts trivially fixes $H^{2,0}(\tilde{Z})^G = H^{2,0}(Z/G)$, which is zero by hypothesis, so it is not $\mathscr{H}_V$. (iii) The centres of the simple factors of $\mathrm{End}_G\mathbb{Q}[G/H]$ are the character fields $\mathbb{Q}(\chi_i)$ of the irreducible constituents of $\mathbb{Q}[G/H]$, generated by sums of roots of unity, so each lies in $\mathbb{Q}(\zeta_{|G|})$ and is abelian; a commutative simple factor is its centre. A non-abelian cubic field is not a subfield of an abelian field. □

For the dihedral group D_n with n odd, $H = \langle\tau\rangle$ of order two and Z the elliptic surfaces $\tilde{\mathscr{E}}_a$ of [88, §4.7], the algebra $\mathrm{End}_{D_n}\mathbb{Q}[D_n/H]$ is $\mathbb{Q} \times \prod_{d|n,\,d>1}\mathbb{Q}(\zeta_d)^+$, and the factor acting on ω is $\mathbb{Q}(\zeta_n)^+$; this is the content of [88, §4.8], where the cycle $\Gamma_1 + \Gamma_{-1}$ induces $\zeta_n + \zeta_n^{-1}$. The components of these correspondences are birational to $\tilde{\mathscr{E}}_a$, which has $p_g \geq 2$, as Proposition 5.11(iv) requires.

Corollary 5.13 (Squares of the dihedral families)**.** *Let X be an elliptic K3 surface of one of the families of* [88, Thm. 1.1]*, with real multiplication by $F = \mathbb{Q}(\zeta_n)^+$ for $n \in \{5,7,9,11\}$, and suppose $E = F$ or E is a CM field; the very general member has $E = F$. Then the Hodge conjecture holds for $X \times X$, and* (F3′) *holds for $X \times X$ and $X^{[2]}$. This covers the cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ on families of dimensions* 3 *and* 2*.*

Real multiplication by $\mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$: moving the surface of Mumford type into the dihedral family

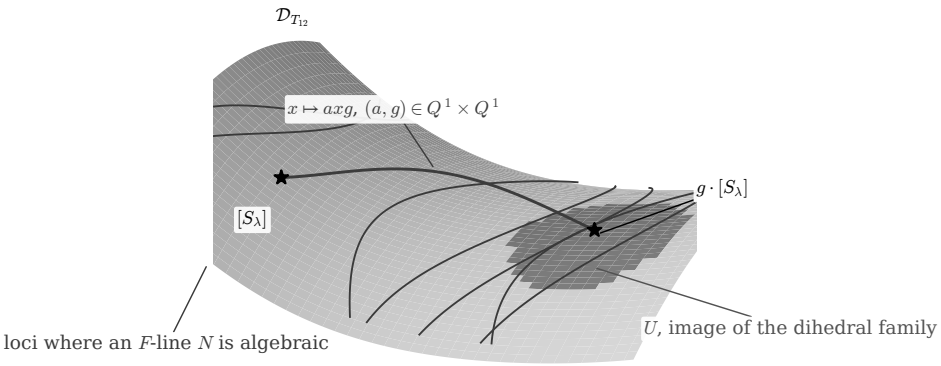


FIGURE 6. The proof of Theorem 5.14. The surface is the two-dimensional period domain of the twelve-dimensional space T_{12} with real multiplication by $F = \mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$; the shaded patch is the image of the two-dimensional dihedral family, on which the multiplication is algebraic. The curves are the loci on which an F-line N becomes algebraic, and the surface of Mumford type is a point on one of them. The rational isometries $x \mapsto axg$ of the quaternion algebra are dense in the real group, so one of them moves the point into the patch; Buskin's theorem turns that isometry into a cycle.

Proof. If $E = F$, E is algebraic by Proposition 5.12(ii) and [88, §4.8]; if E is CM, by Buskin's theorem as in Theorem 5.9(iii). The rest is as there. □

5.6. **The cyclic cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$**

The dihedral families reach the cyclic cubic fields only on their own members. For each of the two cubic fields they contain, $F = \mathbb{Q}(\zeta_n)^+$ with $n \in \{7,9\}$, there is a two-dimensional family on a transcendental lattice of rank twelve: for $n = 9$ the family of [88, §5.6], and for $n = 7$ the Dickson deformation of the one-parameter family of Artebani, Sarti and Taki with Picard lattice $U \oplus E_8$ [7, Ex. 6.1], which [88] do not use, their family for $n = 7$ being the three-dimensional one of Picard number four. Two is the largest dimension a family with this real multiplication on a transcendental space of rank twelve can have, and this maximality lets the cycle travel: every K3 surface of Mumford type for F and for the quaternion algebra ramified only at infinity is a Hodge isometric image of a member of the family, and Buskin's theorem carries the cycle along the isometry. The result settles the frontier case of Section 5.9 for these two fields and this algebra.

Write $\theta = \zeta_n + \zeta_n^{-1}$, so that $F = \mathbb{Q}(\theta)$, $K = \mathbb{Q}(\zeta_n) = F(\sqrt{-\delta})$ with $-\delta = (\zeta_n - \zeta_n^{-1})^2 = \theta^2 - 4$, and $\delta = 4 - \theta^2$ is totally positive. For $n = 7$ one has $\theta^3 + \theta^2 - 2\theta - 1 = 0$ and the three real embeddings send θ to -1.802, -0.445 and 1.247; for $n = 9$ one has $\theta^3 - 3\theta + 1 = 0$ and the embeddings send θ to -1.879, 0.347 and 1.532. Let p_n be the prime dividing n; it is totally ramified in K, so it lies under a single place of F, and $K/\mathbb{Q}$ is unramified outside p_n. The prime 2 has order three in $(\mathbb{Z}/n)^\times/\{\pm 1\} = \mathrm{Gal}(F/\mathbb{Q})$, so it is inert in F. Let D_∞ be the quaternion algebra over F ramified exactly at two real places and at no finite place. Put $\varepsilon = \theta$ for $n = 7$ and $\varepsilon = -\theta$ for $n = 9$; then ε is a unit positive at exactly one real place, and $D_\infty = (-1, \varepsilon)_F$: the symbol is ramified at the two real places where $\varepsilon < 0$, split at every odd finite place because both entries are units there, and split at the one place above 2 by the parity of the number of ramified places. Its corestriction to $\mathbb{Q}$ is unramified everywhere and split at infinity, hence is $M_8(\mathbb{Q})$, so D_∞ is one of Mumford's algebras and the fourfolds of [63] exist for it. For $\mu \in F$ totally positive let q_μ be the quadratic form $\mathrm{Tr}_{F/\mathbb{Q}}(\mu\,\mathrm{trd}(y^2))$ on the nine-dimensional

$\mathbb{Q}$-space D^0_∞ of elements of reduced trace zero, of signature $(2,7)$; the surfaces S_λ of Theorem A.55 for the fourfolds of (F, D_∞) have $T(S_\lambda)_\mathbb{Q} \cong (D^0_\infty, q_\lambda)$.

Theorem 5.14 (Real multiplication by $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ on the surfaces of Mumford type)**.** *Let $n \in \{7,9\}$, $F = \mathbb{Q}(\zeta_n)^+$ and D_∞ as above. Let S be a projective K3 surface such that $T(S)_\mathbb{Q}$ with its intersection form is isometric to (D^0_∞, q_μ) for some totally positive $\mu \in F$, by an isometry under which the multiplication by F on D^0_∞ acts on $T(S)_\mathbb{Q}$ by Hodge endomorphisms. Then $E = F$, the real multiplication by F is induced by algebraic cycles on $S \times S$, and the Hodge conjecture and* (F3′) *hold for $S \times S$ and $S^{[2]}$. This applies to every surface S_λ of a Mumford fourfold of (F, D_∞), at every point of its Shimura curve.*

Proof. Step 1, the family for $n = 9$. Let $\mathfrak{X} \to B$ be the two-dimensional family of [88, §5.6]: elliptic K3 surfaces with very general Picard lattice $U \oplus A_2^4$,[6] deformations of the surfaces $y^2 = x^3 + bx + c_1 t^9 + c_0$ with a non-symplectic automorphism σ of order nine, obtained by replacing t^9 by the Dickson polynomial $p_{9,a}(t)$. The transcendental lattice T_0 of a very general member has rank twelve and discriminant group $(\mathbb{Z}/3)^4$, is a $\mathbb{Z}[\zeta_9]$-module of rank two through σ^*, and carries the flat action of F through $\sigma^* + \sigma^{*-1}$, which on every member of the family is induced by the cycle $\Gamma_1 + \Gamma_{-1}$ of [88, §4.8] on a dihedral cover, as in Proposition 5.12. The period map of B has two-dimensional image [88, §5.6].

Step 1, the family for $n = 7$. The surfaces $X_b\colon\ y^2 = x^3 + bx + c_1 t^7 + c_0$ are the K3 surfaces of [7, Ex. 6.1] with $a = 0$ there: $\sigma(x,y,t) = (x, y, \zeta_7 t)$ is a non-symplectic automorphism of order seven, the fibration has a fibre of type II^* at $t = \infty$ and fourteen fibres of type I_1, and for very general b the Picard lattice is the invariant lattice $U \oplus E_8$ of σ, with $T_0 = U^2 \oplus E_8(-1)$ [7, Table 3]. Replacing t^7 by the Dickson polynomial $p_{7,a}(t) = t^7 - 7at^5 + 14a^2t^3 - 7a^3t$ gives the surfaces[7]

$$\mathfrak{X}_a\colon\ y^2 = x^3 + bx + c_1 p_{7,a}(t) + c_0, \tag{5.3}$$

where $p_{7,a}$ is the Dickson polynomial with $p_{7,a}(v + av^{-1}) = v^7 + a^7 v^{-7}$;[8] these are the deformations (4.4) of [88, Prop. 4.6] with $n = 7$, $\alpha = b$ and $\beta(u) = c_1 u + c_0$; by that proposition each $\mathfrak{X}_a$ is a K3 surface on whose transcendental space F acts by Hodge endomorphisms, induced by the cycle $\Gamma_1 + \Gamma_{-1}$ on the D_7-cover as before. The polynomial $c_1 p_{7,a} + c_0$ has degree seven, so at $t = \infty$ the coefficients of the Weierstrass equation vanish to orders 8 and 5 and the fibre of type II^* survives; hence $\mathrm{Pic}(\mathfrak{X}_a)$ contains the unimodular lattice $U \oplus E_8$ of rank ten, and since the Picard number of a very general member of a family is at most that of a very general member of any subfamily, here the locus $a = 0$, it equals $U \oplus E_8(-1)$, with $T_0 = U^2 \oplus E_8(-1)$, unimodular of rank twelve. The parameters (b, c_1, c_0, a) modulo the scalings $(x,y,t) \mapsto (\lambda^2 x, \lambda^3 y, \mu t)$, which send a to a/μ^2, leave two, and the image of the period map is two-dimensional: otherwise it is the closure of the one-dimensional image of the locus $a = 0$, and a very general $\mathfrak{X}_a$ is isomorphic by the Torelli theorem to some $X_{b'}$, so carries a non-symplectic automorphism σ' of order seven. Such a σ' has no invariant vector in $T_0 \otimes \mathbb{Q}$, so its invariant lattice is $\mathrm{Pic}(\mathfrak{X}_a)$; it fixes the class of a fibre and the class of the zero section O, hence preserves the fibration and O, the only effective curve in its class, and it acts on the base with order seven, since an automorphism over the identity of the base that fixes O is a group automorphism of the fibres, of order dividing six. It fixes $t = \infty$, the only reducible fibre, so it acts by $t \mapsto \zeta t + \beta$ with ζ a primitive seventh root of unity, and since the Weierstrass equation of a fibration with a chosen section is unique up to $(x,y) \mapsto (\lambda^2 x, \lambda^3 y)$, the polynomial $c = c_1 p_{7,a} + c_0$ satisfies $c(\zeta t + \beta) = \lambda^6 c(t)$. The

[6] Families of K3 surfaces polarised by a fixed lattice also occur as the fibres of Calabi–Yau threefolds, the quartic and quintic ones being studied in [12]; the chains of this section act on the fibres only, and the Hodge classes of such a total space are not considered here.

[7] The Dickson polynomial $p_{n,a}$ is the unique polynomial of degree n with $p_{n,a}(v + a/v) = v^n + (a/v)^n$; for $a \neq 0$ the map $t \mapsto p_{n,a}(t)$ from $\mathbb{P}^1$ to itself is a cover of degree n with monodromy group the dihedral group D_n, totally ramified over ∞ [88, Lem. 4.3], since it is the quotient of $v \mapsto v^n + (a/v)^n$, on which D_n acts by $v \mapsto \zeta_n v$ and $v \mapsto a/v$, by the involution. For $a = 0$ it is the cyclic cover $t \mapsto t^n$. The identity $p_{n,a}(\mu t) = \mu^n p_{n,a/\mu^2}(t)$ follows from the defining equation with v replaced by μv.

[8] Explicitly $p_{7,a}(t) = t^7 - 7at^5 + 14a^2t^3 - 7a^3t$, the case $n = 7$ of $p_{n,a}(t) = \sum_{k \le n/2} \frac{n}{n-k}\binom{n-k}{k}(-a)^k t^{n-2k}$; for $a = 0$ it is t^7, so the family $\mathfrak{X}_a$ contains the surfaces X_b.

coefficients of t^7, t^6 and t^5 give $\lambda^6=1$, $\beta=0$ and $7ac_1(\zeta^5-1)=0$, so $a=0$. Thus the direction a is not isotrivial and the image is two-dimensional. If $E\supsetneq F$ then E has degree six or twelve and $\dim_E T_0\otimes\mathbb{Q}\le 2$, which allows at most one modulus by the count of [88, §5.4]; so $E=F$ on the very general member.

In both cases the family is maximal: $\dim_F T_{12}=4$ for $T_{12}=T_0\otimes\mathbb{Q}$, the period domain of Hodge structures of K3 type on T_{12} with F acting by Hodge endomorphisms has dimension $4-2=2$, and the period map has two-dimensional image in its quotient by the arithmetic group, so by the open mapping theorem the image contains a nonempty open set U of one of the two components of that domain.

Step 2, the quadratic structure. Since the intersection form q of T_{12} is σ-invariant there is a unique hermitian form h on the two-dimensional K-space T_{12} with $q=\mathrm{Tr}_{K/\mathbb{Q}}h$, and a unique F-bilinear form $\Phi=\mathrm{Tr}_{K/F}h$ with $q=\mathrm{Tr}_{F/\mathbb{Q}}\Phi$. Diagonalise $h=\langle h_1,h_2\rangle$ with $h_i\in F^\times$. For $x=u+v\sqrt{-\delta}$ one has $\mathrm{Tr}_{K/F}(h_1x\bar{x})=2h_1(u^2+\delta v^2)$, so

$$\Phi\cong 2h_1\langle 1,\delta,h,h\delta\rangle,\qquad h=h_2/h_1,\tag{5.4}$$

and $\langle 1,\delta,h,h\delta\rangle$ is the reduced norm form of the quaternion algebra $Q=(-\delta,-h_1h_2)_F$; thus $(T_{12},\Phi)\cong(Q,2h_1\mathrm{nrd})$ as F-quadratic spaces. The signature of q is $(2,10)$ and splits over the three real places of F as $(2,2)$ at the place σ_1 whose eigenspace carries $H^{2,0}$ and $(0,4)$ at the two others. From (5.4), signature $(2,2)$ at σ_1 forces $h_1h_2<0$ there, and $(0,4)$ at σ_2,σ_3 forces $h_1,h_2<0$ there; so Q is split at σ_1 and ramified at σ_2 and σ_3, and after exchanging h_1 and h_2 we may take $h_1>0$ at σ_1 and $h_1<0$ at the other two places.

Step 3, $Q=D_\infty$. The lattice T_0 is unimodular at every prime $p\neq p_n$, and $K/\mathbb{Q}$ is ramified only at p_n, so for a place v of F not above p_n the ring $\mathcal{O}_K\otimes\mathbb{Z}_p$ is a product of unramified extensions whose trace forms are perfect, and the hermitian $\mathcal{O}_{K,v}$-lattice $T_0\otimes\mathbb{Z}_p$ is unimodular for h; its determinant h_1h_2 is therefore a unit of F_v times a norm from K_v. Units are norms from an unramified extension, so $(-\delta,-h_1h_2)_v=1$ at every finite $v\nmid p_n$, including the place above 2. The ramification set of Q has even cardinality, contains σ_2,σ_3 and is otherwise contained in the single place above p_n; so that place is unramified as well, and $Q\cong D_\infty$. As a check independent of this argument, the rational invariants of $T_0\otimes\mathbb{Q}$, computed from the Picard lattice by Witt cancellation in the K3 lattice, and those of $\mathrm{Tr}_{F/\mathbb{Q}}(\varepsilon\,\mathrm{nrd}_{D_\infty})$ agree for both n: signature $(2,10)$, discriminant 1, Hasse invariant -1 at 2 and $+1$ at every odd prime, while the trace forms of the algebras over F with the same real ramification and ramification at finite places differ from them at 7 or 43 for $n=7$ and at 3 or 19 for $n=9$.

Step 4, the surfaces of Mumford type inside T_{12}. For $n_0\in Q^\times$ the F-line $N=Fn_0$ has $\Phi(n_0,n_0)=2h_1\mathrm{nrd}(n_0)$, and its orthogonal complement is n_0Q^0, with $\Phi(n_0y,n_0y')=-h_1\mathrm{nrd}(n_0)\,\mathrm{trd}(yy')$ for $y,y'\in Q^0$; so $(N^\perp,\Phi)\cong(D_\infty^0,\mu\,\mathrm{trd})$ with $\mu=-h_1\mathrm{nrd}(n_0)$, and N is negative definite exactly when μ is totally positive. By the Hasse–Schilling norm theorem [74, Thm. 33.15] the reduced norms of $Q^\times$ are the elements of $F^\times$ positive at σ_2 and σ_3, and $-h_1$ is negative at σ_1 and positive at the other two places, so every totally positive μ occurs. Given S as in the statement, choose n_0 with $\mu=-h_1\mathrm{nrd}(n_0)$ and an F-linear isometry $T(S)_\mathbb{Q}\cong n_0^\perp$; the period ω_0 of S becomes a point of the period domain of T_{12} at which Fn_0 is algebraic and F acts by Hodge endomorphisms, because ω_0 lies in the σ_1-eigenspace. Replacing S by its complex conjugate, whose period is $\bar{\omega}_0$ and for which the conclusion is equivalent, we may take ω_0 on the component containing U.

Step 5, moving the point. The group $\mathrm{SO}(\Phi)(\mathbb{R})$ at the place σ_1 is $(\mathrm{SL}_2(\mathbb{R})\times\mathrm{SL}_2(\mathbb{R}))/\{\pm1\}$ acting on $Q\otimes_{F,\sigma_1}\mathbb{R}\cong M_2(\mathbb{R})$ by $x\mapsto axg$, and it acts transitively on each component of the two-dimensional period domain. The group $Q^1\times Q^1$ of pairs of elements of reduced norm one is simply connected, so by weak approximation [72, Thm. 7.8] its rational points are dense in its real points; hence there are $a,g\in Q^1(\mathbb{Q})$ with $x\mapsto axg$ carrying ω_0 into U. This map is an F-linear isometry of (T_{12},Φ), it carries $n_0^\perp$ onto $(an_0g)^\perp$ with $\mathrm{nrd}(an_0g)=\mathrm{nrd}(n_0)$, and it carries the Hodge structure of $T(S)$ to the Hodge structure on $(an_0g)^\perp$ determined by the image point, which is the period of a member X_b of the family. The rational classes of type $(1,1)$ in T_{12} at that period are exactly $F\cdot an_0g$, since a rational class orthogonal to the period decomposes into a component in the line and a component in $(an_0g)^\perp$ that is a rational class of type $(1,1)$ in a copy of $T(S)$, which has none. So $T(X_b)_\mathbb{Q}=(an_0g)^\perp$ with the restricted action of F, and $x\mapsto axg$ is a Hodge isometry $\varphi\colon T(S)_\mathbb{Q}\to T(X_b)_\mathbb{Q}$ that commutes with F.

Step 6, the cycle. $H^2(S,\mathbb{Q})$ and $H^2(X_b,\mathbb{Q})$ are both isometric to $\Lambda_{\mathbb{Q}}$, so by Witt cancellation $\mathrm{NS}(S)_{\mathbb{Q}} \cong \mathrm{NS}(X_b)_{\mathbb{Q}}$, and φ extends to a rational Hodge isometry $H^2(S,\mathbb{Q}) \to H^2(X_b,\mathbb{Q})$, which is algebraic by Buskin's theorem [21]. The action of $e \in F$ on $T(X_b)$ is induced by the cycle of Step 1, so $m_e = \varphi^{-1} \circ m_e \circ \varphi$ on $T(S)$ is induced by a cycle on $S \times S$. Since $T(S)$ has rank nine and $F \subseteq E$, E is F or a totally real field of degree nine with $\dim_E T = 1$, which [86] excludes; so $E = F$ is algebraic, and the Hodge conjecture for $S \times S$ and the rest follow as in Theorem 5.9(iii). □

The next proposition says exactly which pairs (F,D) the mechanism reaches and where it stops.

Proposition 5.15 (What the dihedral mechanism reaches)**.** *Let $n \in \{7,9\}$ and $F = \mathbb{Q}(\zeta_n)^+$.*

(i) *Let X be a K3 surface with a purely non-symplectic automorphism of order $n \in \{7,9\}$ whose transcendental lattice has rank twelve and is unimodular at every prime other than p_n. Then $(T(X)_{\mathbb{Q}},\Phi)$ is isometric to $(D_\infty, c\,\mathrm{nrd})$ for some $c \in F^\times$, and every surface of Mumford type obtained from X or from a rational isometric image of X as in Step 4 has the algebra D_∞. The Mumford algebras over F ramified at two finite places are reached by no family of rank twelve with such an automorphism.*

(ii) *Let $F = \mathbb{Q}(\zeta_7)^+$ and let $T_{18} = U^2 \oplus A_6 \oplus E_8(-1)$ be the transcendental lattice of the three-dimensional family of* [88, §5.4]*, with its F-bilinear form Φ_6 of dimension six. For every Mumford algebra D over F and every totally positive λ the space $(D^0, \lambda\,\mathrm{trd})$ embeds F-isometrically into $(T_{18} \otimes \mathbb{Q}, \Phi_6)$, so the four-dimensional period domain of this family contains the Shimura curve of the surfaces S_λ of every pair (F,D). The image of the family is a threefold in it, and Step 5 would need a rational isometry carrying the period of S_λ into that threefold.*

(iii) *A finite group acting on the base $\mathbb{P}^1$ of an elliptic surface is cyclic, dihedral, tetrahedral, octahedral or icosahedral, and the centres of the simple factors of its rational group algebra are the fields $\mathbb{Q}(\zeta_m)$, $\mathbb{Q}(\zeta_m)^+$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$. The only cubic fields among them are $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$, so the base change of elliptic surfaces along covers of $\mathbb{P}^1$ reaches no cyclic cubic field of conductor at least* 13.

Proof. (i) Steps 2 and 3 of the theorem use only the order of the automorphism, the rank, the signature and the unimodularity away from p_n, so they apply to X and give $\Phi \cong 2h_1\mathrm{nrd}_{D_\infty}$. A rational isometry preserves (T,Φ) up to the F-similarity class, which determines the algebra, and Step 4 exhibits the orthogonal complement of every negative definite F-line as $(D^0_\infty, \mu\,\mathrm{trd})$; an algebra $D \neq D_\infty$ has $(D^0, \lambda\,\mathrm{trd}) \not\cong (D^0_\infty, \mu\,\mathrm{trd})$ for all λ,μ, since the even Clifford algebra of the ternary form $\lambda\,\mathrm{trd}$ is D.

(ii) $T_{18} \otimes \mathbb{Q}$ is a three-dimensional hermitian K-space and $\Phi_6 = 2\langle h_1, h_1\delta, h_2, h_2\delta, h_3, h_3\delta\rangle$ has signature $(2,4)$ at σ_1 and $(0,6)$ at σ_2,σ_3, by the same count as in Step 2. The form $q_3 = \lambda\,\mathrm{trd}$ on D^0 has signature $(2,1)$ at σ_1, where D is split, and $(0,3)$ at the two ramified places. The Witt class of $\Phi_6 - q_3$ is represented at every real place by a negative definite ternary form and at every finite place by a ternary form, because a class of odd dimension has an anisotropic representative of dimension one or three; these local ternary forms have the global discriminant $d(\Phi_6)d(q_3)$, Hasse invariants $+1$ at almost every place and satisfying the product formula, since they are products of the invariants of Φ_6 and q_3 and of Hilbert symbols of their discriminants; so by [69, Thm. 72:1] there is a global ternary form q_3' with these localisations, and $\Phi_6 \cong q_3 \perp q_3'$ by the Hasse–Minkowski theorem. The image N of q_3' is a negative definite F-subspace of dimension three with $N^\perp \cong (D^0, \lambda\,\mathrm{trd})$, and the Hodge structures of K3 type on $N^\perp$ with F acting form the Shimura curve of the S_λ of (F,D). The family has three moduli [88, §5.4] in a domain of dimension $6-2=4$.

(iii) The list of finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$ is Klein's. The simple factors of $\mathbb{Q}[G]$ have centres the fields $\mathbb{Q}(\chi)$ generated by the values of the irreducible characters: $\mathbb{Q}(\zeta_m)$ for the cyclic group of order m, $\mathbb{Q}(\zeta_m)^+$ for the dihedral group of order $2m$ (its two-dimensional characters take the values $\zeta_m^k + \zeta_m^{-k}$), $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{-3})$ for A_4, $\mathbb{Q}$ for S_4, and $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$ for A_5. By Proposition 5.12(iii) the correspondences cut out of the action act through these fields; $\mathbb{Q}(\zeta_m)$ is never cubic, and $\mathbb{Q}(\zeta_m)^+$ is cubic exactly for $m \in \{7,9,14,18\}$, where it is $\mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$. □

For the fourfolds X of (F,D_∞) the theorem leaves (F2) where it was: by Theorem 6.47(v) their exceptional classes are algebraic as soon as one algebraic cycle on $S_\lambda \times X^r$ acts nontrivially on $T(S_\lambda)$, and the real multiplication, now algebraic, supplies no such cycle.

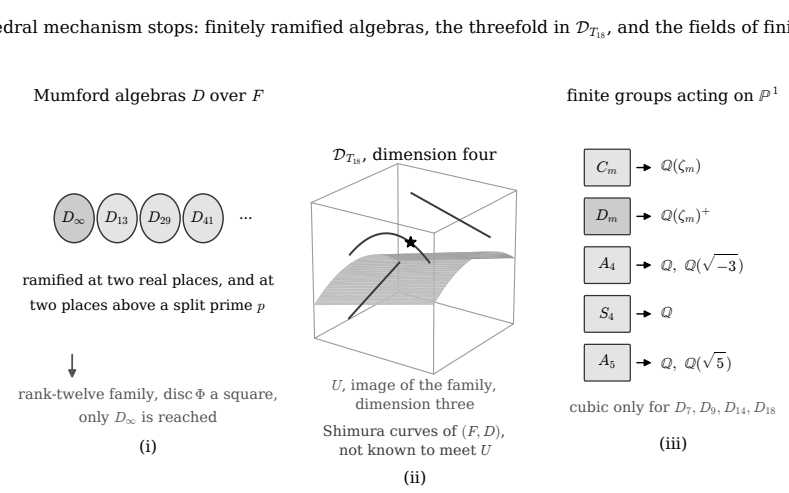


FIGURE 7. Where the dihedral mechanism stops, Proposition 5.15. (i) Over $F = \mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$ the Mumford algebras are D_∞ and the algebras D_p ramified at two real places and at two places above a split prime p; a rank-twelve family unimodular away from p_n has square discriminant and carries only D_∞. (ii) The period domain of the rank-eighteen family is four-dimensional, the image of the family is a threefold, and the Shimura curves of the surfaces of Mumford type for the other algebras lie in the domain; whether the threefold meets them is the open transport condition. (iii) The finite subgroups of PGL_2 acting on the base produce only $\mathbb{Q}(\zeta_m)$, $\mathbb{Q}(\zeta_m)^+$, $\mathbb{Q}$, $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{5})$; a cubic field arises only from the dihedral groups D_7, D_9, D_{14} and D_{18}.

5.7. **Powers of K3 surfaces and Kummer type**

The theorems of this section decide $S \times S$ and $S^{[2]}$. On the higher powers the Hodge classes are described by Varesco [84, Thm. 2.16]. Let E be totally real, $m = \dim_E T$, and for each real embedding σ of E let $V_\sigma = T \otimes_{E,\sigma} \mathbb{C}$, so that $T_\mathbb{C} = \bigoplus_\sigma V_\sigma$. The line $\bigwedge^m V_\sigma \subset T_\mathbb{C}^{\otimes m}$ is spanned by a class $\det V_\sigma$ of type (m,m), and the span of the $\det V_\sigma$ over all σ is defined over $\mathbb{Q}$: it is $\bigwedge^m_E T \subset \bigwedge^m T \subset T^{\otimes m}$, a space of Hodge classes of dimension $[E:\mathbb{Q}]$ in $H^{2m}(S^m)$, the exceptional classes. Varesco's theorem is that every Hodge class in the tensor algebra of T is obtained from the Hodge classes of $T \otimes T$, which are the elements of E, and from the exceptional classes by tensor products, permutations of the factors, linear combinations and contractions with q_S. Each of these operations carries algebraic classes on powers of S to algebraic classes: a tensor product is an exterior product, a permutation is the pull-back along an automorphism of S^k, and the contraction of two factors is the push-forward along the projection $S^k \to S^{k-2}$ after cupping with the pull-back of the class of the diagonal of $S \times S$, since $q_S(a,b) = \int_{S\times S}(a \otimes b) \cup [\Delta_S]$.

Proposition 5.16 (Hodge classes on the powers)**.** *Let S be a projective K3 surface with E totally real and $m = \dim_E T$, so that $m \geq 3$ as in the proof of Theorem 5.5(ii).*

(i) *The Hodge conjecture holds for every power S^n if and only if the elements of E are algebraic on $S \times S$ and the exceptional classes $\bigwedge^m_E T \subset H^{2m}(S^m)$ are algebraic. When $E = \mathbb{Q}$, $m = r = \mathrm{rk}\, T$ and the exceptional class is the determinant $\det T$, which spans $\bigwedge^r T \subset H^{2r}(S^r)$.*

(ii) *For a surface of Theorems 5.5, 5.14 or Corollary 5.13 with E totally real, E is algebraic, and the Hodge conjecture for all powers of S is equivalent to the algebraicity of $\bigwedge^m_E T$, $m = \mathrm{rk}\, T / [E:\mathbb{Q}]$: $m = 3$ for the surfaces S_λ of Theorem 5.14, with $\mathrm{rk}\, T = 9$; $m = 4$ for the very general members of the two-dimensional family of Theorem 5.14, with $\mathrm{rk}\, T = 12$.*

Proof. (i) If the Hodge conjecture holds on all powers, both kinds of classes are algebraic, being Hodge classes on $S \times S$ and on S^m. Conversely, $H^2(S) = \mathrm{NS}(S)_\mathbb{Q} \oplus T$, and $H^0(S)$, $\mathrm{NS}(S)_\mathbb{Q}$ and $H^4(S)$ are spanned by algebraic classes and are Tate twists of trivial Hodge structures. So a Hodge class

of $H^*(S^n) = \bigotimes^n \big(H^0 \oplus \mathrm{NS}_{\mathbb{Q}} \oplus T \oplus H^4\big)$ is a sum of exterior products of algebraic classes with Hodge classes of the tensor powers $T^{\otimes k}$, $k \le n$, pulled back along the projections of S^n onto k of its factors. By Varesco's theorem and the remark preceding the proposition, the Hodge classes of $T^{\otimes k}$ are algebraic as soon as E and the exceptional classes are. When $E = \mathbb{Q}$ there is one embedding, $V_\sigma = T_{\mathbb{C}}$ and $\bigwedge_E^m T = \bigwedge^r T$.

(ii) The theorems cited prove that E is algebraic; $m = \mathrm{rk}\, T/[E:\mathbb{Q}]$ since T is an E-vector space. The ranks are those of the nine-dimensional space D_∞^0 and of the lattice T_0 in Step 1 of the proof of Theorem 5.14. □

Proposition 5.17 (K3 surfaces of Kummer type over $\mathbb{Q}$)**.** *Let S be a projective K3 surface, $r = \mathrm{rk}\, T$, let $\langle a_1, \dots, a_r\rangle$ be a diagonalisation of q_S on T, $d(T) = \prod_i a_i \in \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ its discriminant and $\varepsilon_p(T) = \prod_{i<j}(a_i, a_j)_p$ its Hasse invariant at the prime p. The following are equivalent.*

(a) *T embeds isometrically in $U_{\mathbb{Q}}^3$, the quadratic space $H^2(A,\mathbb{Q})$ of an abelian surface.*

(b) *There is a dominant rational map from S to a Kummer surface.*

(c) *$r \le 3$; or $r = 4$ and $\varepsilon_p(T) = (-1,-1)_p$ at every prime p at which $d(T)$ is a square in $\mathbb{Q}_p$; or $r = 5$ and $\varepsilon_p(T) = (-1,-1)_p$ at every prime p.*

Here $(-1,-1)_p = 1$ for p odd and $(-1,-1)_2 = -1$, so (c) asks for $\varepsilon_p(T) = 1$ at the odd primes concerned and $\varepsilon_2(T) = -1$. If (a)–(c) hold, then S lies in the class $\mathscr{A}$ of Proposition 3.3, (F3′) *holds for every power S^n, and the Hodge conjecture holds for every S^n and every Hilbert scheme $S^{[n]}$.*

The equivalence of (a) and (b) is Ma's theorem [55, Prop. 3.2], where (a) is also shown equivalent to an embedding of lattices $T(S) \hookrightarrow U^3$, and for a Kummer surface itself the Hodge conjecture on all powers follows from that on the powers of the abelian surface [84, §0.2]. What is added here is the form (c) of the condition, which reads it off the invariants of T, and the passage from Ma's rational map to the class $\mathscr{A}$.

Proof. (a)⇔(c). The space $V = U_{\mathbb{Q}}^3 \cong \langle 1,-1\rangle^3$ has signature $(3,3)$, discriminant -1 and Hasse invariants $\varepsilon_p(V) = (-1,-1)_p^3 = (-1,-1)_p$. By Witt's theorem T embeds in V if and only if there is a quadratic space W over $\mathbb{Q}$ with $T \oplus W \cong V$, and by the Hasse–Minkowski theorem [79, Ch. IV, Thm. 9] this is a condition on invariants: W must have dimension $6-r$, signature $(1, 5-r)$, discriminant $d(W) = d(V)d(T)^{-1} = -d(T)$ and, from $\varepsilon_p(T\oplus W) = \varepsilon_p(T)\varepsilon_p(W)(d(T), d(W))_p$ with $(d(T), -d(T))_p = 1$, Hasse invariants $\varepsilon_p(W) = (-1,-1)_p\,\varepsilon_p(T)$. The signature forces $r - 2 \le 3$. A quadratic space over $\mathbb{Q}$ with prescribed dimension k, signature, discriminant and Hasse invariants exists if and only if the invariants satisfy the product formula, the real invariants are those of the signature, and at every prime the local existence condition holds [79, Ch. IV, §2.3, Prop. 6, and §3.3, Prop. 7]; the first two conditions hold here because they hold for T and for V and $T\otimes\mathbb{R} \oplus W\otimes\mathbb{R} \cong V\otimes\mathbb{R}$, and the local condition is empty for $k \ge 3$, is $\varepsilon_p(W) = 1$ at every p at which $d(W) = -1$ in $\mathbb{Q}_p^\times/\mathbb{Q}_p^{\times 2}$ for $k = 2$, and is $\varepsilon_p(W) = 1$ at every p for $k = 1$. With $k = 6-r$, $d(W) = -d(T)$ and $\varepsilon_p(W) = (-1,-1)_p\varepsilon_p(T)$ these are the three cases of (c).

(b) and the class $\mathscr{A}$. Let $f\colon S \dashrightarrow \mathrm{Km}(A)$ be dominant. By Lemma 5.1, $f^* = \Gamma_{f*}^{\mathrm{t}}$ maps $T(\mathrm{Km}(A))_{\mathbb{Q}}$ onto T. Let $\beta\colon \tilde{A} \to A$ be the blow-up of the sixteen points of order two and $\pi\colon \tilde{A} \to \mathrm{Km}(A)$ the quotient by the involution; $\Gamma_A = (\beta\times\pi)_*[\tilde{A}]$ is an algebraic cycle on $A\times\mathrm{Km}(A)$ with $\Gamma_{A*} = \pi_*\beta^*$, and $H^2(\mathrm{Km}(A),\mathbb{Q}) = \pi_*\beta^*H^2(A,\mathbb{Q}) \oplus \bigoplus_{i=1}^{16}\mathbb{Q}[E_i]$ with E_i the exceptional curves. Hence $T \subseteq (\Gamma_f^{\mathrm{t}}\circ\Gamma_A)_*H^2(A,\mathbb{Q})$, and $H^2(S) = \mathrm{NS}(S)_{\mathbb{Q}} \oplus T$ is spanned by this image and by divisor classes, while $H^0(S)$ and $H^4(S)$ are spanned by the class of S and the class of a point. So $S \in \mathscr{A}$, the point being the abelian variety of dimension zero, $S^n \in \mathscr{A}$ and (F3′) holds on S^n by Proposition 3.3. For the Hodge conjecture, the proof of Proposition 3.3(i) writes a Hodge class of S^n as $\sum_j \Gamma_{j*}y_j$ with y_j a Hodge class on one of the abelian varieties spanning $H^*(S^n)$, here the powers A^k with $k \le n$. The Hodge conjecture holds for every product of surfaces with $p_g = 1$ and $q = 2$ [75], in particular for A^k, so every y_j, and with it every Hodge class of S^n, is algebraic. The Chow motive of $S^{[n]}$ is a direct summand of a sum of Tate twists of the motives $h(S^k)$, $k \le n$ [25], so every Hodge class of $S^{[n]}$ is the image under an algebraic correspondence of a Hodge class of some S^k. □

Remark 5.18 (The powers left open)**.** (i) At $\rho(S)\geq 19$ the condition (c) is empty and S has a Shioda–Inose structure [62]. At $\rho(S)=17$ or 18 it is a condition on the Hasse invariants, met by $U_{\mathbb{Q}}^2\oplus\langle -2\rangle$, by $U_{\mathbb{Q}}\oplus\langle 2,-2,-2\rangle$ and by $\langle 2,2,-2,-2\rangle$, and failed at $p=2$ and $p=3$ by $\langle 2,2,-6,-6\rangle$, by $\langle 2,2,-2,-6,-6\rangle$ and by $\langle 2,6,-2,-2,-2\rangle$, the primes of failure coming in pairs by the product formula; all six are transcendental lattices of K3 surfaces of Picard number 18 or 17 by [67] and the surjectivity of the period map. For $r\leq 5$ a totally real E is $\mathbb{Q}$, since $m\geq 3$; so E is $\mathbb{Q}$ or a CM field, and in the CM case the Hodge conjecture holds on all powers of S with or without (c) [75]. At $\rho\geq 17$ the surfaces for which the Hodge conjecture on all powers is not decided by Propositions 5.16 and 5.17 are therefore those with $E=\mathbb{Q}$ and T failing (c), and for them the one class left is $\det T\in H^{2r}(S^r)$, $r\in\{4,5\}$. For $\rho\leq 16$ no surface is of Kummer type.

(ii) The Fricke surfaces have complex multiplication by $K=\mathbb{Q}(\sqrt{-N})$ at every member, and Theorem 5.9(iii) settles the Hodge conjecture on all their powers. Their transcendental spaces $U_{\mathbb{Q}}^2\oplus\langle -1\rangle\oplus\langle -N\rangle$ (Proposition 5.19) have discriminant N, so they are neither of Kummer type nor in the two families of Picard number 16 for which Varesco proves the Hodge conjecture on all powers from the algebraicity of the Kuga–Satake correspondence [84, Thm. 4.3, Exs. 4.4, 4.5], whose transcendental spaces $U_{\mathbb{Q}}^2\oplus\langle -2\rangle^2$ and $U_{\mathbb{Q}}^2\oplus\langle -6\rangle\oplus\langle -2\rangle$ have discriminants 1 and 3 modulo squares; they are settled by the complex multiplication instead. For each surface of Proposition 5.16(ii) the exceptional classes are Hodge classes of degree $2m$ on S^m reached by no correspondence of this section, all of which act on $T\otimes T$; whether they are algebraic, and with it whether (F3′) and the Hodge conjecture hold on S^m, is open. The one construction known to produce such classes is the Kuga–Satake correspondence, through which Varesco proves [84, Thm. 4.3].

Proposition 5.19 (The transcendental lattice and the Kuga–Satake variety of the Fricke surfaces)**.** *Let $N\in\{5,7\}$, $K=\mathbb{Q}(\sqrt{-N})$, let M be the trivial lattice $U\oplus A_4\oplus A_1^2\oplus D_4^2$ or $U\oplus A_6\oplus D_4^2$ of the elliptic fibration of the members of $\mathfrak{A}_N$, and let $T=M_{\mathbb{Q}}^{\perp}\subset H^2(X,\mathbb{Q})$, which is $T(X)_{\mathbb{Q}}$ when $\rho(X)=16$.*

(i) *$T\cong U_{\mathbb{Q}}^2\oplus\langle -1\rangle\oplus\langle -N\rangle$ as a quadratic space over $\mathbb{Q}$, so $d(T)=N$ and the centre of the even Clifford algebra $C^+(T)$ is K.*

(ii) *The Kuga–Satake variety of T is isogenous to B^4 for an abelian fourfold B of K-Weil type with discriminant one. If $\rho(X)=16$ and $\mathrm{End}_{\mathrm{Hdg}}T=K$, then B is isogenous to $E_K\times B_3$, where E_K is an elliptic curve with complex multiplication by K and B_3 is a simple abelian threefold with $\mathrm{End}(B_3)_{\mathbb{Q}}=K$; in particular B is not simple.*

Proof. (i) The root lattices are rationally diagonal: $D_n=\{x\in\mathbb{Z}^n:\sum x_i\in 2\mathbb{Z}\}$ has index two in $\mathbb{Z}^n$, so $D_n\otimes\mathbb{Q}\cong\langle 1\rangle^n$, and $E_8\supset D_8$ with index two gives $E_8\otimes\mathbb{Q}\cong\langle 1\rangle^8$; $A_n=\{x\in\mathbb{Z}^{n+1}:\sum x_i=0\}$ is the orthogonal complement of the vector $(1,\dots,1)$ of norm $n+1$, so $(A_n\oplus\langle n+1\rangle)\otimes\mathbb{Q}\cong\langle 1\rangle^{n+1}$; and $A_1^2=\langle 2,2\rangle\cong\langle 1,1\rangle$ by the vectors $(1,1)$ and $(1,-1)$ of $\mathbb{Z}^2$. Hence, with all forms taken positive definite, $A_4\oplus A_1^2\oplus D_4^2\oplus\langle 1,5\rangle\cong\langle 1\rangle^5\oplus\langle 1\rangle^2\oplus\langle 1\rangle^8\oplus\langle 1\rangle=\langle 1\rangle^{16}$ and $A_6\oplus D_4^2\oplus\langle 1,7\rangle\cong\langle 1\rangle^7\oplus\langle 1\rangle^8\oplus\langle 1\rangle=\langle 1\rangle^{16}$ over $\mathbb{Q}$. The K3 lattice is $\Lambda=U^3\oplus E_8(-1)^2$, so $\Lambda_{\mathbb{Q}}\cong U_{\mathbb{Q}}^3\oplus\langle -1\rangle^{16}$, while $M_{\mathbb{Q}}\oplus U_{\mathbb{Q}}^2\oplus\langle -1,-N\rangle\cong U_{\mathbb{Q}}^3\oplus\langle -1\rangle^{16}$ by the two isometries just written. Thus $M_{\mathbb{Q}}\oplus T\cong\Lambda_{\mathbb{Q}}\cong M_{\mathbb{Q}}\oplus(U_{\mathbb{Q}}^2\oplus\langle -1,-N\rangle)$, and Witt's cancellation theorem gives $T\cong U_{\mathbb{Q}}^2\oplus\langle -1\rangle\oplus\langle -N\rangle$. Its discriminant is $(-1)^2(-1)(-N)=N$, and the centre of $C^+(T)$ is $\mathbb{Q}(\sqrt{(-1)^3d(T)})=\mathbb{Q}(\sqrt{-N})$. The program `julia/fricke_lattice.jl` recomputes these isometries by the invariants of Hasse–Minkowski.

(ii) By Lombardo's theorem [54, Thms. 6.2, 6.4], in the form of [84, Thm. 4.1], the Kuga–Satake variety of a polarised Hodge structure of K3 type on $U_{\mathbb{Q}}^2\oplus\langle a\rangle\oplus\langle b\rangle$ with $a,b<0$ is isogenous to B^4 for an abelian fourfold B of $\mathbb{Q}(\sqrt{-ab})$-Weil type with discriminant one; here $(a,b)=(-1,-N)$, $\mathbb{Q}(\sqrt{-ab})=K$ and $C^+(T)\cong M_4(K)$. Suppose $\rho(X)=16$ and $\mathrm{End}_{\mathrm{Hdg}}T=K$. By Zarhin's theorem [93, Thm. 2.3.1] the Hodge group of T is $G=\mathrm{U}(W,h)$, the unitary group of the hermitian form h on the three-dimensional K-space $W=T$ with $\mathrm{Tr}_{K/\mathbb{Q}}h=q$. Over $\mathbb{C}$ the two eigenspaces W_σ, $W_{\bar\sigma}$ of K on $T_{\mathbb{C}}$ have dimension three and are isotropic for q, since $q(ax,y)=q(x,\bar a y)$, so $T_{\mathbb{C}}=W_\sigma\oplus W_\sigma^*$ and $G_{\mathbb{C}}=\mathrm{GL}(W_\sigma)$ acting on W_σ^* by the dual representation. The Kuga–Satake Hodge structure on $H^1(B,\mathbb{Q})$ is the lift of that of T along $\mathrm{Spin}(T)\to\mathrm{SO}(T)$ [86, §5], so the Hodge group of B is the preimage $\widetilde{G}$ of G in $\mathrm{Spin}(T)$, and $H^1(B,\mathbb{C})=S^+\oplus S^-$ is the sum of the two half-spin representations of $\mathrm{Spin}(T_{\mathbb{C}})\cong\mathrm{SL}_4$, on which the two eigenspaces of the centre K of $C^+(T)$ are S^+ and S^-. For

the isotropic decomposition $T_{\mathbb{C}} = W_\sigma \oplus W_\sigma^*$ the spin representation is $\bigwedge^\bullet W_\sigma \otimes (\det W_\sigma)^{-1/2}$, with S^+ the even and S^- the odd part. Restricted to $\widetilde{G}_{\mathbb{C}}$, $S^+ = (\bigwedge^0 W_\sigma \oplus \bigwedge^2 W_\sigma) \otimes \det^{-1/2}$ is the sum of the character $\chi = \det^{-1/2}$ and of the irreducible representation $W_\sigma^* \otimes \det^{1/2}$ of dimension three, and $S^- = (W_\sigma \oplus \bigwedge^3 W_\sigma) \otimes \det^{-1/2}$ is the sum of $W_\sigma \otimes \det^{-1/2}$ and of χ^{-1}. The four summands are pairwise non-isomorphic, the two characters being distinct and the two three-dimensional representations having different weights, so they are the isotypic components of $H^1(B,\mathbb{C})$ under the connected group $\widetilde{G}_{\mathbb{C}}$. An automorphism of $\mathbb{C}$ over $\mathbb{Q}$ carries an isotypic component to one of the same dimension, since the representation is defined over $\mathbb{Q}$; so $U_{\mathbb{C}} = \chi \oplus \chi^{-1}$ and $V_{\mathbb{C}} = S^+ \oplus S^-$ minus it are stable under all of them and are the complexifications of $\mathbb{Q}$-subspaces U, V with $H^1(B,\mathbb{Q}) = U \oplus V$. Both are stable under $\widetilde{G}$, hence sub-Hodge structures, and under K, which acts by scalars on S^+ and on S^-. By Poincaré reducibility B is isogenous to $E_K \times B_3$ with $H^1(E_K,\mathbb{Q}) = U$ and $H^1(B_3,\mathbb{Q}) = V$. The algebra $\mathrm{End}_{\mathrm{Hdg}}(U) \otimes \mathbb{C} = \mathrm{End}(\chi) \oplus \mathrm{End}(\chi^{-1})$ has dimension two and contains K, so $\mathrm{End}(E_K)_{\mathbb{Q}} = K$; likewise $\mathrm{End}_{\mathrm{Hdg}}(V) \otimes \mathbb{C}$ has dimension two, so $\mathrm{End}(B_3)_{\mathbb{Q}} = K$, a field, and B_3 is simple; and $\mathrm{Hom}(E_K, B_3) = 0$ since U and V share no isotypic component. □

Remark 5.20 (Real multiplication at the fixed points)**.** Theorem 5.9(iii) decides every power of every Fricke surface, so the Kuga–Satake variety of Proposition 5.19(ii) is not needed for them; the proposition records what the splitting of T into the two eigenspaces of K does to the Kuga–Satake construction. For a very general Hodge structure of K3 type on $U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -N\rangle$, with endomorphism field $\mathbb{Q}$, the fourfold B is simple with a definite quaternion algebra as endomorphism algebra [86, Prop. 5.7]; the Fricke families lie in the locus of the four-dimensional period domain where the endomorphism field is K, and there B splits. Real multiplication by $\sqrt{N}$ appears on the two twists at the fixed points of w, the surfaces of [88, §7.6, Prop. 7.7], where $\Lambda^2 = N$ by Lemma 5.8(ii). The fixed points of w are two, so the fixed-point twists are two surfaces; one-dimensional families of transcendental rank six with algebraic real multiplication by $\sqrt{5}$, $\sqrt{7}$ and every other real quadratic field are given by Theorem 5.23, through a correspondence that is not a chain.

5.8. Every real quadratic field in transcendental rank six

Floccari attached to every hyper-Kähler sixfold of generalised Kummer type a K3 surface and proved that its Kuga–Satake correspondence is algebraic [32]. The statement depends only on the rational transcendental space, and that space can be chosen to carry any real multiplication of degree two. Write $\Lambda_{\mathrm{Kum}_3} = U^3 \oplus \langle -8\rangle$ for the second cohomology lattice of a sixfold of generalised Kummer type and

$$V_{\mathrm{F}} = \Lambda_{\mathrm{Kum}_3}(2) \otimes \mathbb{Q} = U(2)_{\mathbb{Q}}^3 \oplus \langle -16\rangle \cong U_{\mathbb{Q}}^3 \oplus \langle -1\rangle,$$

a quadratic space of rank 7, signature $(3,4)$ and discriminant 1; the isometry holds because $U(2)_{\mathbb{Q}} \cong U_{\mathbb{Q}}$ by the basis $e, 2f$, and $\langle -16\rangle \cong \langle -1\rangle$.

Theorem 5.21 (Floccari [32, Thm. 1.3, Cor. 5.8])**.** *Let S be a projective K3 surface such that $T(S)_{\mathbb{Q}}$ embeds isometrically in V_{F}. Then the Kuga–Satake Hodge conjecture holds for S, and the Hodge conjecture holds for S^n for every $n \geq 1$.*

Lemma 5.22 (The space T_d)**.** *Let $d > 1$ be squarefree, $E = \mathbb{Q}(\sqrt{d})$ with real embeddings $\sigma_1(\sqrt{d}) = \sqrt{d}$, $\sigma_2(\sqrt{d}) = -\sqrt{d}$, let W be the three-dimensional E-quadratic space $\langle \sqrt{d}, \sqrt{d}, -\frac{1}{2}\rangle$, and let $T_d = \mathrm{Tr}_{E/\mathbb{Q}} W$, the $\mathbb{Q}$-vector space W with the form $\mathrm{Tr}_{E/\mathbb{Q}} \circ q_W$.*

(i) *$T_d \cong U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -d\rangle$; it has signature $(2,4)$ and discriminant d.*

(ii) *Multiplication M by $\sqrt{d}$ is self-adjoint for the form of T_d and $M^2 = d$; the σ_1-eigenspace $W_{\sigma_1} \subset T_d \otimes \mathbb{R}$ has signature $(2,1)$ and the σ_2-eigenspace is negative definite of rank 3.*

(iii) *$T_d \oplus \langle d\rangle \cong V_{\mathrm{F}}$; in particular T_d embeds isometrically in V_{F}.*

(iv) *The even lattice $L_d = U^2 \oplus \langle -4\rangle \oplus \langle -4d\rangle$ satisfies $L_d \otimes \mathbb{Q} \cong T_d$ and admits a primitive embedding in the K3 lattice Λ, whose orthogonal complement N_d has signature $(1,15)$.*

Proof. (i) For $\alpha \in E^\times$ the binary form $(x,y) \mapsto \mathrm{Tr}_{E/\mathbb{Q}}(\alpha xy)$ on E has, in the basis $1, \sqrt{d}$, the Gram matrix

$$\begin{pmatrix} \mathrm{Tr}\alpha & \mathrm{Tr}\alpha\sqrt{d} \\ \mathrm{Tr}\alpha\sqrt{d} & d\,\mathrm{Tr}\alpha \end{pmatrix}.$$

For $\alpha = \sqrt{d}$ it is $\left(\begin{smallmatrix} 0 & 2d \\ 2d & 0 \end{smallmatrix}\right)$, a hyperbolic plane, and for $\alpha = -\frac{1}{2}$ it is $\mathrm{diag}(-1,-d)$. The signature and the discriminant follow.

(ii) q_W is E-bilinear, so $\mathrm{Tr}(q_W(ex,y)) = \mathrm{Tr}(q_W(x,ey))$ for $e \in E$. Over $\mathbb{R}$, $T_d \otimes \mathbb{R} = W_{\sigma_1} \oplus W_{\sigma_2}$ orthogonally, with the forms $\langle \sigma_i\sqrt{d}, \sigma_i\sqrt{d}, -\frac{1}{2}\rangle$, of signatures $(2,1)$ and $(0,3)$.

(iii) $\langle -d\rangle \oplus \langle d\rangle \cong U_{\mathbb{Q}}$, since it represents zero; so $T_d \oplus \langle d\rangle \cong U_{\mathbb{Q}}^3 \oplus \langle -1\rangle$.

(iv) Over $\mathbb{Q}$, $\langle -4, -4d\rangle \cong \langle -1, -d\rangle$. The lattice L_d is even of signature $(2,4)$ with discriminant group $\mathbb{Z}/4 \oplus \mathbb{Z}/4d$, generated by two elements. Since $3 - 19 \equiv 0 \pmod 8$, $2 \le 3$, $4 \le 19$ and $22 - 6 > 2$, it embeds primitively in the even unimodular lattice Λ of signature $(3,19)$ by Nikulin's criterion [67, Cor. 1.12.3]. □

Theorem 5.23 (Real multiplication by every real quadratic field)**.** *Let $d > 1$ be squarefree and keep the notation of Lemma 5.22. Fix a marking $\Lambda \supset L_d$ and let*

$$\mathscr{D}_d = \{[\omega] \in \mathbb{P}(W_{\sigma_1} \otimes_{\mathbb{R}} \mathbb{C}) \,:\, q(\omega,\omega) = 0,\ q(\omega,\bar\omega) > 0\},$$

the complement of the real points in a smooth conic, a union of two discs.

(i) *For every $[\omega] \in \mathscr{D}_d$ there is a projective K3 surface S_ω with $H^{2,0}(S_\omega) = \mathbb{C}\omega$, $N_d \subseteq \mathrm{NS}(S_\omega)$ and $T(S_\omega) \subseteq L_d$; each isomorphism class occurs for at most countably many ω, so the surfaces form a one-dimensional family.*

(ii) *$T(S_\omega)_{\mathbb{Q}}$ is an E-subspace of T_d, and M restricts to a Hodge endomorphism of it. Outside a countable subset of $\mathscr{D}_d$ one has $T(S_\omega)_{\mathbb{Q}} = T_d$, $\rho(S_\omega) = 16$ and $\mathrm{End}_{\mathrm{Hdg}} T(S_\omega)_{\mathbb{Q}} = \mathbb{Q}(\sqrt{d})$ exactly; at the other points the transcendental rank is 2 or 4 and the endomorphism field is a CM field.*

(iii) *For every $[\omega] \in \mathscr{D}_d$ the Hodge conjecture holds for S_ω^n and for the Hilbert schemes $S_\omega^{[n]}$, for every $n \ge 1$. In particular the real multiplication by $\sqrt{d}$ is algebraic on $S_\omega \times S_\omega$, and the classes $\bigwedge_E^3 T$ of Proposition 5.16 on S_ω^3 are algebraic.*

(iv) *A rational quadratic space T of rank 6 embeds in V_{F} if and only if $T \cong U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -d(T)\rangle$, if and only if $\varepsilon_p(T) = (-1, d(T))_p$ for every prime p. Every projective K3 surface with such a transcendental space satisfies* (iii).

Proof. (i) The point ω is a period of Λ, so the surjectivity of the period map [45, Ch. 7, Thm. 4.1] gives a marked K3 surface S_ω with $H^{2,0} = \mathbb{C}\omega$. Its Néron–Severi group $\omega^\perp \cap \Lambda$ contains $L_d^\perp = N_d$, which contains a class of positive square, so S_ω is projective [15, Thm. IV.6.2], and $T(S_\omega) = \mathrm{NS}(S_\omega)^\perp \subseteq L_d$. Two markings of one surface differ by an element of the countable group $\mathrm{O}(\Lambda)$.

(ii) The space $V = T(S_\omega)_{\mathbb{Q}}$ is the smallest $\mathbb{Q}$-subspace of T_d whose complexification contains ω. As $M\omega = \sqrt{d}\,\omega$, the complexification of $M(V)$ contains ω too, so $V \subseteq M(V)$ and $M(V) = V$: V is an E-subspace of E-dimension 1, 2 or 3. The real operator M is self-adjoint with eigenvectors ω, $\bar\omega$, so it preserves $H^{2,0}$, $H^{0,2}$ and their orthogonal complement in $V_{\mathbb{C}}$, and $M|_V$ is a Hodge endomorphism. There are countably many proper E-subspaces $V' \subset T_d$, and each $V' \otimes_{E,\sigma_1} \mathbb{C}$ is a proper linear subspace of the plane $\mathbb{P}(W_{\sigma_1} \otimes \mathbb{C})$, which meets the smooth conic in at most two points; outside these points $V = T_d$. Then $\rho = 22 - 6 = 16$, and the endomorphism field E' contains $\mathbb{Q}(\sqrt{d})$ and is totally real or CM by Zarhin's theorem [93, Thm. 1.6], of degree dividing 6. If E' is totally real then $\dim_{E'} T \ge 3$ by the first paragraph of the proof of Proposition 5.25, so $E' = \mathbb{Q}(\sqrt{d})$. If E' were CM, of degree 2 it would equal the real field $\mathbb{Q}(\sqrt{d})$, and of degree 6 its cubic totally real subfield would contain $\mathbb{Q}(\sqrt{d})$; both are impossible. At a special point $\dim_E V \le 2$; a totally real field would need $\dim \ge 3$ over itself, so the endomorphism field is CM.

(iii) $T(S_\omega)_{\mathbb{Q}} \subseteq T_d$ embeds in V_{F} by Lemma 5.22(iii), so Theorem 5.21 applies. The classes named are Hodge classes on S_ω^2 and S_ω^3. For the Hilbert schemes, the motive of $S^{[n]}$ is a sum of Tate twists of motives of symmetric products of S [25], so the Hodge conjecture for $S^{[n]}$ follows from that for the powers of S, as in Theorem 5.5. The real multiplication also has a second proof that uses only the

Kuga–Satake part of Theorem 5.21: $q(Mx,My) = q(M^2x,y) = d\,q(x,y)$, so M is a Hodge similarity, and a Hodge similarity between K3 surfaces satisfying the Kuga–Satake Hodge conjecture is algebraic by Varesco's theorem [83, Thm. 4.5, Cor. 4.6], the Lefschetz standard conjecture in degree two being trivial for surfaces.

(iv) If $T \subset V_{\mathrm{F}}$ then $V_{\mathrm{F}} \cong T \oplus \langle c\rangle$ with $c \equiv d(V_{\mathrm{F}})d(T) = d(T)$ modulo squares, and comparing signatures $c > 0$. Since also $V_{\mathrm{F}} \cong U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -c\rangle \oplus \langle c\rangle$, Witt cancellation gives $T \cong U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -c\rangle$; the converse is Lemma 5.22(iii) with d replaced by c. By the Hasse–Minkowski theorem two rational quadratic spaces with the same rank, discriminant and signature are isometric if and only if their Hasse invariants agree at every prime [79, Ch. IV, Thm. 9], and $\varepsilon_p(\langle 1,-1,1,-1,-1,-c\rangle) = (-1,-1)_p^3(-1,-c)_p^3 = (-1,c)_p$, using $(-1,-1)_p(-1,-c)_p = (-1,c)_p$. □

TABLE 4. Families of projective K3 surfaces on which a real multiplication by $\mathbb{Q}(\sqrt{d})$ is known to be algebraic.

d	source	dimension	rk T	all powers S^n
2	[83]; [88, Prop. 6.2]	4	12	open
3	[83]; [88, Prop. 7.2]	2; 3	8; 12	open
5	[88, Thm. 1.1]	7	20	open
6	Theorem 5.5	2	8	open
sums of two squares	[76], with [86]	1	6	proved [84]
every squarefree $d > 1$	Theorem 5.23	1	6	proved

For $d = N \in \{5,7\}$ the space $T_N = U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -N\rangle$ is exactly the transcendental space of the Fricke families of Proposition 5.19(i). So the families $\mathfrak{A}_N$, with complex multiplication by $\mathbb{Q}(\sqrt{-N})$, and the curve $\mathscr{D}_N$ of Theorem 5.23, with real multiplication by $\mathbb{Q}(\sqrt{N})$, lie in the same four-dimensional period domain, and every K3 surface whose transcendental space lies in T_N satisfies the Hodge conjecture on all its powers by Theorem 5.21. The theorem uses neither chains of symplectic quotients nor fibrewise isogenies: the correspondence realising $\sqrt{d}$ is built from the Kuga–Satake cycle, so Proposition 5.25 is not contradicted, and Proposition 5.11 is consistent with it, since M is a similarity of multiplier d. What Theorem 5.23 does not give is a family of dimension at least two: the Kuga–Satake Hodge conjecture is known for K3 surfaces only through embeddings in V_{F}, of rank seven, so the transcendental rank is at most six, and a totally real field of degree e acting on a space of rank six needs $3e \le 6$.

Remark 5.24 (Picard number at least seventeen)**.** If $\rho(S) \ge 18$ then $\operatorname{rk} T(S) \le 4$ and $T(S)_{\mathbb{Q}}$ embeds in V_{F}, because a rational quadratic space embeds in another of rank at least three more and of dominating signature; so the Hodge conjecture holds for all powers of every projective K3 surface of Picard number at least 18. For $\rho = 17$ the same holds when $\varepsilon_p(T) = (-1,-1)_p$ at every prime p at which $-d(T)$ is a square in $\mathbb{Q}_p$, the condition for $T \oplus \langle c\rangle \oplus \langle c'\rangle \cong V_{\mathrm{F}}$ to be solvable [79, Ch. IV, §2].

5.9. **Where this stops**

The frontier case of (F3′) in Theorem A.55 and Remark A.56 is a K3 surface S_λ of Picard number 13 with $T(S_\lambda)_{\mathbb{Q}} \cong T$, the nine-dimensional representation of a Mumford fourfold, and $E = F$ a totally real cubic field, which may have Galois group S_3. Theorem 5.14 settles it for the two cyclic fields $F = \mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ with the algebra D_∞; the frontier is now S_λ for every other pair (F,D): for these two fields with the algebras ramified at two finite places, for the cyclic cubic fields of conductor at least 13, and for every field that is not cyclic. There, by Proposition 5.11(v), no composite of similarities reaches F,[9] so Theorem 5.3, the Fricke isogenies and every correspondence of the kinds listed are

[9]Reaching F means producing, by algebraic correspondences from S_λ to itself, Hodge endomorphisms of $T(S_\lambda)_{\mathbb{Q}}$ that generate the cubic field $E = F$, which is what (F3′) asks of this surface, since the class of a generator of F in $H^4(S_\lambda \times S_\lambda)$ is a Hodge class reached from no abelian variety as long as it is not algebraic. The correspondences built in this section, and every correspondence whose components have geometric genus at most one, act on $T(S_\lambda)$ as similarities, and a composite of similarities is a similarity; by Proposition 5.11(ii) a similarity lies in $\mathbb{Q}$ when $[E:\mathbb{Q}]$ is odd, so it generates nothing. The

blind to it; by Proposition 5.11(iv) a correspondence that does reach it has an irreducible component whose resolution has $p_g \geq 2$, transcendental rank at least 18 and $b_2 \geq 31$; and by Proposition 5.12(iii) that component is not the graph of an element of a finite group acting on a cover of S_λ when F is not cyclic. When F is cyclic the dihedral families are the natural source, as van Geemen observed for $\mathbb{Q}(\zeta_9)^+$ [86, Rem. 7.9]; Theorem 5.14 is the statement that for the two fields of conductor 7 and 9 and the algebra D_∞ the source actually reaches S_λ, and Proposition 5.15 says that it reaches nothing else. The only correspondences of this size that we know to act on a transcendental space of K3 type are the dihedral ones, which give abelian fields, and the Kuga–Satake correspondence into the cohomology of an abelian variety of dimension 2^7, whose algebraicity is open [27, 85]; in Section 6 we replace its abelian variety by a sixfold and show what the replacement does and does not give. Within quadratic fields the method reaches $\sqrt{d}$ for $d \in \{2,3,6\}$ in families, and the Fricke isogenies give complex multiplication by $\mathbb{Q}(\sqrt{-5})$ and $\mathbb{Q}(\sqrt{-7})$ on one-dimensional families, where Theorem 5.9 settles the Hodge conjecture on every power; real multiplication by $\sqrt{5}$ and $\sqrt{7}$ is algebraic on the two surfaces of [88] and on no family known to us (Remark 5.20); for $\mathbb{Q}(\sqrt{10})$, $\mathbb{Q}(\sqrt{14})$ and the other composite discriminants no family is known on which the chain of Theorem 5.3 can be started. On the powers beyond the square, Proposition 5.16 reduces the Hodge conjecture for every surface of this section with totally real endomorphism field to its exceptional classes, Proposition 5.17 settles it with (F3′) for the surfaces of Kummer type over $\mathbb{Q}$, Theorem 5.9 settles it for every member of the Fricke families, and Remark 5.18 records what is left.

Proposition 5.25 (Chains of symplectic quotients reach exactly $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$)**.** *Let S be a projective K3 surface whose endomorphism field E is totally real and different from $\mathbb{Q}$, and let $S = S_0, \dots, S_r$ be a chain as in Theorem 5.3 in which every rational map is the quotient by a finite group of symplectic automorphisms. Then no degree n_i is divisible by 5 or by 7, every n_i divides a power of 24, and the multiplier $\prod n_i^{\pm 1}$ lies in $\{1,2,3,6\}$ modulo squares. In particular no such chain makes $\sqrt{10}$, $\sqrt{14}$, $\sqrt{15}$ or $\sqrt{21}$ algebraic, and the surfaces with real multiplication by these fields are not reached by Theorem 5.3 in this form.*

Proof. Along the chain the transcendental spaces are isomorphic up to scaling of the form, by Lemma 5.1 and Lemma 5.2, so every S_i has the same transcendental rank t and the same endomorphism field E as S. By Zarhin [93, Thms. 2.2.1 and 2.3.1] the Hodge group of $T(S)_\mathbb{Q}$ is the group of E-linear special isometries of the E-quadratic space W with $T(S)_\mathbb{Q} = \mathrm{Tr}_{E/\mathbb{Q}} W$, of dimension $m = t/[E:\mathbb{Q}]$ over E. If $m \leq 2$ this group is commutative (trivial for $m = 1$, a torus of dimension $[E:\mathbb{Q}]$ for $m = 2$). Then every $g \in \mathrm{Hdg}(\mathbb{Q})$ commutes with the Hodge group and is an endomorphism of the Hodge structure, so $g \in E$. The elements of the totally real field E are self-adjoint for the form on $T(S)_\mathbb{Q}$ and g is an isometry, so $g = g^* = g^{-1}$ and $g^2 = 1$. As $\mathbb{Q}$-points are Zariski dense in a connected linear algebraic group over $\mathbb{Q}$, the Hodge group would be trivial, which is impossible because $h^{2,0}(S) = 1$. So $m \geq 3$ and $t \geq 3[E:\mathbb{Q}] \geq 6$.

Let G be a finite group of symplectic automorphisms of a K3 surface X and $\Omega_G \subset \mathrm{NS}(X)$ its coinvariant lattice, negative definite because G fixes the transcendental lattice and an ample class. If 5 divides $|G|$, then G contains an element of order five, whose coinvariant lattice has rank 16 [68, § 5] and lies in Ω_G, so $\mathrm{rk}\,\Omega_G \geq 16$; if 7 divides $|G|$ the same argument with the rank 18 of an element of order seven gives $\mathrm{rk}\,\Omega_G \geq 18$. The transcendental lattice $T(X)$ lies in $\Omega_G^\perp \subset H^2(X,\mathbb{Z})$, which has rank $22 - \mathrm{rk}\,\Omega_G \leq 6$ and signature $(3, 19 - \mathrm{rk}\,\Omega_G)$, and $T(X)$ has signature $(2, \mathrm{rk}\,T(X) - 2)$; so $\mathrm{rk}\,T(X) \leq 5$ when 5 divides $|G|$ and $\mathrm{rk}\,T(X) \leq 3$ when 7 does. Every symplectic automorphism of finite order has order at most eight [68], so if some $n_i = |G_i|$ is not of the form $2^a 3^b$ then 5 or 7 divides it, the source of that quotient has transcendental rank at most five, and so has S; this contradicts $t \geq 6$. Hence every n_i is of the form $2^a 3^b$, and the multiplier is $2^a 3^b$ modulo squares, that is 1, 2, 3 or 6. A multiplier n makes only e with $e^2 \equiv n$ algebraic, which gives the list. □

correspondences coming from finite group actions act through abelian fields by Proposition 5.12(iii), and a cubic field whose Galois closure has group S_3 is not abelian. The two propositions together say what a correspondence realising F must look like: an irreducible component with $p_g \geq 2$ and transcendental rank at least eighteen, not arising from a finite group; no construction in the literature produces one.

Remark 5.26. The proposition settles the lattice question of whether a chain $S \to Y \leftarrow S'$ through a symplectic involution and a symplectic automorphism of order five, of multiplier $2/5 \equiv 10$, can start on a surface with real multiplication by $\mathbb{Q}(\sqrt{10})$: it cannot, since S' would have transcendental rank at most five and real multiplication by a real quadratic field needs rank at least six. The same bound shows that the surfaces of transcendental rank at most five with a symplectic automorphism of order five or seven have endomorphism field $\mathbb{Q}$ or a CM field; in the CM case the Hodge group is a torus and the chain closes only through the complex multiplication itself. Real multiplication by $\sqrt{10}$, $\sqrt{14}$ or any field outside $\{\mathbb{Q}(\sqrt{2}),\mathbb{Q}(\sqrt{3}),\mathbb{Q}(\sqrt{5}),\mathbb{Q}(\sqrt{6}),\mathbb{Q}(\sqrt{7})\}$ therefore needs a correspondence that is not a composite of symplectic quotients, Hodge isometries and fibrewise isogenies; in transcendental rank six the Kuga–Satake correspondence of Theorem 5.23 is one, and in higher rank none is known.

6. Constructions at (F2)

The first classes (F2) has to reach are the Weil classes of a very general abelian $2n$-fold of Weil type with $n \geq 3$, and the exceptional classes of a Mumford fourfold. We build an object at each.

6.1. The polarised semiregularity criterion

Let A be a very general member of the polarised family $\mathscr{D}_{n,n}$ of abelian $2n$-folds of Weil type for an imaginary quadratic field K, with polarisation class θ and Weil plane $W(A)$. Then $\mathrm{Hdg}^*(A) = \mathbb{Q}[\theta] \oplus W(A)$ and $W(A)\cdot\theta = 0$ Theorem A.3. A class $\gamma \in H^{\mathrm{ev}}(A)$ has the polarised shape if

$$\gamma = N\omega + \sum_{k=0}^{2n} c_k \theta^k, \qquad N \neq 0,\ 0 \neq \omega \in W(A),\ c_k \in \mathbb{Q}.$$

Its Hankel rank $\rho \in \{0,1,2,3\}$ is the rank of the $(2n-1)\times 3$ Hankel matrix $[\mu_{i+j}]_{0\leq i\leq 2n-2, 0\leq j\leq 2}$, $\mu_k = k!\,c_k$ (Setting A.14); it is the number of distinct exponentials $e^{a\theta}$ needed to write the polynomial part when that number is at most two, and 3 otherwise.[10]

Theorem 6.1 (The polarised criterion)**.** *Let $n \geq 3$ and let $\mathscr{E}$ be a perfect complex on A with $\mathrm{ch}(\mathscr{E}) = \gamma$ of polarised shape and Hankel rank ρ. Then*

$$\dim \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) \ \geq\ r(\gamma) := (4+\rho)n^2 - 2n,$$

and if equality holds the semiregularity map[11] of $\mathscr{E}$ is injective, $\mathscr{E}$ deforms with A along the whole of $\mathscr{D}_{n,n}$, and the Weil classes are algebraic on every member of the family with full Hodge group. If some $c_k \neq 0$ with $1 \leq k \leq 2n-1$, the first-order Hodge locus of γ in $H^1(A,T_A)$ is the tangent space of $\mathscr{D}_{n,n}$, of dimension n^2. A perfect complex whose Chern character is a nonzero multiple of a Weil class plus classes in degrees 0 and $4n$ never has injective semiregularity map (Theorem A.21), so the polynomial part is forced.

The number $r(\gamma)$ is the rank of the contraction $HT^2(A) \to H^*(A)$, $\xi \mapsto \xi \lrcorner \gamma$, on the Hochschild cohomology $HT^2 = \bigoplus_{p+q=2} H^q(A, \bigwedge^p T_A)$, of dimension $\binom{4n}{2}$; its kernel is the annihilator $\mathrm{Ann}(\gamma)$, which meets $H^1(A,T_A)$ in the first-order Hodge locus. The semiregularity map $\sigma_{\mathscr{E}}$ of a perfect complex

[10] The Hankel matrix of a sequence $(a_k)_{k\geq 0}$ has entries a_{i+j}. If a_k is a sum $\sum_j \lambda_j b_j^k$ of m terms with distinct b_j and nonzero λ_j, the matrix factors as $V^{\mathsf{T}}\,\mathrm{diag}(\lambda_j)V$ with V the Vandermonde matrix of the b_j, so its rank is m; conversely a sequence whose Hankel matrix has finite rank m satisfies a linear recurrence of order m and is a sum of m exponentials with polynomial coefficients. For the polynomial part $\sum_k c_k\theta^k$ of a Chern character, writing it as $\sum_j \lambda_j e^{b_j\theta}$ means $k!\,c_k = \sum_j \lambda_j b_j^k$, so the number of exponentials needed is the Hankel rank of $(k!\,c_k)$, provided the degree $2n$ allows the recurrence to be read off, which it does for $m \leq 2$; the cap at three records that the criterion of Theorem 6.1 does not distinguish further.

[11] For a perfect complex $\mathscr{E}$ on a smooth projective variety the semiregularity map of Buchweitz and Flenner [10] is the linear map $\sigma_{\mathscr{E}}\colon \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) \to \prod_q H^{q+1}(X,\Omega_X^{q-1})$ sending ξ to the trace of $\xi \circ \exp\bigl(-\mathrm{At}(\mathscr{E})\bigr)$, where $\mathrm{At}(\mathscr{E}) \in \mathrm{Ext}^1(\mathscr{E},\mathscr{E}\otimes \Omega_X^1)$ is the Atiyah class; it generalises Bloch's map for a subvariety [17], which is the case of the structure sheaf of the subvariety. Its value is that the obstruction to deforming $\mathscr{E}$ along a first-order deformation of X lies in $\mathrm{Ext}^2(\mathscr{E},\mathscr{E})$ and is sent by $\sigma_{\mathscr{E}}$ to the obstruction to keeping $\mathrm{ch}(\mathscr{E})$ of Hodge type; so when $\sigma_{\mathscr{E}}$ is injective, $\mathscr{E}$ deforms wherever its Chern character stays a Hodge class, and the Chern character of the deformed complex is algebraic there. The dimension count of the theorem is the arithmetic of this statement: the rank of $\sigma_{\mathscr{E}}$ is at least the rank of the contraction into $\mathrm{ch}(\mathscr{E})$, so $\dim \mathrm{Ext}^2$ is at least $r(\gamma)$, and equality forces injectivity.

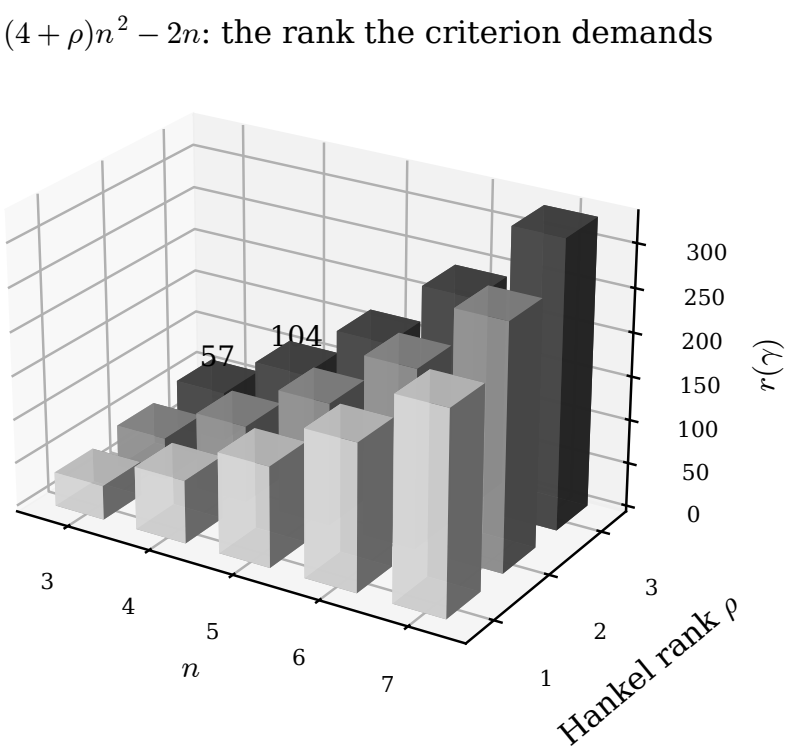


FIGURE 8. The lower bound $r(\gamma) = (4+\rho)n^2 - 2n$ of Theorem 6.1 for $3 \le n \le 7$ and Hankel rank $\rho = 1,2,3$. The two labelled bars are the product cases of Theorem 6.3.

composes with the characteristic map $\mathrm{ev}_{\mathscr{E}} : HT^2 \to \mathrm{Ext}^2(\mathscr{E},\mathscr{E})$, $\xi \mapsto \xi \lrcorner \mathrm{At}(\mathscr{E})$, to the contraction into $\mathrm{ch}(\mathscr{E})$; this is how the lower bound arises, and it is the form in which the criterion is used below. Figure 8 shows $r(\gamma)$; the two values that matter below are $r = 57$ at $n = 3$, $\rho = 3$ and $r = 104$ at $n = 4$, $\rho = 3$. On the product E_0^6 of six copies of the elliptic curve $\mathbb{C}/\mathbb{Z}[i]$ every twisted complex of line bundles has $\dim \mathrm{Ext}^2 \ge 69 > 57$ Theorem A.32; the criterion is not met there by any such object.

6.2. **The Weil plane of a product**

Lemma 6.2. *Let A, A' be of Weil type for the same field K, of dimensions $2m$, $2m'$, with the diagonal K-action on $A \times A'$. Then $W(A \times A')$ is a two-dimensional rational subspace of the four-dimensional space $W(A) \otimes W(A') \subseteq H^{2n}(A \times A')$, $n = m + m'$, and every rational $w \in W(A \times A')$ is $\sum_{i,j} a_{ij}\, \omega_i \otimes \omega'_j$ with $a_{ij} \in \mathbb{Q}$ for rational bases (ω_i), (ω'_j) of $W(A)$, $W(A')$.*

Proof. Over $\mathbb{C}$, $H^1(A \times A') = V \oplus V'$ splits into eigenspaces $V_\pm \oplus V'_\pm$ of K, and $\bigwedge_K^{2n}$ of the sum is spanned by $\det V_+ \otimes \det V'_+$ and $\det V_- \otimes \det V'_-$; these lie in the span of the four products $\omega_\pm \otimes \omega'_\pm$ of the eigenvectors of $W(A) \otimes \mathbb{C}$ and $W(A') \otimes \mathbb{C}$. Both $W(A \times A')$ and $W(A) \otimes W(A')$ are defined over $\mathbb{Q}$, so the inclusion holds over $\mathbb{Q}$. The explicit formula is (A.3): $\omega_1(A \times A') = \omega_1 \otimes \omega'_1 - d\, \omega_2 \otimes \omega'_2$ for $K = \mathbb{Q}(\sqrt{-d})$. □

6.3. **An object with the polarised Chern character**

Theorem 6.3 (The product object)**.** *Let A, A' be very general in their polarised Weil families, with ample classes θ_A, $\theta_{A'}$, $\theta = \theta_A + \theta_{A'}$, and suppose that $W(A)$ and $W(A')$ are algebraic. Then there is a perfect complex $\mathscr{E}$ on $A \times A'$, a finite direct sum with shifts of exterior products*

$$F_i \boxtimes G'_j, \quad F_i \boxtimes L'^k, \quad L^l \boxtimes G'_j, \quad L^l \boxtimes L'^k,$$

with F_i, G'_j coherent sheaves on A, A' and L, L' the ample bundles, such that

$$\mathrm{ch}(\mathscr{E}) = Nw + \sum_{k=0}^{2n} c_k \theta^k, \qquad N \neq 0,\ 0 \neq w \in W(A \times A'),$$

has the polarised shape with Hankel rank $\rho = 3$, so that $r(\mathrm{ch}\mathscr{E}) = 7n^2 - 2n$, and the first-order Hodge locus of $\mathrm{ch}(\mathscr{E})$ in $H^1(A \times A', T)$ is the tangent space of $\mathscr{D}_{n,n}$.

The hypotheses hold in the two open cases nearest to the known ones: A, A' very general Weil fourfolds of trivial discriminant, where the Weil classes are algebraic by Markman's theorem for discriminant one [57] and the specialisation of algebraic classes, so that $A \times A'$ is an eightfold with $n = 4$; and A such a fourfold and $A' = B$ a Weil surface of any discriminant d, whose Weil plane is of type $(1,1)$ and algebraic by Lefschetz, so that $A \times B$ is a sixfold of discriminant d with $n = 3$. For $d = -1$ the sixfold is of split Weil type and its Weil classes are algebraic by Markman's secant-sheaf theorem [59]; the open sixfolds are those with $d \neq -1$ in $\mathbb{Q}^\times / N(k^\times)$.

Proof. Since $\mathrm{Hdg}^*(A) = \mathbb{Q}[\theta_A] \oplus W(A)$, algebraic classes on A are rational combinations of Chern characters of coherent sheaves, and the Chern character of a coherent sheaf is a Hodge class, the algebraicity of $W(A)$ gives coherent sheaves F_1, F_2 with $\mathrm{ch}(F_i) = N_i\omega_i + p_i(\theta_A)$, $N_i \neq 0$, for a basis (ω_1, ω_2) of $W(A)$; likewise G'_1, G'_2 on A' with $\mathrm{ch}(G'_j) = N'_j\omega'_j + p'_j(\theta_{A'})$. Fix $0 \neq w \in W(A \times A')$ rational, $w = \sum a_{ij}\omega_i \otimes \omega'_j$ by Lemma 6.2, and a common denominator M of the numbers $Ma_{ij}/(N_iN'_j)$, so that these are integers. Negative multiplicities are realised by the shift $[1]$, which changes the sign of ch.

Step 1. $\mathscr{E}_0 = \bigoplus_{i,j} (F_i \boxtimes G'_j)^{\oplus Ma_{ij}/(N_iN'_j)}$ has, by multiplicativity,

$$\mathrm{ch}(\mathscr{E}_0) = Mw + \sum_i \omega_i \otimes q'_i(\theta_{A'}) + \sum_j q_j(\theta_A) \otimes \omega'_j + s, \tag{6.1}$$

with q'_i, q_j polynomials and $s \in \mathbb{Q}[\theta_A] \otimes \mathbb{Q}[\theta_{A'}]$.

Step 2. The classes $e^{k\theta_{A'}} = \mathrm{ch}(L'^k)$, $k = 0, \ldots, 2m'$, form a basis of $\mathbb{Q}[\theta_{A'}]/(\theta_{A'}^{2m'+1})$: the matrix $(k^j/j!)_{j,k}$ has determinant $\prod_j (1/j!)$ times the Vandermonde determinant of $0, 1, \ldots, 2m'$, which is nonzero. Write $q'_i = \sum_k b_{ik} e^{k\theta_{A'}}$ and add $(F_i \boxtimes L'^k)^{\oplus -Mb_{ik}/N_i}$, after enlarging M to clear denominators; this removes $\omega_i \otimes q'_i$ and adds only terms $p_i(\theta_A) \otimes e^{k\theta_{A'}}$, which lie in $\mathbb{Q}[\theta_A] \otimes \mathbb{Q}[\theta_{A'}]$. Symmetrically remove $q_j \otimes \omega'_j$ with summands $L^l \boxtimes G'_j$.

Step 3. The classes $e^{l\theta_A + k\theta_{A'}} = \mathrm{ch}(L^l \boxtimes L'^k)$ form a basis of $\mathbb{Q}[\theta_A]/(\theta_A^{2m+1}) \otimes \mathbb{Q}[\theta_{A'}]/(\theta_{A'}^{2m'+1})$ by the same determinant, so the remaining part in $\mathbb{Q}[\theta_A] \otimes \mathbb{Q}[\theta_{A'}]$ is cancelled by line bundles with integer multiplicities and shifts. Finally add $(L \boxtimes L')^{\otimes k}$, $k = 0, 1, 2$, with generic nonzero multiplicities μ_k. The result is

$$\mathrm{ch}(\mathscr{E}) = Mw + \sum_{k=0}^{2} \mu_k e^{k\theta},$$

a polarised shape whose polynomial part is a sum of three distinct exponentials with nonzero coefficients. Three exponentials are not one of the shapes of Hankel rank at most two, so $\rho = 3$; and some c_k with $1 \leq k \leq 2n-1$ is nonzero, so the first-order Hodge locus is the tangent space of $\mathscr{D}_{n,n}$ by Theorem 6.1. □

The construction was carried out in the exterior-algebra model of $H^*(A \times A')$ for $(m, m') = (2,1)$ and $(2,2)$: the cross terms of $\mathrm{ch}(\mathscr{E}_0)$ vanish, nothing is left outside $W(A \times A') \oplus \mathbb{Q}[\theta]$ after Steps 2 and 3, the Hankel rank is 3, and the first-order Hodge locus has dimension 9, resp. 16, equal to n^2 (Appendix C).

6.4. **What a direct sum cannot do**

The tangent space of $\mathscr{D}_{n,n}$ at a product point contains the tangent space of the product locus $\mathscr{D}_{m,m} \times \mathscr{D}_{m',m'}$, of dimension $m^2 + m'^2$, and $n^2 - m^2 - m'^2 = 2mm'$ mixed directions. In the same model, each exterior product $F \boxtimes G'$ of sheaves of polarised shape has first-order Hodge locus inside $T\mathscr{D}_{n,n}$ equal to the tangent space of the product locus: all $2mm'$ mixed directions obstruct it, four for $A \times B$ and

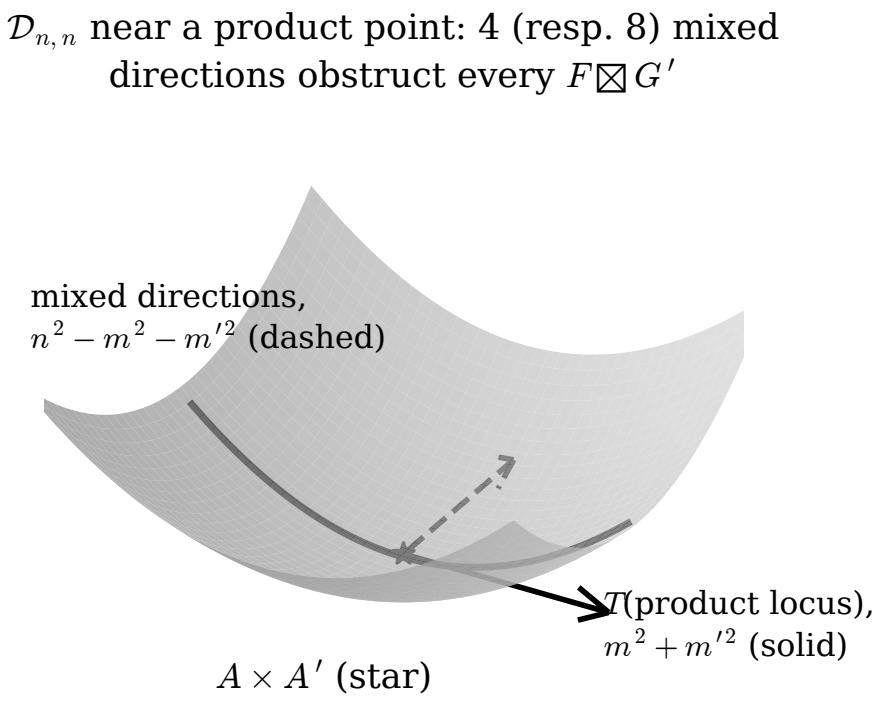


FIGURE 9. The domain $\mathscr{D}_{n,n}$ near a product point $A \times A'$. The curve is the product locus, whose tangent space has dimension $m^2 + m'^2$; the $2mm'$ mixed directions obstruct every exterior product $F \boxtimes G'$, and only the combined object of Theorem 6.3 has the full tangent space as first-order Hodge locus.

eight for $A \times A'$ (Fig. 9). The object $\mathscr{E}$ of Theorem 6.3 has the whole tangent space as its first-order locus because its Chern character is a single polarised class and not a product. Semiregularity is a different matter.

Proposition 6.4 (Direct sums are never semiregular)**.**

(i) *Let $\mathscr{E} = \bigoplus_k \mathscr{E}_k$ be a direct sum of at least two nonzero complexes. Then the semiregularity map $\sigma_{\mathscr{E}}$ vanishes on $\bigoplus_{k \neq l} \operatorname{Ext}^2(\mathscr{E}_k, \mathscr{E}_l)$. So $\sigma_{\mathscr{E}}$ is injective only if $\operatorname{Ext}^2(\mathscr{E}_k, \mathscr{E}_l) = 0$ for all $k \neq l$.*

(ii) *For every direct-sum realisation $\mathscr{E}$ of Theorem 6.3, $\sigma_{\mathscr{E}}$ is not injective and $\dim \operatorname{Ext}^2(\mathscr{E}, \mathscr{E}) > r(\operatorname{ch}\mathscr{E})$.*

Proof. (i) The semiregularity map of Buchweitz and Flenner is built from the traces $\xi \mapsto \operatorname{tr}\big(\xi \circ \operatorname{At}(\mathscr{E})^p\big)$ of compositions with powers of the Atiyah class [10, Sect. 4]. The Atiyah class of a direct sum is block diagonal, so for ξ in an off-diagonal block $\operatorname{Ext}^2(\mathscr{E}_k, \mathscr{E}_l)$ the composition $\xi \circ \operatorname{At}(\mathscr{E})^p$ is off-diagonal, and the trace of an off-diagonal block is zero.

(ii) The summands of $\mathscr{E}$ include $F_1 \boxtimes L'^k$ and $F_1 \boxtimes L'^l$ for two exponents $k < l$, because Step 2 of the proof uses every power L'^k with $0 \leq k \leq 2m'$ (the exponentials form a basis and q'_1 is a nonzero polynomial of positive degree for a sheaf of polarised shape, which has p_1 of positive degree by Theorem 6.1). By the Künneth formula

$$\operatorname{Ext}^2(F_1 \boxtimes L'^k, F_1 \boxtimes L'^l) \supseteq \operatorname{Ext}^2(F_1, F_1) \otimes H^0(A', L'^{l-k}), \tag{6.2}$$

and $\dim \operatorname{Ext}^2(F_1, F_1) \geq r(\operatorname{ch} F_1) \geq 12$ by Theorem 6.1 on the fourfold A ($n = 2$, $\rho \geq 0$), while H^0 of an ample bundle is nonzero. So an off-diagonal block is nonzero, $\sigma_{\mathscr{E}}$ has a kernel by (i), and $\dim \operatorname{Ext}^2(\mathscr{E}, \mathscr{E}) > r$ because equality would force injectivity. □

Proposition 6.5 (Convolutions)**.** *Let $\mathscr{E}_\delta$ be a one-sided twisted complex on pieces $P_1,\dots,P_N$ with shifts s_a, ordered so that the differential $\delta=(\delta_{ab})_{a<b}$, $\delta_{ab}\in\operatorname{Ext}^{1+s_a-s_b}(P_a,P_b)$, is strictly upper triangular; let $\gamma=\operatorname{ch}(\mathscr{E}_\delta)=\sum_a(-1)^{s_a}\operatorname{ch}(P_a)$.*

(i) *The semiregularity map $\sigma_{\mathscr{E}_\delta}$ vanishes on every class of $\operatorname{Ext}^2(\mathscr{E}_\delta,\mathscr{E}_\delta)$ represented by a strictly upper triangular cocycle. So the criterion holds for $\mathscr{E}_\delta$ only if every strictly upper triangular cocycle of degree two is a coboundary.*

(ii) *Suppose no term of the twisted differential carries a class of degree one into a diagonal block $\operatorname{Ext}^*(P_c,P_c)$. Let $v\in T\mathscr{D}_{n,n}$ have a nonzero mixed component. If some piece $P=X\boxtimes Y$ has $\operatorname{rk}Y\,c_1(X)/\theta_A\neq\operatorname{rk}X\,c_1(Y)/\theta_{A'}$, then $\operatorname{ev}_{\mathscr{E}_\delta}(v)\neq0$ and $\sigma_{\mathscr{E}_\delta}(\operatorname{ev}_{\mathscr{E}_\delta}(v))=0$: the criterion fails.*

(iii) *Suppose the pieces split into two nonempty groups with $\operatorname{Ext}^*(P_a,P_b)=0$ in both directions for a, b in different groups. Then $\mathscr{E}_\delta=\mathscr{E}'\oplus\mathscr{E}''$ is the direct sum of the convolutions of the groups, and the criterion holds for $\mathscr{E}_\delta$ only if every $v\in T\mathscr{D}_{n,n}$ annihilates $\operatorname{ch}(\mathscr{E}')$ and $\operatorname{ch}(\mathscr{E}'')$ separately.*

Proof. (i) The components of $\sigma_{\mathscr{E}_\delta}(\xi)$ are the traces $\operatorname{tr}(\xi\circ\operatorname{At}(\mathscr{E}_\delta)^p)/p!$ [10, Sect. 4]. In the harmonic model of a twisted complex the Atiyah class is represented by $[\nabla,\bar\partial+\delta]$ with ∇ the direct sum of the Chern connections of the pieces; it is block upper triangular with diagonal blocks $\operatorname{At}(P_a)$ and off-diagonal blocks $\nabla\delta_{ab}$. The homotopy transfer $\xi\mapsto\sum_k(-h[\delta,\cdot])^k\xi$ that produces the chain-level representative of ξ raises or preserves the block position $b-a$, so a strictly upper triangular cocycle has a strictly upper triangular representative. The product of an upper triangular block matrix with a strictly upper triangular one is strictly upper triangular, and its trace, the sum of the traces of its diagonal blocks, is zero. The trace is a chain map for $\bar\partial+[\delta,\cdot]$ by cyclicity, so the vanishing holds in cohomology. Since $\sigma_{\mathscr{E}_\delta}\circ\operatorname{ev}_{\mathscr{E}_\delta}$ is contraction into γ, of rank $r(\gamma)$, the kernel of $\sigma_{\mathscr{E}_\delta}$ contains the image of the strictly upper triangular classes and $\dim\operatorname{Ext}^2\geq r(\gamma)+\dim(\text{that image})$.

(ii) The Atiyah class of $X\boxtimes Y$ is $\operatorname{At}(X)\otimes1+1\otimes\operatorname{At}(Y)$, so the diagonal component of $\operatorname{ev}_{\mathscr{E}_\delta}(v)$ at P is $v\lrcorner\operatorname{At}(P)$, with trace $v\lrcorner\operatorname{ch}_1(P)$. Write $v=v_A+v_{A'}+z\otimes d'+z'\otimes d$ with $v_A\in T\mathscr{D}_{m,m}$, $v_{A'}\in T\mathscr{D}_{m',m'}$, $z\in H^{0,1}(A)$, $d'\in H^0(T_{A'})$, $z'\in H^{0,1}(A')$, $d\in H^0(T_A)$. Since v kills $\theta=\theta_A+\theta_{A'}$ and $v_A\lrcorner\theta_A=v_{A'}\lrcorner\theta_{A'}=0$, we have $v\lrcorner\theta_{A'}=-v\lrcorner\theta_A=-z'\wedge(d\lrcorner\theta_A)$, nonzero when $d\neq0$ because θ_A is nondegenerate and $z'\wedge$ is injective on $H^{0,1}(A)$; the case $d=0$, $d'\neq0$ is symmetric. On a very general member $c_1(X)$ and $c_1(Y)$ are multiples of the polarisations, so $v\lrcorner\operatorname{ch}_1(P)=\big(\operatorname{rk}Y\,c_1(X)/\theta_A-\operatorname{rk}X\,c_1(Y)/\theta_{A'}\big)\,v\lrcorner\theta_A\neq0$ under the hypothesis on P. Under the assumption on the differential, the diagonal components of a class in $\operatorname{Ext}^2(\mathscr{E}_\delta,\mathscr{E}_\delta)$ are well defined, so $\operatorname{ev}_{\mathscr{E}_\delta}(v)\neq0$, while $\sigma_{\mathscr{E}_\delta}(\operatorname{ev}_{\mathscr{E}_\delta}(v))=v\lrcorner\gamma=0$ because v is tangent to the Hodge locus of γ.

(iii) With no Ext groups between the groups there are no components of δ between them, so the twisted complex is block diagonal,

$$\begin{aligned}\operatorname{Ext}^2(\mathscr{E}_\delta,\mathscr{E}_\delta)&=\operatorname{Ext}^2(\mathscr{E}',\mathscr{E}')\oplus\operatorname{Ext}^2(\mathscr{E}'',\mathscr{E}''),\\ \dim\operatorname{Ext}^2(\mathscr{E}_\delta,\mathscr{E}_\delta)&\geq\dim HT^2-\dim\big(\operatorname{Ann}(\operatorname{ch}\mathscr{E}')\cap\operatorname{Ann}(\operatorname{ch}\mathscr{E}'')\big),\end{aligned}\tag{6.3}$$

with $\sigma_{\mathscr{E}_\delta}=\sigma_{\mathscr{E}'}\oplus\sigma_{\mathscr{E}''}$; the inequality holds because for each summand the contraction into its Chern character factors through ev and σ, so that $\dim\operatorname{Ext}^2(\mathscr{E}',\mathscr{E}')$ is at least the rank of $\xi\mapsto\xi\lrcorner\operatorname{ch}\mathscr{E}'$, and likewise for $\mathscr{E}''$. Here $r(\gamma)=\dim HT^2-\dim\operatorname{Ann}(\gamma)$ with $\operatorname{Ann}(\gamma)=T\mathscr{D}_{n,n}$. Equality $\dim\operatorname{Ext}^2=r(\gamma)$ forces the common annihilator to be all of $T\mathscr{D}_{n,n}$. □

For the pieces of Theorem 6.3 the splitting in (iii) that one would like to use separates the pieces $F_i\boxtimes Y$ carrying a secant sheaf on A from the pieces $L^l\boxtimes Y'$. Two statements settle what that splitting gives. The first computes the annihilator of the secant pieces by hand; the second shows that the splitting is never available.

Theorem 6.6 (The annihilator of the secant pieces)**.** *Let $\mathscr{E}'$ be the sum, with the multiplicities and shifts of Theorem 6.3, of the pieces $F_i\boxtimes G'_j$ and $F_i\boxtimes L'^k$. Then*

$$\operatorname{ch}(\mathscr{E}')=Mw+\sum_i\frac{M}{N_i}\,p_i(\theta_A)\otimes R_i,\qquad R_i=\sum_j a_{ij}\omega'_j,\tag{6.4}$$

where (a_{ij}) is the matrix of w in the bases (ω_i), (ω'_j) and is invertible, and the annihilator of $\mathrm{ch}(\mathscr{E}')$ *in* $T\mathscr{D}_{n,n}$ *is the tangent space* $T\mathscr{D}_{m,m}\oplus T\mathscr{D}_{m',m'}$ *of the product locus, of dimension* $m^2+m'^2$. *If* $\mathrm{Ext}^*(F_i,L^l)=0$ *for every power* L^l *among the pieces, every convolution* $\mathscr{E}_\delta$ *of the pieces has* $\dim\mathrm{Ext}^2(\mathscr{E}_\delta,\mathscr{E}_\delta)\geq r(\gamma)+2mm'$, *which is* 61 *at* $n=3$ *and* 120 *at* $n=4$.

Proof. Steps 1 and 2 of Theorem 6.3 add and subtract, for each i, the classes

$$\sum_j \frac{Ma_{ij}}{N_iN'_j}\,\mathrm{ch}(F_i)\mathrm{ch}(G'_j)\quad\text{and}\quad \frac{M}{N_i}\,\mathrm{ch}(F_i)\otimes q'_i,$$

where $q'_i=\sum_j(a_{ij}/N'_j)p'_j$; the difference is $\mathrm{ch}(F_i)\otimes R_i$, and summing over i with $\mathrm{ch}(F_i)=N_i\omega_i+p_i(\theta_A)$ gives (6.4), the terms $\omega_i\otimes R_i$ adding up to Mw by the definition of a. By Lemma 6.2 $w=x\,\omega_+\otimes\omega'_+ + \bar{x}\,\omega_-\otimes\omega'_-$ with $x\neq0$, a tensor of rank two, so a is invertible and R_1, R_2 form a basis of the Weil plane of A'.

Write $V=H^1(A,\mathbb{C})=V_+\oplus V_-$ for the eigenspaces of K, $\omega_\pm=\det V_\pm$, and $V'_\pm$, $\omega'_\pm$ for their analogues on A'. A tangent vector $v\in H^1(T_{A\times A'})$ acts on cohomology as a derivation and decomposes as $v=v_A+v_{A'}+\mu+\mu'$ with $v_A\in H^{0,1}(A)\otimes H^0(T_A)$, $v_{A'}$ its analogue on A', and the mixed parts $\mu\in H^{0,1}(A)\otimes H^0(T_{A'})$, $\mu'\in H^{0,1}(A')\otimes H^0(T_A)$. Let $v\in T\mathscr{D}_{n,n}$. The three Künneth components of $v\lrcorner\theta$ vanish separately: $v_A\lrcorner\theta_A=0$, $v_{A'}\lrcorner\theta'=0$ and $\mu\lrcorner\theta'+\mu'\lrcorner\theta_A=0$. Contraction into a nondegenerate form is an isomorphism from $H^0(T)$ onto $H^{0,1}$, so $\mu\mapsto\mu\lrcorner\theta'$ and $\mu'\mapsto\mu'\lrcorner\theta_A$ are injective into $H^{0,1}(A)\otimes H^{0,1}(A')$, and $\mu=0$ exactly when $\mu'=0$. Next, $v\lrcorner(\omega_+\otimes\omega'_+)=0$. The term $(v_A\lrcorner\omega_+)\otimes\omega'_+$ has no component in $\Lambda^{2m}V_+\otimes\Lambda^{2m'}V'_+$, because that component is the trace of v_A on V_+ and v_A sends $V_+^{1,0}$ into $V_+^{0,1}$ and kills $V_+^{0,1}$; it lies in $\Lambda^{2m-1}V_+\otimes V_-\otimes\Lambda^{2m'}V'_+$, while the other three terms lie in $\Lambda^{2m}V_+\otimes\Lambda^{2m'-1}V'_+\otimes V'_-$, in $\Lambda^{2m-1}V_+\otimes\Lambda^{2m'}V'_+\otimes V'_-$ and in $\Lambda^{2m}V_+\otimes V_-\otimes\Lambda^{2m'-1}V'_+$, four distinct weight spaces for $GL(V_+)\times GL(V_-)\times GL(V'_+)\times GL(V'_-)$. Hence $v_A\lrcorner\omega_+=0$, and in the same way $v_A\lrcorner\omega_-=0$ and $v_{A'}\lrcorner\omega'_\pm=0$. A vector of $T\mathscr{D}_{n,n}$ with vanishing mixed part therefore lies in $T\mathscr{D}_{m,m}\oplus T\mathscr{D}_{m',m'}$, and conversely every such sum kills θ and the Weil plane; the dimension $m^2+m'^2$ is that of the two Weil families [92].

Now let $v\in T\mathscr{D}_{n,n}$ have $\mu\neq0$. It kills Mw; v_A kills $p_i(\theta_A)$ and $v_{A'}$ kills R_i; so $v\lrcorner\mathrm{ch}(\mathscr{E}')=(\mu+\mu')\lrcorner\sum_k\theta_A^k\otimes R^{(k)}$, where $p_i=\sum_k c_{ik}\theta_A^k$ and $R^{(k)}=\sum_i(Mc_{ik}/N_i)R_i$. Write $\mu=\sum_s z_s\otimes d'_s$ and $\mu'=\sum_s z'_s\otimes d_s$. The contraction of $\theta_A^k\otimes R^{(k)}$ against μ' is $k\sum_s\theta_A^{k-1}(d_s\lrcorner\theta_A)\otimes(z'_s\wedge R^{(k)})$, of Künneth bidegree $(2k-1,2m'+1)$, and against μ it is $\sum_s(z_s\wedge\theta_A^k)\otimes(d'_s\lrcorner R^{(k)})$, of bidegree $(2k+1,2m'-1)$. These bidegrees are pairwise distinct over all k and both types, so every term vanishes on its own. Some $R^{(k)}$ is nonzero, since p_1, p_2 are not both zero and R_1, R_2 are a basis. For $R=\sigma\omega'_+ + \bar\sigma\omega'_-$ with $\sigma\neq0$ the map $z'\mapsto z'\wedge R$ is injective on $H^{0,1}(A')$, because $z'\wedge\omega'_+=z'_-\wedge\omega'_+$ and $z'\wedge\omega'_-=z'_+\wedge\omega'_-$ lie in independent weight spaces and vanish only with $z'_\mp$; dually $d'\mapsto d'\lrcorner R$ is injective on $H^0(T_{A'})$. If the nonzero $R^{(k)}$ has $k\geq1$, hard Lefschetz makes θ_A^{k-1} injective on $H^1(A)$, since $k-1\leq2m-1$, so the μ'-term vanishes only if $\sum_s(d_s\lrcorner\theta_A)\otimes z'_s=0$, that is $\mu'=0$, and then $\mu=0$. If $k=0$ the μ-term is $(\mathrm{id}\otimes(\cdot\lrcorner R^{(0)}))(\mu)$, which vanishes only if $\mu=0$. Either way the mixed part is zero, against the assumption.

The bound is Proposition 6.5(iii): the common annihilator of $\mathrm{ch}\mathscr{E}'$ and $\mathrm{ch}\mathscr{E}''$ in $T\mathscr{D}_{n,n}$ lies in that of $\mathrm{ch}\mathscr{E}'$, so $\dim\mathrm{Ext}^2\geq\dim HT^2-m^2-m'^2=r(\gamma)+n^2-m^2-m'^2$, and $n=m+m'$. □

Proposition 6.7 (Secant and line-bundle pieces are never orthogonal)**.** *For every realisation of Theorem 6.3 there are an index i and a power L^l among the pieces with $\chi(F_i,L^l)\neq0$. Hence* $\mathrm{Ext}^*(F_i,L^l)\neq0$ *and* $\mathrm{Ext}^*(L^l,F_i)\neq0$, *and the hypothesis of the last sentence of Theorem 6.6 fails.*

Proof. The Todd class of A is 1, $\omega_i\theta_A=0$ and $\int\theta_A^4=24$, so Riemann–Roch gives $\chi(F_i,L^l)=\int p_i(-\theta_A)e^{l\theta_A}=P_i(l)$ with

$$P_i(l)=c_{i0}l^4-4c_{i1}l^3+12c_{i2}l^2-24c_{i3}l+24c_{i4},\tag{6.5}$$

a nonzero polynomial because $p_i\neq0$. Expand $p_i=\sum_l b_{il}e^{l\theta_A}$, $0\leq l\leq4$, modulo θ_A^5, which is possible since the five exponentials span $\mathbb{Q}[\theta_A]/(\theta_A^5)$, and call the set S_i of l with $b_{il}\neq0$ the support of p_i. Then $P_i(s)=\sum_{l\in S_i}b_{il}(s-l)^4$. If P_i vanished on all of S_i with $|S_i|\geq2$, the vector $(b_{il})_{l\in S_i}$ would lie in

the kernel of the symmetric matrix $\big((s-l)^4\big)_{s,l\in S_i}$. That matrix is nonsingular: for two elements its determinant is $-(s-l)^8$, for three it is $2\prod(s-l)^4$, for the five four-element subsets of $\{0,\dots,4\}$ it is 29952, 1032192, 6054912, 1032192, 29952, and for the whole set it is 7962624. So a support with at least two elements contains some l with $P_i(l)\neq 0$.

Step 2 of Theorem 6.3 uses the pieces $L^l\boxtimes G'_j$ for every l in the support of $q_j=\sum_i(Ma_{ij}/N_i)p_i$, and the last step uses $L^k\boxtimes L'^k$ for three values of k whose net multiplicities are nonzero. If p_1 and p_2 are independent, q_1 and q_2 span the same plane because a is invertible, so the union of their supports is $S_1\cup S_2$; if p_2 is a multiple of p_1, both q_j are multiples of p_1, not both zero, with the same support. In either case every $l\in S_1\cup S_2$ gives a piece $L^l\boxtimes G'_j$. If $|S_1|\geq 2$ or $|S_2|\geq 2$ we are done. Otherwise $p_i=b_ie^{l_i\theta_A}$ with $b_i\neq 0$. If $l_1\neq l_2$ then $L^{l_2}\boxtimes G'_j$ is a piece and $P_1(l_2)=b_1(l_2-l_1)^4\neq 0$. If $l_1=l_2=l$ the residue of Steps 1 to 3 lies in $e^{l\theta_A}\otimes\mathbb{Q}[\theta']$, so at least two of the three final exponents k differ from l, and $P_i(k)=b_i(k-l)^4\neq 0$. The vanishing of $\mathrm{Ext}^*(F_i,L^l)$ would force $\chi(F_i,L^l)=0$; Serre duality gives $\mathrm{Ext}^*(L^l,F_i)\neq 0$ as well. □

Theorem 6.6 is therefore a conditional statement whose condition never holds: Proposition 6.5(iii) cannot be applied to the splitting between secant pieces and line-bundle pieces, and no bound on $\dim\mathrm{Ext}^2$ of a convolution follows from the Chern characters alone. The hypothesis of part (ii) can fail with line bundles only. The cone of a section of $L^l\boxtimes L'^{k'-k}$, viewed as a map $L^l\boxtimes L'^k\to L^l\boxtimes L'^{k'}$ with $k'>k$, is a twisted complex of two pieces in which the Serre dual of the section, a class of degree one, enters the diagonal block of the piece $L^l\boxtimes L'^k$; the only terms that can enter a diagonal block of a line-bundle piece are of this form, with the same power on A. Part (ii) applies to every piece $L^l\boxtimes L'^k$ with $l\neq k$ whose block receives no such term, which is how Step 3 of Theorem 6.3 uses it.

6.5. **Where the product object stops**

Every test that depends on $\mathrm{ch}(\mathscr{E})$ alone is passed: polarised shape, $N\neq 0$, Hankel rank 3, first-order Hodge locus equal to $T\mathscr{D}_{n,n}$, lower bound 57 or 104. The criterion then asks for $\dim\mathrm{Ext}^2(\mathscr{E},\mathscr{E})=r$, Proposition 6.4 shows that no direct sum achieves it, and Proposition 6.5 says what a convolution of the same pieces must do: kill every strictly upper triangular class of degree two, carry classes of degree one from the lower triangle into the diagonal blocks of every piece with mismatched slopes, and, since by Proposition 6.7 the secant pieces and the line-bundle pieces are never orthogonal, carry classes between the two groups in a way that cancels the mixed directions, which by Theorem 6.6 the secant pieces alone leave uncancelled. All three demands are statements about the groups $\mathrm{Ext}^*(F_1,F_2)$, $\mathrm{Ext}^*(F_i,L^l)$, $\mathrm{Ext}^*(L^l,F_i)$ and their analogues on A'. Those groups are not computed in [59, 60, 61], where the only object with known Ext groups on a very general Weil fourfold of trivial discriminant is the twisted transform $\Phi(F\boxtimes F^\vee)$ with $\dim\mathrm{Ext}^k=1,8,18,8,1$, whose Brauer class is half the Poincaré class and which therefore cannot be mixed with untwisted line bundles; and we have not computed them. Riemann–Roch fixes $\chi(F_i,L^l)=P_i(l)$ and the Künneth formula fixes everything on the line-bundle side, so the unknown is exactly the thirteen graded groups $\mathrm{Ext}^*(F_i,F_j)$ and $\mathrm{Ext}^*(F_i,L^l)$. This is the exact stopping point: the polarised criterion at a product point is a finite question about the Ext algebra of the secant sheaves, and Proposition 6.5 lists what the answer must contain for the route to stay open. The analogous question on E_0^6, for line bundles, closed negatively with $\dim\mathrm{Ext}^2\geq 69$ Theorem A.32; that is an analogy and not a proof of the answer here.

At a point of complex multiplication the thirteen groups are known. We record them, since they turn the stopping point into a statement about one explicit variety.

Lemma 6.8 (The Ext groups at a point of complex multiplication)**.** *Let $K=\mathbb{Q}(i)$, $E_0=\mathbb{C}/\mathbb{Z}[i]$, and let $A_0=E_0^6$ carry the K-action by i on three factors and by $-i$ on the other three; it is a member of $\mathscr{D}_{3,3}$ for every discriminant, since the K-hermitian forms $\sum_{j\leq 3}a_jz_j\bar w_j-\sum_{j>3}a_jz_j\bar w_j$ with $a_j>0$ are polarisations of every discriminant class. Let L be ample on A_0 with class θ, let $Z,Z'\subseteq A_0$ be abelian subvarieties of dimension three with $Z\cap Z'$ finite, and put $\chi_Z=(\theta|_Z)^3/6$, a positive integer.*

(i) *Every Hodge class on A_0 is a polynomial in divisor classes, and every product of three divisor classes is a rational combination of classes $[Z]$ of abelian threefolds. The Weil classes are such*

combinations, and no Weil class is a multiple of a single $[Z]$. *For every* Z, $\mathrm{ch}(\mathscr{O}_Z) = [Z]$ *and* $\mathrm{ch}(\mathscr{O}_Z \otimes L^a) = [Z]\, e^{a\theta}$.

(ii) $\mathrm{Ext}^k(\mathscr{O}_Z \otimes L^a, \mathscr{O}_{Z'} \otimes L^b) = 0$ *for* $k \neq 3$, *and* $\dim \mathrm{Ext}^3 = |Z \cap Z'| = (Z \cdot Z')$, *for all* a, b.

(iii) *The groups* $\mathrm{Ext}^k(\mathscr{O}_Z \otimes L^a, \mathscr{O}_Z \otimes L^b)$ *are given by* (6.6) *below, with* $N_Z \cong \mathscr{O}_Z^3$. *For* $a = b$ *it has dimension* $\binom{6}{k}$ *and the characteristic map* $HT^2(A_0) \to \mathrm{Ext}^2$ *is onto; for* $b > a$ *it lives in degrees* 0 *to* 3, *of dimension* $\binom{3}{k}(b-a)^3\chi_Z$; *for* $b < a$ *in degrees* 3 *to* 6, *of dimension* $\binom{3}{k-3}(a-b)^3\chi_Z$.

(iv) *For* $b > a$, $\mathrm{Ext}^k(\mathscr{O}_Z \otimes L^a, L^b)$ *is concentrated in degree* 3 *and* $\mathrm{Ext}^k(L^a, \mathscr{O}_Z \otimes L^b)$ *in degree* 0, *both of dimension* $(b-a)^3\chi_Z$; *for* $b < a$ *the first is concentrated in degree* 6 *and the second in degree* 3, *both of dimension* $(a-b)^3\chi_Z$; *for* $a = b$ *the dimensions are* $\binom{3}{k-3}$ *and* $\binom{3}{k}$.

(v) *For* $b > a$, $\mathrm{Ext}^k(L^a, L^b) = H^k(A_0, L^{b-a})$ *is concentrated in degree* 0, *of dimension* $(b-a)^6\theta^6/720$; *for* $b < a$ *in degree* 6; *for* $a = b$ *it is* $H^k(\mathscr{O}_{A_0})$, *of dimension* $\binom{6}{k}$.

Proof. (i) The Hodge group of A_0 is the norm-one torus K^1 acting by z on V_+ and $\bar{z}$ on V_-, so over $\mathbb{C}$ the Hodge classes of degree $2k$ are $\bigwedge^k V_+ \otimes \bigwedge^k V_-$, which is spanned by the products of k elements of $V_+ \otimes V_- = \mathrm{Hdg}^2(A_0)_{\mathbb{C}}$; since $\mathrm{Hdg}^2 = \mathrm{NS}_{\mathbb{Q}}$ by Lefschetz, the $\mathbb{Q}$-span of the products of k divisor classes has full dimension $\binom{6}{k}^2$. The classes $f^*[0]$, $f \in \mathrm{Hom}(E_0^6, E_0) = \mathbb{Z}[i]^6$, are the rank-one hermitian forms $v\bar{v}^{\mathsf{T}}$ and span $\mathrm{NS}_{\mathbb{Q}}$. For f_1, f_2, f_3 independent over K the map $f = (f_1, f_2, f_3)\colon A_0 \to E_0^3$ is a surjective homomorphism, hence flat, and $f_1^*[0]\, f_2^*[0]\, f_3^*[0] = f^*[0] = [f^{-1}(0)]$ is a positive multiple of the class of the abelian threefold $\ker(f)^\circ$; for dependent f_i the product is zero. A Weil class w satisfies $w\theta = 0$, while $[Z]\theta = i_*(\theta|_Z) \neq 0$ because $(\theta|_Z)^3 > 0$. Since Z is a translate of a subgroup, N_Z is trivial and $\mathrm{td}(N_Z) = 1$, so Grothendieck–Riemann–Roch gives $\mathrm{ch}(i_*\mathscr{O}_Z) = i_*(1) = [Z]$ and $\mathrm{ch}(\mathscr{O}_Z \otimes L^a) = [Z]e^{a\theta}$.

(ii) Since $\dim Z \cap Z' = 0$ and the tangent spaces are translation invariant, $T_Z \oplus T_{Z'} = T_{A_0}$ at every point of $Z \cap Z'$, so Z and Z' are locally transverse complete intersections and $Z \cap Z'$ is reduced. In local coordinates with $Z = \{x_1 = x_2 = x_3 = 0\}$ and $Z' = \{y_1 = y_2 = y_3 = 0\}$, the Koszul resolution of $\mathscr{O}_Z$ gives $\mathscr{E}xt^k(\mathscr{O}_Z, \mathscr{O}_{Z'})$ as the cohomology of the Koszul complex of x_1, x_2, x_3 on $\mathscr{O}_{Z'}$, a regular sequence there; so $\mathscr{E}xt^k = 0$ for $k \neq 3$ and $\mathscr{E}xt^3 = \mathscr{O}_{Z \cap Z'}$. The local-to-global spectral sequence has one row, $H^{k-3}(Z \cap Z', \mathscr{O})$, and twisting by line bundles does not change a sheaf on a finite reduced scheme; the intersection number equals $|Z \cap Z'|$ by transversality.

(iii) Let $\pi\colon A_0 \to A_0/Z$ be the quotient. The normal bundle N_Z is the restriction of the trivial bundle $\pi^* T_{A_0/Z}$, so it extends to all of A_0, and by the theorem of Arinkin and Căldăraru [1] the derived restriction $i^* i_* \mathscr{O}_Z$ is formal, $i^* i_* \mathscr{O}_Z \cong \bigoplus_p \bigwedge^p N_Z^\vee[p]$. Hence

$$\mathrm{Ext}^k(\mathscr{O}_Z \otimes L^a, \mathscr{O}_Z \otimes L^b) = \mathrm{Ext}^k_Z\big(i^* i_* \mathscr{O}_Z, L^{b-a}|_Z\big) = \bigoplus_{p+q=k} H^q\big(Z, L^{b-a}|_Z\big) \otimes \bigwedge^p N_Z. \tag{6.6}$$

For $a = b$ the dimension is $\sum_{p+q=k} \binom{3}{q}\binom{3}{p} = \binom{6}{k}$; for $b \neq a$ the cohomology of $L^{b-a}|_Z$ on the abelian threefold Z sits in degree 0 or 3 according to the sign of $b-a$, with dimension $|b-a|^3\chi_Z$ by Riemann–Roch, which gives the two ranges of degrees and their dimensions. The characteristic map sends $H^2(\mathscr{O}_{A_0})$ onto $H^2(\mathscr{O}_Z)$ by restriction, $H^1(T_{A_0}) = H^1(\mathscr{O}) \otimes T_0$ onto $H^1(\mathscr{O}_Z) \otimes N_{Z,0}$ through the quotient $T_0 \to N_{Z,0}$, and $H^0(\bigwedge^2 T_{A_0}) = \bigwedge^2 T_0$ onto $\bigwedge^2 N_{Z,0}$; all three are surjective, the first because $H^*(A_0) \to H^*(Z)$ is onto for an abelian subvariety.

(iv) By Grothendieck duality for the regular embedding, whose relative dualising sheaf $\det N_Z$ is trivial, and by adjunction,

$$\mathrm{Ext}^k(\mathscr{O}_Z \otimes L^a, L^b) = H^{k-3}\big(Z, L^{b-a}|_Z\big),$$
$$\mathrm{Ext}^k(L^a, \mathscr{O}_Z \otimes L^b) = H^k\big(Z, L^{b-a}|_Z\big).$$

On Z the cohomology of an ample bundle is concentrated in degree 0 and that of an anti-ample one in degree 3, of dimension $\chi = c_1^3/6$ by Riemann–Roch, and $c_1(L^m|_Z)^3 = m^3(\theta|_Z)^3$; for $m = 0$ one gets $H^*(\mathscr{O}_Z)$.

(v) is the same statement on A_0, with $\chi(L^m) = m^6\theta^6/720$. □

Lemma 6.9 (Threefolds that meet along a positive-dimensional set)**.** *Let $Z, Z' \subseteq A_0$ be distinct abelian threefolds, let $W = Z \cap Z'$, of dimension $d \in \{0,1,2\}$, and put $c = 3-d$ and $m = b-a$.*

(i) *$\mathscr{H}om(\mathscr{O}_Z, \mathscr{O}_{Z'}) = 0$, and $\mathscr{E}xt^q(\mathscr{O}_Z, \mathscr{O}_{Z'}) \cong \mathscr{O}_W^{\binom{d}{q-c}}$ for $c \le q \le 3$, zero otherwise.*

(ii) *$\mathrm{Ext}^k(\mathscr{O}_Z \otimes L^a, \mathscr{O}_{Z'} \otimes L^b) = 0$ unless $c \le k \le 3$ when $m > 0$, $3 \le k \le 3+d$ when $m < 0$, and $c \le k \le 3+d$ when $m = 0$. In particular the group vanishes in degrees $k \le 2$ when $m < 0$, and $\mathrm{Ext}^1 \neq 0$ only when $d = 2$ and $m \ge 0$, where $\mathrm{Ext}^1 = H^0(W, L^m|_W)$.*

(iii) *For the threefolds Z_ζ, $Z_{\zeta'}$ of Appendix B, d is the number of indices j with $\zeta_j = \zeta'_j$, W is the product of the shared curves with $\prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$ points, and for $d = 2$ and $m = 0$, $\dim \mathrm{Ext}^1 = |\zeta_l - \zeta'_l|^2 \in \{2,4\}$, l being the index at which they differ.*

Proof. (i) W is a closed subgroup scheme of A_0, reduced in characteristic zero, so a disjoint union of translates of an abelian variety of dimension d, and it is nonempty since Z and Z' contain the origin. At a point of W choose translation-invariant linear coordinates in which $Z = \{x_1 = x_2 = x_3 = 0\}$. The restrictions of x_1, x_2, x_3 to Z' are linear forms cutting out W, of codimension c in Z'; after a linear change $x_1, \dots, x_c$ restrict to a regular sequence on Z' with zero locus W and $x_{c+1}, \dots, x_3$ restrict to zero. The Koszul resolution of $\mathscr{O}_Z$ computes $\mathscr{E}xt^q(\mathscr{O}_Z, \mathscr{O}_{Z'})$ as the cohomology of the Koszul complex of these restrictions on $\mathscr{O}_{Z'}$, the tensor product of the Koszul complex of the regular sequence, with cohomology $\mathscr{O}_W$ in degree c, and of d complexes $\mathscr{O}_{Z'} \xrightarrow{0} \mathscr{O}_{Z'}$; so $\mathscr{E}xt^q = \bigwedge^{q-c}(\mathscr{O}_W^d)$. Since $c \ge 1$, $\mathscr{H}om = 0$.

(ii) The local-to-global spectral sequence has $E_2^{p,q} = H^p(W, \mathscr{E}xt^q \otimes L^m|_W)$ with $0 \le p \le d$ and $c \le q \le 3$. On each component of W the bundle $L^m|_W$ is ample for $m > 0$, with cohomology in degree 0 only, anti-ample for $m < 0$, with cohomology in degree d only, and trivial for $m = 0$. This gives the three ranges. For $k = 1$ the only possible term is $E_2^{0,1}$, nonzero only when $c = 1$ and $m \ge 0$, and it survives to E_∞ because $E_2^{1,0} = E_2^{2,0} = 0$.

(iii) In the coordinates of Appendix B, $Z_\zeta = \{u_j + \zeta_j v_j = 0,\ j = 1,2,3\}$; a shared index gives a form vanishing on $Z_{\zeta'}$, and an index with $\zeta_j \neq \zeta'_j$ gives the form $(\zeta_j - \zeta'_j)v_j$ on $Z_{\zeta'}$, whose zero locus in $C_{\zeta'_j}$ has $|\zeta_j - \zeta'_j|^2$ points by Lemma B.1(ii). So W is as stated, and $\mathrm{Ext}^1 = H^0(W, \mathscr{O}_W)$ has dimension the number of components of W. □

Theorem 6.1 applies at A_0 with the same number $r = 57$: the rank of the contraction of HT^2 into $\mathrm{ch}(\mathscr{E})$ is invariant under $\mathrm{SU}(V,H)(\mathbb{R})$, which fixes the Weil classes and θ and acts transitively on $\mathscr{D}_{3,3}$, so it is the rank computed at a very general point, and the argument of Buchweitz and Flenner is local on the base. An object $\mathscr{E}$ with $\dim \mathrm{Ext}^2(\mathscr{E}, \mathscr{E}) = 57$ at A_0 would therefore deform along all of $\mathscr{D}_{3,3}$ and prove the Weil classes algebraic on every member with full Hodge group. By (i) every Chern character of polarised shape is realised at A_0 by a convolution of pieces $\mathscr{O}_{Z_a} \otimes L^{k_a}[s_a]$ and $L^k[t]$, with at least two subvariety pieces, since w is not a multiple of one $[Z]$, and at least three powers of L, since the Hankel rank is three; by (ii) to (v) every group between two pieces is known. Appendix B writes the classes, the intersection numbers and every dimension of the lemma in coordinates for the threefolds that are products of elliptic curves of the three Weil surfaces, and exhibits the signed sum of thirty-two of them equal to $32(\alpha + \bar{\alpha})$. What the lemma decides is how the diagonal classes can disappear.

Proposition 6.10 (No cancellation in two steps)**.** *Let $\mathscr{E}$ be a convolution of pieces $F_a[s_a]$ of the types of Lemma 6.8, whose threefolds meet pairwise in finite sets, and no two of which are isomorphic sheaves placed at shifts differing by at most one. A class of $\mathrm{Ext}^2(F_a, F_a)$ is cancelled in the spectral sequence of the convolution only through a closed chain $a = a_0, a_1, \dots, a_m = a$ of pieces with $\mathrm{Ext}^{k_i}(F_{a_{i-1}}, F_{a_i}) \neq 0$ and $\sum_i k_i = m$, and no such chain has $m = 2$. The chains with $m = 3$ are exactly those through three twists $k < k' < k''$ of one threefold, $\mathscr{O}_Z \otimes L^k[s] \to \mathscr{O}_Z \otimes L^{k'}[s+1] \to \mathscr{O}_Z \otimes L^{k''}[s+2] \to \mathscr{O}_Z \otimes L^k[s]$, with degrees $0,0,3$. Hence every cancellation among the diagonal classes of degree two is a Massey product of length at least three.*

Proof. The Maurer–Cartan components of the twisted complex and the class η whose differential is computed all have total degree one, that is, a component in $\mathrm{Ext}^k(F_a, F_b)$ has $k + s_b - s_a = 1$. A product of m such components along a closed chain lands in $\mathrm{Ext}^{\sum k_i - m + 2}(F_a, F_a)$, since the shifts cancel around

the chain and the m-fold product lowers the degree by $m-2$; so it reaches degree two exactly when $\sum k_i = m$. For $m=2$ both components have degree at most two. Between two pieces that are not isomorphic sheaves, Lemma 6.8 gives the degrees: 3 and 3 between $\mathcal{O}_Z$ and $\mathcal{O}_{Z'}$; 0 to 3 one way and 3 to 6 the other between two twists of one $\mathcal{O}_Z$; 3 and 3 between $\mathcal{O}_Z \otimes L^a$ and L^b with $a<b$, 6 and 0 with $a>b$, and 3 to 6 against 0 to 3 for $a=b$; 0 and 6 between two powers of L. In every case the two degrees add up to at least three. Between two copies of one sheaf at shifts s, s' the two degrees are $1-s'+s$ and $1-s+s'$, which are both at most two only when $|s-s'|\le 1$, the case excluded. For $m=3$ the edges of degree at most two are, by the same list, the edges that raise the twist and the edges $L^a \to \mathcal{O}_Z \otimes L^a$; a closed chain cannot consist of them alone, so one edge has degree at least three and the other two have degree zero. Two edges of degree zero from a line bundle lead to a power of L or to a subvariety piece of strictly larger twist, from which every edge back to the line bundle has degree six; from $\mathcal{O}_Z \otimes L^k$ they lead only to $\mathcal{O}_Z \otimes L^{k''}$ with $k''>k$ through an intermediate twist k', and the edge back has degree three, through $H^3(Z, L^{k-k''}|_Z) \otimes \bigwedge^0 N_Z$. These are the chains displayed, and their three groups are nonzero.

□

For line-bundle pieces alone the same bookkeeping holds on E_0^6, and there the Massey products of length four are the ones that act, with the result $\dim \mathrm{Ext}^2 \ge 69$ Theorem A.32. With subvariety pieces and three powers of L no closed chain of any length exists, and the criterion fails for a reason that involves no product at all.

Proposition 6.11 (Convolutions with three line bundles fail)**.** *Let $\mathscr{E}$ be a convolution at A_0 of pieces $\mathcal{O}_{Z_a} \otimes L^{k_a}[s_a]$ and $L^{l_b}[t_b]$ in which no sheaf occurs twice, the threefolds Z_a are pairwise distinct, there are at most three line bundles, and $\mathrm{Ext}^1(\mathcal{O}_{Z_a} \otimes L^{k_a}, \mathcal{O}_{Z_b} \otimes L^{k_b}) = 0$ whenever $s_a = s_b$. Suppose $\mathrm{ch}(\mathscr{E})$ has the polarised shape. Then no term of the twisted differential carries a class of total degree one into a diagonal block, and*

$$\dim \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) \;>\; r(\mathrm{ch}\mathscr{E}) :$$

the criterion fails for $\mathscr{E}$. The hypothesis on Ext^1 holds when the threefolds meet pairwise in finite sets, and for the thirty-two threefolds Z_ζ of Appendix B placed at one twist with shifts of the parity of $\zeta_1\zeta_2\zeta_3$.

Proof. Step 1, no closed chain. As in the proof of Proposition 6.10, a term of the differential landing in a diagonal block runs along a closed chain $a_0, a_1, \dots, a_m = a_0$ of pieces with a nonzero component in $\mathrm{Ext}^{k_i}(F_{a_{i-1}}, F_{a_i})$, $k_i = 1 + s_{a_{i-1}} - s_{a_i}$, so that $\sum_i k_i = m$; the pieces of the chain are distinct, because the differential is strictly upper triangular in the one-sided order. We show that no such chain exists. Between two distinct threefold pieces $\mathrm{Hom} = 0$ by Lemma 6.9(i), and a component of degree one needs $s_a = s_b$, which the hypothesis excludes; so every edge between two threefold pieces has degree at least two. By Lemma 6.8(iv) and (v), an edge $L^l \to \mathcal{O}_Z \otimes L^k$ has degree 0 when $k>l$, 3 when $k<l$ and 0 to 3 when $k=l$, so a degree at most one forces $k \ge l$; an edge $\mathcal{O}_Z \otimes L^k \to L^l$ has degree 3 when $l>k$, 6 when $l<k$ and 3 to 6 when $l=k$, so a degree at most four forces $l \ge k$; and an edge between two line bundles has degree 0 when it raises the twist and 6 when it lowers it. A chain of threefold pieces alone has $\sum_i k_i \ge 2m > m$. Otherwise the chain visits $\ell \le 3$ line-bundle pieces; between consecutive visits it either passes directly, by an edge that raises the twist (u such edges in all) or lowers it (w in all), or runs through $r \ge 1$ threefold pieces, entered with degree e and left with degree x; let R be the number of runs. Then $R+u+w=\ell$, $m = \ell + \sum_{\mathrm{runs}} r$, a run contributes at least $e+x+2(r-1)$ to $\sum_i k_i$, and so

$$\sum_i k_i - m \;\ge\; \sum_{\mathrm{runs}} \big(e+x+r-3\big) + 5w - u,$$

where each bracket is at least $0+3+1-3=1$. If $w \ge 1$ then $u \le \ell - R - w \le 2-R$ and the right side is at least $R+5-(2-R) > 0$. If $w=0$ then $u = \ell - R \le 3-R$ and the right side is at least $2R-3$, so $R \le 1$; $R=0$ is a cycle of edges that all raise the twist, which cannot close; and for $R=1$ the single bracket must equal u, so $r \le 2$ and $e+x \le 5-r$, that is $(e,x) \in \{(0,3),(0,4),(1,3)\}$ for $r=1$ and $(e,x)=(0,3)$ for $r=2$. In both cases the run enters the first threefold, of twist k, from L^l with $k \ge l$, leaves the last one, of twist k', for $L^{l'}$ with $l' \ge k'$, and returns from $L^{l'}$ to L^l by $u \ge 1$ edges that raise

the twist, so $l' < l$ and $k' < k$. For $r = 1$ this reads $k < k$. For $r = 2$ the internal edge goes from the twist k to the strictly smaller twist k' and has degree two, which Lemma 6.9(ii) forbids. So there is no closed chain.

Step 2, the class $\mathrm{ev}_{\mathscr{E}}(v)$. By Step 1 the diagonal components of a class in $\mathrm{Ext}^2(\mathscr{E},\mathscr{E})$ are well defined, as in Proposition 6.5(ii), and for $v \in H^1(A_0, T_{A_0})$ the diagonal components of $\mathrm{ev}_{\mathscr{E}}(v)$ are the classes $\mathrm{ev}_{F_a}(v) = v \lrcorner \mathrm{At}(F_a)$ of the pieces, the Atiyah class of the twisted complex being block upper triangular with diagonal blocks $\mathrm{At}(F_a)$. For a piece $F = \mathscr{O}_Z \otimes L^k$ the semiregularity map sends $\mathrm{ev}_F(v)$ to $v \lrcorner \mathrm{ch}(F) = v \lrcorner [Z] + k\, v \lrcorner ([Z]\theta) + \cdots$, whose component of lowest degree is $v \lrcorner [Z]$. There is at least one threefold piece, since the Weil component of $\mathrm{ch}(\mathscr{E})$ is nonzero and powers of L contribute none. Step 3 gives $v \in T\mathscr{D}_{3,3}$ with $v \lrcorner [Z_a] \neq 0$ for such a piece, so $\mathrm{ev}_{\mathscr{E}}(v) \neq 0$; and $\sigma_{\mathscr{E}}(\mathrm{ev}_{\mathscr{E}}(v)) = v \lrcorner \mathrm{ch}(\mathscr{E}) = 0$, because $T\mathscr{D}_{3,3}$ annihilates θ and the Weil plane. So $\sigma_{\mathscr{E}}$ has a nonzero kernel, while its rank is at least the rank $r(\mathrm{ch}\mathscr{E})$ of $\sigma_{\mathscr{E}} \circ \mathrm{ev}_{\mathscr{E}}$; hence $\dim \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) \geq r(\mathrm{ch}\mathscr{E}) + 1$.

Step 3, no $[Z]$ is annihilated by $T\mathscr{D}_{3,3}$. The tangent space $T\mathscr{D}_{3,3} \subseteq H^1(A_0, T_{A_0}) = \mathrm{Hom}(V^{1,0}, V^{0,1})$ consists of the K-linear maps that are skew for the polarisation; it is the part $\mathfrak{g}^{-1,1}$ of the Lie algebra $\mathfrak{g}$ of $\mathrm{SU}(V,H)$ in the Hodge structure that A_0 induces on it, and the contraction $v \lrcorner$ on $H^*(A_0) = \bigwedge^* V_{\mathbb{C}}$ is the derivation by which $\mathfrak{g}_{\mathbb{C}}$ acts. Suppose $v \lrcorner [Z] = 0$ for all $v \in \mathfrak{g}^{-1,1}$. Since $[Z]$ is real it is annihilated by the conjugate space $\mathfrak{g}^{1,-1}$ as well, and $\mathfrak{g}^{-1,1} \oplus \mathfrak{g}^{1,-1}$, the off-diagonal blocks of $\mathfrak{sl}(V_+ \oplus V_-)$, generates $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}_6$; so $[Z]$ is invariant under the connected group $\mathrm{SU}(V,H)$, in particular under its rational points. Every $g \in \mathrm{SU}(V,H)(\mathbb{Q}) \subseteq \mathrm{GL}_K(V) = \mathrm{End}^0(A_0)^\times$ is a quasi-isogeny of A_0, because E_0 has complex multiplication by K; for an integer n with $\varphi = ng$ an isogeny, $\varphi^*[Z] = [Z]$ gives $\varphi_*[Z] = \varphi_*\varphi^*[Z] = \deg(\varphi)[Z]$, so the class of the abelian threefold $\varphi(Z)$, whose tangent space is $g\,T_Z$, is a positive multiple of $[Z]$. The group $\mathrm{SU}(V,H)(\mathbb{C}) \cong \mathrm{SL}_6(\mathbb{C})$ acts transitively on the three-dimensional subspaces of V_+, which determine the K-subspaces of V, the condition $g\,T_Z \cap T_Z = 0$ is Zariski open and nonempty in g, and $\mathrm{SU}(V,H)(\mathbb{Q})$ is Zariski dense in the connected group $\mathrm{SU}(V,H)$ [18, Cor. 18.3]. So some g makes $Z \cap \varphi(Z)$ finite; it contains the origin, and $(Z \cdot \varphi(Z)) = |Z \cap \varphi(Z)| \geq 1$ by Lemma 6.8(ii). But $(Z \cdot \varphi(Z))$ is a positive multiple of $[Z]^2$, and $[Z]^2$ is a multiple of $f^*([0]^2) = 0$ for the map $f\colon A_0 \to E_0^3$ of Lemma 6.8(i), since $[0]^2 = 0$ on E_0^3. This contradiction proves the claim, and the proposition.

For the last sentence of the statement: when the threefolds meet in finite sets every group between two of them sits in degree three by Lemma 6.8(ii); and two of the Z_ζ at the same shift have the same value of $\zeta_1\zeta_2\zeta_3$, hence differ in at least two coordinates, so $d \leq 1$ and $\mathrm{Ext}^1 = 0$ by Lemma 6.9(ii). □

Proposition 6.11 settles the configuration that Appendix B writes down, the thirty-two threefolds Z_ζ at one twist with three powers of L: no convolution of these thirty-five pieces meets the criterion. By Lemma 6.8(i) every object meeting the criterion at A_0 is a convolution of pieces $\mathscr{O}_Z \otimes L^k[s]$ and $L^l[t]$, so what remains are the convolutions with four or more line bundles, with a sheaf at two shifts, with a threefold at several twists, or with two threefold pieces at the same shift meeting along an abelian surface, as two of the Z_ζ with two equal coordinates do. Closed chains exist among them. With the threefolds at twist 0 and the line bundles of positive twist, every edge from a line bundle into a threefold piece and every edge out of one has degree three, so the shortest closed chain through a threefold piece runs through five line bundles at five consecutive shifts, $L^{l_1}[\sigma-2] \to L^{l_2}[\sigma-1] \to L^{l_3}[\sigma] \to L^{l_4}[\sigma+1] \to L^{l_5}[\sigma+2] \to \mathscr{O}_Z[\sigma] \to L^{l_1}[\sigma-2]$ with $0 < l_1 \leq \cdots \leq l_5$ and degrees $0,0,0,0,3,3$; one threefold at three twists carries the chain of Proposition 6.10; and two pieces $\mathscr{O}_{Z_\zeta} \otimes L^k[s]$, $\mathscr{O}_{Z_{\zeta'}} \otimes L^k[s]$ with two equal coordinates carry the chain of length two with degrees $1,1$, through the groups $H^0(W, \mathscr{O}_W)$ of Lemma 6.9(iii). The question at the stopping point is whether the diagonal components $\big(\mathrm{ev}_{F_a}(v)\big)_a$, $v \in T\mathscr{D}_{3,3}$, lie in the image of the products along such chains; the Hodge locus of the obstruction, the only other way the criterion could fail, is settled by Theorem 6.1 itself. The next subsection answers the question twice without computing a product: by counting the blocks that no chain reaches, and by observing that one threefold piece outside every closed chain decides it.

6.6. **Counting the remaining configurations at E_0^6**

The four kinds of convolution left open by Proposition 6.11 can be tested by a bound that uses no product at all. Fix pieces F_a with shifts s_a and a minimal A_∞-model of their full subcategory, so that $\mathrm{Hom}^\bullet(\mathscr{E},\mathscr{E})$ for a convolution $\mathscr{E}$ of the F_a is the direct sum of the blocks $\mathrm{Ext}^k(F_a,F_b)$ placed in total

TABLE 5. The isolated-block bound of Lemma 6.12 over the configurations at $A_0 = E_0^6$. "Polarised" counts the configurations of the family in the window whose Chern character has the polarised shape; "least" is the smallest bound among them, with the configuration attaining it. The criterion needs $\dim \mathrm{Ext}^2 = 57$.

family (least bound attained at)	polarised	least
thirty-five pieces, L^1, L^2, L^3 at shifts $0,0,0$	1	7213
the same at shifts $2,3,4$ or $-3,-2,-1$	2	2605
(a) four line bundles ($L^1[-2]$, $L^2[-1]$, $L^3[0]$, $L^4[1]$)	2401	0
(b) one threefold at a second shift	0	
(c) one threefold at three twists ($Z_{(1,1,1)} \otimes L^1$ at $[-1]$, $[0]$ and $\otimes L^2$ at $[1]$, $[2]$)	432	0
(d) a second threefold meeting Z_ζ along a surface, at the shift of Z_ζ and a neighbouring one ($Z_{(1,1,-1)}$ at $[0]$, $[1]$)	6	5325

degree $k + s_b - s_a$, with the differential $x \mapsto \sum_{p,q} m_{p+q+1}(\delta, \dots, \delta, x, \delta, \dots, \delta)$ built from the twisting cochain δ, whose components lie in the blocks of total degree one. Call a block $\mathrm{Ext}^k(F_a, F_b)$ of total degree two *isolated* when no block of total degree one maps to it and it maps to no block of total degree three along such a term: that is, there is no chain $a \to \cdots \to a'$ and $b' \to \cdots \to b$ of pieces with every step a nonzero block of total degree one and $(a', b') \neq (a, b)$ carrying a block of total degree one, and no such chain $a' \to \cdots \to a$, $b \to \cdots \to b'$ with (a', b') carrying a block of total degree three.

Lemma 6.12 (Isolated blocks)**.** *For every twisting cochain δ on the pieces F_a, the isolated blocks of total degree two inject into $\mathrm{Ext}^2(\mathscr{E}, \mathscr{E})$; so $\dim \mathrm{Ext}^2(\mathscr{E}, \mathscr{E})$ is at least the sum of their dimensions.*

Proof. An element of an isolated block is a cycle, since every term of its differential lands in a block of total degree three reached from it along a chain of components of δ, and there is none; and it is not a boundary, since every term of the differential of a class of total degree one lands in a block reached from that class along such a chain, and none reaches an isolated block. The blocks are direct summands, so the cycles of the isolated blocks meet the boundaries only in zero. □

The bound depends only on the Ext groups of Lemmas 6.8 and 6.9 and on the shifts, not on δ, so it can be computed over all configurations of a family at once. We did this for the configurations of the four kinds within a window of twists 1 to 4 and shifts -3 to 3 around the thirty-five pieces, keeping those whose Chern character has the polarised shape, and recorded for each the sums C^1, C^2, C^3 of the block dimensions in total degrees one to three, the Euler characteristic $C^2 - C^1 - C^3$ and the isolated blocks (Table 5). The programs are in the archive of the Data availability section.

Proposition 6.13 (What counting decides)**.** *At $A_0 = E_0^6$, with the pieces of Appendix B:*

(i) *For the thirty-five pieces of Proposition 6.11 with the three line bundles at any shifts in the window, $\dim \mathrm{Ext}^2(\mathscr{E}, \mathscr{E}) \geq 2605$ for every twisting cochain, by isolated blocks alone.*

(ii) *Adding to them a threefold $Z_{\zeta'}$ meeting Z_ζ along an abelian surface, at the shift of Z_ζ and at a neighbouring shift so that the Chern character keeps the polarised shape, gives $\dim \mathrm{Ext}^2 \geq 5325$: the configuration of two threefold pieces at the same shift meeting along a surface fails for every twisting cochain.*

(iii) *Adding one further copy of a threefold at a second shift within the window never gives the polarised shape.*

(iv) *For four line bundles and for one threefold at three twists the isolated blocks are 0 at the configurations of Table 5 and the Euler characteristic is negative throughout the window: the bound decides nothing there.*

Proof. (i), (ii) and (iv) are Lemma 6.12 applied to the block dimensions of Lemmas 6.8 and 6.9, enumerated over the window; (iii) is the enumeration of Chern characters, since a second copy of $\mathscr{O}_{Z_\zeta}$ at a shift of the other parity changes the Weil component by $\pm 2[Z_\zeta]$, which no choice of line bundles compensates. □

Proposition 6.13 closes the configuration (d) of two threefolds at one shift and every placement of the thirty-five pieces. For the configurations (a) and (c) the bound is silent, and a different argument

decides them: it suffices that one threefold piece lies outside every closed chain. The configuration (a) with more line bundles is decided in Section 6.7, by hand up to five and by the enumeration of the archive up to seven.

Let Γ be the directed graph whose vertices are the pieces F_a and which has an edge $a \to b$, for $a \neq b$, when the block of total degree one, $\mathrm{Ext}^{1+s_a-s_b}(F_a,F_b)$, is nonzero. The chains of Lemma 6.12 are its paths, a closed chain in the sense of Proposition 6.10 is one of its directed cycles, and the components of every twisting cochain δ sit on its edges. Call a piece *chain-free* when it lies on no directed cycle of Γ; this is a property of the pieces and their shifts alone.

Proposition 6.14 (A piece outside every closed chain)**.** *Let $\mathscr{E}$ be a convolution at A_0 of pieces $\mathscr{O}_{Z_a} \otimes L^{k_a}[s_a]$ and $L^{l_b}[t_b]$ whose Chern character has the polarised shape, and suppose that some threefold piece $F_a = \mathscr{O}_{Z_a} \otimes L^{k_a}[s_a]$ is chain-free. Then, for every twisting cochain δ,*

$$\dim \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) \;\geq\; r(\mathrm{ch}\mathscr{E}) + 1,$$

and the criterion fails for $\mathscr{E}$.

Proof. Let $\eta \in \mathrm{Hom}^1(\mathscr{E},\mathscr{E})$ have components η_{bc} in the blocks of total degree one, and consider the component of its differential in the block $\mathrm{Ext}^2(F_a,F_a)$. It is a sum of terms $m_{p+q+1}(\delta,\dots,\delta,\eta_{bc},\delta,\dots,\delta)$ with $p+q \geq 1$, and such a term is nonzero only if its arguments are composable and run from F_a back to F_a: the p components of δ on one side of η_{bc} along edges of Γ from a to b, the q components on the other side along edges from c to a, and $\eta_{bc} \neq 0$. If $b = c$, the $p+q \geq 1$ components of δ form a closed walk of positive length, which is impossible because δ is strictly upper triangular in the one-sided order. If $b \neq c$, then $b \to c$ is an edge of Γ, and $a \to \cdots \to b \to c \to \cdots \to a$ is a closed walk through a with at least two edges, whose consecutive vertices are distinct; its first return to a, with repeated vertices shortcut, is a directed cycle of Γ through a. Since F_a is chain-free, no term survives: the differential of every element of total degree one has zero component in $\mathrm{Ext}^2(F_a,F_a)$. Hence the map $\pi_a\colon \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) \to \mathrm{Ext}^2(F_a,F_a)$ that reads off the component of a cocycle in that block is well defined on cohomology.

As in the proof of Proposition 6.5(ii), the Atiyah class of the twisted complex is block upper triangular with diagonal blocks $\mathrm{At}(F_b)$, and the homotopy transfer to the minimal model only raises the block position, so it preserves the diagonal blocks of an upper triangular element; hence for $v \in H^1(A_0,T_{A_0})$ the class $\mathrm{ev}_{\mathscr{E}}(v)$ satisfies $\pi_a(\mathrm{ev}_{\mathscr{E}}(v)) = \mathrm{ev}_{F_a}(v) = v \lrcorner \mathrm{At}(F_a)$. By Step 3 of the proof of Proposition 6.11 there is $v \in T\mathscr{D}_{3,3}$ with $v\lrcorner[Z_a] \neq 0$, and then $\mathrm{ev}_{F_a}(v) \neq 0$, because the semiregularity map of F_a sends it to $v\lrcorner \mathrm{ch}(F_a)$, whose component of lowest degree is $v\lrcorner[Z_a]$. So $\mathrm{ev}_{\mathscr{E}}(v) \neq 0$, while $\sigma_{\mathscr{E}}(\mathrm{ev}_{\mathscr{E}}(v)) = v\lrcorner \mathrm{ch}(\mathscr{E}) = 0$, since $T\mathscr{D}_{3,3}$ annihilates θ and the Weil plane. The kernel of $\sigma_{\mathscr{E}}$ is nonzero and its rank is at least $r(\mathrm{ch}\mathscr{E})$, the rank of $\sigma_{\mathscr{E}} \circ \mathrm{ev}_{\mathscr{E}}$; the bound follows, and equality in Theorem 6.1 is excluded. □

Corollary 6.15 (Configurations with a chain-free piece)**.** *Let $\mathscr{E}$ be a convolution at A_0 whose Chern character has the polarised shape and whose pieces are the thirty-two threefolds $\mathscr{O}_{Z_\zeta}$ of Appendix B at twist 0, at the shifts s and $s+1$ of the parity of $\zeta_1\zeta_2\zeta_3$, together with further pieces, line bundles $L^l[t]$ and threefolds $\mathscr{O}_Z \otimes L^k[t]$, all of positive twist. The criterion fails for $\mathscr{E}$, for every twisting cochain, in each of the following cases.*

(i) *The further pieces are at most four line bundles, of pairwise distinct twists: then Γ has no directed cycle.*

(ii) *Every further piece sits at a shift at most $s+2$: then no edge of Γ ends at a piece of shift $s+1$, and the sixteen pieces there are chain-free.*

(iii) *The further pieces are at most five line bundles, at any shifts: then a cycle through a piece at shift σ needs five line bundles at the shifts $\sigma-2,\dots,\sigma+2$, and the sixteen pieces at one of the two shifts are chain-free.*

Both bounds are sharp: $L^l[s+3]$ enters every piece at shift $s+1$, and six line bundles $L^{l_1}[s-2],\dots,L^{l_6}[s+3]$ with $0 < l_1 \leq \cdots \leq l_6$ put each of the thirty-two pieces on a cycle of length six.

Proof. Along a directed cycle $a_0,a_1,\dots,a_m = a_0$ of Γ write $k_i = 1 + s_{a_{i-1}} - s_{a_i}$ for the degree of the block on the edge $a_{i-1} \to a_i$; the shifts cancel around the cycle, so $\sum_i k_i = m$. The degrees available

on the edges are read off from Lemmas 6.8 and 6.9. Two of the thirty-two pieces at one shift have the same value of $\zeta_1\zeta_2\zeta_3$, hence differ in at least two coordinates, so $d \le 1$ and the degrees between them are 2 and 3; as Hom $= 0$ between distinct threefolds and a degree one would need equal shifts, every edge between two of the thirty-two has degree at least 2. An edge from a line bundle of positive twist into a piece of twist 0 has degree 3, and so has an edge from a piece of twist 0 into a line bundle of positive twist; an edge into a piece of twist 0 from a threefold piece of positive twist has degree at least 3; an edge between two line bundles has degree 0 when it raises the twist, 6 when it lowers it, and $1+t-t' \ge 0$ between two copies of one line bundle at shifts t and t'. Parts (i) and (iii) use of the thirty-two pieces only that they have one twist, pairwise distinct threefolds and no Ext^1 at equal shifts, as in Proposition 6.11.

(i) A cycle among pieces of twist 0 alone has $\sum_i k_i \ge 2m$, so a cycle visits $\ell' \ge 1$ line-bundle pieces, $\ell' \le 4$. Between two consecutive visits it either runs through $r \ge 1$ pieces of twist 0, entered and left with degree 3 and with $r-1$ internal edges of degree at least 2, which contributes at least $2r+4$ to $\sum_i k_i$ and r to m, or it passes directly along an edge that raises the twist, u such edges in all, or lowers it, w in all. With R runs, $R+u+w=\ell'$ and

$$0 = \sum_i k_i - m \;\ge\; \sum_{\text{runs}} (r+4) + 6w - \ell' \;\ge\; 4R + 5w - u.$$

If $R \ge 1$ this gives $u \ge 4$ and $\ell' \ge 5$. If $R = 0$ it gives $w = 0$, and a cycle whose edges all raise the twist does not close. So Γ has no cycle; every threefold piece is chain-free, there is one, and Proposition 6.14 applies.

(ii) An edge $u \to v$ into a piece v at shift $s+1$ has degree $1 + s_u - (s+1) = s_u - s$, at most 2 when $s_u \le s+2$. No line bundle and no threefold piece of positive twist has an edge of degree at most 2 into a piece of twist 0, and an edge into v from one of the thirty-two would have degree 0 or 1, which do not occur. So no edge ends at v.

(iii) Let a cycle pass through a piece v of twist 0 at shift σ, and let q count its direct edges between two copies of one line bundle, so that $R+u+w+q = \ell' \le 5$ and, the q edges contributing at least 0, $0 \ge 4R + 5w - u - q$ as in (i). Since $R \ge 1$, this forces $R = 1$, $w = 0$, $u+q = 4$, $\ell' = 5$ and equality throughout: the only run is v itself, and the four direct edges have degree 0, so each raises the shift by one, while the two edges of degree 3 at v lower it by two. The five line-bundle pieces therefore sit at the shifts $\sigma-2, \dots, \sigma+2$, one at each, with twists that do not decrease along the shifts. Five line bundles occupy at most five shifts, and the five shifts around s differ from the five around $s+1$; so the pieces at one of the two shifts are chain-free.

For the sharpness, $\mathrm{Ext}^3(L^l, \mathscr{O}_{Z_\zeta}) \ne 0$ is the edge $L^l[s+3] \to \mathscr{O}_{Z_\zeta}[s+1]$; and the six line bundles carry the cycle $L^{l_1}[s-2] \to \cdots \to L^{l_5}[s+2] \to \mathscr{O}_{Z_\zeta}[s] \to L^{l_1}[s-2]$ through every piece at shift s and the cycle $L^{l_2}[s-1] \to \cdots \to L^{l_6}[s+3] \to \mathscr{O}_{Z_\zeta}[s+1] \to L^{l_2}[s-1]$ through every piece at shift $s+1$, each edge being one of the nonzero groups listed. □

Remark 6.16 (The configurations of Table 5)**.** Corollary 6.15(i) covers the configuration (a) of the table, (ii) covers (c), whose further twists sit at shifts at most $2 = s+2$, and (iii) covers every configuration of the thirty-two pieces with at most five line bundles. The programs of the archive rebuild Γ from the dimensions of Lemmas 6.8 and 6.9 and find, in the window of the table, no cycle in any of the 2401 configurations of (a); at least fifteen chain-free threefold pieces in each of the 3024 configurations of (c) with further twists $1 \le k < k' \le 3$, twenty-two at the configuration of the table; no edge into the sixteen pieces at shift 1 from the thirty-two or from the 1560 pieces of positive twist at shifts -3 to 2, while $L^l[3]$ and $\mathscr{O}_{Z_{(1,1,1)}} \otimes L^1[3]$ do enter; in each of the 28 812 configurations of four or five line bundles of twists 1 to 5 at shifts -3 to 3 one class of sixteen pieces chain-free, the other class, when it lies on cycles, having line bundles at its five shifts; and with $L^1[-2], \dots, L^6[3]$ every piece on a cycle. The Lean file checks the counts 32 and 22 of chain-free pieces at the configurations (a) and (c) of the table, the absence of entering edges from the line bundles L^1 to L^4 and the twists 1 to 3 of the sixty-four threefolds Z_ζ at shifts -3 to 2, the entering edges at shift 3 and the six-cycles. No statement above rests on these computations.

6.7. **Chains through line bundles**

Corollary 6.15 needs the threefolds at one twist and the further pieces of positive twist, and it stops at five line bundles because six of them put every threefold piece on a cycle. This subsection removes both restrictions. It allows the threefolds any twists and any shifts, it works for every polarisation of A_0 and hence at every discriminant, and in place of the absence of cycles it uses a property of the cycles that the Maurer–Cartan equation of the twisted complex forces. By hand this excludes every convolution of distinct threefolds and at most five line bundles; the enumeration of Remark 6.28, a computation of the archive, extends the exclusion to six and seven line bundles and to all but two arrangements of eight. The next subsection turns from counting pieces to the support of any object meeting the criterion.

Setting 6.17 (Threefolds and line bundles at A_0)**.** Throughout this subsection $\mathscr{E} = \mathscr{E}_\delta$ is a convolution at $A_0 = E_0^6$, in the sense of Proposition 6.5, of threefold pieces $\mathscr{O}_{Z_\zeta} \otimes L^{k_\zeta}[s_\zeta]$, $\zeta \in \mu_4^3$, and line-bundle pieces $L^{l_i}[t_i]$, where L now denotes a line bundle of class θ_h (defined below), such that (S1) each threefold Z_ζ occurs in at most one piece, (S2) the exponents l_i are pairwise distinct, and (S3) two threefold pieces at one shift differ in at least two coordinates of ζ, so that $\mathrm{Ext}^1 = 0$ between them by Lemma 6.9(ii) and (iii). The twists k_ζ, l_i and the shifts are arbitrary integers. For $h \in \mathbb{Z}^6_{>0}$ let $\theta_h = \sum_j (h_j a_j + h_{3+j} b_j)$ be the class of the hermitian form of Lemma 6.8 with coefficients h, so that $\theta_{(1,\dots,1)} = \theta$ is the class of L, let $\mathscr{D}_h$ be the family $\mathscr{D}_{3,3}$ of the discriminant of θ_h, which contains (A_0, θ_h), and suppose that $\mathrm{ch}(\mathscr{E})$ has the polarised shape with respect to θ_h: $\mathrm{ch}(\mathscr{E}) = N\omega + \sum_k c_k \theta_h^k$ with $N \neq 0$ and $0 \neq \omega \in W$. The Weil plane W of (B.4) does not depend on h. The class θ_h is that of a positive hermitian form, so a line bundle L of class θ_h is ample, and for $h = (1,\dots,1)$ it is the line bundle of Lemma 6.8. The proofs of Lemmas 6.8 and 6.9 use only that L is ample and Serre duality, so their statements on vanishing and nonvanishing hold for L of class θ_h word for word.

Remark 6.18 (Why the twist follows the polarisation)**.** If the pieces were twisted by powers of the bundle of class θ while the polarised shape is taken with respect to θ_h with h not constant, the shape would force $c_1 = \cdots = c_5 = 0$. Indeed, by Lemma 6.23 the part of weight zero of $\mathrm{ch}(\mathscr{E})$ in H^{2k} is $P_k \theta^k + R_{k-3} X_0 \theta^{k-3}$ with rational P_k, R_{k-3} ($R_{k-3} = 0$ for $k < 3$), and it must equal $c_k \theta_h^k$. In the basis of squarefree monomials $u_S = \prod_{x \in S} x$ in $a_1, a_2, a_3, b_1, b_2, b_3$ one has $\theta_h^k = k! \sum_{|S|=k} h_S u_S$, $h_S = \prod_{x \in S} h_x$, and $X_0 \theta^{k-3} = (k-3)! \sum_{|S|=k} t(S) u_S$, with $t(S)$ the number of three-element subsets of S meeting each pair $\{a_j, b_j\}$ once. Suppose $c_k \neq 0$. For $k = 1, 2$ all h_S are equal, so h is constant. For $k = 3$ the sets $\{a_j, b_j, x\}$ have $t = 0$, so $h_j h_{3+j} h_x$ does not depend on $x \notin \{a_j, b_j\}$, and h is constant. For $k = 4$ the three sets $\{a_j, b_j, a_l, b_l\}$ have $t = 0$, so $h_1 h_4 = h_2 h_5 = h_3 h_6 =: p$; the sets $\{a_j, b_j, x, y\}$ with x, y from the two other pairs have $t = 2$, so $p h_x h_y$ is independent of the choice, which gives $h_l = h_{3+l}$ for all l and then $h_l^2 = p$. For $k = 5$ every S has $t(S) = 4$, so $h_S = h_1 \cdots h_6 / h_x$, $x \notin S$, is independent of x. Then $\gamma = N\omega + c_0 + c_6 \theta_h^6$, the contraction has rank at most 48 and the first-order Hodge locus is larger than $T\mathscr{D}_h$. With L of class θ_h the same weight-zero parts read $P_k \theta_h^k + R_{k-3} X_0 \theta_h^{k-3}$, and the shape no longer forces $c_k = 0$: it asks $R_{k-3} X_0 \theta_h^{k-3} \in \mathbb{Q}\, \theta_h^k$, and then $c_k = P_k + \lambda R_{k-3}$ with the constant of proportionality λ. For $k = 4$ the condition is $R_1 = 0$, because the coefficient of $a_1 b_1 a_2 b_2$ is 0 in $X_0 \theta_h$ and $24 h_1 h_4 h_2 h_5 \neq 0$ in θ_h^4. The program `julia/chern_h.jl` of Appendix C checks both statements for $h \in \{1,2,3\}^6$ and computes the ranks 57, 48, 39, 30 of the contraction for γ of polarised shape, for $c_1 = \cdots = c_5 = 0$, and with in addition $c_0 = 0$ and $c_6 = 0$.

Remark 6.19 (The discriminant of θ_h)**.** The hermitian form of θ_h is $H_h = \mathrm{diag}(h_1, h_2, h_3, -h_4, -h_5, -h_6)$ up to a rational scalar, so the discriminant of $\mathscr{D}_h$ is $-h_1 \cdots h_6$ in $\mathbb{Q}^\times / N(\mathbb{Q}(i)^\times)$. When $h_1 \cdots h_6$ is a sum of two rational squares, in particular for $h = (1,\dots,1)$, $\mathscr{D}_h$ is the family of abelian sixfolds of Weil type for $\mathbb{Q}(i)$ of discriminant -1, on which Koike proved the Weil classes algebraic [49, Cor. 2.1]; an object at (A_0, θ_h) meeting the criterion would then give a new proof of a known case. Discriminant -1 is exactly the split Weil type for sixfolds, so Markman's theorem for abelian sixfolds of split Weil type [59, 60] covers the same families and no others. The cases not covered by any theorem known to us are those with $h_1 \cdots h_6$ not a sum of two squares, for instance $h = (3,1,1,1,1,1)$. Every exclusion of this subsection holds for every h.

Theorem 6.1 applies at (A_0,θ_h) with $r=r(\gamma)$ for every h, by the argument given before Proposition 6.10 with $\mathrm{SU}(V,H_h)$ in place of $\mathrm{SU}(V,H)$: the group fixes W and θ_h and acts transitively on $\mathscr{D}_h$. The tangent space $T\mathscr{D}_h\subseteq H^1(A_0,T_{A_0})$ consists of the K-linear maps skew for θ_h; it annihilates θ_h and W, and Step 3 of the proof of Proposition 6.11 holds for it word for word, $\mathrm{SU}(V,H_h)(\mathbb{Q})$ being again a group of quasi-isogenies of A_0 Zariski dense in $\mathrm{SU}(V,H_h)$: for every abelian threefold Z there is $v\in T\mathscr{D}_h$ with $v\lrcorner[Z]\neq 0$. Suppose now that the criterion holds for $\mathscr{E}$, that is $\dim\mathrm{Ext}^2(\mathscr{E},\mathscr{E})=r(\gamma)$; for γ of Hankel rank three, $r(\gamma)=57$. Then $\sigma_{\mathscr{E}}$ is injective, and for every $v\in T\mathscr{D}_h$ the class $\mathrm{ev}_{\mathscr{E}}(v)\in\mathrm{Ext}^2(\mathscr{E},\mathscr{E})$, which $\sigma_{\mathscr{E}}$ sends to $v\lrcorner\mathrm{ch}(\mathscr{E})=0$, vanishes. Everything below draws consequences from this vanishing.

Let Γ be the graph of Section 6.6 on the pieces F_a of $\mathscr{E}$, and let $\Gamma_\delta\subseteq\Gamma$ be the subgraph of the edges $a\to b$ on which the component δ_{ab} is nonzero. Since δ is strictly upper triangular in the one-sided order, Γ_δ has no directed cycle, and its walks are paths. The twisted complex lives over the minimal A_∞-model of the pieces, whose underlying graded space is $\bigoplus_{a,b}\mathrm{Ext}^*(F_a,F_b)$ with the block $\mathrm{Ext}^k(F_a,F_b)$ in total degree $k+s_b-s_a$. The cochain δ satisfies the Maurer–Cartan equation $\sum_{k\geq 2}m_k(\delta,\dots,\delta)=0$, and the differential of the twisted complex is $D(\eta)=\sum_{p,q\geq 0}m_{p+q+1}(\delta,\dots,\delta,\eta,\delta,\dots,\delta)$ with p components of δ before η and q after it. A term of either sum is nonzero only if its arguments are composable: along a path of Γ_δ for the Maurer–Cartan equation, and, for a term of $D(\eta)_{ad}$ with the argument η_{bc}, along a path of Γ_δ from a to b, the edge $b\to c$ of Γ, and a path of Γ_δ from c to d.

Lemma 6.20 (The cocycle of a tangent vector)**.** *For $v\in H^1(A_0,T_{A_0})$ the class $\mathrm{ev}_{\mathscr{E}}(v)$ is represented in the minimal model by a cocycle $c(v)$ whose component in the diagonal block $\mathrm{Ext}^2(F_a,F_a)$ is $\mathrm{ev}_{F_a}(v)=v\lrcorner\mathrm{At}(F_a)$ and whose component in a block (a,c) with $a\neq c$ is a sum of terms each of which is a composite along a path of Γ_δ from a to c. In particular $c(v)_{ac}=0$ when Γ_δ has no path from a to c.*

Proof. As in the proof of Proposition 6.5(i), the Atiyah class of the twisted complex is represented in the harmonic model by $[\nabla,\bar\partial+\delta]$, block upper triangular with diagonal blocks $\mathrm{At}(F_a)$ and off-diagonal blocks $\nabla\delta_{ab}$, which vanish off the edges of Γ_δ; so $v\lrcorner[\nabla,\bar\partial+\delta]$ has diagonal blocks $v\lrcorner\mathrm{At}(F_a)$ and off-diagonal blocks on the edges of Γ_δ only. The homotopy transfer $\xi\mapsto\sum_k(-h[\delta,\cdot])^k\xi$ to the minimal model composes each block of ξ with components of δ on either side, the homotopy h acting block by block; so every term of the component (a,c) of the transferred cocycle is a composite of one block of $v\lrcorner[\nabla,\bar\partial+\delta]$, sitting on a vertex or an edge of Γ_δ, with components of δ along edges of Γ_δ leading from a to it and from it to c. For $a=c$ only the term with $k=0$ occurs, since a composite of $k\geq 1$ components of δ from a back to a would be a cycle of Γ_δ. □

Call a directed cycle of Γ *effective* for δ when exactly one of its edges lies outside Γ_δ.

Proposition 6.21 (Every threefold piece lies on an effective cycle)**.** *Suppose $\mathscr{E}=\mathscr{E}_\delta$ meets the criterion. Then for every $v\in T\mathscr{D}_h$ there is η of total degree one with $D(\eta)=c(v)$, and for every piece F_a*

$$\mathrm{ev}_{F_a}(v)=\sum m_{p+q+1}(\delta,\dots,\delta,\eta_{bc},\delta,\dots,\delta),$$

the sum running over the effective cycles $a\to\cdots\to b\to c\to\cdots\to a$ of Γ through a, with $b\to c$ the edge outside Γ_δ and p, q the numbers of edges before and after it. In particular every threefold piece lies on an effective cycle whose edge outside Γ_δ carries a nonzero component of η.

Proof. The class $\mathrm{ev}_{\mathscr{E}}(v)$ vanishes, so its representative $c(v)$ of Lemma 6.20 is a coboundary $D(\eta)$. The diagonal component $D(\eta)_{aa}$ is the sum of the terms $m_{p+q+1}(\delta,\dots,\eta_{bc},\dots,\delta)$ whose arguments are composable from a to a: a path of Γ_δ from a to b, the edge $b\to c$ with $\eta_{bc}\neq 0$, and a path of Γ_δ from c to a. If $b=c$ the term is $m_1(\eta_{aa})=0$ when $p=q=0$, and otherwise its components of δ form a cycle of Γ_δ; so $b\neq c$. The two paths are increasing in the one-sided order, so they meet only in a; the edge $b\to c$ is not in Γ_δ, as otherwise Γ_δ would contain a cycle; and the walk is a directed cycle of Γ with exactly one edge outside Γ_δ. By Lemma 6.20 the left side is $c(v)_{aa}=\mathrm{ev}_{F_a}(v)$. For a threefold piece choose $v\in T\mathscr{D}_h$ with $v\lrcorner[Z_a]\neq 0$; then $\mathrm{ev}_{F_a}(v)\neq 0$, as in the proof of Proposition 6.14, so some term of the sum is nonzero. □

Lemma 6.22 (Two rules)**.** *Let $a = L^{l_a}[t]$, $b = L^{l_b}[t+1]$ and $c = L^{l_c}[t+2]$ be line-bundle pieces of $\mathscr{E}$ with $l_a < l_b < l_c$, so that $a \to b \to c$ is a path of Γ whose edges carry sections, $\delta_{ab} \in H^0(L^{l_b - l_a})$ and $\delta_{bc} \in H^0(L^{l_c - l_b})$.*

(i) *The path $a \to b \to c$ is not the only path of Γ_δ from a to c: either δ vanishes on one of its two edges, or Γ_δ has another path from a to c.*

(ii) *Suppose $a \to b \to c$ is the only path of Γ from a to c, let $v \in T\mathscr{D}_h$ and let η satisfy $D(\eta) = c(v)$. If $\delta_{bc} = 0 \neq \delta_{ab}$ then $\eta_{bc} = 0$, and if $\delta_{ab} = 0 \neq \delta_{bc}$ then $\eta_{ab} = 0$.*

Proof. (i) The component (a,c) of the Maurer–Cartan equation is the sum, over the paths of Γ_δ from a to c, of the products $m_k(\delta,\dots,\delta)$ along them. If $a \to b \to c$ were the only such path it would read $m_2(\delta_{ab},\delta_{bc}) = 0$, and m_2 on sections is their product, which is nonzero for two nonzero sections of line bundles on the integral variety A_0.

(ii) A term of $D(\eta)_{ac}$ runs along a walk of Γ from a to c made of a path of Γ_δ, one edge carrying η and a path of Γ_δ. If the walk repeats a vertex, the repetition is a vertex common to the two paths of Γ_δ, and shortcutting it leaves a path of Γ_δ from a to c; under the hypothesis this path is $a \to b \to c$ with both components nonzero, which is excluded in either case. So every term runs along a path of Γ from a to c, that is along $a \to b \to c$, and $D(\eta)_{ac} = \pm m_2(\delta_{ab},\eta_{bc}) \pm m_2(\eta_{ab},\delta_{bc})$. The right side $c(v)_{ac}$ vanishes by Lemma 6.20, since Γ_δ has no path from a to c. When $\delta_{bc} = 0 \neq \delta_{ab}$ this gives $\delta_{ab}\eta_{bc} = 0$, the product of the nonzero section δ_{ab} with $\eta_{bc} \in \mathrm{Ext}^0(L^{l_b},L^{l_c}) = H^0(L^{l_c - l_b})$, so $\eta_{bc} = 0$; the other case is the same with the roles exchanged. □

Lemma 6.23 (Both parities occur)**.** *Grade $H^*(A_0,\mathbb{C})$ by $\mathbb{Z}^3$, giving dz_j and $d\bar{z}_{3+j}$ the weight e_j and $d\bar{z}_j$, dz_{3+j} the weight $-e_j$. Then a_j, b_j, θ and every θ_h have weight 0, c_j has weight $2e_j$, $\bar{c}_j$ has weight $-2e_j$, W lies in the weights $\pm 2(e_1+e_2+e_3)$, and the weight-zero part of $[Z_\zeta]$ is $X_0 = \prod_j (a_j + b_j)$ for every ζ. Consequently $\sum_\zeta (-1)^{s_\zeta} = 0$ over the threefold pieces of $\mathscr{E}$: there are threefold pieces at even shifts and at odd shifts.*

Proof. The weights are read off from (B.1), (B.2) and (B.4), and in $[Z_\zeta] = \prod_j (a_j + b_j + \bar{\zeta}_j c_j + \zeta_j \bar{c}_j)$ a product of one term from each factor has weight zero only if it takes a_j or b_j from every factor, the e_j being independent. The part of $\mathrm{ch}(\mathscr{E})$ in H^6 is $\sum_\zeta (-1)^{s_\zeta}[Z_\zeta] + P\theta_h^3$ with $P = \sum_i (-1)^{t_i} l_i^3/6$, by Lemma 6.8(i) for L of class θ_h, and it equals $N\omega + c_3\theta_h^3$. Taking the parts of weight zero, with $Q = \sum_\zeta (-1)^{s_\zeta}$,

$$QX_0 + P\theta_h^3 = c_3\theta_h^3.$$

The squarefree cubic monomials in $a_1,b_1,\dots,a_3,b_3$ are linearly independent. The coefficient of $a_1b_1a_2$ is 0 in X_0 and $6h_1h_4h_2 \neq 0$ in θ_h^3, so $P = c_3$. The coefficient of $a_1a_2a_3$ is 1 in X_0 and $6h_1h_2h_3$ in θ_h^3, so $Q + 6Ph_1h_2h_3 = 6c_3h_1h_2h_3$. Hence $Q = 0$. There is a threefold piece, since $N\omega \neq 0$ and the line bundles contribute nothing to the Weil component; so there are pieces of both parities. □

The edges of Γ between line-bundle pieces are of two kinds, by Lemma 6.8(v): an *up* edge $L^l[s] \to L^{l'}[s+1]$ with $l < l'$, of degree 0, and a *down* edge $L^l[s] \to L^{l'}[s-5]$ with $l > l'$, of degree 6; in general an edge of degree k joins the shifts s and $s-(k-1)$. By Lemma 6.8(iv) an edge $L^l \to \mathscr{O}_Z \otimes L^k$ has degree 0 when $k > l$, 3 when $k < l$ and 0 to 3 when $k = l$, an edge $\mathscr{O}_Z \otimes L^k \to L^l$ has degree 3 when $l > k$, 6 when $l < k$ and 3 to 6 when $l = k$, and by Lemma 6.9(ii) and (S3) an edge between two threefold pieces has degree at least 2, and at least 3 when it lowers the twist. A *run* is a maximal path of Γ through threefold pieces from a line-bundle piece $L^l[s]$ to a line-bundle piece $L^{l'}[s-\beta]$; its *excess* is β, and it is *decreasing* when $l' < l$.

Lemma 6.24 (The shape of a cycle)**.** *Let C be a directed cycle of Γ through a threefold piece, with ℓ line-bundle pieces, $R \geq 1$ runs, u up edges and w down edges, so that $\ell = R + u + w$.*

(i) *A run through r threefold pieces has excess at least r, with equality only when the twists do not decrease along it. A decreasing run has excess at least 3, with equality only for $L^l \to \mathscr{O}_Z \otimes L^k \to \mathscr{O}_{Z'} \otimes L^{k'} \to L^{l'}$ with $k' \leq l' < l \leq k$ and degrees $0,3,3$, whose two threefold pieces have the parity of shift of $L^{l'}$.*

(ii) *$u = \sum_{\text{runs}} \beta + 5w$. If $\ell \leq 7$ then $w = 0$, exactly one run is decreasing, and $\ell \geq 2R+2$.*

(iii) *If $\ell \leq 5$ then $R = 1$: the line-bundle pieces of C sit at ℓ consecutive shifts, joined by up edges in the order of their exponents, and the run leads from the highest back to the lowest. If $\ell = 4$ the run is the one of (i) with excess 3, and both its threefold pieces have the parity of the lowest shift.*

Proof. (i) An edge of degree k lowers the shift by $k-1$. Along a run the entering edge has degree $e \geq 0$, the $r-1$ internal edges have degree at least 2 and the leaving edge has degree $x \geq 3$, so the excess is at least $-1+(r-1)+2 = r$, with equality only for $e = 0$, internal degrees 2 and $x = 3$, that is $l \leq k_1 \leq \cdots \leq k_r \leq l'$ for the twists. Let $l' < l$. With $r = 1$ the excess $(e-1)+(x-1)$ is at least 4: $e = 0$ forces $k \geq l > l'$, hence $x = 6$; $e = 3$ gives $2+(x-1) \geq 4$; and $e \in \{1,2\}$ forces $k = l > l'$, hence $x = 6$. With $r = 2$ and degrees e,d,x the excess $e+d+x-3$ equals 3 only for $(e,d,x) = (0,3,3)$: $e = 0$ forces $k_1 \geq l > l'$, and then either $k_2 \geq k_1 > l'$, so $x = 6$, or $k_2 < k_1$, so $d \geq 3$ and $x \geq 3$, with $x = 3$ exactly when $k_2 \leq l'$; $e = 3$ gives at least $3+2+3$; $e = 1$ forces $k_1 = l$, and then $k_2 \geq l > l'$ gives $x = 6$ while $k_2 < k_1$ gives $d \geq 3$, each at least 7; $e = 2$ gives at least $2+2+3$. With $r \geq 3$ the excess is at least $r+1 \geq 4$. Along the run with $(e,d,x) = (0,3,3)$ from L^l at shift s the shifts are $s+1$, $s-1$ and $s-3$.

(ii) The shifts return to their value around C: each up edge adds 1, each down edge subtracts 5 and each run subtracts its excess. Each run has excess at least 1, so $\ell = R+u+w = R+\sum\beta+6w \geq 2R+6w$, and $w \geq 1$ forces $\ell \geq 8$. Let $w = 0$. The exponents also return to their value, and the up edges raise them, so some run is decreasing. With $D \geq 1$ decreasing runs, of excess at least 3, $\ell = R+\sum\beta \geq R+3D+(R-D) = 2R+2D$; so $\ell \leq 7$ forces $D = 1$, and $\ell \geq 2R+2$.

(iii) $\ell \leq 5$ gives $R = 1$ by (ii), and $u = \ell-1$: the line-bundle pieces form a chain of $\ell-1$ up edges, each raising the shift by one and the exponent, closed by one run from the last to the first, which is decreasing. For $\ell = 4$ the run has excess $u = 3$, so it is the run of (i), and the parity of its threefold pieces is that of its exit, the lowest line bundle. □

Theorem 6.25 (Convolutions with at most five line bundles fail)**.** *In the Setting 6.17, if $\mathscr{E}$ has at most five line-bundle pieces then* $\dim \mathrm{Ext}^2(\mathscr{E},\mathscr{E}) > r(\gamma)$*: the criterion fails for $\mathscr{E}$, for every twisting cochain, for every twist of the threefolds and for every polarisation θ_h.*

Proof. Suppose the criterion holds. By Lemma 6.23 there are threefold pieces of both parities of shift, and by Proposition 6.21 each lies on an effective cycle carrying a nonzero component of η, in particular on a directed cycle of Γ. A cycle through a threefold piece has at least four line-bundle pieces by Lemma 6.24(ii), so with at most three there is none. Let there be four or five, $L_1,\ldots,L_n$ in the order of their exponents.

With four, every cycle through a threefold piece uses all four, at consecutive shifts in the order of the exponents, and its threefold pieces have the parity of the lowest shift, by Lemma 6.24(iii); so the threefold pieces on cycles have one parity, and a threefold piece of the other parity lies on no cycle.

With five, a cycle through a threefold piece has four or five line bundles, at consecutive shifts in the order of the exponents. If the five are not at five consecutive shifts in that order, no cycle uses all five, and two cycles of four, omitting L_m and $L_{m'}$ with $m \neq m'$, have the same lowest shift: the three line bundles common to both occupy three of four consecutive shifts in each; if these are consecutive, $x, x+1, x+2$, the omitted bundles sit at $x-1$ or at $x+3$, and at the same one of the two, since L_m at $x-1$ and $L_{m'}$ at $x+3$ would put the five at five consecutive shifts in the order of the exponents; and if they leave a gap, both omitted bundles fill it. So again the threefold pieces on cycles have one parity, and a threefold piece of the other parity lies on no cycle.

It remains to treat L_i at the shift $\sigma-3+i$, $i = 1,\ldots,5$. The cycles through threefold pieces are C_5 through all five, C_{1234} and C_{2345}, each a chain of up edges followed by one run. The only path of Γ from L_i to L_{i+2} is $L_i \to L_{i+1} \to L_{i+2}$: a path leaving the chain of up edges enters a run, which ends at least three shifts below its start, at L_1 or L_2, and from there the up edges lead only through line bundles already visited. By Lemma 6.22(i) applied to the three triples, δ vanishes on one of the edges $L_1 \to L_2$, $L_2 \to L_3$, on one of $L_2 \to L_3$, $L_3 \to L_4$, and on one of $L_3 \to L_4$, $L_4 \to L_5$. An effective cycle has exactly one edge outside Γ_δ. So C_5, with two such edges, is not effective; C_{1234} is effective only when its edge outside Γ_δ is $L_2 \to L_3$, and then $\delta_{12} \neq 0$, so that $\eta_{23} = 0$ by Lemma 6.22(ii); and C_{2345}

is effective only when that edge is $L_3 \to L_4$, with $\delta_{23} \neq 0$, so that $\eta_{34} = 0$. No effective cycle carries a nonzero component of η on its edge outside Γ_δ.

Each case contradicts Proposition 6.21. So the criterion fails, and since Theorem 6.1 gives $\dim \mathrm{Ext}^2 \geq r(\gamma)$ with equality excluded, $\dim \mathrm{Ext}^2 > r(\gamma)$. □

Lemma 6.26 (The shifts along a cycle form an interval)**.** *The shifts of the line-bundle pieces along a directed cycle of* Γ*, or along a path of* Γ *between two line-bundle pieces the second of which is at the higher shift, take every value between their least and their greatest. Hence, if the shifts of the line-bundle pieces of* $\mathscr{E}$ *do not form an interval of integers, the line bundles split along a gap into two groups such that every cycle of* Γ *through a threefold piece, and every path* $a \to b \to c$ *of Lemma 6.22, has its line bundles in one group.*

Proof. Between consecutive line-bundle pieces of a cycle or a path the shift rises by exactly 1, along an up edge, or falls, along a down edge or a run; from the least shift to the greatest it therefore passes through every intermediate value. The three line bundles of Lemma 6.22 are at three consecutive shifts. □

Lemma 6.27 (A sharper second rule)**.** *Let* $a = L^{l_a}[t]$, $b = L^{l_b}[t+1]$, $c = L^{l_c}[t+2]$ *be line-bundle pieces with* $l_a < l_b < l_c$*, suppose that exactly one of* δ_{ab}, δ_{bc} *vanishes, and let* N *be a set of edges of* Γ *on which every* η *with* $D(\eta) = c(v)$, $v \in T\mathscr{D}_h$*, vanishes. Suppose that* (H1) Γ_δ *has no path from* a *to* c*, and* (H2) *every path of* Γ *from* a *to* c *other than* $a \to b \to c$ *has at least two edges outside* Γ_δ*, or exactly one, lying in* N*. Then* η *vanishes on the edge of* $a \to b \to c$ *outside* Γ_δ*.*

Proof. By (H1) and Lemma 6.20, $c(v)_{ac} = 0$, so $D(\eta)_{ac} = 0$. A term of $D(\eta)_{ac}$ runs along a path of Γ_δ from a to x, the component η_{xy}, and a path of Γ_δ from y to c. If $x = y$, or $x \to y$ lies in Γ_δ, or the walk repeats a vertex, the walk contains a path of Γ_δ from a to c, which (H1) excludes; Γ_δ is acyclic, so a repeated vertex shortcuts to such a path. So every nonzero term runs along a path of Γ from a to c with exactly one edge outside Γ_δ, carrying η, and by (H2) the terms along paths other than $a \to b \to c$ vanish. What is left is $\pm m_2(\delta_{ab}, \eta_{bc})$ if $\delta_{bc} = 0$, or $\pm m_2(\eta_{ab}, \delta_{bc})$ if $\delta_{ab} = 0$, a product of two sections of line bundles on the integral variety A_0 one of which is nonzero; so the other vanishes. □

Lemma 6.22(ii) is the case $N = \emptyset$ in which $a \to b \to c$ is the only path of Γ. Edges found by Lemma 6.27 may be added to N and the lemma applied again.

Remark 6.28 (Six, seven and eight line bundles)**.** The rules used in the proof of Theorem 6.25 depend on the shifts of the line bundles and the order of their exponents alone. The program of the archive described in Appendix C runs through every arrangement of $n \leq 8$ line bundles whose shifts form an interval, builds the graph of up edges, down edges and runs, admitting every excess and every parity of threefold shift that the degrees of Lemmas 6.8 and 6.9 allow for a run, and, for every set of up edges on which δ is taken to vanish that Lemma 6.22(i) admits, records the parities of the threefold pieces on the cycles with at most one edge in that set whose edge in the set is not forced to carry $\eta = 0$ by Lemma 6.22(ii). Runs and down edges are taken to carry nonzero components, and the condition that the edge outside Γ_δ point downwards in the one-sided order is dropped; both choices enlarge the set of cycles counted, so an arrangement for which no choice reaches both parities is one for which Proposition 6.21 and Lemma 6.23 exclude every twisting cochain. The counts are: 75 arrangements of four line bundles and 541 of five, none reaching a parity; 4683 of six and 47 293 of seven, of which 4 and 140 reach one parity and none both; and 545 835 of eight, of which 31 reach both. By Lemma 6.26 an arrangement whose shifts do not form an interval splits into arrangements of fewer line bundles, each contributing the parities it reaches, and with at most eight line bundles at most one part has six or more, so at most one parity is reached. Hence, in the Setting 6.17, every convolution with at most seven line-bundle pieces fails the criterion for every polarisation θ_h, and so does every convolution with eight line bundles outside the thirty-one arrangements; those have their eight line bundles at four to seven consecutive shifts with at most three at one shift, the simplest being $L^{l_1}[0], L^{l_2}[1], L^{l_3}[2], L^{l_4}[3], L^{l_5}[3], L^{l_6}[4], L^{l_7}[5], L^{l_8}[6]$ with $l_1 < \cdots < l_8$. These two statements are the output of the enumeration and rest on it; Theorem 6.25 rests on nothing computed.

A finer enumeration reduces the thirty-one to two. It replaces Lemma 6.22(ii) by Lemma 6.27 and treats the support of δ on runs as a variable. Call an up edge, down edge or run between two line

bundles *live* when δ is nonzero on it, on every edge of the run in the last case, and let G be the set of live edges. If $\mathscr{E}$ meets the criterion then: (C1) G has no directed cycle, because Γ_δ has none; (C2) for line bundles a, c two shifts apart the paths of G from a to c are not exactly one path $a \to b \to c$ of up edges, by Lemma 6.22(i); (C3) the least set $\mathscr{Z}$ of dead up edges closed under Lemma 6.27, read on G, carries $\eta = 0$; and (C4) for each parity of shift some cycle through a run admitting that parity has exactly one edge outside G, and that edge is not in $\mathscr{Z}$, by Lemma 6.23 and Proposition 6.21. Each condition follows from the lemma cited, since paths of Γ_δ between line bundles project to paths of G and paths of G lift to walks of Γ_δ. Running through all 2^m sets G, $15 \le m \le 22$, for each of the thirty-one arrangements, twenty-nine admit no G satisfying (C1) to (C4), and two remain:

$$L^{l_1}[0], L^{l_2}[1], L^{l_3}[1], L^{l_4}[2], L^{l_5}[1], L^{l_6}[2], L^{l_7}[2], L^{l_8}[3] \quad \text{and}$$
$$L^{l_1}[0], L^{l_2}[1], L^{l_3}[1], L^{l_4}[2], L^{l_5}[2], L^{l_6}[3], L^{l_7}[3], L^{l_8}[4],$$

$l_1 < \cdots < l_8$, with 165 and 269 admissible sets G. Both have rank $\sum(-1)^{t_i} = 0$, so the eight arrangements of rank ± 2, where Proposition 6.39 below leaves only the products, are excluded, and so is the arrangement displayed above. The same procedure for nine line bundles, over 7 087 261 arrangements whose shifts form an interval, leaves 101, each one of the two above with a ninth line bundle added; by Lemma 6.26 a non-interval arrangement of nine leaves only one of the two with a ninth bundle across a gap. So the rules of this subsection cannot bound the number of line bundles: in both remaining arrangements every two line bundles two shifts apart are joined by two or more paths of up edges, Lemma 6.27 never applies, and a typical admissible G is a complex of sections $L^{l_1} \to L^{l_2} \oplus L^{l_3} \to L^{l_4} \oplus L^{l_5} \to L^{l_6} \oplus L^{l_7} \to L^{l_8}$ with the threefold pieces on runs $L^{l_6}, L^{l_7} \to \mathscr{O}_Z \otimes L^k \to \mathscr{O}_{Z'} \otimes L^{k'} \to L^{l_1}$ and $L^{l_8} \to \mathscr{O}_Z \otimes L^k \to \mathscr{O}_{Z'} \otimes L^{k'} \to L^{l_2}, L^{l_3}$, each with one dead edge carrying η. Deciding them needs the products $H^0(A_0, L^m) \otimes H^0(Z, L^k) \to H^0(Z, L^{m+k})$ and the higher products along these runs. Two independent implementations in `julia/line_bundle_fine.jl`, sharing only the table of runs, give the same counts.

6.8. **The support of an object meeting the criterion**

The results of this subsection do not count pieces. They start from one consequence of the criterion and read off where an object meeting it can be supported. Let $\mathscr{E}$ be a perfect complex on A_0 whose Chern character has the polarised shape with respect to θ_h and which meets the criterion, $\dim \mathrm{Ext}^2(\mathscr{E}, \mathscr{E}) = 57$, so that $\sigma_{\mathscr{E}}$ is injective. For $v \in T\mathscr{D}_h$ the obstruction $\mathrm{ob}_{\mathscr{E}}(v) = v \lrcorner \mathrm{At}(\mathscr{E}) \in \mathrm{Ext}^2(\mathscr{E}, \mathscr{E})$ to deforming $\mathscr{E}$ along v is sent by $\sigma_{\mathscr{E}}$ to $v \lrcorner \mathrm{ch}(\mathscr{E})$ [10], which vanishes because $T\mathscr{D}_h$ annihilates θ_h and W; so

$$\mathrm{ob}_{\mathscr{E}}(v) = 0 \qquad \text{for every } v \in T\mathscr{D}_h. \tag{6.7}$$

Everything below is drawn from (6.7). The open subsets $U \subseteq A_0$ are Zariski open; inside the proofs we pass to the analytic spaces, which changes no Ext group and no Atiyah class on the projective A_0 [78], and the cohomology with supports and the Chern characters with supports are those of the analytic spaces. For a closed subset Z of an open U we write $H^q_Z(U, F)$ and $\mathscr{H}^q_Z(F)$ for the cohomology and the local cohomology sheaves with supports in Z [41]. Two facts are used throughout: if Z is a submanifold of codimension c and F is locally free then $\mathscr{H}^q_Z(F) = 0$ for $q \ne c$ (depth below c, the Čech complex of c local equations above c), so that $H^{c+p}_Z(U, F) = H^p(U, \mathscr{H}^c_Z(F))$; and if $C \subseteq Z$ is closed of codimension at least $c+1$ then $\mathscr{H}^0_C(\mathscr{H}^c_Z(F)) = 0$, because this sheaf embeds into $\mathscr{H}^c_C(F)$, which vanishes. So a section of $\mathscr{H}^c_Z(F)$ that vanishes outside C vanishes.

For the threefolds we use the coordinates of Appendix B. Fix $\zeta \in \mu_4^3$ and $T = Z_\zeta$. The linear forms $n_m = z_m + \zeta_m z_{3+m}$ on the universal cover cut out the lift of T; their differentials dn_m, and the dz_m for $m \le 3$, are translation-invariant forms on A_0 and together a frame of $\Omega^1_{A_0}$. On a tubular neighbourhood N of T the n_m themselves are single-valued, and $\pi(z) = z - \sum_m \bar{\zeta}_m n_m(z) \partial_{3+m}$ retracts N onto T with $z_m \circ \pi = z_m$; the pair (n, π) identifies N with $T \times D^3$, and a holomorphic function on $V \times (D^*)^3$, for $V \subseteq T$ open and D^* the punctured disc, is a Laurent series in n_1, n_2, n_3 whose coefficients are holomorphic on V. The local cohomology $\mathscr{H}^3_T(\mathscr{O})$ is computed by the Čech complex of the cover of $N \setminus T$ by the three open sets $\{n_m \ne 0\}$, all of whose intersections are Stein; so on T the sheaf $\mathscr{H}^3_T(\mathscr{O})$ is the sheaf of Laurent tails $\sum_{a,b,c \ge 1} g_{abc} n_1^{-a} n_2^{-b} n_3^{-c}$ with $g_{abc} \in \mathscr{O}_T$, and $\mathscr{H}^3_T(F) = \mathscr{H}^3_T(\mathscr{O}) \otimes F$ for F locally free.

The part of $\mathscr{H}^3_T(\mathscr{O})$ annihilated by the ideal I_T is $\mathscr{O}_T\cdot[1/n_1n_2n_3]$. We write $\rho_T\colon \mathscr{H}^3_T(\mathscr{O})\to\mathscr{O}_T$ for the map sending the class of $\sum g_{abc}n_1^{-a}n_2^{-b}n_3^{-c}$ to g_{111}, a morphism of sheaves of vector spaces on T with $\rho_T(g\,[1/n_1n_2n_3])=g|_T$ for g holomorphic near T, extended coefficientwise to $\mathscr{H}^3_T(\Omega^q)$ in the frame above. For a coherent sheaf G on U supported in T its multiplicity $\mu_T(G)$ is the length of G at the generic point of T, the sum over $k\ge 0$ of the ranks of the $\mathscr{O}_T$-modules $I_T^kG/I_T^{k+1}G$; and $\mu_T(\mathscr{E}|_U)=\sum_q(-1)^q\mu_T(H^q(\mathscr{E}|_U))$ when every $H^q(\mathscr{E}|_U)$ is supported in T.

Lemma 6.29 (The Chern character with supports)**.** *Let U be open, $Z\subseteq U$ closed, and $\mathscr{F}$ a perfect complex on U whose cohomology sheaves are supported in Z.*

(i) The trace $\mathrm{R}\mathscr{H}om(\mathscr{F},\mathscr{F})\to\mathscr{O}_U$ factors through a unique morphism $\mathrm{tr}^Z\colon \mathrm{R}\mathscr{H}om(\mathscr{F},\mathscr{F})\to \mathrm{R}\Gamma_Z\mathscr{O}_U$. Setting $\mathrm{ch}^Z_k(\mathscr{F})=\mathrm{tr}^Z(\mathrm{At}(\mathscr{F})^k)/k!\in H^k_Z(U,\Omega^k)$ defines classes that forget to $\mathrm{ch}_k(\mathscr{F})$, are compatible with restriction to open subsets and with enlarging Z, are additive in exact triangles, and satisfy

$$v\lrcorner\mathrm{ch}^Z_{k+1}(\mathscr{F})=\frac{1}{k!}\,\mathrm{tr}^Z\big(\mathrm{ob}_{\mathscr{F}}(v)\circ\mathrm{At}(\mathscr{F})^k\big)\in H^{k+2}_Z(U,\Omega^k)$$

for $v\in H^1(U,T_U)$, where $\mathrm{ob}_{\mathscr{F}}(v)=v\lrcorner\mathrm{At}(\mathscr{F})$. In particular $\mathrm{ob}_{\mathscr{F}}(v)=0$ implies $v\lrcorner\mathrm{ch}^Z_{k+1}(\mathscr{F})=0$ for every k.

(ii) For $T=Z_\zeta$ and every open U meeting T the class $[T]:=\mathrm{ch}^T_3(\mathscr{O}_{T\cap U})$ in $H^0(U,\mathscr{H}^3_T(\Omega^3))$ is

$$[T]=c\Big[\frac{dn_1\wedge dn_2\wedge dn_3}{n_1n_2n_3}\Big],$$

with one constant $c\neq 0$, the same for all sixty-four threefolds.

(iii) If the cohomology sheaves of $\mathscr{F}$ are supported in $T\cap U$ then $\mathrm{ch}^T_3(\mathscr{F})=\mu_T(\mathscr{F})\,[T]$. If they are supported in a union Z of threefolds T_i of the sixty-four, any two meeting in dimension at most two, then $\mathrm{ch}^Z_3(\mathscr{F})=\sum_i\mu_{T_i}(\mathscr{F})[T_i]_Z$, where $[T_i]_Z$ is the image of $[T_i]$ in $H^3_Z(U,\Omega^3)$.

Proof. (i) Let $j\colon U\setminus Z\to U$. In the triangle $\mathrm{R}\Gamma_Z\mathscr{O}\to\mathscr{O}\to\mathrm{R}j_*\mathscr{O}_{U\setminus Z}$ the third term receives no nonzero morphism from a complex K whose cohomology sheaves are supported in Z, since $\mathrm{Hom}(K,\mathrm{R}j_*\mathscr{O}_{U\setminus Z})=\mathrm{Hom}(K|_{U\setminus Z},\mathscr{O}_{U\setminus Z})=0$; so $\mathrm{Hom}(K,\mathrm{R}\Gamma_Z\mathscr{O})\to\mathrm{Hom}(K,\mathscr{O})$ is bijective. The complex $\mathrm{R}\mathscr{H}om(\mathscr{F},\mathscr{F})$ has cohomology supported in Z, which gives tr^Z and its uniqueness; the same bijection applied to the sources $\mathscr{F}^\vee\otimes\mathscr{F}\otimes\Omega^k$ transports every identity of the trace to tr^Z: its compatibility with restriction to open subsets, with the morphism $\mathrm{R}\Gamma_Z\to\mathrm{R}\Gamma_{Z'}$ for $Z\subseteq Z'$, its additivity in triangles, and its graded cyclicity. The Atiyah class $\mathrm{At}(\mathscr{F})\in\mathrm{Ext}^1(\mathscr{F},\mathscr{F}\otimes\Omega^1)$ is compatible with restriction and with triangles, so the classes ch^Z_k have the listed properties, and cyclicity applied to the $k+1$ factors of $\mathrm{tr}^Z(\mathrm{At}^{k+1})$ gives $v\lrcorner\mathrm{tr}^Z(\mathrm{At}^{k+1})=(k+1)\,\mathrm{tr}^Z((v\lrcorner\mathrm{At})\circ\mathrm{At}^k)$, which is the displayed identity.

(ii) The trace is $\mathscr{O}_U$-linear and $\mathrm{At}(\mathscr{O}_T)^3$ is a morphism between $\mathscr{O}_T$-modules, so $[T]$ is annihilated by I_T on every open subset and is a section of $\mathscr{O}_{T\cap U}\,[1/n_1n_2n_3]\otimes\Omega^3$. By the compatibility with restriction it suffices to take $U=A_0$, where the coefficients are holomorphic functions on the compact connected T, hence constants: $[T]=\sum_a g_a[1/n_1n_2n_3]\,\omega_a$ with $g_a\in\mathbb{C}$ and ω_a running through the wedge products of three elements of the frame $dn_1,dn_2,dn_3,dz_1,dz_2,dz_3$. For $0<\lambda<1$ the map $\varphi_\lambda(z)=z-(1-\lambda)\sum_m\bar\zeta_m n_m(z)\partial_{3+m}$ is a biholomorphism of N onto a smaller tubular neighbourhood of T, fixes T pointwise, and satisfies $n_m\circ\varphi_\lambda=\lambda n_m$, $z_m\circ\varphi_\lambda=z_m$ and $\varphi_\lambda^*\mathscr{O}_T=\mathscr{O}_T$. The Atiyah class, the trace and the local cohomology are natural under biholomorphisms onto open subsets, and a section of $\mathscr{H}^3_T(\Omega^3)$ is determined by its restriction to any neighbourhood of T; so $\varphi_\lambda^*[T]=[T]$. If ω_a contains a of the dn_m, then φ_λ^* multiplies $g_a[1/n_1n_2n_3]\,\omega_a$ by λ^{a-3}; so $g_a=0$ unless $a=3$, and $[T]=c\,[dn_1\wedge dn_2\wedge dn_3/n_1n_2n_3]$. The automorphisms $z_{3+m}\mapsto i^{e_m}z_{3+m}$ of A_0 permute the sixty-four threefolds transitively and carry the forms n_m, dn_m of one to those of another, so c does not depend on ζ. Forgetting supports sends $[T]$ to $\mathrm{ch}_3(\mathscr{O}_T)\in H^3(A_0,\Omega^3)$, which is the class of T by the Riemann–Roch theorem for the embedding of T [33, Ch. 15] (the Chern character defined by the Atiyah class is the usual one on the projective A_0 [10]); and $[Z_\zeta]\theta^3=48\neq 0$ by Lemma B.1, so $c\neq 0$.

(iii) Both identities are between sections of a sheaf, so they may be checked locally on U and, by the second fact recalled above, off a closed subset of codimension at least four. For the first, ch^T_3

is additive and $\mathscr{F}$ is an iterated extension of its shifted cohomology sheaves, so let G be a coherent sheaf supported in T; near a point of T some $I_T^k G$ vanishes, and G is an iterated extension of the $\mathscr{O}_T$-modules $G_k = I_T^k G/I_T^{k+1}G$, of ranks r_k with $\sum_k r_k = \mu_T(G)$. Each G_k is locally free of rank r_k off a proper closed analytic subset of T, where it is locally $\mathscr{O}_T^{r_k}$ and $\mathrm{ch}_3^T(G_k) = r_k[T]$ by additivity and (ii). For the second, let C be the union of the pairwise intersections of the T_i, of codimension at least four. On $U \setminus C$ the T_i are disjoint, $\mathrm{R}\Gamma_Z = \bigoplus_i \mathrm{R}\Gamma_{T_i}$ by the Mayer–Vietoris triangle for supports, and $\mathscr{F} \cong \mathrm{R}\Gamma_Z\mathscr{F} = \bigoplus_i \mathrm{R}\Gamma_{T_i}\mathscr{F}$, a sum of perfect complexes with cohomology supported in the T_i, whose multiplicities along T_i are those of $\mathscr{F}$; so there $\mathrm{ch}_3^Z(\mathscr{F}) = \sum_i \mu_{T_i}(\mathscr{F})[T_i]_Z$ by additivity and the first identity. □

Theorem 6.30 (Locality)**.** *Let $\mathscr{E}$ be as above, let $T = Z_\xi$, and let $U \subseteq A_0$ be open with $T \cap U \neq \emptyset$, every cohomology sheaf of $\mathscr{E}|_U$ supported in T, and $\mu_T(\mathscr{E}|_U) \neq 0$. Then the union S of the components of $T \setminus U$ of codimension one in T is a nonempty ample divisor on T. In particular, if T is an irreducible component of the support of $\mathscr{E}$ along which $\mathscr{E}$ has nonzero multiplicity, then the other components of the support cut T in a set whose divisorial part is a nonempty ample divisor.*

The case of a shifted line bundle on $T \cap U$ is the case $\mu_T = \pm 1$ with one cohomology sheaf.

Proof. Let $v \in T\mathscr{D}_h$. Since T is a subgroup, its normal bundle is $\mathscr{O}_T \otimes N_0$ with $N_0 = T_0A_0/T_0T$, and we write $\kappa_T(v) \in H^1(T, \mathscr{O}_T) \otimes N_0$ for the image of v under $H^1(A_0, T_{A_0}) \to H^1(T, T_{A_0}|_T) \to H^1(T, N_T)$, and $\kappa_T(v)_m \in H^1(T, \mathscr{O}_T)$ for its image under $dn_m \colon N_0 \to \mathbb{C}$; on a cover of A_0 by open sets U_a with v represented by vector fields v_{ab}, the class $\kappa_T(v)_m$ is represented by the functions $dn_m(v_{ab})|_T$. By (6.7) and Lemma 6.29(i), (iii),

$$0 = v \lrcorner \mathrm{ch}_3^T(\mathscr{E}|_U) = \mu_T(\mathscr{E}|_U)\, v \lrcorner [T]$$

in $H_T^4(U, \Omega^2) = H^1(T \cap U, \mathscr{H}_T^3(\Omega^2))$. The right-hand side is represented by the cocycle $v_{ab} \lrcorner [T] = c\,[(\nu_1\, dn_2 \wedge dn_3 - \nu_2\, dn_1 \wedge dn_3 + \nu_3\, dn_1 \wedge dn_2)/n_1n_2n_3]$ with $\nu_m = dn_m(v_{ab})$, and ρ_T sends it to $c\,(\kappa_T(v)_1\, dn_2 \wedge dn_3 - \kappa_T(v)_2\, dn_1 \wedge dn_3 + \kappa_T(v)_3\, dn_1 \wedge dn_2)$ restricted to $T \cap U$. As $\mu_T(\mathscr{E}|_U)c \neq 0$, every $\kappa_T(v)_m$ restricts to zero on $T \cap U$; the dn_m are a basis of $N_0^\vee$, so $\kappa_T(v)$ lies in the kernel of the restriction $H^1(T, \mathscr{O}_T) \otimes N_0 \to H^1(T \cap U, \mathscr{O}_T) \otimes N_0$, for every $v \in T\mathscr{D}_h$.

We compute that kernel. Let C be the union of the components of $T \setminus U$ of codimension at least two in T, so that $T \cap U = (T \setminus S) \setminus C$. Since T is smooth, $\mathscr{O}_T$ has depth at least two along C, the local cohomology $H_C^1(T \setminus S, \mathscr{O})$ vanishes, and $H^1(T \setminus S, \mathscr{O}) \to H^1(T \cap U, \mathscr{O})$ is injective; so the kernel is that of $H^1(T, \mathscr{O}) \to H^1(T \setminus S, \mathscr{O})$, which is zero when $S = \emptyset$. Let $S \neq \emptyset$, let $H \subseteq T$ be the connected component of the stabiliser $\{x \in T : S + x = S\}$, an abelian subvariety, and let $q \colon T \to T' = T/H$. Each component of S is H-invariant, H being connected, so $S = q^{-1}(S')$ for the divisor $S' = q(S)$ of T', whose stabiliser is the finite group of components of the stabiliser of S; hence S' is ample [64, §6, Application 1] and $T' \setminus S'$ is affine. By Poincaré reducibility there is an abelian subvariety $H' \subseteq T$ with $H \times H' \to T$ an isogeny, and the composite $H \times H' \to T \to T'$ is the projection to H' followed by the isogeny $H' \to T'$. So the preimage of $T \setminus S$ in $H \times H'$ is $H \times U'$ with $U' \subseteq H'$ the preimage of the affine $T' \setminus S'$ under a finite map, which is affine. Pullback along the finite étale map $H \times U' \to T \setminus S$ is injective on $H^1(\mathscr{O})$, since composed with the trace it is multiplication by the degree, and by the Künneth formula and $H^1(U', \mathscr{O}) = 0$,

$$H^1(H \times U', \mathscr{O}) = H^1(H, \mathscr{O}) \otimes H^0(U', \mathscr{O}).$$

The composite $H^1(T, \mathscr{O}) \to H^1(H \times U', \mathscr{O})$ factors through $H^1(H \times H', \mathscr{O}) = H^1(H, \mathscr{O}) \oplus H^1(H', \mathscr{O})$, an isomorphism for the isogeny, followed by the restriction to $H \times U'$, which is injective on the first summand and zero on the second. So the kernel is the second summand, which is $q^*H^1(T', \mathscr{O})$: the classes of the invariant $(0,1)$-forms on T vanishing on the tangent space of H. Hence $\kappa_T(v)(\bar{u}) = 0$ for every $v \in T\mathscr{D}_h$ and every $u \in T_0H$, where $H = T$ when $S = \emptyset$.

Finally we compute $\kappa_T(v)$ in the coordinates of Appendix B. Write $v = \sum_{k,l} v_{lk}\, d\bar{z}_k \otimes \partial_l$, so that v replaces dz_l by $\sum_k v_{lk} d\bar{z}_k$. The K-linear maps have $v_{lk} \neq 0$ only when one of k, l is at most 3 and the other at least 4, and

$$v \lrcorner \theta_h = \tfrac{i}{2} \sum_{k<l} (h_l v_{lk} - h_k v_{kl})\, d\bar{z}_k \wedge d\bar{z}_l$$

vanishes exactly when $h_l v_{lk} = h_k v_{kl}$ for all $k < l$; so $T\mathscr{D}_h$ is the nine-dimensional space of the v with $v_{3+m,j} = Y_{mj}$ and $v_{j,3+m} = (h_{3+m}/h_j)Y_{mj}$ for $j,m \le 3$, Y any complex 3×3 matrix, and the other entries zero. The threefold $T = Z_\zeta$ is $\{z_j + \zeta_j z_{3+j} = 0,\ j = 1,2,3\}$, with tangent space spanned by $w_j = \zeta_j\partial_j - \partial_{3+j}$ and normal space N_0 with basis $f_m = [\partial_{3+m}]$, so that $[\partial_m] = \bar\zeta_m f_m$. The class $\kappa_T(v)$ is that of the invariant form $\sum_{k,l} v_{lk}\, d\bar z_k|_T \otimes [\partial_l]$, and on $\bar w_j = \bar\zeta_j\bar\partial_j - \bar\partial_{3+j}$, where $d\bar z_j(\bar w_j) = \bar\zeta_j$ and $d\bar z_{3+j}(\bar w_j) = -1$,

$$\kappa_T(v)(\bar w_j) = \sum_{m=1}^{3}\Big(\bar\zeta_j Y_{mj} - \frac{h_{3+j}}{h_m}\,\bar\zeta_m Y_{jm}\Big) f_m.$$

Let $0 \ne u = \sum_j u_j w_j \in T_0T$ with $u_a \ne 0$, choose $b \ne a$, and let Y have the entry $Y_{ab} = 1$ and all others 0. Then

$$\kappa_T(v)(\bar u) = \bar\zeta_b\Big(\bar u_b f_a - \frac{h_{3+a}}{h_b}\,\bar u_a f_b\Big) \ne 0.$$

So $T_0H = 0$: S is nonempty, H is trivial, $S = S'$ has finite stabiliser, and S is ample. □

Theorem 6.30 says where the rest of the support must cut a threefold of nonzero multiplicity; the next results say where it can lie. Fix ζ with $\zeta_1\zeta_2\zeta_3 = \pm 1$, let $T = Z_\zeta$, fix $j \in \{1,2,3\}$ with $\{j,k,l\} = \{1,2,3\}$, and let $\zeta^{(j)}$ be ζ with the sign of the j-th coordinate changed, $T' = Z_{\zeta^{(j)}}$. On the j-th factor B_j the curves C_{ζ_j} and $C_{-\zeta_j}$ meet in the set P_j of the four points of order two of C_{ζ_j} ($|2\zeta_j|^2 = 4$ by Lemma B.1), so

$$S := T \cap T' = P_j \times C_{\zeta_k} \times C_{\zeta_l}$$

is a disjoint union of four surfaces, and $Z = T \cup T'$ is cut out near S by $n_j n_j' = n_k = n_l = 0$, where $n_j' = z_j - \zeta_j z_{3+j}$ is the j-th linear form of T'. For $y \in C_{\zeta_k} \times C_{\zeta_l}$ let $\Gamma_y = C_{\zeta_j} \times \{y\} \subseteq T$ and $\Gamma_y' = C_{-\zeta_j} \times \{y\} \subseteq T'$; each meets S in the four points $P_j \times \{y\}$. On C_{ζ_j} one has $z_{3+j} = -\bar\zeta_j z_j$, so $n_j'|_{\Gamma_y} = 2z_j$ is a coordinate vanishing at the four points and $dn_j'|_{\Gamma_y} = 2\,dz_j$ is a nonzero invariant form; on $C_{-\zeta_j}$ likewise $n_j|_{\Gamma_y'} = 2z_j$ and $dn_j|_{\Gamma_y'} = 2\,dz_j$.

Lemma 6.31 (The curves through the four points)**.** *Let $y = (y_k, y_l) \in C_{\zeta_k} \times C_{\zeta_l}$ with $y_k \notin C_\xi$ for $\xi \ne \zeta_k$ and $y_l \notin C_\xi$ for $\xi \ne \zeta_l$, which excludes finitely many values of each coordinate. The threefolds Z_ξ, $\xi \in \mu_4^3$, meeting Γ_y or Γ_y' are those with $\xi_k = \zeta_k$ and $\xi_l = \zeta_l$; among the thirty-two with $\xi_1\xi_2\xi_3 = \pm 1$ they are T and T' alone.*

Proof. $Z_\xi \cap \Gamma_y = (C_{\xi_j} \cap C_{\zeta_j}) \times (\{y_k\} \cap C_{\xi_k}) \times (\{y_l\} \cap C_{\xi_l})$; the first factor contains the origin, and $y_k \in C_{\xi_k}$ for $\xi_k \ne \zeta_k$ only for the finitely many $y_k \in C_{\xi_k} \cap C_{\zeta_k}$. The same holds for Γ_y'. The condition $\xi_k = \zeta_k$, $\xi_l = \zeta_l$, $\xi_1\xi_2\xi_3 = \pm 1$ leaves $\xi_j = \pm\zeta_j$. □

Theorem 6.32 (Two threefolds meeting along a surface)**.** *Let $\mathscr{E}$ be as above and let $U \subseteq A_0$ be open such that every cohomology sheaf of $\mathscr{E}|_U$ is supported in $Z = T \cup T'$ and $\Gamma_y \cup \Gamma_y' \subseteq U$ for general $y \in C_{\zeta_k} \times C_{\zeta_l}$. Then $\mu_T(\mathscr{E}|_U) = \mu_{T'}(\mathscr{E}|_U) = 0$.*

Proof. Let $\mu = \mu_T(\mathscr{E}|_U)$ and $\mu' = \mu_{T'}(\mathscr{E}|_U)$, so that by Lemma 6.29(iii)

$$\mathrm{ch}_3^Z(\mathscr{E}|_U) = \mu[T]_Z + \mu'[T']_Z.$$

Let $v \in T\mathscr{D}_h$ be the vector whose matrix Y, in the notation of the proof of Theorem 6.30, is E_{jk}, the matrix with the single entry $Y_{jk} = 1$. By (6.7) and Lemma 6.29(i) the class $v \lrcorner \mathrm{ch}_3^Z(\mathscr{E}|_U)$ vanishes in $H_Z^4(U,\Omega^2)$. The Mayer–Vietoris sequence for supports

$$H_S^4(U,\Omega^2) \to H_T^4(U,\Omega^2) \oplus H_{T'}^4(U,\Omega^2) \to H_Z^4(U,\Omega^2)$$

is exact, so some $s \in H_S^4(U,\Omega^2)$ maps to the pair

$$(\mu\, v\lrcorner[T],\ \pm\mu'\, v\lrcorner[T']).$$

Near S the forms $dn_j, dn_j', dn_k, dn_l, dz_k, dz_l$ are a frame of Ω^1, the functions n_j, n_j', n_k, n_l are transverse coordinates to S, and the coordinates z_k, z_l of T restrict to coordinates on each component of S; as S is

smooth of codimension four, $H^4_S(U,\Omega^2) = H^0(S\cap U, \mathscr{H}^4_S(\Omega^2))$, and s is a Laurent tail

$$s = \sum_{\alpha,\beta,\gamma,\delta\geq 1} c_{\alpha\beta\gamma\delta}\, n_j^{-\alpha}\, n_j'^{-\beta}\, n_k^{-\gamma}\, n_l^{-\delta}$$

in each coefficient of the frame of Ω^2, the $c_{\alpha\beta\gamma\delta}$ holomorphic functions on $S\cap U$. The map $H^4_S(U,\Omega^2)\to H^4_T(U,\Omega^2) = H^1(T\cap U, \mathscr{H}^3_T(\Omega^2))$ forgets the support: since $\mathrm{R}\Gamma_S = \mathrm{R}\Gamma_S\mathrm{R}\Gamma_T$ it is the connecting map of $0\to\mathscr{H}^3_T(\Omega^2)\to j_*\mathscr{H}^3_T(\Omega^2)|_{T\setminus S}\to\mathscr{H}^1_S(\mathscr{H}^3_T(\Omega^2))\to 0$, which sends the tail s, read as a section over $(T\cap U)\setminus S$ of $\mathscr{H}^3_T(\Omega^2)$ whose coefficients have poles along S, to its class. Applying ρ_T to that class gives the class in $H^1(T\cap U,\mathscr{O}_T)$ of the principal parts $\sum_\beta c_{1\beta11}\, n_j'^{-\beta}$ along S, and symmetrically for T' with $\sum_\alpha c_{\alpha111}\, n_j^{-\alpha}$.

We compare the coefficients of $dn_j\wedge dn_l$. In $v\lrcorner[T']$ this coefficient is zero, because $[T'] = c\,[dn_j'\wedge dn_k\wedge dn_l/n_j'n_kn_l]$ contains no dn_j; in $v\lrcorner[T]$ it is $\pm c\,[dn_k(v)/n_jn_kn_l]$, and ρ_T sends $\mu\, v\lrcorner[T]$ to $\pm\mu c\,\kappa_T(v)_k$. Let $c_{\alpha\beta\gamma\delta}$ be the coefficients of $dn_j\wedge dn_l$ in s. Restricting to the curve $\Gamma'_y\subseteq U$, y general, the T' side gives that the class of the principal parts $\sum_\alpha c_{\alpha111}(p)\, n_j^{-\alpha}$ at the four points $p\in P_j\times\{y\}$ vanishes in $H^1(\Gamma'_y,\mathscr{O})$; by Serre duality its pairing with the holomorphic form $dn_j|_{\Gamma'_y}$, the sum of the residues $\sum_p c_{1111}(p)$, is zero. Restricting to $\Gamma_y\subseteq U$, the T side gives that the class of the principal parts $\sum_\beta c_{1\beta11}(p)\, n_j'^{-\beta}$ equals $\pm\mu c\,\kappa_T(v)_k|_{\Gamma_y}$ in $H^1(\Gamma_y,\mathscr{O})$. Pair both with $dn_j'|_{\Gamma_y}$. With the normalisation in which a class represented by a Dolbeault form ϕ pairs with ω to $(2\pi i)^{-1}\int\phi\wedge\omega$, a class given by principal parts g_p pairs to $\sum_p \mathrm{Res}_p(g_p\omega)$ up to a sign independent of the class; so the left side pairs to $\pm\sum_p c_{1111}(p) = 0$. The class $\kappa_T(v)_k$ is represented by the invariant form $\sum_b (v_{kb}+\zeta_k v_{3+k,b})\, d\bar z_b|_T$, since $dn_k(\partial_k) = 1$, $dn_k(\partial_{3+k}) = \zeta_k$ and dn_k kills the other ∂_m; on Γ_y only $d\bar z_j$ and $d\bar z_{3+j} = -\zeta_j d\bar z_j$ survive, and the entries of v for $Y = E_{jk}$ are $v_{k,3+j} = h_{3+j}/h_k$ and $v_{3+k,j} = 0$ among those that occur, so

$$\kappa_T(v)_k|_{\Gamma_y} = -\zeta_j\frac{h_{3+j}}{h_k}\, d\bar z_j|_{\Gamma_y},$$

$$\frac{1}{2\pi i}\int_{\Gamma_y}\kappa_T(v)_k\wedge dn_j' = -\frac{\zeta_j h_{3+j}}{\pi i\, h_k}\int_{C_{\zeta_j}} d\bar z_j\wedge dz_j \neq 0.$$

Hence $\mu = 0$. The coefficients of $dn_j'\wedge dn_l$ give $\mu' = 0$ in the same way, with the roles of T and T' exchanged: there $v\lrcorner[T]$ has coefficient zero, the restriction to Γ_y paired with dn_j' gives $\sum_p c'_{1111}(p) = 0$ for the coefficients c' of that component, and on Γ'_y, where $d\bar z_{3+j} = \zeta_j d\bar z_j$, the class $\kappa_{T'}(v)_k|_{\Gamma'_y} = \zeta_j(h_{3+j}/h_k)\, d\bar z_j$ pairs with dn_j to a nonzero number. □

Proposition 6.33 (Where the rest of the support lies)**.** *Let $\mathscr{E}$ be as above, let $\zeta_1\zeta_2\zeta_3 = \pm 1$, and suppose that $T = Z_\zeta$ lies in the support of $\mathscr{E}$ with $\mu_T(\mathscr{E}|_{U_T})\neq 0$, where U_T is the complement of the components of the support other than T. Then for each $j\in\{1,2,3\}$ there is an irreducible component R_j of the support of $\mathscr{E}$ of dimension at least three, not one of the thirty-two threefolds Z_ξ with $\xi_1\xi_2\xi_3 = \pm 1$, which meets $\Gamma_y\cup\Gamma'_y$ for every $y\in C_{\zeta_k}\times C_{\zeta_l}$; and $R_j\cap(T\cup T')$ has an irreducible component that maps onto $C_{\zeta_k}\times C_{\zeta_l}$.*

Proof. Let R be the union of the components of the support not contained in $T\cup T'$ and $U = A_0\setminus R$. Then every cohomology sheaf of $\mathscr{E}|_U$ is supported in $T\cup T'$ and $\mu_T(\mathscr{E}|_U) = \mu_T(\mathscr{E}|_{U_T})\neq 0$, so by Theorem 6.32 the curves $\Gamma_y\cup\Gamma'_y$ are not contained in U for general y: the closed set $R\cap(T\cup T')$ projects onto a dense, hence the whole, subset of $C_{\zeta_k}\times C_{\zeta_l}$ under the proper projection (z_k,z_l). Some irreducible component R_j of R then has $R_j\cap(T\cup T')$ projecting onto $C_{\zeta_k}\times C_{\zeta_l}$, so one of its components does, and R_j meets every $\Gamma_y\cup\Gamma'_y$. If $\dim R_j\leq 2$ then $R_j = R_j\cap(T\cup T')\subseteq T\cup T'$, which is excluded; so $\dim R_j\geq 3$. By Lemma 6.31 no Z_ξ with $\xi_1\xi_2\xi_3 = \pm 1$ other than T and T' meets Γ_y or Γ'_y for general y. □

Corollary 6.34 (Objects supported on the thirty-two threefolds)**.** *No perfect complex $\mathscr{E}$ on A_0 whose support lies in the union of the thirty-two threefolds Z_ξ, $\xi_1\xi_2\xi_3 = \pm 1$, has a Chern character of polarised shape with respect to some θ_h and meets the criterion. In particular no convolution of*

pieces $\mathscr{O}_{Z_\xi}\otimes L^k[s]$ over these threefolds, at any twists and shifts, does so; the same holds for the thirty-two threefolds with $\xi_1\xi_2\xi_3=\pm i$.

Proof. Suppose $\mathscr{E}$ has polarised shape and meets the criterion. For each ξ with $\xi_1\xi_2\xi_3=\pm1$ and each j, let U be the complement of the thirty threefolds other than Z_ξ and $Z_{\xi(j)}$. Every cohomology sheaf of $\mathscr{E}|_U$ is supported in $Z_\xi\cup Z_{\xi(j)}$, and $\Gamma_y\cup\Gamma'_y\subseteq U$ for general y by Lemma 6.31; so $\mu_{Z_\xi}(\mathscr{E})=0$ by Theorem 6.32. Two distinct threefolds meet in dimension at most two, so Lemma 6.29(iii) with $U=A_0$ and Z the union of the thirty-two gives $\mathrm{ch}_3^Z(\mathscr{E})=\sum_\xi\mu_{Z_\xi}(\mathscr{E})[Z_\xi]_Z=0$, and forgetting supports, $\mathrm{ch}_3(\mathscr{E})=0$. But $\mathrm{ch}_3(\mathscr{E})=N\omega+c_3\theta_h^3$ with $N\omega\neq0$, and $\omega\in W$ and θ_h^3 are linearly independent, W having the weights $\pm2(e_1+e_2+e_3)$ and θ_h^3 weight zero (Lemma 6.23). A convolution of pieces over the threefolds is supported in their union. The automorphism $z_4\mapsto iz_4$ of A_0 preserves θ, every θ_h and W, and exchanges the two families of thirty-two threefolds. □

Corollary 6.34 treats one family at a time; the threefolds of both families together need one more relation, read off from a line of four of them. Fix $j\in\{1,2,3\}$ with $\{j,k,l\}=\{1,2,3\}$ and $\xi_k,\xi_l\in\mu_4$. For $a\in\mu_4$ let $T_a=Z_x$ with $x_j=a$, $x_k=\xi_k$, $x_l=\xi_l$; two of the four lie in each family. Let $C_a=\{z_j+az_{3+j}=0\}\subseteq B_j$, let $Q=\bigcup_a C_a$, a divisor on B_j, and let $\Pi=C_{\xi_k}\times C_{\xi_l}\subseteq B'=B_k\times B_l$, a surface, so that $A_0=B_j\times B'$ and

$$Z:=\bigcup_{a\in\mu_4}T_a=Q\times\Pi.$$

Two of the T_a meet in $(C_a\cap C_b)\times\Pi$, a union of surfaces, so Lemma 6.29(iii) applies to Z. The divisor Q is ample: the connected component H of its stabiliser $\{x\in B_j:Q+x=Q\}$ is an abelian subvariety which, being connected, leaves each C_a invariant, so $H\subseteq C_1\cap C_i$, a finite set, and Q has finite stabiliser [64, §6, Application 1]. Hence $B_j\setminus Q$ is affine.

Lemma 6.35 (The four threefolds through a surface)**.** *Let $\mathscr{E}$ be as above and let $U\subseteq A_0$ be open such that every cohomology sheaf of $\mathscr{E}|_U$ is supported in Z and $Q\times\{y\}\subseteq U$ for general $y\in\Pi$. Then the multiplicities $\mu_a=\mu_{T_a}(\mathscr{E}|_U)$ satisfy*

$$\sum_{a\in\mu_4}\mu_a=0,\qquad\sum_{a\in\mu_4}a\,\mu_a=0,$$

that is, $\mu_a=ma^2$ for one integer m: $\mu_1=\mu_{-1}=-\mu_i=-\mu_{-i}$. In particular, if one of the four threefolds carries no cohomology sheaf of $\mathscr{E}|_U$, none of them has nonzero multiplicity.

Proof. Let $n_a=z_j+az_{3+j}$, $n_k=z_k+\xi_kz_{3+k}$, $n_l=z_l+\xi_lz_{3+l}$ and $f=\prod_a n_a$, so that near Z the set Z is cut out by $f=n_k=n_l=0$, a regular sequence, $Q=\{f=0\}$ in B_j and $\Pi=\{n_k=n_l=0\}$ in B'. As for one threefold, the pair (n_k,n_l) and the retraction along $\partial_{3+k},\partial_{3+l}$ identify a tubular neighbourhood of Π in B' with $\Pi\times D^2$, and the local cohomology $\mathscr{H}^3_Z(F)$ of a locally free sheaf F, computed by the Čech complex of the cover of the complement of Z by $\{f\neq0\}$, $\{n_k\neq0\}$, $\{n_l\neq0\}$, is the sheaf on Z of Laurent tails $\sum_{\beta,\gamma\geq1}g_{\beta\gamma}n_k^{-\beta}n_l^{-\gamma}$ whose coefficients $g_{\beta\gamma}$ are sections of $\mathscr{H}^1_Q(F)$: over $W\times V$, with $W\subseteq B_j$ and $V\subseteq\Pi$ open and Stein, a coefficient is a section of F on $(W\setminus Q)\times V$ modulo the sections on $W\times V$. In particular $\mathscr{H}^1_{C_a}(F)\subseteq\mathscr{H}^1_Q(F)$, and $[T_a]=c\,[dn_a\wedge dn_k\wedge dn_l/n_an_kn_l]_Z$ has the single tail $n_k^{-1}n_l^{-1}$ with coefficient $c\,[dn_a/n_a]\wedge dn_k\wedge dn_l$, where $[dn_a/n_a]\in H^0(C_a,\mathscr{H}^1_{C_a}(\Omega^1_{B_j}))$ is the class of the form dn_a/n_a with a pole along C_a.

Let $v\in T\mathscr{D}_h$ have the matrix $Y=E_{jk}$, as in the proof of Theorem 6.32: then v replaces dz_{3+j} by $d\bar z_k$ and dz_k by $(h_{3+j}/h_k)\,d\bar z_{3+j}$ and kills the other dz_m, so $dn_a(v)=a\,d\bar z_k$, $dn_k(v)=(h_{3+j}/h_k)\,d\bar z_{3+j}$ and $dn_l(v)=0$. By (6.7) and Lemma 6.29(i), (iii),

$$\begin{aligned}0=v\lrcorner\mathrm{ch}_3^Z(\mathscr{E}|_U)&=\sum_a\mu_a\,v\lrcorner[T_a]_Z\\&=c\sum_a\mu_a\Big[\frac{a\,d\bar z_k\,dn_k\wedge dn_l-(h_{3+j}/h_k)\,d\bar z_{3+j}\,dn_a\wedge dn_l}{n_an_kn_l}\Big]_Z\end{aligned}$$

in $H^4_Z(U,\Omega^2)=H^1(Z\cap U,\mathscr{H}^3_Z(\Omega^2))$, the classes of the invariant $(0,1)$-forms $d\bar z_k$, $d\bar z_{3+j}$ in $H^1(U,\mathscr{O})$ acting by cup product. The decomposition $\Omega^2_{A_0}=\Omega^2_{B_j}\oplus(\Omega^1_{B_j}\boxtimes\Omega^1_{B'})\oplus\Omega^2_{B'}$ into locally free summands

splits the local cohomology; $dn_k \wedge dn_l$ lies in the third summand and $dn_a \wedge dn_l$ in the second, so the second summand of the identity reads

$$[d\bar z_{3+j}] \cup \sum_a \mu_a \Big[\frac{dn_a \wedge dn_l}{n_a n_k n_l}\Big]_Z = 0$$

in $H^1\big(Z\cap U, \mathscr{H}^3_Z(\Omega^1_{B_j} \boxtimes \Omega^1_{B'})\big)$. Reading off the coefficient of $dn_l\, n_k^{-1} n_l^{-1}$ in the frame dn_k, dn_l, dz_k, dz_l of $\Omega^1_{B'}$ is a morphism of sheaves of vector spaces $\mathscr{H}^3_Z(\Omega^1_{B_j} \boxtimes \Omega^1_{B'}) \to \mathscr{H}^1_Q(\Omega^1_{B_j}) \boxtimes \mathscr{O}_\Pi$ on Z, linear over the functions pulled back from B_j; restricting to $Q \times \{y\} \subseteq Z \cap U$ for general y, where the $(0,1)$-form $d\bar z_{3+j}$ restricts to the same form on B_j, gives

$$[d\bar z_{3+j}] \cup \sum_a \mu_a\, [dn_a/n_a] = 0$$

in $H^1(Q, \mathscr{H}^1_Q(\Omega^1_{B_j})) = H^2_Q(B_j, \Omega^1_{B_j})$. Since $B_j \setminus Q$ is affine, $H^1(B_j \setminus Q, \Omega^1) = 0$, and the exact sequence $H^1(B_j \setminus Q, \Omega^1) \to H^2_Q(B_j, \Omega^1) \to H^2(B_j, \Omega^1)$ shows that forgetting the support is injective on $H^2_Q(B_j, \Omega^1)$; it is compatible with the cup product by $H^1(B_j, \mathscr{O})$. The argument of Lemma 6.29(ii) in codimension one, with the scaling $z \mapsto z - (1-\lambda)\bar a\, n_a(z)\partial_{3+j}$ of B_j, which fixes C_a, multiplies n_a by λ and preserves z_j, shows that $\mathrm{ch}_1^{C_a}(\mathscr{O}_{C_a}) = c_1[dn_a/n_a]$ in $H^0(C_a, \mathscr{H}^1_{C_a}(\Omega^1))$ with a constant c_1, the same for the four curves since $z_{3+j} \mapsto i z_{3+j}$ permutes them and carries n_a to n_{ia}, and $c_1 \neq 0$ since forgetting the support gives $\mathrm{ch}_1(\mathscr{O}_{C_a}) = [C_a] \neq 0$. So the image of $[dn_a/n_a]$ in $H^1(B_j, \Omega^1)$ is $c_1^{-1}[C_a]$, and

$$[d\bar z_{3+j}] \cup \sum_a \mu_a [C_a] = 0 \quad \text{in } H^2(B_j, \Omega^1) = H^{1,2}(B_j).$$

By Lemma B.1(i) with $(u,v) = (z_j, z_{3+j})$, $[C_a] = \frac{i}{2}(dz_j + a\, dz_{3+j}) \wedge (d\bar z_j + \bar a\, d\bar z_{3+j})$, so

$$[C_a] \wedge d\bar z_{3+j} = \tfrac{i}{2}\big(dz_j + a\, dz_{3+j}\big) \wedge d\bar z_j \wedge d\bar z_{3+j},$$

and the invariant forms $dz_j \wedge d\bar z_j \wedge d\bar z_{3+j}$ and $dz_{3+j} \wedge d\bar z_j \wedge d\bar z_{3+j}$ are linearly independent in $H^{1,2}(B_j)$. Their coefficients $\sum_a \mu_a$ and $\sum_a a\mu_a$ vanish. Separating real and imaginary parts of the second relation gives $\mu_1 = \mu_{-1}$ and $\mu_i = \mu_{-i}$, and the first then gives $\mu_i = -\mu_1$; so $\mu_a = \mu_1 a^2$, and if some μ_a vanishes all do. □

Theorem 6.36 (Objects supported on the sixty-four threefolds)**.** *No perfect complex $\mathscr{E}$ on A_0 whose support lies in the union of the sixty-four threefolds Z_x, $x \in \mu_4^3$, has a Chern character of polarised shape with respect to some θ_h and meets the criterion. In particular no convolution of pieces $\mathscr{O}_{Z_x} \otimes L^k[s]$ over threefolds of both families, at any twists and shifts, does so.*

Proof. Suppose $\mathscr{E}$ has polarised shape and meets the criterion, and write $\mu_x = \mu_{Z_x}(\mathscr{E})$. Fix j and (ξ_k, ξ_l) as above and let U be the complement of the sixty threefolds $Z_{x'}$ with $(x'_k, x'_l) \neq (\xi_k, \xi_l)$. Every cohomology sheaf of $\mathscr{E}|_U$ is supported in $Z = \bigcup_a T_a$, and $\mu_{T_a}(\mathscr{E}|_U) = \mu_x$ for $x_j = a$, the generic point of T_a lying in U. For $y = (y_k, y_l) \in \Pi$ with $y_k \notin C_\xi$ for $\xi \neq \xi_k$ and $y_l \notin C_\xi$ for $\xi \neq \xi_l$, which excludes finitely many values of each coordinate, $Z_{x'} \cap (Q \times \{y\}) = (C_{x'_j} \cap Q) \times (\{y_k\} \cap C_{x'_k}) \times (\{y_l\} \cap C_{x'_l})$ is empty unless $(x'_k, x'_l) = (\xi_k, \xi_l)$; so $Q \times \{y\} \subseteq U$ for general y, and Lemma 6.35 gives that μ_x / x_j^2 does not depend on x_j when x_k, x_l are fixed. This holds for $j = 1,2,3$, so $\mu_x/(x_1^2x_2^2x_3^2) = \mu_x/\sigma(x)^2$ is independent of each coordinate of x separately, hence constant: $\mu_x = m\,\sigma(x)^2$ for one integer m. Two distinct threefolds meet in dimension at most two, so Lemma 6.29(iii) with $U = A_0$ and Z the union of the sixty-four, followed by forgetting supports and (B.5) with $m = 2$ there, gives

$$\mathrm{ch}_3(\mathscr{E}) = \sum_x \mu_x [Z_x] = m \sum_{\omega \in \mu_4} \omega^2 S(\omega) = m \prod_{j=1}^3 T_{2,j} = 0.$$

But $\mathrm{ch}_3(\mathscr{E}) = N\omega + c_3\theta_h^3 \neq 0$, as in the proof of Corollary 6.34. A convolution of pieces over the threefolds is supported in their union. □

The theorem contains Corollary 6.34, since a complex supported on one family has $\mu_x = 0$ on the other, hence $m = 0$; the corollary is kept for its proof, which localises to two threefolds and is what Proposition 6.33 uses. The relations of Lemma 6.35 hold along every line of four threefolds in the

support, whatever else the support contains, as long as the rest of the support misses $Q \times \{y\}$ for general y.

Theorem 6.36 removes every convolution built from threefold pieces alone, of one family or of both, at any twists and shifts; it leaves the convolutions whose support reaches outside the sixty-four threefolds. For these the next two results say what the rest of the support can be. By (B.4) the Weil class is $\omega = (p+qi)\alpha + (p-qi)\bar{\alpha}$ with $p,q \in \mathbb{Q}$; for $x \in \mu_4^3$ let $\sigma(x) = x_1x_2x_3$ and $c(x) = (x_1^2, x_2^2, x_3^2) \in \{\pm 1\}^3$. The fibres of c are the eight orbits of the group $\{\pm 1\}^3$ acting on μ_4^3 by sign changes; $\sigma(x)^2 = \prod_j c(x)_j$, so the four orbits with $\prod_j c_j = 1$ make up the family $\sigma = \pm 1$ and the other four the family $\sigma = \pm i$.

Lemma 6.37 (The four orbits)**.** *Let $\nu\colon \mu_4^3 \to \mathbb{Q}$ satisfy $\sum_x \nu_x[Z_x] \equiv N\omega$ modulo $\theta_h H^4(A_0, \mathbb{C})$. Then for every $c \in \{\pm 1\}^3$*

$$\sum_{c(x)=c} \nu_x \sigma(x) = \begin{cases} Np/4, & \prod_j c_j = 1, \\ -iNq/4, & \prod_j c_j = -1. \end{cases}$$

So if $Np \neq 0$ each orbit of the family $\sigma = \pm 1$ carries an x with $\nu_x \neq 0$, and if $Nq \neq 0$ each orbit of the family $\sigma = \pm i$ does.

Proof. For $\varepsilon \in \{\pm 1\}^3$ let $X_\varepsilon = \prod_j (c_j \text{ if } \varepsilon_j = 1,\ \bar{c}_j \text{ if } \varepsilon_j = -1)$, of weight 2ε; X_ε spans the part of $H^6(A_0, \mathbb{C})$ of that weight, $H^4(A_0, \mathbb{C})$ has no class of weight $\pm 2\varepsilon$, and θ_h has weight zero, so the component of weight 2ε of $\theta_h H^4$ is zero and $\alpha = X_{(1,1,1)}$, $\bar{\alpha} = X_{(-1,-1,-1)}$. In $[Z_x] = \prod_j (a_j + b_j + \bar{x}_j c_j + x_j \bar{c}_j)$ the coefficient of X_ε is $\prod_{\varepsilon_j=1} \bar{x}_j \prod_{\varepsilon_j=-1} x_j = \sigma(x) \prod_{\varepsilon_j=1} \bar{x}_j^2 = \sigma(x) \prod_{\varepsilon_j=1} c(x)_j$, as $x_j^4 = 1$. Writing $W_c = \sum_{c(x)=c} \nu_x \sigma(x)$, the coefficient of X_ε in $\sum_x \nu_x [Z_x]$ is $\sum_c W_c \chi_\varepsilon(c)$ with $\chi_\varepsilon(c) = \prod_{\varepsilon_j=1} c_j$, and the χ_ε are the eight characters of $\{\pm 1\}^3$. The hypothesis says that this coefficient is $N(p+qi)$ for $\varepsilon = (1,1,1)$, $N(p-qi)$ for $\varepsilon = (-1,-1,-1)$ and 0 otherwise. Fourier inversion gives $8W_c = N(p+qi)c_1c_2c_3 + N(p-qi)$, which is the claim. □

Proposition 6.38 (A smooth complete intersection cannot carry the rest)**.** *Let $\mathscr{E}$ be as above and suppose that its support lies in $Y \cup \bigcup_{\sigma(x)=\pm 1} Z_x$, where $Y \subseteq A_0$ is a smooth complete intersection of dimension $d \leq 5$ of divisors in the linear systems $|L_h^{m_i}|$ of powers of the polarisation L_h. Then:*

- *(i) $Np \neq 0$, and every orbit of the family $\sigma = \pm 1$ contains a threefold $Z_x \not\subseteq Y$ along which $\mathscr{E}$ has nonzero multiplicity.*
- *(ii) If Y meets every Z_x in dimension at most one, then $\mathscr{E}$ does not exist: no such complex has polarised shape and meets the criterion.*

Proof. (i) Let $Y_1, \dots, Y_c$ be the divisors, $c = 6 - d \geq 1$. Each $A_0 \setminus Y_i$ is affine and smooth, as is every intersection of them, so has vanishing homology above degree six [2]; the Mayer–Vietoris spectral sequence of the cover of $A_0 \setminus Y$ by these c open sets gives $H_q(A_0 \setminus Y) = 0$ for $q > 5 + c$, and Lefschetz duality $H^q(A_0, Y) \cong H_{12-q}(A_0 \setminus Y)$ gives $H^q(A_0, Y) = 0$ for $q \leq d$. So $H^{2d-6}(A_0, \mathbb{Q}) \to H^{2d-6}(Y, \mathbb{Q})$ is onto, as $2d - 6 < d$. For $\eta \in H^{2d-6}(A_0)$ the projection formula gives $i_* i^* \eta = \eta \cup [Y] = \left(\prod_i m_i\right) \theta_h^c \eta$, so $i_* H^{2d-6}(Y) \subseteq \theta_h H^4(A_0)$. Now let $\Sigma = Y \cup \bigcup_{x \in I} Z_x$, I the set of x with $\sigma(x) = \pm 1$ and $Z_x \subseteq \operatorname{Supp} \mathscr{E}$, $Z_x \not\subseteq Y$. By the Riemann–Roch theorem for singular varieties [33, Ch. 18], applied to the cohomology sheaves of $\mathscr{E}$, which are coherent sheaves on infinitesimal neighbourhoods of Σ, the class $\operatorname{ch}_3(\mathscr{E}) \in H^6(A_0, \mathbb{Q})$ is the image of a class $\tau \in A_3(\Sigma)_\mathbb{Q}$ (the Todd class of A_0 being 1). The localisation sequence $A_3(Y) \to A_3(\Sigma) \to A_3(\Sigma \setminus Y) \to 0$ is exact, $A_3(\Sigma \setminus Y)$ is free on the classes of the three-dimensional components $Z_x \setminus Y$, $x \in I$, and the component of τ there is $\sum_{x \in I} \mu_{Z_x}(\mathscr{E})[Z_x \setminus Y]$, the top-dimensional part of the Riemann–Roch class of a sheaf being its cycle. Hence

$$\operatorname{ch}_3(\mathscr{E}) - \sum_{x \in I} \mu_{Z_x}(\mathscr{E})[Z_x] \in i_* H^{2d-6}(Y) \subseteq \theta_h H^4(A_0),$$

and $\operatorname{ch}_3(\mathscr{E}) = N\omega + c_3 \theta_h^3$ with $\theta_h^3 \in \theta_h H^4$. Lemma 6.37 applies to $\nu_x = \mu_{Z_x}(\mathscr{E})$ for $x \in I$ and $\nu_x = 0$ otherwise: since ν vanishes on the family $\sigma = \pm i$, $Nq = 0$, so $Np \neq 0$ as $N\omega \neq 0$, and each orbit of the family $\sigma = \pm 1$ contains $x \in I$ with $\mu_{Z_x}(\mathscr{E}) \neq 0$.

(ii) Take x from (i), $T = Z_x$. The multiplicity of $\mathscr{E}$ along T on the complement of the other components equals $\mu_{Z_x}(\mathscr{E}) \neq 0$, so Proposition 6.33 gives a component R_1 of the support, of dimension at

least three and not among the thirty-two threefolds, hence contained in Y, such that $R_1 \cap (Z_x \cup Z_{x^{(1)}})$ contains a surface. But $Y \cap (Z_x \cup Z_{x^{(1)}})$ has dimension at most one. □

The hypotheses of Proposition 6.38(ii) hold, for instance, when the rest of the support is a smooth curve or a smooth surface cut out by divisors of the polarisation and meeting the threefolds in finitely many points; the smooth divisors Y meet every Z_x in a surface and are only constrained by (i) and by Theorem 6.30. When the support is large the localised obstruction says nothing at all:

Proposition 6.39 (Where the localised obstruction stops)**.** *Let $\mathscr{E}$ be as above and suppose that its support contains an ample effective divisor Y. Then $v \lrcorner \mathrm{ch}^Z_{k+1}(\mathscr{E}|_U) = 0$ for every $v \in T\mathscr{D}_h$, every $k \geq 0$, every open $U \subseteq A_0$ and every closed $Z \subseteq U$ containing the support of $\mathscr{E}|_U$. In particular a convolution of line-bundle pieces $L^{l_i}[t_i]$ and threefold pieces with $\sum_i (-1)^{t_i} \neq 0$ has support A_0, and the criterion for it is decided by its products alone.*

Proof. Let Σ be the support of $\mathscr{E}$ and $X = A_0 \setminus Y$, an affine variety, so that $H^1(X, T_X) = 0$ and $H^{k+1}(X, \Omega^k) = 0$ for $k \geq 0$. The restriction of v to X vanishes, so the image of $v \lrcorner \mathrm{ch}^\Sigma_{k+1}(\mathscr{E})$ under $H^{k+2}_\Sigma(A_0, \Omega^k) \to H^{k+2}_{\Sigma\setminus Y}(X, \Omega^k)$, which is $v|_X \lrcorner \mathrm{ch}^{\Sigma\setminus Y}_{k+1}(\mathscr{E}|_X)$, vanishes; by the exact sequence

$$H^{k+2}_Y(A_0, \Omega^k) \to H^{k+2}_\Sigma(A_0, \Omega^k) \to H^{k+2}_{\Sigma\setminus Y}(X, \Omega^k)$$

the class $v \lrcorner \mathrm{ch}^\Sigma_{k+1}(\mathscr{E})$ comes from a class $b \in H^{k+2}_Y(A_0, \Omega^k)$. The sequence

$$H^{k+1}(X, \Omega^k) \to H^{k+2}_Y(A_0, \Omega^k) \to H^{k+2}(A_0, \Omega^k)$$

shows that b is determined by its image in $H^{k+2}(A_0, \Omega^k)$, which is $v \lrcorner \mathrm{ch}_{k+1}(\mathscr{E}) = 0$. So $v \lrcorner \mathrm{ch}^\Sigma_{k+1}(\mathscr{E}) = 0$, and its restriction to U, followed by the enlargement of supports from $\Sigma \cap U$ to Z, is $v \lrcorner \mathrm{ch}^Z_{k+1}(\mathscr{E}|_U)$. A complex of rank $\sum_i (-1)^{t_i} \neq 0$ has a cohomology sheaf of positive rank, whose support is A_0. □

What remains at E_0^6. By Theorem 6.36 every convolution of threefold pieces fails, of one family or of both, at any twists and shifts, which covers the threefolds at several twists of Setting 6.17; Theorem 6.25 and Remark 6.28 cover the convolutions with at most seven line bundles, eight line bundles outside two arrangements, and nine outside the arrangements containing one of the two. Left are the convolutions in those arrangements and with ten or more line bundles, with or without threefold pieces at any twists. Both remaining eight-bundle arrangements have rank zero, so the support is a proper closed subset containing every threefold piece; Theorem 6.30 and Propositions 6.33 and 6.38 constrain it, and the products decide the rest. The products along the chains of Proposition 6.10 and along the cycles through the line bundles are computable in the theta-function model of Section A.4, where the fourfold products of line-bundle pieces at distance at least two were found to give $\dim \mathrm{Ext}^2 \geq 69$; we have not computed them, and (F2) at E_0^6 stands or falls with them.

6.9. **Seven consecutive shifts at E_0^8**

At $n = 4$ the same route was taken in Section A.4 on E_0^8, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, with the 128 line bundles L_ζ, $\zeta \in \mu_4^4$, $\prod \zeta = \pm 1$, whose signed Chern characters add up to a multiple of the Weil class, and three multiples M_1, M_2, M_3 of the polarisation; the criterion asks for $\dim \mathrm{Ext}^2(E, E) = 104$. In the setting of Lemma A.33 the pieces of one parity, $\prod \zeta = \varepsilon$, have shifts of one parity and the pieces of the other parity have shifts of the other, and Corollary A.45 excludes every convolution whose 128 shifts lie in six consecutive integers. We extend this to seven. The groups used below are the groups $\mathscr{H}^k(W, V)$ between two pieces in the harmonic model of the convolution Lemma A.23, as in Theorem A.34 and Lemma A.36: for pieces W, V that differ in exactly three coordinates and have $d_W = d_V + 1$, the groups $\mathscr{H}^3(W, V)$, of total dimension 640 over the 28 ratios W/V of this kind, are reached by no coboundary, and a term of d_E on them passes through a piece at the shift $d_W + 1$, $d_W + 2$, $d_V - 1$ or $d_V - 2$.

Proposition 6.40 (Two separated adjacent pairs)**.** *Let E be a convolution as in Lemma A.33, let $g \geq 4$, and suppose that the shifts of the pieces L_ζ take exactly the four values s, $s-1$, $s-g$ and $s-g-1$. Then for every order, every choice of the shifts of the M_i and every placement of the t_i, $\dim \mathrm{Ext}^2(E, E) \geq 640$, and E does not meet the polarised criterion.*

Proof. Let P, Q, R, T be the sets of pieces at the shifts s, $s-1$, $s-g$, $s-g-1$; all four are nonempty. Two shifts that differ by one belong to pieces of opposite parity, so P and Q have opposite parities, and so do R and T; P and R have the same parity when g is even and opposite parities when g is odd. Let W and V differ in exactly three coordinates with $d_W = d_V + 1$. Then $W \in P$ and $V \in Q$, or $W \in R$ and $V \in T$, since these are the only pairs of occupied shifts at distance one. A term of d_E on $\mathscr{H}^3(W,V)$ passes through a piece at one of the shifts $s+1$, $s+2$, $s-2$, $s-3$ when $W \in P$, and $s-g+1$, $s-g+2$, $s-g-2$, $s-g-3$ when $W \in R$. Since $g \geq 4$ none of these is among s, $s-1$, $s-g$, $s-g-1$, so no term acts and the groups $\mathscr{H}^3(W,V)$ inject into $\mathrm{Ext}^2(E,E)$.

Let Γ be the group of ratios of two pieces, of order 128, acting simply transitively on the pieces, and let $S_3 \subset \Gamma$ be the 28 ratios that move exactly three coordinates, with the weight given by the dimensions of the groups $\mathscr{H}^3$; every ratio in S_3 changes the parity. Put $X = P \cup T$ if g is even and $X = P \cup R$ if g is odd. In both cases X is a union of one set of each parity, and its complement is the union of the other two. The edges of the Cayley graph of (Γ, S_3) join pieces of opposite parity, so the edges between X and its complement join P to Q and R to T, and these are exactly the pairs (W,V) above. Hence the dimensions of the surviving groups add up to the weight of the cut of X. The set X is neither empty nor everything, and the proof of Proposition A.44 shows, through Lemma A.37, that every such cut weighs at least 640. □

For $g = 4$ this is Proposition A.44. Together with the results of Section A.4 it settles most arrangements in seven consecutive integers.

Corollary 6.41 (Seven consecutive shifts)**.** *Let E be a convolution as in Lemma A.33 whose* 128 *pieces L_ζ have their shifts in $\{s, s-1, \dots, s-6\}$. Let $U \subseteq \{s, s-2, s-4, s-6\}$ and $U' \subseteq \{s-1, s-3, s-5\}$ be the values taken by the two parities. Then $\dim \mathrm{Ext}^2(E,E) \geq 640$ for every order, every choice of the shifts of the M_i and every placement of the t_i, except possibly when U is one of*

$$\{s, s-2, s-6\}, \quad \{s, s-4, s-6\}, \quad \{s, s-2, s-4, s-6\},$$

U' is one of

$$\{s-1\}, \quad \{s-3\}, \quad \{s-5\}, \quad \{s-1, s-5\},$$

and (U, U') is neither $(\{s, s-2, s-6\}, \{s-5\})$ nor $(\{s, s-4, s-6\}, \{s-1\})$.

Proof. If the shifts do not take both values s and $s-6$ they lie in six consecutive integers and Corollary A.45 applies; so $s, s-6 \in U$. If U' contains two values two apart it is $\{s-1, s-3\}$, $\{s-3, s-5\}$ or $\{s-1, s-3, s-5\}$, and Theorem A.38 applies to the parity at U' for every U. Otherwise U' is $\{s-1\}$, $\{s-3\}$, $\{s-5\}$ or $\{s-1, s-5\}$. Let $U = \{s, s-6\}$. For $U' = \{s-3\}$ the two parities sit at $s-3$ and at $s-3 \pm 3$, and Theorem A.39 gives $\dim \mathrm{Ext}^2 \geq 67584$. For $U' = \{s-1\}$, Proposition A.40 with $y = s-1$ applies, since the pieces at $y+1 = s$ exist and no piece has the shift $s-2$ or $s+2$; for $U' = \{s-5\}$ its second form with $y = s-5$ applies, since the pieces at $s-6$ exist and no piece has the shift $s-4$ or $s-8$. For $U' = \{s-1, s-5\}$ the shifts take exactly the values s, $s-1$, $s-5$, $s-6$, and Proposition 6.40 applies with $g = 5$. The same two applications of Proposition A.40 settle $U = \{s, s-4, s-6\}$ with $U' = \{s-1\}$, where no piece has the shift $s-2$ or $s+2$, and $U = \{s, s-2, s-6\}$ with $U' = \{s-5\}$, where no piece has the shift $s-4$ or $s-8$. □

6.10. **Weil sixfolds from cyclic triple covers of curves of genus four**

Abelian covers of curves give explicit Weil families. For an étale cyclic cover of degree $\ell \in \{3, 4, 6\}$ of a curve B of genus four, defined by a line bundle η of order ℓ, the pairs (B, η) have nine moduli, as many as the dimension of the domain $\mathscr{D}_F$, $F = \mathbb{Q}(\zeta_\ell)$, $n = 3$, that receives the period map, and the image of this map is open near every point at which the multiplication map

$$\mu_\eta : H^0(K_B \otimes \eta) \otimes H^0(K_B \otimes \eta^{-1}) \longrightarrow H^0(K_B^{\otimes 2}) \tag{6.8}$$

is surjective (proof of Proposition 6.42(iv)). For $\ell = 3$ the Prym map of these covers is generically finite, by Faber [29], and surjectivity of μ_η at a general point follows. We exhibit an explicit point at which μ_η is an isomorphism.

Proposition 6.42 (An explicit cover)**.** *For $\lambda \in \mathbb{C} \setminus \{0, 1\}$ let $B_\lambda \subset \mathbb{P}^1 \times \mathbb{P}^1$ be the curve $x^3 + y^3 + 1 + \lambda x^3 y^3 = 0$, in affine coordinates x, y, and $\eta = \mathscr{O}(1, -1)|_{B_\lambda}$.*

(i) *B_λ is a smooth curve of genus four with $K_{B_\lambda} = \mathscr{O}(1,1)|_{B_\lambda}$.*

(ii) *η has order exactly three.*

(iii) *$H^0(K\otimes\eta)$ and $H^0(K\otimes\eta^{-1})$ are spanned by $1,x,x^2$ and by $1,y,y^2$, and μ_η is an isomorphism.*

(iv) *The period map of the étale cyclic triple covers of curves of genus four, with values in the domain $\mathscr{D}_F$ of $(\mathbb{Q}(\sqrt{-3}),3)$-Weil type that contains the χ-part of H^1 of the cover $C_\lambda \to B_\lambda$ defined by η, is a local isomorphism at (B_λ,η). Its image contains an open subset of $\mathscr{D}_F$, and the Weil classes are algebraic on every member of $\mathscr{D}_F$ as soon as they are algebraic on the abelian sixfolds $H_\chi(C_t)$ for the covers C_t of a non-meagre subset of a neighbourhood of (B_λ,η).*

Proof. (i) In bihomogeneous coordinates the curve is $F = x_1^3y_0^3+x_0^3y_1^3+x_0^3y_0^3+\lambda x_1^3y_1^3$, of bidegree $(3,3)$. In the chart $x_0=y_0=1$ the partial derivatives are $3x^2(1+\lambda y^3)$ and $3y^2(1+\lambda x^3)$. At a singular point with $x=0$ the equation gives $y^3=-1$ and then $\partial_y F = 3y^2 \neq 0$; similarly $y\neq 0$. So $x^3=y^3=-1/\lambda$, and the equation becomes $1-1/\lambda=0$, excluded. The equation is symmetric in x and y; the substitution $(x,y)=(1/x',\lambda^{-1/3}y')$, after multiplication by x'^3, carries it to the same form in x',y' with λ replaced by $1/\lambda$; and the substitution $(x,y)=(\lambda^{-1/3}/x',\lambda^{-1/3}/y')$, after multiplication by $\lambda x'^3y'^3$, carries it to itself. So the other three charts reduce to the first. By adjunction $K_B = (K_{\mathbb{P}^1\times\mathbb{P}^1}+B)|_B = \mathscr{O}(1,1)|_B$, of degree $(1,1)\cdot(3,3)=6=2g-2$, so $g=4$.

(ii) Let D_x and D_y be the intersections of B with the lines $x=\infty$ and $y=\infty$. At $x=\infty$ the equation reads $1+\lambda y^3=0$, so D_x consists of three points (∞,c) with $c^3=-1/\lambda$, and at each of them $u=1/x$ is a local parameter, since the derivative of $1+u^3y^3+u^3+\lambda y^3$ in y is $3\lambda c^2\neq 0$; along B, $1+\lambda y^3 = -u^3(1+y^3)$, so $\varphi = y^3+1/\lambda$ vanishes to order three there. Its other zeros would have $y^3=-1/\lambda$ and $x\neq\infty$, hence $x^3(1+\lambda y^3)=0=1+y^3$, which forces $\lambda=1$. Its poles are at D_y, three points at each of which $1/y$ is a local parameter by the same computation, of order three. So $\operatorname{div}\varphi = 3D_x-3D_y$, and since $D_x\in|\mathscr{O}(1,0)|_B|$ and $D_y\in|\mathscr{O}(0,1)|_B|$, $\eta^{\otimes 3}\cong\mathscr{O}_B$. If η were trivial, D_y would lie in $|\mathscr{O}_B(1,0)|$. From $0\to\mathscr{O}(-2,-3)\to\mathscr{O}(1,0)\to\mathscr{O}_B(1,0)\to 0$ and $H^0(\mathscr{O}(-2,-3))=H^1(\mathscr{O}(-2,-3))=0$ by the Künneth formula, every section of $\mathscr{O}_B(1,0)$ is $a+bx$, so every divisor in that system is a fibre $x=c$ of B. The three points of D_y have $x^3=-1/\lambda$, three distinct values of x, so D_y is not such a fibre.

(iii) $K\otimes\eta = \mathscr{O}(2,0)|_B$ and $K\otimes\eta^{-1}=\mathscr{O}(0,2)|_B$. The kernels $\mathscr{O}(-1,-3)$ and $\mathscr{O}(-3,-1)$ of the restrictions have no cohomology in degrees 0 and 1, again by the Künneth formula, so restriction gives $H^0(K\otimes\eta)=H^0(\mathbb{P}^1\times\mathbb{P}^1,\mathscr{O}(2,0))$, spanned by $1,x,x^2$, and $H^0(K\otimes\eta^{-1})=H^0(\mathscr{O}(0,2))$, spanned by $1,y,y^2$. The product of the two is $H^0(\mathscr{O}(2,2))$, of dimension nine, and μ_η is its restriction to B, with values in $H^0(\mathscr{O}_B(2,2))=H^0(K^{\otimes 2})$, of dimension $3g-3=9$. The kernel of the restriction is $H^0(\mathscr{O}(-1,-1))=0$, so μ_η is injective between spaces of dimension nine, hence an isomorphism.

(iv) Write $\pi\colon C_\lambda\to B=B_\lambda$ for the cover defined by η, so that $\pi_*\mathscr{O}_{C_\lambda}=\bigoplus_{i=0}^2\eta^i$ and μ_3 acts on η^i through the character $\zeta\mapsto\zeta^i$. The eigenspace of $H^1(C_\lambda,\mathbb{C})$ for the character χ of exponent one is $H^1(B,\mathbb{L})$ for the unitary local system $\mathbb{L}$ of rank one underlying η, with Hodge pieces $H^0(K\otimes\eta)$ of type $(1,0)$ and $H^1(\eta)\cong H^0(K\otimes\eta^{-1})^\vee$ of type $(0,1)$. Since $\deg\eta=0$ and $\eta\neq\mathscr{O}_B$, $h^0(\eta)=h^0(\eta^{-1})=0$, and Riemann–Roch gives $h^0(K\otimes\eta)=h^0(K\otimes\eta^{-1})=g-1=3$. The sum H_χ of the χ- and $\bar\chi$-eigenspaces is defined over $\mathbb{Q}$, carries the polarisation induced by the intersection form and an action of $F=\mathbb{Q}(\sqrt{-3})$, and the Hermitian form on the χ-eigenspace has signature $(3,3)$; so H_χ is of $(F,3)$-Weil type and the abelian sixfold $H_\chi(C_\lambda)$ lies in a domain $\mathscr{D}_F\cong U(3,3)/(U(3)\times U(3))$ of dimension nine. A line bundle of finite order deforms uniquely with the curve, so the pairs (B,η) are locally parametrised by $H^1(T_B)$, of dimension $3g-3=9$. By Griffiths' description of the differential of the period map [90, Ch. 10], the differential at (B,η) is the map $H^1(T_B)\to\operatorname{Hom}(H^0(K\otimes\eta),H^1(\eta))$ given by cup product, and after Serre duality, $H^1(T_B)^\vee=H^0(K^{\otimes 2})$ and $H^1(\eta)^\vee=H^0(K\otimes\eta^{-1})$, its transpose is μ_η. By (iii) the differential is an isomorphism and the period map is a local isomorphism at (B_λ,η); its image contains an open set $\Omega\subseteq\mathscr{D}_F$. A local biholomorphism sends a non-meagre subset of a neighbourhood to a non-meagre subset of Ω, so under the hypothesis of the last assertion the algebraic locus of the Weil classes in $\mathscr{D}_F$ is not meagre, and by Lemma 4.2 it is all of $\mathscr{D}_F$. □

The curve B_λ is a cyclic cover of $\mathbb{P}^1\times\mathbb{P}^1$ in disguise: it is invariant under μ_3^2 acting on x and y, and its quotient is the line $X+Y+1+\lambda XY=0$ in the coordinates $X=x^3$, $Y=y^3$. For $\ell=4$ and $\ell=6$

the same argument needs a curve of genus four with a line bundle of order four or six for which μ_η is surjective, and we have not found one.

6.11. **Integral characters at sextic CM fields**

For a sextic CM field $F = F_0(\sqrt{-q})$, with F_0 totally real cubic and $q \in F_0$ totally positive, Proposition A.47 shows that a perfect complex $\mathscr{F}$ with $\mathrm{ch}(\mathscr{F}) = v$ in the flat secant space $S(0,q)$ and the least possible Ext^0, Ext^1, Ext^2 exists only if $\chi(v,v)$ lies below an explicit threshold, and Proposition A.48 shows that the condition of the count holds whenever $\chi(v,v) \le -26$, computes $\chi(v,v) = -8\big(\mathrm{Nm}(q)c_0^2 + \mathrm{Tr}(\mathrm{Nm}(q)q^{-1}f^2) + \mathrm{Tr}(qg^2) + c_3^2\big)$ on the rational points $v(c_0,f,g,c_3)$, and exhibits integral points with $\chi = -32$ for $q = 1$.

Remark 6.43 (No exclusion over twelve cubic fields)**.** For the twelve totally real cubic fields of discriminant 49, 81, 148, 169, 229, 257, 316, 321, 361, 404, 469, 473, whose rings of integers are generated by one element, and for $q = 1$ and $q = k + \alpha$ with α the generator and k the least integer making q totally positive, the lattice of integral points of $S(0,q)$ was computed exactly, in the model of Proposition A.48, with the Gram matrix of $-\chi/8$, which is integral. No nonzero integral point has $-\chi \le 24$, and since $\chi \in 8\mathbb{Z}$ on the lattice, every nonzero integral point has $\chi \le -32$ and meets the condition of Proposition A.47(iii). For $q = 1$ the least value $-\chi = 32$ is attained by four points of shape $N_w = 2$, $r^2 = 30$, $r^3 = 40$; for $q = k + \alpha$ it is 32 for the discriminants 49, 81 and 288, 384, 912, 352, 480, 600, 960, 544, 2400, 1208 for the other ten. So over these fields the count of Proposition A.47 excludes no integral character of a minimal object, and the route through it is closed there. This is a computation of the archive [8], and nothing in the paper depends on it.

6.12. **The exceptional classes of a Mumford fourfold**

Keep Setting 3.6 and write ψ for the polarisation form on V, $T = D^0$, $q_\lambda(x,y) = \mathrm{Tr}_{F/\mathbb{Q}}(\lambda\,\mathrm{trd}(xy))$ for $\lambda \in F$, and m_a for multiplication by $a \in F$ on T. The copy $T_X(-1) \subseteq V \otimes V$, $x \mapsto \big((v,w) \mapsto \psi(xv,w)\big)$, is symmetric and carries the form induced by $\psi \otimes \psi$, equal to $-4\,\mathrm{trd}(xy)$ summed over the three real places, that is q_1 up to the scalar -4. The exceptional classes of $X \times X$ are the elements m_a, $a \in F \setminus \mathbb{Q}$, of $\mathrm{End}_{\mathrm{Hdg}} T_X(-1) = F$ Proposition A.53, and in Theorem A.55 their algebraicity is derived from the Kuga–Satake class of a K3 surface S_λ with $T(S_\lambda)_{\mathbb{Q}} \cong (T, q_\lambda)$. We replace S_λ by an abelian variety.

Setting 6.44 (The sixfold Z_K)**.** Let $K = F(\sqrt{-\delta}) \subseteq D$ be a CM quadratic extension of F inside D, $\delta \in F$ totally positive; every maximal commutative CM subfield of D is of this form since D is definite at two real places. Let D carry the left action of $G = D^1$ and the right action of the torus $K^1 = \ker(N_{K/F})$. Over $\bar{\mathbb{Q}}$, $D = D_1 \oplus D_2 \oplus D_3$ with $D_i = V_i \otimes W_i$, V_i the standard representation of the i-th factor of G and $W_i = \chi_i \oplus \chi_i^{-1}$ that of the i-th factor of K^1. Fix a point t of the Shimura curve C of X, with Hodge cocharacter μ_X nontrivial only at the split place σ_3, and let $\mu_Z = \mu'\mu''$, with μ' the action of μ_X by left multiplication and μ'' the cocharacter of K^1 of weight $\pm\frac{1}{2}$ on $\chi_1^{\pm1}$, $\chi_2^{\pm1}$ and 0 on $\chi_3^{\pm1}$. Then $D_{\mathbb{C}}$ has type $(1,0)$ on $V_1\chi_1$, $V_2\chi_2$, $V_3^{1,0}W_3$ and $(0,1)$ on the complements, so D carries a rational Hodge structure of weight one with $h^{1,0} = 6$ and Hodge group inside $G \cdot K^1$. For $c \in K$ with $\bar c = -c$ put $\psi_c(x,y) = \mathrm{Tr}_{F/\mathbb{Q}}\mathrm{trd}(\bar x y c)$. The abelian sixfold $Z_K = Z_{K,t}$ is the one with $H^1(Z_K) = D$ and polarisation ψ_c for c in the cone where ψ_c is positive; over a finite cover of C these form an abelian scheme [80], and we use only the fibre at t.

Lemma 6.45. *ψ_c is alternating, invariant under left multiplication by D^1 and right multiplication by K^1, and the ψ_c, $c \in K$ with $\bar c = -c$, fill the three-dimensional space of all $G \times K^1$-invariant alternating forms on D. Hence $\mathrm{NS}(Z_K)_{\mathbb{Q}}$ has dimension at least three at every point t of C, and exactly three at every t outside a countable subset of C.*

Proof. The reduced trace is invariant under conjugation and cyclic, so $\psi_c(y,x) = \mathrm{Trtrd}(\bar y x c) = \mathrm{Trtrd}(\bar c \bar x y) = \mathrm{Trtrd}(\bar x y \bar c)$, and ψ_c is alternating exactly when $\bar c = -c$. For g with $\mathrm{nrd}(g) = 1$, $\mathrm{trd}(\overline{gx}gyc) = \mathrm{trd}(\bar x \bar g g y c) = \mathrm{trd}(\bar x y c)$. For $k \in K^1$, $\mathrm{trd}(\overline{xk}ykc) = \mathrm{trd}(\bar k \bar x y k c) = \mathrm{trd}(\bar x y k c \bar k) = \mathrm{trd}(\bar x y c)$ because K is commutative and $k\bar k = 1$. Over $\bar{\mathbb{Q}}$ the invariant alternating forms live in the three summands $\bigwedge^2(V_i \otimes W_i)^{G \times K^1} = \bigwedge^2 V_i \otimes (\chi_i\chi_i^{-1})$, one dimension each, and the cross terms $D_i \otimes D_j$ carry the nontrivial characters $\chi_i^{\pm1}\chi_j^{\pm1}$; so the space has dimension $3 = \dim_{\mathbb{Q}} K^-$, which the ψ_c fill since

$c \mapsto \psi_c$ is injective. The Hodge group H_t of $Z_{K,t}$ lies in $G \cdot K^1$, so these three forms are Hodge classes and $\dim \mathrm{NS}(Z_{K,t})_{\mathbb{Q}} \geq 3$; a smaller group can have more invariants, so this is only a lower bound.[12] For the upper bound at a very general t, recall that the Hodge group at a very general point of a family is the generic Hodge group H, and that by Deligne and André [4] the connected monodromy group is a normal subgroup of its derived group. The monodromy of the family over C acts through left multiplication by an arithmetic subgroup of G, Zariski dense in G because G is $\mathbb{Q}$-simple and isotropic at the split place, so $G \subseteq H \subseteq G \cdot K^1$. The image of H in the torus $G \cdot K^1/G$, a quotient of the three-dimensional torus $\mathrm{Res}_{F/\mathbb{Q}}K^1$ by a finite group, is the smallest $\mathbb{Q}$-subtorus through which the image of μ'' factors. The cocharacter group of $\mathrm{Res}_{F/\mathbb{Q}}K^1$, tensored with $\mathbb{Q}$, is $\mathbb{Q}^3$, one coordinate for each real place of F, and the Galois group permutes the coordinates transitively up to sign; the orbit of $\mu'' = (\frac{1}{2}, \frac{1}{2}, 0)$ contains its three cyclic shifts, which are linearly independent. So $H = G \cdot K^1$ at a very general t, and its invariant alternating forms are exactly the three forms ψ_c, all of type $(1,1)$. □

Lemma 6.46 (Where T sits in Z_K)**.**

(i) *The K^1-invariants of $\bigwedge^2 D_i$ form a four-dimensional space: $S^2V_i \otimes \bigwedge^2 W_i$, the adjoint representation of $\mathfrak{sl}(V_i)$, and $\bigwedge^2 V_i \otimes \mathbf{1}$. The sum T_Z of the three adjoint pieces is defined over $\mathbb{Q}$, is a sub-Hodge structure of $H^2(Z_K)$, and is isomorphic, as a Hodge structure with G-action, to $T_X(-1)$.*

(ii) *The form induced on $\bigwedge^2 D$ by ψ_c restricts on T_Z to q_{c^2}, up to a rational scalar independent of the place; here $c^2 = -N_{K/F}(c) = -f^2\delta \in F$ for $c = f\sqrt{-\delta}$.*

Proof. (i) $\bigwedge^2(V_i \otimes W_i) = S^2V_i \otimes \bigwedge^2 W_i \oplus \bigwedge^2 V_i \otimes S^2W_i$; the torus acts trivially on $\bigwedge^2 W_i = \chi_i\chi_i^{-1}$ and on one summand of $S^2W_i = \chi_i^2 \oplus \mathbf{1} \oplus \chi_i^{-2}$. The cross terms carry nontrivial characters. The Galois group permutes the places, so T_Z is defined over $\mathbb{Q}$; as a G-representation it is the adjoint representation T. The cocharacter μ'' acts trivially on K^1-invariants and the centre acts on $\bigwedge^2 D$ with weight two, so the Hodge structure on T_Z is the one induced by the adjoint action of μ', twisted to weight two; this is the recipe that gives $T_X(-1)$ at the same point t.

(ii) By Schur's lemma the induced form is, on each irreducible T_i, a multiple of $\mathrm{trd}(xy)$. In the split model at the place i the element c acts on D_i by $\sigma_i(c)\,\mathrm{diag}(1,-1)$, whose square is $\sigma_i(c^2)$; the multiple is $\sigma_i(c)^2$ times a rational constant that is the same at every place, since the computation in the weight basis is identical at the three places. The sum over the places is q_{c^2} up to that constant. □

Theorem 6.47 (The sixfold link)**.** *Let $t \in C$ be a point at which $X = X_t$ has Hodge group G, and let ξ be a nonzero element of* $\mathrm{Hom}_{\mathrm{Hdg}}(T_X(-1), T_Z) \subseteq H^4(X \times X \times Z_K)$.

(i) $\mathrm{Hom}_{\mathrm{Hdg}}(T_X(-1), T_Z) = F \cdot \iota$ *for the identification ι of Lemma 6.46, so $\xi = m_e \circ \iota$ for a unique $e \in F^\times$. The $G \cdot K^1$-invariant classes of degree four on $X \times X \times Z_K$ have dimension 26; they are Hodge classes at every t, at a very general t they are all the Hodge classes of degree four, and there ξ and $\psi \otimes \mathrm{NS}(Z_K)$ are the only ones in the Künneth component $H^1(X) \otimes H^1(X) \otimes H^2(Z_K)$.*

(ii) *If ξ is algebraic, then $\xi^\dagger\xi$, the composite with the adjoint for the Lefschetz pairing on $X \times X$ and the intersection pairing on Z_K, is an algebraic endomorphism of $T_X(-1)$ equal to $m_{a(e)}$ with $a(e) = \kappa e^2 c^2 \in F$, $\kappa \in \mathbb{Q}^\times$.*

(iii) *If $a(e) \notin \mathbb{Q}$, the exceptional classes of $X \times X$ are algebraic at t, and so is every Hodge class on every power of X.*

(iv) *$a(e) \notin \mathbb{Q}$ for every $e \in F^\times$ if and only if K contains no imaginary quadratic field. If $K = F \cdot k$ with $k = \mathbb{Q}(\sqrt{-d})$ and $c = f\sqrt{-d}$, then $a(e) \in \mathbb{Q}$ exactly for $e \in \mathbb{Q}f^{-1}$.*

(v) *Conversely, if the exceptional classes are algebraic and some algebraic cycle on $X^r \times Z_K$ acts nonzero on T_Z, then every ξ is algebraic.*

Proof. (i) The Hodge group of $X \times X \times Z_K$ lies in $G \cdot K^1$ and maps onto $\mathrm{Hg}(X) = G$; it acts on $T_X(-1)$ and on T_Z through that image, since K^1 acts trivially on both. Both are the irreducible G-representation

[12] The direction matters: if $H \subseteq H'$ then every H'-invariant is H-invariant, so containment of the Hodge group in a known group bounds the Hodge classes from below, never from above.

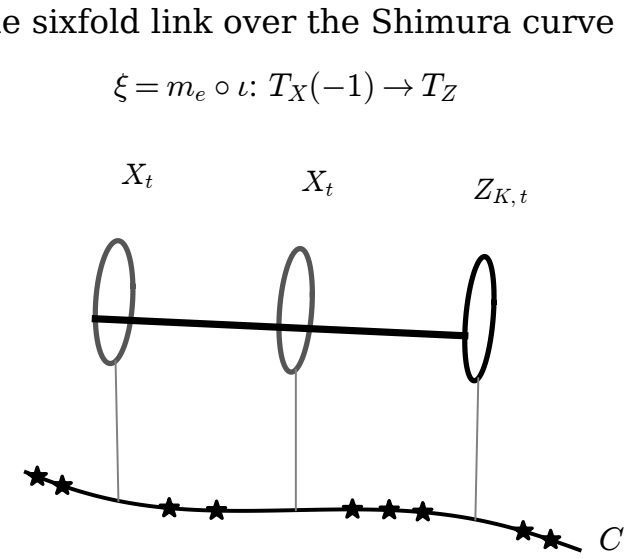


FIGURE 10. The sixfold link. Two copies of the Mumford fourfold X_t and the sixfold $Z_{K,t}$ sit over the same point t of the Shimura curve C; the class ξ_t in the mixed Künneth component identifies the two copies of T and is flat along C. The stars mark the dense set Σ_K of points of complex multiplication, at which ξ_t is algebraic by Theorem 6.51.

T with $\mathrm{End}_G T = F$ Proposition A.53, so $\mathrm{Hom}_{\mathrm{Hdg}} = F\iota$. The count is the dimension of the kernel of the six raising and lowering operators of $\mathfrak{sl}_2^3$ on the torus-weight-zero part of $\bigwedge^4(V \oplus V \oplus D)$, computed by Künneth type: $(1,1,2)$ gives $3+3$, the line $F\iota$ and $\psi \otimes \mathrm{NS}(Z_K)$; $(2,2,0)$ gives $4 = \mathbb{Q} \oplus F$, the Hodge classes of $X \times X$ in $H^2 \otimes H^2$; $(0,0,4)$ gives $6 = S^2\mathrm{NS}(Z_K)$; $(2,0,2)$ and $(0,2,2)$ give 3 each; and $(4,0,0)$, $(0,4,0)$, $(3,1,0)$, $(1,3,0)$ one each, products of divisor classes. The sum is 26. The Hodge group of the product lies in $G \cdot K^1$, so these 26 classes are Hodge classes at every t, and 26 is a lower bound for the Hodge classes of degree four. At a very general t the argument of Lemma 6.45, applied to the family $X \times X \times Z_K$ over C, whose monodromy is again Zariski dense in G, gives Hodge group exactly $G \cdot K^1$, and then the 26 classes are all of them; the description of the Künneth component $(1,1,2)$ follows.

(ii) The adjoint of a correspondence for the Lefschetz pairing on $H^2(X \times X)$ and the polarised intersection pairing on Z_K is the transposed cycle composed with cup products with powers of the polarisations and the inverse Lefschetz isomorphism, which is algebraic on an abelian variety [48, §2]; so $\xi^\dagger \xi$ is algebraic. By Lemma 6.46, $T_X(-1)$ carries $-4q_1$ and T_Z carries a rational multiple of q_{c^2}, and for $\xi = m_e$, $q_{c^2}(m_e x, m_e y) = \mathrm{Tr}(c^2 e^2 \mathrm{trd}(xy)) = q_1(c^2 e^2 x, y)$, so $\xi^\dagger \xi = \kappa m_{e^2 c^2}$.

(iii) An irrational $a \in F$ generates the cubic field, so every m_b, $b \in F$, is a polynomial in $\xi^\dagger \xi$, algebraic on $T_X(-1)$. Compose with the product map $\mu \colon H^2(X \times X) \supseteq T_X(-1) \to H^2(X)$ and its adjoint, both built from ψ, the diagonal and θ and hence algebraic Lemma A.54: the composite $\mu \circ (m_b \otimes 1) \circ \mu^\dagger$ acts on the primitive part U of $H^2(X)$ by the element $\nu(\mathrm{Tr} b - b)/2$ of F, irrational when b is. With the identity and the projector onto $\mathbb{Q}\theta$ this generates $\mathbb{Q} \times F = \mathrm{End}_{\mathrm{Hdg}} H^2(X)$, so by Proposition A.53 the exceptional classes of $X_t \times X_t$ are algebraic, and then every Hodge class on every power of X_t is algebraic Proposition A.58. The classes are flat along C, but by Lemma 4.2 algebraicity at t gives algebraicity on the stratum through t and, through the Hecke correspondences of C, on a countable dense set; whether some stratum is all of C is not decided by this argument, so the conclusion is stated at t.

(iv) $a(e)=\kappa e^2f^2\delta$ is rational iff $e^2f^2\delta\in\mathbb{Q}$. If $-\delta\notin\mathbb{Q}^\times(F^\times)^2$ then $\delta\notin\mathbb{Q}^\times(F^\times)^2$ and $e^2f^2\delta\in\mathbb{Q}$ is impossible. If $-\delta=-dg^2$ with $d\in\mathbb{Q}^\times$, $g\in F^\times$, then with $c=f\sqrt{-d}$ one gets $a(e)=\kappa e^2f^2d$, rational iff $(ef)^2\in\mathbb{Q}$ iff $ef\in\mathbb{Q}$, a cubic field having no irrational element with rational square. And $-\delta\in\mathbb{Q}^\times(F^\times)^2$ exactly when $K\supseteq\mathbb{Q}(\sqrt{-d})$ for some d.

(v) By Theorem A.57 the Hodge conjecture holds for all powers of X once the exceptional classes are algebraic. A cycle on $X^r\times Z_K$ acting nonzero on the irreducible T_Z carries it isomorphically onto a summand of $H^{2k}(X^r)(k-1)$; projecting onto that summand and composing with an isomorphism to $T_X(-1)\subseteq H^2(X\times X)$, a Hodge class on X^{r+2} and algebraic, gives an algebraic nonzero element of $\mathrm{Hom}_{\mathrm{Hdg}}(T_Z,T_X(-1))=F$; composing with the algebraic m_a gives every element of that line, and transposing gives every ξ. □

Proposition 6.48 (Hyper-Kähler carriers give ±1)**.** *Let $Y_1,\dots,Y_m$ be projective hyper-Kähler varieties of $K3^{[n_j]}$ type whose transcendental spaces are Hodge isometric to $(T,r_jq_{\lambda_j})$, $r_j\in\mathbb{Q}^\times$, $\lambda_j\in F$ totally positive. Every composite of rational Hodge isometries between them that returns to Y_j acts on T by ±1, and every composite of Hodge similarities by an element of $\mathbb{Q}$.*

Proof. A Hodge isometry $(T,r_jq_{\lambda_j})\to(T,r_kq_{\lambda_k})$ is a Hodge endomorphism of T, hence m_a for some $a\in F$, and it is an isometry iff $r_j\lambda_j=r_k\lambda_ka^2$. A composite returning to j is m_a with $a^2=1$, and a composite of similarities is m_a with $a^2\in\mathbb{Q}$, which in a cubic field forces $a\in\mathbb{Q}$. The isometries are algebraic by [21, 58], which is why the statement matters: they realise m_b between S_λ and $S_{\lambda b^{-2}}$ and between their moduli spaces of sheaves, but every return is trivial. □

6.13. The link at the points of complex multiplication

The class ξ is flat along C, so the natural first test of Theorem 6.47 is the countable set of special points. When K contains an imaginary quadratic field the test succeeds on a dense set, with published inputs only.

Setting 6.49 (CM conventions)**.** Let $K=F\cdot k$ with $k=\mathbb{Q}(\sqrt{-d})$ imaginary quadratic. Fix $\iota\colon k\to\mathbb{C}$; the embeddings of K are τ_i, restricting to σ_i on F and to ι on k, and their conjugates $\bar\tau_i$, $i=1,2,3$. Let E be an elliptic curve with complex multiplication by k, of type $\{\iota\}$, and A an abelian threefold with complex multiplication by K of type $\Phi_A=\{\bar\tau_1,\bar\tau_2,\tau_3\}$. The type Φ_A is not lifted from k, so A is simple; E and A are determined up to isogeny by their types. Let $\Sigma_K\subseteq C$ be the set of points whose CM field, the F-subalgebra of D generated by the stabiliser torus, is F-isomorphic to K.

Theorem 6.50 (Hodge classes on $E^a\times A^b$)**.** *Let k act on $B=A\times E$ through $k\subseteq K$ on A and through the complex multiplication on E.*

(i) *(B,k) is an abelian fourfold of Weil type: the ι-eigenspace of k on $H^1(B)\otimes\bar{\mathbb{Q}}$ is spanned by x_{τ_1},x_{τ_2} of type $(0,1)$ and x_{τ_3},y_ι of type $(1,0)$, and its Weil plane $W_B\subseteq H^4(B,\mathbb{Q})$ satisfies $W_B\otimes\bar{\mathbb{Q}}=\bar{\mathbb{Q}}w\oplus\bar{\mathbb{Q}}\bar w$ with $w=x_{\tau_1}\wedge x_{\tau_2}\wedge x_{\tau_3}\wedge y_\iota$.*

(ii) *For all $a,b\geq0$ the ring of Hodge classes of $E^a\times A^b$ is generated by divisor classes and the classes φ^*w', $\varphi\in\mathrm{Hom}(E^a\times A^b,B)\otimes\mathbb{Q}$, $w'\in W_B$.*

(iii) *B carries a k-compatible polarisation of discriminant one. Hence W_B is algebraic* [57, Thm. 1.5]*, and the Hodge conjecture holds for every $E^a\times A^b$.*

Proof. (i) A vector x_τ of the τ-eigenline of $H^1\otimes\bar{\mathbb{Q}}$ of a CM abelian variety has type $(1,0)$ exactly when τ lies in the CM type; read this off Φ_A and $\{\iota\}$. The Rosati involution of any polarisation of a CM abelian variety is complex conjugation on its CM field, so the product polarisation is k-compatible.

(ii) $H^1(E^a\times A^b)\otimes\bar{\mathbb{Q}}$ has the basis $x_\tau^{(c)}$ ($c\leq b$, τ an embedding of K) and $y_\varepsilon^{(c)}$ ($c\leq a$, $\varepsilon\in\{\iota,\bar\iota\}$), eigenvectors of the torus $(K^\times)^b\times(k^\times)^a$ with pairwise distinct characters on wedge monomials. The Hodge classes are stable under this torus, hence spanned by the monomials they contain, and a monomial is a Hodge class iff all its Galois conjugates have type (p,p) [71]. For a monomial with $n_{i,\pm}$ factors x_{τ_i} resp. $x_{\bar\tau_i}$ and $m_\pm$ factors y_ι resp. $y_{\bar\iota}$ put $d_i=n_{i,+}-n_{i,-}$ and $d_E=m_+-m_-$. The conjugates of Φ_A are $\{\bar\tau_i,\bar\tau_j,\tau_l\}$, $\{i,j,l\}=\{1,2,3\}$, and their complements; the type condition for a conjugate and its complement gives by subtraction $d_l-d_i-d_j+d_E=0$ for $l=1,2,3$, hence $d_1=d_2=d_3=d_E$. So a Hodge monomial is, up to sign, a product of conjugate pairs $x_\tau^{(c)}\wedge x_{\bar\tau}^{(c')}$, $y_\iota^{(c)}\wedge y_{\bar\iota}^{(c')}$ and of quadruples

$q = x_{\tau_1}^{(c_1)} x_{\tau_2}^{(c_2)} x_{\tau_3}^{(c_3)} y_\iota^{(c_4)}$ or $\bar{q}$. For fixed c, c' the span of the pairs is Galois stable and of type $(1,1)$, hence spanned by divisor classes. For $\alpha \in K^b$, $\beta \in k^a$ the homomorphism $\varphi_{\alpha,\beta}(a_1,\dots,e_a) = (\sum \alpha_i a_i, \sum \beta_j e_j)$ has $\varphi^* x_\tau = \sum_i \tau(\alpha_i) x_\tau^{(i)}$ and $\varphi^* y_\iota = \sum_j \iota(\beta_j) y_\iota^{(j)}$, so the coefficient of the quadruple with indices (c_1,c_2,c_3,c_4) in $\varphi^* w$ is $\tau_1(\alpha_{c_1})\tau_2(\alpha_{c_2})\tau_3(\alpha_{c_3})\iota(\beta_{c_4})$. These are distinct monomials in the coordinates of $(K^b \times k^a) \otimes \bar{\mathbb{Q}}$, linearly independent as functions on its Zariski dense set of rational points; so the $\varphi^* w$, $\varphi^* \bar{w}$ span all quadruples.

(iii) $\mathrm{NS}(A)_{\mathbb{Q}} = \{\mathrm{Tr}_{K/\mathbb{Q}}(a\sqrt{-d}\,x\bar{y}) : a \in F\}$ and $\mathrm{NS}(E)_{\mathbb{Q}} = \mathbb{Q}\,\mathrm{Tr}_{k/\mathbb{Q}}(\sqrt{-d}\,x\bar{y})$, all k-compatible; positivity is a sign condition on a at the three real places. The hermitian form of $h_{a,b} = (a,b)$ on the k-space $H_1(A \times E, \mathbb{Q}) = K \oplus k$ is, up to a rational multiple, $\mathrm{Tr}_{K/k}(a x\bar{y}) \oplus b x\bar{y}$; on the k-basis of K given by a $\mathbb{Q}$-basis f_i of F the first block is the rational matrix $\mathrm{Tr}_{F/\mathbb{Q}}(a f_i f_j)$, of determinant $N_{F/\mathbb{Q}}(a)\, d_F$ up to a square. So the discriminant is $b N_{F/\mathbb{Q}}(a)\, d_F \in \mathbb{Q}^\times / N(k^\times)$, and $b = N_{F/\mathbb{Q}}(a)\, d_F$, which has the sign required for positivity, makes it trivial. The hypotheses of [57, Thm. 1.5] are signature $(2,2)$, compatibility and discriminant one; they hold. Pull-backs and products of algebraic classes are algebraic, so (ii) gives the rest. □

Theorem 6.51 (The link on a dense set of special points)**.** *Let $K = F \cdot k \subseteq D$ as in Setting 6.49.*

(i) *Σ_K is countable and dense in $C(\mathbb{C})$.*

(ii) *For $t \in \Sigma_K$, $X_t \sim E \times A$ and $Z_{K,t} \sim E^3 \times A$. In particular* $\mathrm{Hom}(X_t, Z_{K,t}) \neq 0$.

(iii) *For $t \in \Sigma_K$ and all $a, b \geq 0$ the Hodge conjecture holds for $X_t^a \times Z_{K,t}^b$. Every link class ξ_t and every exceptional class of $X_t \times X_t$ is algebraic.*

Proof. (ii) First let $t = t_0$ be a fixed point of K^1. Write $D = K \oplus Kj$ with $jx = \bar{x}j$. Left multiplication by $x \in K^1$ and right multiplication by $y \in K$ act on K by xy and on Kj by $x\bar{y}$, so both summands are sub-Hodge structures stable under the right action of K, and $Z_{K,t_0} \sim B_1 \times B_2$ with B_1, B_2 CM threefolds for K. In the notation of Setting 6.44 the right τ_i-eigenline of K is $\lambda_i \chi_i$ and that of Kj is $\lambda_i^{-1}\chi_i$, where λ_i is the character τ_i of the left torus. Type $(1,0)$ sits on $V_1\chi_1$, $V_2\chi_2$ and $V_3^{1,0}W_3$, with $V_3^{1,0}$ the λ_3^η-line, $\eta = \pm 1$ according to the fixed point. So the CM types are $\{\tau_1, \tau_2, \tau_3^\eta\}$ and $\{\tau_1, \tau_2, \tau_3^{-\eta}\}$ ($\tau^+ = \tau$, $\tau^- = \bar{\tau}$). One is the lift of $\{\iota\}$, giving $E \otimes_k K \sim E^3$; the other is the conjugate of Φ_A, giving A after twisting the action by complex conjugation. For X_{t_0}: the torus $S = \mathrm{Res}_{F/\mathbb{Q}} K^1$ acts on $V = \mathrm{Cor}_{F/\mathbb{Q}} D$ through G with the eight distinct characters $\lambda^s = \prod \lambda_i^{s_i}$, $s \in \{\pm 1\}^3$, so $\mathrm{End}_S V$ is a commutative étale algebra of degree eight commuting with the Hodge structure. The Galois orbits of the sign vectors are $\{\pm(1,1,1)\}$, with stabiliser $\mathrm{Gal}(\tilde{K}/k)$, and the other six, with the stabiliser of $(-1,-1,1)$ fixing the index 3; so $\mathrm{End}_S V = k \times K$ and $V = V_E \oplus V_A$. The character λ^s has type $(1,0)$ iff $s_3 = \eta$; for $\eta = 1$ the type of V_E is $\{\iota\}$ and that of V_A is $\{\tau_3, \bar{\tau}_2, \bar{\tau}_1\} = \Phi_A$, and $\eta = -1$ gives the conjugate types and the same isogeny classes. For general $t \in \Sigma_K$ the CM field is gKg^{-1}, $g \in D^\times$, by Skolem–Noether; left multiplication by g^{-1} commutes with the right K-action and carries h_t to h_{t_0} for a fixed point t_0, and the representation of G on V extends to $D^\times$, so both $H^1(Z_{K,t})$ and $H^1(X_t)$ are isomorphic to their values at t_0.

(i) By Skolem–Noether every F-embedding of K in D is conjugate to the given one, and K^1 has exactly one fixed point t_0 in the upper half plane, since $K \otimes_{\sigma_3} \mathbb{R} = \mathbb{C}$ makes K^1 a compact torus in $\mathrm{GL}_2(\mathbb{R})$. So Σ_K consists of images of fixed points of the countably many tori gK^1g^{-1}, $g \in D^\times(F)$, and is countable; it contains the images of the points $g \cdot t_0$ with $g \in D^\times(F)$ and $\sigma_3(\mathrm{nrd} g) > 0$, the fixed points of gK^1g^{-1} in the upper half plane. $D^\times(F)$ is the set of F-points of a Zariski open subset of affine four-space and $\mathbb{Q} \subseteq F$ is dense in $\mathbb{R}$, so these g are dense in $\mathrm{GL}_2^+(\mathbb{R}) = (D \otimes_{\sigma_3} \mathbb{R})^{\times,+}$, which acts transitively and continuously on the upper half plane; the orbit of t_0 is dense, and so is its image in C.

(iii) By (ii), $X_t^a \times Z_{K,t}^b \sim E^{a+3b} \times A^{a+b}$; apply Theorem 6.50. The classes ξ_t and the exceptional classes are Hodge classes at every t (Theorem 6.47(i)). □

Proposition 6.52 (Weil fourfolds do not reach ξ at a very general point)**.** *Let t be very general, so that the Hodge group of $X \times Z_K$ is $G \cdot K^1$. Every homomorphism from $X^a \times Z_K^b$ to an abelian fourfold of Weil type is zero; so the algebra generated by divisor classes and pull-backs of Weil classes of such fourfolds is the algebra of divisor classes, and ξ is not in it.*

TABLE 6. The link class ξ_t along the Shimura curve C. "Algebraic" means proved here with published inputs; the alternative of Lemma 4.2 says the algebraic locus of ξ is C or countable.

Points $t \in C$	Decomposition of X_t, $Z_{K,t}$	Status of ξ_t
$t \in \Sigma_K$, $K = F \cdot k$	$E \times A$, $E^3 \times A$	algebraic (Thm. 6.51)
CM, field $F \cdot k'$, $k' \neq k$	$X_t \sim E' \times A'$	powers of X_t algebraic; ξ_t^1 open
CM, field without imag. quadr. subfield	simple, octic $\mathrm{End}^0 X_t$	ξ_t^1 open (degree-12 Weil class)
very general	simple, $\mathrm{End}^0 X = \mathbb{Q}$	open; not in the Weil-fourfold algebra (Prop. 6.52)

Proof. X is simple with $\mathrm{End}^0 X = \mathbb{Q}$ [63]. $\mathrm{End}^0 Z_K$ is the commutant of $G \cdot K^1$ in $\mathrm{End}_{\mathbb{Q}} D$: the commutant of G is right multiplication by D, and the commutant of K^1 in it is the field K, so Z_K is simple of dimension six. A nonzero image of $X^a \times Z_K^b$ of dimension at most four is isogenous to X, and $\mathrm{End}^0 X = \mathbb{Q}$ contains no imaginary quadratic field. In the Künneth component $H^1(X) \otimes H^1(X) \otimes H^2(Z_K)$ the divisor algebra contributes $\psi \otimes \mathrm{NS}(Z_K)$, while ξ has its component in $T \otimes T_Z$ and $T \cap \mathbb{Q}\psi = 0$. □

Remark 6.53 (Every special point, and the residual class)**.** At any CM point t with CM field $L = F(\ell) \subseteq D$, $T = F\ell \oplus Lj'$ as Hodge structures, $F\ell(-1)$ consists of Hodge classes, and $\xi_t = \xi_t^0 + \xi_t^1$ with ξ_t^0 a sum of products of divisor classes and $\xi_t^1 \in \mathrm{Hom}_{\mathrm{Hdg}}((Lj')_X, (Lj')_Z)$; so ξ_t is algebraic iff ξ_t^1 is. The fourfold X_t is always of Weil type: $\mathrm{End}^0 X_t$ contains $k^* = \mathbb{Q}(\sqrt{-N_{F/\mathbb{Q}}\delta})$ when $L = F(\sqrt{-\delta})$, with signature $(2,2)$, and the computation of Theorem 6.50(ii) shows that the Hodge ring of every power of X_t is generated by divisors and pull-backs of its Weil plane. The Hodge conjecture for all powers of X_t then follows from [57, Thm. 1.5] whenever X_t carries a k^*-compatible polarisation of discriminant one, which `julia/mumford_residual.jl` exhibits for $F = \mathbb{Q}(\zeta_7)^+$, $\mathbb{Q}(\zeta_9)^+$ and the cubic field of discriminant 148, and at every point of complex multiplication from Markman's theorem for all abelian fourfolds of Weil type [59, 60]. When L has no imaginary quadratic subfield, ξ_t^1 is equivalent to a single Hodge line over the degree-twelve CM field $M = L \cdot k^*$, the pull-back of a Weil class for M on an abelian variety isogenous to $X_t^3 \times B_1^2 \times B_2^2$ with $\dim_M H^1 = 4$; no algebraicity theorem covers Weil classes for CM fields of degree larger than two, and [61] sets up this case without closing it.

6.14. **Where the link stops**

Theorem 6.47 replaces the Kuga–Satake variety of Theorem A.55, of dimension 2^7, by an abelian sixfold over the same curve, and it replaces the irrationality that had to be extracted from anticommuting Clifford blocks by the arithmetic of K: for every K that is not a compositum $F \cdot k$ every link closes the target. Its logical status is the same as the Kuga–Satake link: by (v) it is equivalent to the target together with one algebraic link, not a reduction to something smaller.

The missing step is the algebraicity of one class $\xi \in H^4(X \times X \times Z_K)$ at a very general point of C. It is exceptional: its component in $V \otimes V$ is the nine-dimensional T, not a multiple of ψ, so it is not invariant under $\mathrm{Sp}(V)$ and not in the Lefschetz algebra of $X \times X \times Z_K$ (Lemma 3.7); the theorems that prove the conjecture when the Hodge ring is generated by divisors do not apply, and by Proposition 6.52 the Weil fourfold theorem cannot be fed in through homomorphisms. When $K = F \cdot k$, Theorem 6.51 gives ξ_t on the dense set Σ_K, so by Lemma 4.2 the algebraic locus of ξ is either countable and dense or all of C; deciding between the two is the variational Hodge conjecture for one flat section over a curve, and it is open. When K contains no imaginary quadratic field, the smallest open instance is the residual class ξ^1 of Remark 6.53 at the fixed point of K^1: one Hodge line over a CM field of degree twelve on a tenfold. This is the price of staying inside abelian varieties, and it is the exact statement that (F2) needs at its smallest instance.

TABLE 7. Every statement of the paper with its status. "Proved" means proved in the text from the cited results; "open" means not proved here or elsewhere to our knowledge.

statement	where	status	what it needs
HC $\Leftrightarrow$ (F2) $\wedge$ (F3′)	Thm. 1.1	proved	–
(F3′) for fourfolds with zero-cycles on a threefold	Prop. 3.4	proved from [19]	–
(F2) $\Leftrightarrow$ **AB**	Thm. 3.8	proved	–
six minimal sufficient sets	Prop. 3.9	proved, checked in Lean	–
dichotomy of the algebraic locus	Lem. 4.2	proved	–
(F2) as propagation from CM points	Thm. 4.5	proved	–
Weil loci dense with CM base point	Prop. 4.6	proved	–
chains make e with $e^2 \in \mathbb{Q}$ algebraic	Thm. 5.3	proved	a chain of the right multiplier
symplectic chains reach exactly $\sqrt{2}, \sqrt{3}, \sqrt{6}$	Prop. 5.25	proved	–
$\sqrt{6}$ algebraic on two-dimensional families	Thm. 5.5	proved	–
$\sqrt{-5}$, $\sqrt{-7}$ algebraic on one-dimensional families	Thm. 5.9	proved	–
HC on all powers and Hilbert schemes of the Fricke surfaces	Thm. 5.9	proved	–
similarities act by multipliers $d_1 d_2$	Prop. 5.11	proved	–
finite-group correspondences give abelian E	Prop. 5.12	proved	–
HC for $X \times X$, X in the dihedral families	Cor. 5.13	proved	–
(F3′) for S_λ, $F = \mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$, $D = D_\infty$	Thm. 5.14	proved	–
where the dihedral mechanism stops	Prop. 5.15	proved	–
(F3′) for S_λ, other cubic F or other D	Sect. 5.9	open	a non-similarity correspondence
every $\mathbb{Q}(\sqrt{d})$ in transcendental rank six, **HC** on all powers	Thm. 5.23	proved from [32]	–
Hodge classes on all powers of S: E and $\bigwedge_E^m T$	Prop. 5.16	proved from [84]	–
K3 surfaces of Kummer type over $\mathbb{Q}$: (F3′) and **HC** on all powers	Prop. 5.17	proved	–
Fricke Kuga–Satake fourfold: $B \sim E_K \times B_3$	Prop. 5.19	proved	–
$\bigwedge_E^m T$ for the other surfaces of Sect. 5	Rem. 5.18	open	a cycle on S^m
(F3′) in general	–	open	–
product object of polarised shape	Thm. 6.3	proved	–
direct sums never semiregular	Prop. 6.4	proved	–
annihilator of the secant pieces is the product locus	Thm. 6.6	proved	–
secant and line-bundle pieces orthogonal	Prop. 6.7	false for every realisation	–
(P2′) at a product point	Sect. 6.5	open	Ext* of secant sheaves
the Ext groups at the CM point E_0^6	Lem. 6.8	proved	–
no two-step cancellation of diagonal classes	Prop. 6.10	proved	–
three line bundles at E_0^6	Prop. 6.11	proved, fail	–
isolated Ext blocks at E_0^6	Prop. 6.13	proved, fail	–
a threefold piece outside every closed chain	Prop. 6.14	proved, fail	–
further pieces within one shift of the thirty-two	Cor. 6.15	proved, fail	–
twists along θ_h; discriminant of $\mathscr{D}_h$	Rems. 6.18, 6.19	proved	–
at most five line bundles, any twists, every polarisation	Thm. 6.25	proved, fail	–
six or seven line bundles; eight outside two arrangements	Lem. 6.27, Rem. 6.28	enumeration, fail	–
the rest of the support cuts a threefold in an ample divisor	Thm. 6.30	proved	–
two threefolds meeting along a surface carry nothing	Thm. 6.32	proved	–
objects on the thirty-two threefolds of one family	Cor. 6.34	proved, fail	–
objects on the sixty-four threefolds of both families	Thm. 6.36	proved, fail	–
a smooth complete intersection cannot carry the rest	Prop. 6.38	proved, fail	–
an ample divisor in the support empties the localised obstruction	Prop. 6.39	proved	–
(P2′) at E_0^6: two arrangements of eight, or ten or more line bundles	Rem. 6.28	open	the chain products
E_0^8: shifts in seven consecutive integers	Cor. 6.41	18 of 28 proved	–
an étale Prym map of triple covers at genus four	Prop. 6.42	proved	–
sextic CM fields: the count excludes nothing	Rem. 6.43	enumeration	–
sixfold link $\Rightarrow$ exceptional classes	Thm. 6.47	proved	–
link ξ_t on the dense set Σ_K, $K = F \cdot k$	Thm. 6.51	proved from [57]	–
Weil fourfolds do not reach ξ at a very general point	Prop. 6.52	proved	–
the link ξ at a very general point	Sect. 6.14	open	a cycle on $X \times X \times Z_K$
dominant maps, blow-ups, derived equivalences preserve the defect	Prop. 7.2	proved	–
residues of complete intersections	Prop. 7.3	proved	rational residues
HC for α as a vanishing mass gap	Prop. 7.4	proved	–
exceptional classes of $X \times X$	–	open	–
(F2) in general	–	open	–
the Hodge conjecture	–	open	(F2) and (F3′)

7. What is proved and what is open

Table 7 lists the statements. Three remarks on what the table says.

The two open statements are not independent problems that happen to add up to the conjecture; they are two halves of one phenomenon. Both say that the algebraic locus of a flat Hodge class, a countable union of closed strata through a dense set of points of complex multiplication, is not meagre. For (F2) the dense set is supplied by explicit cycles at CM points and the first instance is a single class on an abelian fourteenfold or a single Ext group on an eightfold; for (F3′) the dense set is supplied by the K3 surfaces with complex multiplication and by the chains of Section 5, and the first instance is a cubic real multiplication on one K3 surface of Picard number 13, which Theorem 5.14 supplies for the fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ with the algebra ramified at two real places, and which stays open for every other pair of a cubic field and an algebra.

Each construction of this paper ends at a question with a definite answer. Whether some convolution of the pieces of Theorem 6.3 has $\dim \mathrm{Ext}^2 = 57$ is a finite computation in the Ext algebra of two sheaves on a fourfold; at the point E_0^6 of complex multiplication every group entering it is written down in Lemma 6.8, Proposition 6.11 answers it in the negative for every convolution with three powers of L, Proposition 6.13 answers it in the negative for two threefolds at one shift and for every placement of the thirty-five pieces, Proposition 6.14 with Corollary 6.15 for every further piece of positive twist within one shift of the thirty-two, Theorem 6.25 for at most five line bundles at any twists and every polarisation, the enumeration of Remark 6.28, with Lemma 6.27, for six and seven and for eight outside two arrangements, Corollary 6.34 and Theorem 6.36 for every convolution of threefold pieces, of one family or of both, at any twists and shifts, and for the configurations left, the two arrangements of eight and ten or more line bundles, with or without threefolds, it is the rank of the products along the chains of Proposition 6.10 and along the cycles through the line bundles, the support of the object being constrained by Theorem 6.30 and Propositions 6.33 and 6.38. Whether the link ξ is algebraic is a question about one Hodge class on one abelian variety, equivalent by Theorem 6.47(v) to the exceptional classes together with one more cycle; Theorem 6.51 answers it on a dense set of special points when K contains an imaginary quadratic field, so that what remains is whether the algebraic locus of a flat class over a curve, known to be dense, is the whole curve. Whether a correspondence acting on $T(S_\lambda)$ otherwise than by a similarity exists is a question about one K3 surface; for $F = \mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$ and $D = D_\infty$ Theorem 5.14 answers it, and for every other pair (F, D) it is open, with Proposition 5.15 marking the three places where the present mechanism fails. A negative answer to the first closes that route, as the analogous question on E_0^6 closed; a positive answer to any of them proves the corresponding statement at its first instance.

7.1. Specialisation, degeneration and blow-ups

Three devices are often proposed for passing from the members where a class is known to be algebraic to the others, and we record exactly what each one proves.

Proposition 7.1. *Let $\pi : \mathscr{X} \to B$ be a smooth projective family over an irreducible base and ξ a flat Hodge class on its fibres.*

(i) *If ξ_t is algebraic at the generic point of a curve $C \subseteq B$ then it is algebraic at every point of C. The converse implication, from one point of C to its generic point, holds when ξ_{t_0} is the Chern character of a perfect complex whose semiregularity map is injective, and in no greater generality is it known.*

(ii) *Let $\tilde{X}$ be the blow-up of a smooth projective variety X along a smooth closed subvariety Z of codimension c. Then $\mathrm{Hdg}^k(\tilde{X})/\mathrm{Alg}^k(\tilde{X}) \cong \mathrm{Hdg}^k(X)/\mathrm{Alg}^k(X) \oplus \bigoplus_{i=1}^{c-1} \mathrm{Hdg}^{k-i}(Z)/\mathrm{Alg}^{k-i}(Z)$. In particular the Hodge conjecture for $\tilde{X}$ is equivalent to the Hodge conjecture for X and for Z, and a blow-up never reduces the defect of a variety.*

(iii) *For the Weil classes of $\mathscr{D}_{n,n}$ and the exceptional class of a Mumford fourfold, every degeneration to a fibre with a semistable model on which the limit class is algebraic has its lift to the generic fibre as its missing step, and for the Mumford fourfold no degeneration keeping the class of Hodge type exists, since the Hodge locus of the class is a compact Shimura curve.*

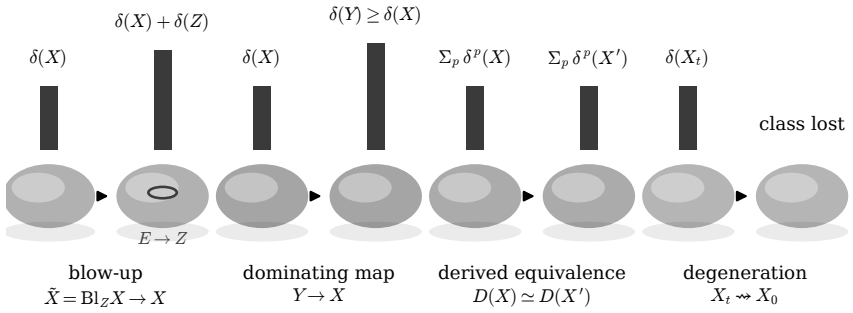


FIGURE 11. Where the defect goes under surgery. A blow-up along Z carries the Hodge classes of X to $\tilde{X}$ and adds those of Z, so the quotient $\mathrm{Hdg}/\mathrm{Alg}$ grows by the defect of the centre; a dominating map $Y \to X$ carries the defect of X into Y; a Fourier–Mukai equivalence permutes degrees and preserves the total defect; a degeneration to X_0 loses the class only in the special fibre, from which no cycle lifts.

Proof. (i) Specialisation of cycles along a curve [33, Ch. 20] sends an algebraic cycle on the generic fibre to one on each special fibre with the same flat cohomology class; this is the direction used by Schoen's degeneration and by Proposition 4.6. In the other direction the algebraic locus is a countable union of closed sets by Lemma 4.2, and a point in one stratum says nothing about the stratum being the whole base; the one proved lift is the semiregularity theorem of Bloch, Buchweitz and Flenner [17, 10], which is the content of Theorem 6.1. (ii) Manin's decomposition of the motive of a blow-up [56], $h(\tilde{X}) = h(X) \oplus \bigoplus_{i=1}^{c-1} h(Z)(-i)$, is realised by the algebraic correspondences π^* and $j_* \circ p^* \circ$ (powers of the relative hyperplane class), which are compatible with Hodge classes and with algebraic classes; so both the Hodge classes and the algebraic classes of $\tilde{X}$ decompose accordingly, and the quotients add. Since every summand is non-negative, the defect of $\tilde{X}$ is at least that of X, and a blow-down along a smooth centre Z lowers the defect by exactly that of Z in the shifted degrees. (iii) A flat class has a limit in the limit mixed Hodge structure of a semistable fibre, and on a totally degenerate fibre the graded pieces are combinatorial and every class is algebraic on the components; the return to the generic fibre is the lift of (i), with the same unproved step. For a Mumford fourfold the Hodge locus of the exceptional class is the compact Shimura curve of Proposition A.58, which has no boundary, so no fibre at which the class is still of Hodge type degenerates. □

7.2. **Dominating maps, birational models and derived equivalences**

Write $\delta^p(X) = \dim \mathrm{Hdg}^p(X) - \dim \mathrm{Alg}^p(X)$ for the defect of the conjecture in degree $2p$, $\delta^p_{\mathrm{ab}}(X) = \dim \mathrm{Hdg}^p(X) - \dim \mathrm{Ab}^p(X)$ for the defect of (F3′), and $\delta(X) = \sum_p \delta^p(X)$. The conjecture for X is $\delta(X) = 0$. The following proposition records how the surgeries of birational geometry move these numbers; Figure 11 draws it.

Proposition 7.2 (Defects under surgery)**.** *Let X, Y be smooth complex projective varieties.*

(i) *If $f\colon Y \to X$ is surjective and generically finite, then $\delta^p(Y) \geq \delta^p(X)$ and $\delta^p_{\mathrm{ab}}(Y) \geq \delta^p_{\mathrm{ab}}(X)$ for every p. In particular no blow-up along a singular centre, weighted blow-up, resolution of a modification or finite cover of X has a smaller defect than X.*

(ii) *If X and Y are birational, then $\delta^2(X) = \delta^2(Y)$ and $\delta^{d-1}(X) = \delta^{d-1}(Y)$, $d = \dim X$; for $3 \leq p \leq d-2$ the difference $\delta^p(X) - \delta^p(Y)$ is a signed sum of defects $\delta^{p-j}(Z)$, $j \geq 1$, of smooth projective varieties Z of dimension less than d.*

(iii) *If $\Gamma \in \mathrm{CH}^*(X \times Y)_{\mathbb{Q}}$ induces an isomorphism from $H^*(X,\mathbb{Q})$ onto $H^*(Y,\mathbb{Q})$ whose inverse is induced by a cycle, for instance the kernel of an equivalence between the derived categories of X and Y, then $\delta(X) = \delta(Y)$ and $\delta_{\mathrm{ab}}(X) = \delta_{\mathrm{ab}}(Y)$.*

(iv) *Let $\mathscr{X} \to \Delta$ be a projective family, smooth over the punctured disc, with a flat Hodge class ξ on the smooth fibres that is algebraic on a crepant resolution $\hat{X}$ of the special fibre X_0, or on the components of a semistable model of X_0. Then ξ is algebraic on X_t for $t \neq 0$ if and only if the specialisation $H^{2p}(X_0) \to H^{2p}(X_t)$ sends an algebraic class to an algebraic class, which is the statement (V) for the family; the monodromy plays no role, since it fixes ξ.*

Proof. (i) f_*f^* is multiplication by the degree, so f^* is an injective morphism of Hodge structures and $f^*\mathrm{Hdg}^p(X) \subseteq \mathrm{Hdg}^p(Y)$. If $f^*\alpha$ is algebraic then so is $\deg(f)\,\alpha = f_*f^*\alpha$; hence $\mathrm{Alg}^p(Y) \cap f^*\mathrm{Hdg}^p(X) = f^*\mathrm{Alg}^p(X)$ and f^* embeds $\mathrm{Hdg}^p(X)/\mathrm{Alg}^p(X)$ into $\mathrm{Hdg}^p(Y)/\mathrm{Alg}^p(Y)$. For Ab the same argument applies because f_* of a class $\Gamma_*\gamma$ reached from an abelian variety is $(\Gamma_f \circ \Gamma)_*\gamma$, reached from the same one. The coarse space of a weighted blow-up has quotient singularities and $f_*f^* = \mathrm{id}$ with rational coefficients, so the argument is unchanged there.

(ii) By weak factorisation [3] a birational map between smooth projective varieties is a composite of blow-ups and blow-downs along smooth centres with smooth projective intermediate varieties, and Proposition 7.1(ii) gives the change of δ^p at each step as $\pm\sum_{j=1}^{c-1}\delta^{p-j}(Z)$. For $p=2$ the terms are $\delta^1(Z)$ and $\delta^0(Z)$, which vanish by the Lefschetz $(1,1)$ theorem; for $p = d-1$ they are $\delta^{d-1-j}(Z)$ with $\dim Z \leq d-2$, hence $\delta^q(Z)$ with $q \geq \dim Z - 1$ after $j \geq 1$, and these vanish by hard Lefschetz and Lefschetz $(1,1)$ on Z.

(iii) By the projection formula Γ_* sends $\bigoplus_p \mathrm{Hdg}^p(X)$ into $\bigoplus_p \mathrm{Hdg}^p(Y)$, since $[\Gamma]$ is a sum of classes of type (i,i) and cup product and push-forward are morphisms of Hodge structures up to Tate twists; it sends algebraic classes to algebraic classes because $[\Gamma] \cup \mathrm{pr}_X^*[Z]$ is the class of a cycle and the push-forward of a cycle is a cycle; and it sends Ab to Ab by composing correspondences. The inverse does the same in the other direction, so the two total quotients are isomorphic. The kernel of a Fourier–Mukai equivalence and the kernel of its inverse are cycles by Mukai's and Orlov's description of the inverse functor [70], which is the equivalence with kernel the dual of the original one shifted by the dimension.

(iv) ξ is monodromy invariant, so by the local invariant cycle theorem it is the specialisation of a class in $H^{2p}(X_0)$, which pulls back to $\hat{X}$ or to the components of the semistable model; the hypothesis says that the pulled back class is algebraic there, and push-forward keeps it algebraic on X_0. Cycles specialise from the generic fibre to the special fibre [33, Ch. 20] and not the other way, so the remaining step is exactly the one in the statement. If the limit cycle on X_0 lifts to the family, its restriction to X_t has class ξ_t; conversely if ξ_t is algebraic the specialisation of a representing cycle is a cycle on X_0 with class the limit of ξ, and the specialisation map sends that class to ξ_t. The vanishing cycles, on which the monodromy acts, are orthogonal to the invariant part, and ξ is invariant, so no condition on them enters. □

Items (i) to (iii) say that every surgery available in birational and derived geometry carries the defect of X into its output as long as the open class survives there, and loses the class only in outputs of lower dimension, the centres, where it has become a defect of those varieties. Both open statements of this paper sit in degree four, (F2) for the Weil classes and the exceptional classes, (F3′) for the classes of $H^4(S \times S)$ on a K3 surface, and in degree four the defect is a strict birational invariant by (ii) and a derived invariant by (iii); so the open classes are present, and not known to be algebraic, on every birational model and every Fourier–Mukai partner of the varieties that carry them. Item (iv) is the reason that degenerations to products of elliptic curves, where every Hodge class is algebraic [66], do not close (F2): the class returns to the smooth fibre only through (V). Flows of Kähler metrics change no complex structure between their surgeries, and the surgeries themselves are the contractions and flips of the minimal model programme, covered by (i) and (ii); the surgery of Perelman has no counterpart for Hodge structures, since it cuts a manifold along spheres and a Hodge structure has no local pieces to cut.

The devices therefore relocate the two open statements without weakening them. The one true reduction in shape is Theorem 4.5: the whole of (F2) is equivalent to a lift from CM points along polarised abelian schemes, and the lift along each scheme is a question about one semiregular object at one point.

7.3. **Residues and mass-minimising currents**

Two analytic descriptions of algebraic classes are exact, and we record what each reduces the conjecture to. The first is Griffiths' residue calculus on hypersurfaces [37]. For a smooth hypersurface $X=\{F=0\}\subset\mathbb{P}^{n+1}$ of even dimension $n=2m$ and degree d, with Jacobian ring $R(F)$ and $\Omega=\sum_i(-1)^i x_i\,dx_0\wedge\ldots\widehat{dx_i}\cdots\wedge dx_{n+1}$, the residues of the forms $P\Omega/F^{q+1}$ identify $H^{n-q,q}_{\mathrm{prim}}(X)$ with $R(F)_{(q+1)d-n-2}$.

Proposition 7.3 (Residues of complete intersections)**.** *Let $X=\{F=0\}$ be as above, and suppose that $F=\sum_{i=1}^{m+1}f_ig_i$ with $Z=\{f_1=\cdots=f_{m+1}=0\}$ a complete intersection of codimension $m+1$ in $\mathbb{P}^{n+1}$ whose ideal is generated by the f_i.*

(i) *Let θ be the hyperplane class and J_H the Jacobian determinant of $(f_1,g_1,\ldots,f_{m+1},g_{m+1})$, a polynomial of degree $(m+1)(d-2)$. Then*

$$[Z]=\frac{\deg Z}{d}\,\theta^m-\frac{m!}{d}\,\operatorname{res}\Big(\frac{J_H\,\Omega}{F^{m+1}}\Big)^{m,m}$$

[89, Thm. 1.1].

(ii) *Let $J(F)\subseteq R(F)_{(m+1)(d-2)}$ be the span of the classes of the determinants J_H over all such decompositions of F. If the rational span of the residues of $J(F)$ and θ^m contains $\mathrm{Hdg}^m(X)$, the Hodge conjecture holds for X.*

Proof. (i) is the cited theorem; the degree of the determinant is $\sum_i(\deg f_i+\deg g_i)-(n+2)=(m+1)d-2m-2$. (ii) By (i) every residue of an element of $J(F)$ is a rational combination of θ^m and of classes of complete intersections in X, so it is algebraic. The Hodge conjecture for a hypersurface holds outside the middle degree by the Lefschetz hyperplane theorem, and in the middle degree it is the statement that $\mathrm{Hdg}^m(X)$ consists of algebraic classes. □

The proposition replaces a cycle by a polynomial, and the open step becomes the statement that the polynomials of $J(F)$ have enough rational residues. Deciding whether a given element of $R(F)$ has a rational residue requires its periods, so the residue calculus moves the conjecture from cycles to transcendental numbers without closing it. On a Fermat hypersurface $\sum_i x_i^d=0$ the linear subspaces cut out by $x_{2i}=\zeta_i x_{2i+1}$, $\zeta_i^d=-1$, are of this form, since $x_{2i}^d+x_{2i+1}^d$ factors into linear forms; whether their residues and those of the other decompositions exhaust $\mathrm{Hdg}^m(X)$ is again a question about periods.

The second description uses currents and is the analogue for cycles of the flows of geometric analysis. Let X be a smooth complex projective variety of dimension N, h an ample class with Kähler form ω, and $0<p<N$. For $\beta\in H_{2(N-p)}(X,\mathbb{Z})$ let $M(\beta)$ be the infimum of the masses of the integral currents in β, put $\mathrm{vol}(\beta)=\int_\beta\omega^{N-p}/(N-p)!$, and call $g(\beta)=M(\beta)-\mathrm{vol}(\beta)$ the gap of β. We identify $H^{2p}(X,\mathbb{Z})$ with $H_{2(N-p)}(X,\mathbb{Z})$ by Poincaré duality.

Proposition 7.4 (Mass-minimising currents)**.**

(i) *Every class β contains an integral current of mass $M(\beta)$, and $g(\beta)\geq 0$, with equality if and only if β is the class of an effective algebraic cycle with integer coefficients.*

(ii) *Let $\alpha\in H^{2p}(X,\mathbb{Z})$. The image of α in $H^{2p}(X,\mathbb{Q})$ is algebraic if and only if $g(e\alpha+kh^p)=0$ for some integers $e\geq 1$ and $k\geq 0$.*

(iii) *There are X and α with $g(\alpha+kh^p)>0$ for every $k\geq 0$ although $g(e\alpha)=0$ for some $e\geq 1$: for a very general hypersurface $X\subset\mathbb{P}^4$ of degree D divisible by ℓ^3 for an integer $\ell>1$ prime to 6, $p=2$, and the generator α of $H^4(X,\mathbb{Z})$.*

Proof. (i) By the compactness theorem of Federer and Fleming [30] the integral currents in β of mass at most $M(\beta)+1$ form a compact set for the flat topology, and mass is lower semicontinuous, so the infimum is attained. For an integral current T in β, Wirtinger's inequality, that $\omega^{N-p}/(N-p)!$ has comass one with equality exactly on the complex $(N-p)$-planes with their canonical orientation, gives $M(T)\geq T\big(\omega^{N-p}/(N-p)!\big)=\mathrm{vol}(\beta)$, the last value depending only on β because ω is closed [42]. If equality holds, T is calibrated by $\omega^{N-p}/(N-p)!$, hence a closed positive current of bidimension $(N-p,N-p)$ [42], and by King's theorem [47] a closed positive integral current of that bidimension

is $\sum_i n_i[V_i]$ with n_i positive integers and V_i irreducible analytic subvarieties, which are algebraic by Chow's theorem. Conversely, for an effective cycle $Z=\sum n_iV_i$ the tangent planes of the smooth locus of each V_i are complex, so the current $[Z]$ has mass $\mathrm{vol}([Z])$ and $g([Z])=0$.

(ii) If $g(e\alpha+kh^p)=0$, then $e\alpha+kh^p$ is the class of an effective cycle by (i), and $\alpha=\big([Z]-kh^p\big)/e$ in $H^{2p}(X,\mathbb{Q})$. Conversely let α be rationally algebraic, so that $e_0\alpha-[Z^+]+[Z^-]$ is a torsion class for an integer $e_0\geq 1$ and effective cycles $Z^{\pm}$; multiplying by its order, $e\alpha=[Z^+]-[Z^-]$ in $H^{2p}(X,\mathbb{Z})$. Write $Z^-=\sum_i c_iV_i$. For r large the ideal sheaf of each V_i twisted by $\mathscr{O}(rh)$ is globally generated, so p general members of $|rh|$ containing V_i meet in a complete intersection Y_i of codimension p containing V_i, and $[Y_i]=r^ph^p$; the cycle Z_i' of Y_i minus V_i is effective. Then $e\alpha+\sum_i c_ir^ph^p=[Z^+ +\sum_i c_iZ_i']$, effective, and (i) gives $g=0$ with $k=\sum_i c_ir^p$.

(iii) By the Lefschetz hyperplane theorem $H^4(X,\mathbb{Z})\cong\mathbb{Z}\alpha$ with $\int_X\alpha\cdot h=1$, and $h^2=D\alpha$, the class of a complete intersection, so $g(D\alpha)=0$. By Kollár's theorem [50], in the form of [82, Thm. 2], every curve on X has degree divisible by ℓ. The class $\alpha+kh^2$ has degree $1+kD$, prime to ℓ, so it is not the class of an effective cycle, and $g(\alpha+kh^2)>0$ by (i). □

The analytic part of the variational approach is therefore complete. A minimiser exists in every class without any flow, so a mass-decreasing flow of a cycle, with whatever surgeries along its singularities, can at best reach it; what such a flow cannot provide is the equality case, and the equality case is the whole conjecture: by (ii) the Hodge conjecture for a class α is the vanishing of the gap of $e\alpha+kh^p$ for one pair (e,k). Part (iii) shows that the multiplier e cannot be omitted, since integral minimisers of a class that is rationally algebraic need not be calibrated for any k, and so no argument by integral currents alone, including one modelled on Ricci flow with surgery, proves the conjecture without passing to rational coefficients. For the Weil classes of $\mathscr{D}_{n,n}$ and the exceptional classes of a Mumford fourfold the gap of $e\alpha+kh^p$ is not known to vanish for any (e,k).

Nothing in the paper assumes an open statement, and no section depends on a computation that is not also proved by hand in the text. The machine checks of Appendix C recompute the finite quantities; they do not carry any proof.

Appendix A. Weil classes, the polarised criterion and Mumford fourfolds: statements and proofs

This appendix states and proves the results on Weil classes, on the polarised semiregularity criterion, on convolutions of line bundles, on the closure graph, on Mumford fourfolds and on K3 surfaces with real multiplication that the body of the paper uses. It is self-contained apart from the published literature it cites. The numbering is that of the appendix; the notation is that of the paper, with the additions fixed in each subsection.

A.1. **Abelian varieties of Weil type for a CM field**

Setting A.1 (Weil type for a CM field)**.** Let F be a CM field of degree $2m$ with maximal totally real subfield F_0, and write $\alpha\mapsto\overline{\alpha}$ for the nontrivial automorphism of F/F_0, which is complex conjugation under every embedding. The real places of F_0 are $\tau_1,\dots,\tau_m$, and each extends to a conjugate pair $\sigma_\tau,\overline{\sigma}_\tau$ of embeddings of F into $\mathbb{C}$. Let V be an F-vector space of dimension $2n$, let $H\colon V\times V\to F$ be a nondegenerate hermitian form, F-linear in the first variable, and fix a totally imaginary $\xi\in F^\times$, that is $\overline{\xi}=-\xi$. Put

$$E(x,y)=\mathrm{Tr}_{F/\mathbb{Q}}\big(\xi\,H(x,y)\big),$$

a nondegenerate alternating $\mathbb{Q}$-bilinear form with $E(\alpha x,y)=E(x,\overline{\alpha}y)$. For each τ let $V_\tau=V\otimes_{F_0,\tau}\mathbb{R}$, a vector space of dimension $2n$ over $F\otimes_{F_0,\tau}\mathbb{R}\cong\mathbb{C}$, and let (p_τ,q_τ) be the signature of the hermitian form H_τ induced by H on it. We say that (V,H) is of (F,n)-*Weil type* if $p_\tau=q_\tau=n$ for every τ, and we call the class of $(-1)^n\det H$ in $F_0^\times/\mathrm{N}_{F/F_0}(F^\times)$ its discriminant δ; it is totally positive. A polarised abelian variety (A,ι,E) of dimension $2mn$ with $\iota\colon F\to\mathrm{End}^0(A)$ and $H^1(A,\mathbb{Q})\cong V$ as F-modules with polarisation E is of (F,n,δ)-Weil type when (V,H) is. We work throughout with $V=H^1$; a complex structure on $V_\mathbb{R}$ is the same datum as one on the tangent space. The Weil classes of A are

$$W_F(A)=\textstyle\bigwedge_F^{2n}V\ \subseteq\ \bigwedge_\mathbb{Q}^{2n}V=H^{2n}(A,\mathbb{Q}),$$

the $\mathbb{Q}$-subspace whose complexification is $\bigoplus_\sigma \bigwedge^{2n} V_\sigma$, where $V_\sigma = V \otimes_{F,\sigma} \mathbb{C}$ runs over the $2m$ eigenspaces of F on $V_\mathbb{C}$; it is a one-dimensional F-vector space, of dimension $2m$ over $\mathbb{Q}$. Write

$$\mathbf{W}(F,n,\delta)\text{: every abelian variety of } (F,n,\delta)\text{-Weil type has algebraic Weil classes.}$$

The period domain $\mathscr{D}_F$ is the set of complex structures J on $V_\mathbb{R}$ commuting with F and polarised by E, and the group $G_F = \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V,H)$ acts on it. We fix an arithmetic subgroup $\Gamma \subset G_F(\mathbb{Q})$ and write $\mathscr{S}_F = \Gamma\backslash\mathscr{D}_F$ for the quotient, $p\colon \mathscr{D}_F \to \mathscr{S}_F$ for the projection, $\pi\colon \mathscr{A} \to \mathscr{S}_F$ for the universal family and $\Sigma_F \subseteq \mathscr{D}_F$ for the set of s at which $W_F(A_s)$ is algebraic.

Definition A.2 (The imaginary quadratic case)**.** For $m = 1$ the field $F = K = \mathbb{Q}(\sqrt{-d})$ is imaginary quadratic and $F_0 = \mathbb{Q}$. We then write $V_\mathbb{C} = V_+ \oplus V_-$ for the two eigenspaces of K, $W(A) = W_K(A)$ for the two-dimensional rational space of Weil classes, $\theta \in \bigwedge^2 V$ for the class of the polarisation E, and $\mathscr{D}_{n,n} = \mathscr{D}_K$. We fix generators $\alpha_\pm$ of the lines $\bigwedge^{2n} V_\pm$, so that $W(A)\otimes\mathbb{C} = \mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$; a real class of $W(A)\otimes\mathbb{R}$ is $u\alpha_+ + \overline{u}\alpha_-$ once $\alpha_- = \overline{\alpha_+}$. Since $E(\alpha x, y) = E(x, \overline{\alpha} y)$, the form E vanishes on $V_+ \times V_+$ and on $V_- \times V_-$, so

$$\theta \in V_+ \otimes V_- \subseteq \bigwedge^2 V_\mathbb{C}, \tag{A.1}$$

and H identifies V_- with the dual of V_+ as a representation of $\mathrm{SU}(V,H)\otimes\mathbb{C} = \mathrm{SL}(V_+)$.

Theorem A.3 (The Hodge ring in the imaginary quadratic case)**.** *Let A be an abelian $2n$-fold of Weil type for the imaginary quadratic field K with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$. Then*

$$\mathrm{Hdg}^*(A) = \mathbb{Q}[\theta] \oplus W(A), \qquad W(A)\cdot\theta = 0,$$

that is, the space of Hodge classes of degree $2k$ is $\mathbb{Q}\theta^k$ for $k \neq n$ and $\mathbb{Q}\theta^n \oplus W(A)$ for $k = n$. The equality $W(A)\cdot\theta = 0$ holds for every abelian variety of Weil type for K.

Proof. The Hodge classes are the invariants of $\mathrm{Hg}(A)$ in $\bigwedge^* V$, and taking invariants commutes with extension of scalars, so we compute the invariants of $\mathrm{SL}(V_+) \cong \mathrm{SL}_{2n}(\mathbb{C})$ in

$$\bigwedge^k V_\mathbb{C} = \bigoplus_{r+s=k} \bigwedge^r V_+ \otimes \bigwedge^s V_+^\vee.$$

An invariant is a vector of weight zero. Fix a basis of V_+, the diagonal torus, and for $T,U \subseteq \{1,\dots,2n\}$ the monomials $v_T \otimes v_U^\vee$ in the basis and its dual. The weight of $v_T \otimes v_U^\vee$ is $\mathbf{1}_T - \mathbf{1}_U$ modulo $(1,\dots,1)$, with entries in $\{-1,0,1\}$, so it is zero only for $T = U$, for $(T,U) = (\{1,\dots,2n\},\emptyset)$ and for $(T,U) = (\emptyset,\{1,\dots,2n\})$. The last two occur for $(r,s) = (2n,0)$ and $(0,2n)$ and give the invariant lines $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$, which span $W(A)\otimes\mathbb{C}$. For $r = s$ the invariants in $\bigwedge^r V_+ \otimes \bigwedge^r V_+^\vee \cong \mathrm{End}(\bigwedge^r V_+)$ form the line of scalars, by Schur's lemma, since $\bigwedge^r V_+$ is irreducible. By (A.1) the class θ^r lies in this summand; it is invariant, and nonzero for $r \leq 2n$ because θ is a nondegenerate form on a space of dimension $4n$. This gives the list. Finally $\alpha_\pm\theta \in \bigwedge^{2n+1} V_\pm \otimes V_\mp = 0$ by (A.1), with no hypothesis on the Hodge group. □

Proposition A.4 (The Weil classes of a CM field)**.** *Keep Setting A.1, with (V,H) of (F,n)-Weil type.*

(i) *A polarised complex structure J is the same as an H_τ-orthogonal decomposition $V_\tau = V_\tau^+ \oplus V_\tau^-$ for every τ, with H_τ definite of the sign of $\mathrm{Im}\,\sigma_\tau(\xi)$ on V_τ^+ and of the opposite sign on V_τ^-; then $\dim V_{\sigma_\tau}^{1,0} = \dim V_\tau^+ = n$ and $\dim V_{\overline{\sigma}_\tau}^{1,0} = \dim V_\tau^- = n$, so every class in $W_F(A_J)$ is of type (n,n). The spaces $W_F(A_J)$ form a flat subsystem W_F of $R^{2n}\pi_*\mathbb{Q}$ of rank $2m$ on which G_F acts trivially, and $\mathscr{D}_F \cong \prod_\tau \mathscr{D}_{n,n}$ is connected of dimension mn^2.*

(ii) *If one nonzero element of $W_F(A)$ is algebraic, every element is.*

(iii) *If $\mathrm{Hg}(A_s) = G_F$, the Hodge ring of A_s is generated by $\mathrm{NS}(A_s)_\mathbb{Q}$ and by W_F, so the Hodge conjecture for A_s in every degree is equivalent to the algebraicity of W_F. For $n \geq 2$ the space $\mathrm{NS}(A_s)_\mathbb{Q}$ is the m-dimensional space of classes $\eta_t = [E(t\cdot,\cdot)]$, $t \in F_0$.*

(iv) *Two members of (F,n,δ)-Weil type have isomorphic (V,H), so every one of them is a fibre of π.*

Proof. (i) A complex structure commuting with F preserves each V_τ, since $F_0\otimes\mathbb{R}$ acts diagonally, and is $\mathbb{C}$-linear on it for the structure $F\otimes_{F_0,\tau}\mathbb{R} \cong \mathbb{C}$, so it is $\pm\sqrt{-1}$ on two complementary $\mathbb{C}$-subspaces $V_\tau^\pm$. Write $\sigma_\tau(\xi) = \sqrt{-1}\,r_\tau$ with $r_\tau \in \mathbb{R}^\times$. For $x \in V_\tau^\varepsilon$ and $y \in V_\tau^{\varepsilon'}$ one has $H_\tau(Jx,Jy) = \varepsilon\varepsilon' H_\tau(x,y)$, so $E(Jx,Jy) = E(x,y)$ forces $E(x,y) = 0$ for $\varepsilon \neq \varepsilon'$, and replacing x by $\sqrt{-1}\,x$ then forces $H_\tau(x,y) = 0$:

the decomposition is orthogonal. For $x \in V_\tau^\varepsilon$, $E(x,Jx) = 2\mathrm{Re}\big(\sqrt{-1}\, r_\tau \overline{\varepsilon\sqrt{-1}} H_\tau(x,x)\big) = 2\varepsilon r_\tau H_\tau(x,x)$, so positivity is the sign condition stated. Conversely such a decomposition defines a polarised J. Under $V_\tau \otimes_{\mathbb{R}} \mathbb{C} = V_{\sigma_\tau} \oplus V_{\overline{\sigma}_\tau}$ the space $V^{1,0}$, the $\sqrt{-1}$-eigenspace of J on $V_{\mathbb{C}}$, meets V_{σ_τ} in the image of V_τ^+ and $V_{\overline{\sigma}_\tau}$ in the conjugate of the image of V_τ^-, which gives the two equalities of dimensions. Since H_τ has signature (n,n) and is definite, of opposite signs, on the complementary orthogonal subspaces $V_\tau^\pm$, both have dimension n. So each V_σ is the sum of its intersections with $V^{1,0}$ and $V^{0,1}$, both of dimension n, and a generator α_σ of $\bigwedge^{2n} V_\sigma$ is of type (n,n). The subspace $\bigwedge_F^{2n} V$ of $H^{2n}(A_s,\mathbb{Q})$ is defined by the action of F on V alone and does not depend on s, so it is flat; $\mathrm{SU}(V,H)$ acts on $\bigwedge_F^{2n} V$ by the F-determinant, which is 1. Finally, the choice of V_τ^+ is a point of the bounded symmetric domain of $\mathrm{SU}(V_\tau,H_\tau) \cong \mathrm{SU}(n,n)$, of dimension n^2, and $\mathscr{D}_F$ is the product over τ.

(ii) For $\alpha \in \mathscr{O}_F$ the endomorphism $\iota(\alpha)$ acts on $\bigwedge_F^{2n} V$ by α^{2n}, through an algebraic correspondence, so the algebraic classes in W_F form a $\mathbb{Q}$-subspace stable under multiplication by every α^{2n}. By the polarisation identity, in characteristic zero every product $x_1 \cdots x_{2n}$ of elements of a field is a rational combination of $2n$-th powers of sums of the x_i; taking $x_2 = \cdots = x_{2n} = 1$ shows that the $2n$-th powers span F over $\mathbb{Q}$, and since every element of F is a rational multiple of an element of $\mathscr{O}_F$, so do the α^{2n} with $\alpha \in \mathscr{O}_F$. Hence the $\mathbb{Q}$-span of $\{\alpha^{2n} w\}$ is $Fw = W_F$ for every $w \neq 0$.

(iii) Hodge classes are the invariants of the Hodge group, which here is G_F. Over $\mathbb{C}$ it is $\prod_\tau \mathrm{SL}(U_\tau)$ with $U_\tau = V_{\sigma_\tau}$, and H identifies $V_{\overline{\sigma}_\tau}$ with $U_\tau^\vee$, so $\bigwedge^* V_{\mathbb{C}} = \bigotimes_\tau \bigwedge^*(U_\tau \oplus U_\tau^\vee)$ with the τ-th factor of the group acting on the τ-th tensor factor alone. The invariants of a product group in a tensor product of representations are the tensor product of the invariants, and the invariants of $\mathrm{SL}(U)$ in $\bigwedge^*(U \oplus U^\vee)$ are generated by the pairing $\psi \in U \otimes U^\vee$ and the two determinants, by the weight computation in the proof of Theorem A.3. The classes η_t are Hodge classes: J commutes with t and preserves E, hence preserves the F_0-twisted form $E_t = E(t\cdot,\cdot) = \mathrm{Tr}(\xi t H)$, which for totally positive t is again a polarisation of A_s. The complexified span of the η_t is $\bigoplus_\tau \mathbb{C}\psi_\tau$, and the complexification of W_F contains all the determinants. So the $\mathbb{Q}$-algebra generated by $\mathrm{NS}_{\mathbb{Q}}$ and W_F has complexification equal to the ring of invariants, hence equals $\mathrm{Hdg}^*(A_s)$. Since divisor classes are algebraic, the conjecture reduces to W_F, and by (ii) to one class. For the last sentence, the invariants in $\bigwedge^2 V_{\mathbb{C}}$ are $\bigoplus_\tau \mathbb{C}\psi_\tau$ when $n \geq 2$: a summand $\bigwedge^2 U_\tau$ or $\bigwedge^2 U_\tau^\vee$ has an $\mathrm{SL}(U_\tau)$-invariant only if $\dim U_\tau = 2n$ equals 2, and a summand mixing two different τ has none. So $\mathrm{NS}(A_s)_{\mathbb{Q}}$ has dimension m, and it contains the classes η_t, which form an m-dimensional space because $t \mapsto \eta_t$ is injective on F_0, E being nondegenerate.

(iv) Two hermitian forms over F/F_0 of the same rank, the same discriminant and the same signature at every real place of F_0 are isomorphic [77, Ch. 10, §6]. So the datum (F,n,δ) determines (V,H) up to isomorphism, and the Hodge structure of any member is, by (i), a polarised complex structure on that $V_{\mathbb{R}}$, that is a point of $\mathscr{D}_F$. □

Lemma A.5 (The Hodge group of a very general member)**.** *Keep Setting A.1. For every $s \in \mathscr{D}_F$ one has $\mathrm{Hg}(A_s) \subseteq G_F$, and there is a countable union Z of proper closed analytic subsets of $\mathscr{D}_F$ such that $\mathrm{Hg}(A_s) = G_F$ for every $s \notin Z$.*

Proof. The Hodge group of A_s commutes with $\mathrm{End}^0(A_s) \supseteq F$ and preserves E, so it lies in $\mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{U}(V,H)$: an F-linear map preserving $E = \mathrm{Tr}_{F/\mathbb{Q}}(\xi H)$ preserves H, because H is recovered from E and the action of F: the trace form of F is nondegenerate, so $H(x,y)$ is determined by the values $E(\alpha x,y) = \mathrm{Tr}_{F/\mathbb{Q}}(\xi \alpha H(x,y))$, $\alpha \in F$. It fixes every Hodge class, in particular the Weil classes, which are Hodge classes by Proposition A.4(i); an element g of $\mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{U}(V,H)$ acts on $\bigwedge_F^{2n} V$ by $\det_F(g)$, so $\det_F(g) = 1$ and $g \in G_F$.

For the second statement let T be the set of rational tensors $t \in \bigoplus_{a,b} V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ that are not fixed by G_F, a countable set, and for $t \in T$ let Z_t be the set of s at which t is a Hodge class, of type (p,p) with $2p = a - b$, for the Hodge structure induced by A_s. The condition is that t lies in the p-th step of the Hodge filtration, which varies holomorphically with s, so Z_t is a closed analytic subset of $\mathscr{D}_F$. It is proper. Otherwise t is fixed by $h_s(\mathrm{U}(1))$ for every $s \in \mathscr{D}_F$, where h_s is the Hodge representation, which acts on the (p,q) part through z^{q-p}, and hence by the subgroup N of $G_F(\mathbb{R})$ that these tori generate. Since $\mathscr{D}_F$ is one orbit of $G_F(\mathbb{R})^+$ and $h_{gs} = g h_s g^{-1}$, the group N is a connected normal subgroup of $G_F(\mathbb{R})^+ = \prod_\tau \mathrm{SU}(V_\tau,H_\tau)$, so its Lie algebra is an ideal of $\bigoplus_\tau \mathfrak{su}(V_\tau,H_\tau)$, a sum of some of these simple summands. It contains every summand, because $h_s(\sqrt{-1})$ acts on V_τ as the complex structure

J, which is $\sqrt{-1}$ on V_τ^+ and $-\sqrt{-1}$ on V_τ^- by Proposition A.4(i), and is therefore not central in the factor $\mathrm{SU}(V_\tau, H_\tau)$. So $N = G_F(\mathbb{R})^+$, which is Zariski dense in the connected group G_F, and t would be fixed by G_F, contrary to $t \in T$.

Put $Z = \bigcup_{t\in T} Z_t$. The Hodge group of A_s is the subgroup of $\mathrm{GL}(V)$ fixing every rational Hodge class in the spaces $V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ [28, §3]. For $s \notin Z$ each such class is fixed by G_F, so $G_F \subseteq \mathrm{Hg}(A_s)$, and equality follows from the first statement. □

Theorem A.6 (A base point in every family, for every CM field)**.** *Let (V,H) be of (F,n)-Weil type with discriminant δ. Then $\mathscr{D}_F$ contains a point s_0 with the following properties.*

(i) *A_{s_0} is isogenous to $B_\Phi^n \times B_{\overline{\Phi}}^n$, where B_Φ is an abelian m-fold with complex multiplication by F of type $\Phi = \{\sigma : \mathrm{Im}\,\sigma(\xi) > 0\}$ and $B_{\overline{\Phi}}$ its conjugate; there is an H-orthogonal basis $e_1, \dots, e_{2n}$ of V with $H(e_i, e_i)$ totally positive for $i \le n$ and totally negative for $i > n$, and $Fe_i = H^1(B_\Phi, \mathbb{Q})$ for $i \le n$, $Fe_i = H^1(B_{\overline{\Phi}}, \mathbb{Q})$ for $i > n$.*

(ii) *For $i \le n$ put $i' = i+n$ and let $D_{ii'} \subseteq H^2(A_{s_0}, \mathbb{Q})$ be the F-balanced part of $Fe_i \wedge Fe_{i'}$, spanned by the classes*

$$\delta_i(f) = \sum_k (\omega_k f e_i) \wedge (\omega_k^* e_{i'}), \qquad f \in F,$$

for dual $\mathbb{Q}$-bases (ω_k), (ω_k^*) of F with respect to the trace form. Every $\delta_i(f)$ is a Hodge class of type $(1,1)$, hence a divisor class, and $D_{ii'}$ is a one-dimensional F-vector space with $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$.

(iii) *The Weil classes are the products*

$$w(f) = \sum_{k_1,\dots,k_{n-1}} \delta_1(\omega_{k_1} f) \cdot \delta_2(\omega_{k_1}^* \omega_{k_2}) \cdots \delta_{n-1}(\omega_{k_{n-2}}^* \omega_{k_{n-1}}) \cdot \delta_n(\omega_{k_{n-1}}^*), \qquad f \in F,$$

which lie in $W_F(A_{s_0})$, span it as f runs over F, and are polynomials with rational coefficients in divisor classes. In particular $W_F(A_{s_0})$ is algebraic, and $\Sigma_F \neq \emptyset$ for every F, every $n \ge 1$ and every δ.

Proof. (i) Since H_τ has signature (n,n) at every τ, the determinant $\det H$ has sign $(-1)^n$ at every real place, so $b = (-1)^n \det H$, a representative of δ, is totally positive, and the diagonal form $H' = \mathrm{diag}(b, 1, \dots, 1, -1, \dots, -1)$ with n positive and n negative entries has the same dimension, the same discriminant and the same signatures as H. Hermitian forms over F/F_0 are classified by these invariants [77, Ch. 10, §6], so $(V,H) \cong (F^{2n}, H')$ and the standard basis is the required orthogonal basis e_i, with $a_i = H(e_i,e_i)$. Write $\sigma_\tau(\xi) = \sqrt{-1}\, r_\tau$ and define J on $V_\tau = \bigoplus_i \mathbb{C}e_i$ to be $\varepsilon_{i,\tau}\sqrt{-1}$ on $\mathbb{C}e_i$, with $\varepsilon_{i,\tau} = \mathrm{sign}(r_\tau)\,\mathrm{sign}(\tau(a_i))$. The decomposition $V_\tau^+ = \bigoplus_{\varepsilon_{i,\tau}=+} \mathbb{C}e_i$, $V_\tau^- = \bigoplus_{\varepsilon_{i,\tau}=-} \mathbb{C}e_i$ is H_τ-orthogonal, with H_τ of sign $\mathrm{sign}(r_\tau)$ on V_τ^+ and of the opposite sign on V_τ^-, so Proposition A.4(i) makes J a point s_0 of $\mathscr{D}_F$. On $Fe_i \otimes \mathbb{R} = \prod_\tau \mathbb{C}e_i$ the structure J is multiplication by the element $(\varepsilon_{i,\tau}\sqrt{-1})_\tau$ of $F \otimes_{\mathbb{Q}} \mathbb{R}$, so each Fe_i is a rational sub-Hodge structure of weight one with F-action, polarised by the restriction of E; that is, it is the H^1 of an abelian m-fold with complex multiplication by F, of CM type $\Phi_i = \{\sigma : (Fe_i)_\sigma \subseteq V^{1,0}\}$, which by the computation in Proposition A.4(i) is $\{\sigma_\tau^{\varepsilon_{i,\tau}}\}$, where $\sigma^+ = \sigma$ and $\sigma^- = \overline{\sigma}$. For $i \le n$ the a_i are totally positive, so $\varepsilon_{i,\tau} = \mathrm{sign}(r_\tau)$ and $\Phi_i = \{\sigma : \mathrm{Im}\,\sigma(\xi) > 0\} = \Phi$; for $i > n$ the signs are reversed and $\Phi_i = \overline{\Phi}$. So $V = \bigoplus_i Fe_i$ splits A_{s_0} up to isogeny as stated.

(ii) Write $L_{i,\sigma} = (Fe_i)_\sigma$, a line, so that $V_{\mathbb{C}} = \bigoplus_{i,\sigma} L_{i,\sigma}$ and $L_{i,\sigma}$ has type $(1,0)$ exactly when $\sigma \in \Phi_i$. For dual bases, $\sum_k \sigma(\omega_k)\sigma'(\omega_k^*) = 1$ if $\sigma = \sigma'$ and 0 otherwise, because the matrices $(\sigma(\omega_k))$ and $(\sigma(\omega_k^*))$ are inverse transposes by the definition of the trace form. Let $\ell_{i,\sigma}$ be the component of e_i in $L_{i,\sigma}$, a generator of that line, so that $\omega e_i = \sum_\sigma \sigma(\omega)\,\ell_{i,\sigma}$ for $\omega \in F$. Hence in $Fe_i \otimes Fe_{i'} \otimes \mathbb{C} = \bigoplus_{\sigma,\sigma'} L_{i,\sigma} \otimes L_{i',\sigma'}$ the class $\delta_i(f)$ has component $\sigma(f)\,\ell_{i,\sigma} \wedge \ell_{i',\sigma}$ on the diagonal $\sigma = \sigma'$ and zero off it. In particular it is F-balanced: $\alpha \in F$ acting through the first factor and through the second agree on it, both multiplying the component at σ by $\sigma(\alpha)$, and this is the identity $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$. Since $\Phi_{i'} = \overline{\Phi}_i$, exactly one of $L_{i,\sigma}, L_{i',\sigma}$ has type $(1,0)$, so every component, hence $\delta_i(f)$, has type $(1,1)$; and $\delta_i(f)$ is rational, being a sum of wedges of rational vectors. By the Lefschetz theorem on $(1,1)$ classes it is a divisor class. As f varies the components $\sigma(f)$ vary independently, so $D_{ii'} \otimes \mathbb{C} = \bigoplus_\sigma L_{i,\sigma} \otimes L_{i',\sigma}$ and $D_{ii'} \cong F$.

(iii) The complexification of W_F is spanned by $\alpha_\sigma = \pm\prod_{i\le n}\ell_{i,\sigma}\wedge\ell_{i',\sigma}$, a generator of $\bigwedge^{2n}V_\sigma = \bigwedge^{2n}\big(\bigoplus_i L_{i,\sigma}\big)$. In the product $w(f)$, the component indexed by $(\sigma_1,\dots,\sigma_n)$, with σ_i the embedding chosen in the i-th factor, carries the coefficient $\sigma_1(f)\prod_{j=1}^{n-1}\sum_{k_j}\sigma_j(\omega_{k_j})\sigma_{j+1}(\omega_{k_j}^*)$, which vanishes unless $\sigma_1=\cdots=\sigma_n$ and then equals $\sigma_1(f)$. So $w(f)=\sum_\sigma\sigma(f)\,\alpha_\sigma$ up to the sign of the reordering, an element of $W_F\otimes\mathbb{C}$ which is rational because it is a rational polynomial in rational classes; as f runs over F the vectors $(\sigma(f))_\sigma$ span $\mathbb{C}^{2m}$, so the $w(f)$ span W_F. Each $w(f)$ is a rational combination of n-fold products of the divisor classes $\delta_i(\cdot)$, and a product of divisor classes is algebraic. □

A.2. **The algebraic locus of the Weil classes**

Theorem A.7 (Density and strata of the algebraic locus)**.** *Fix (F,n,δ) with $n\ge 2$ and keep Setting A.1.*

(i) *$\mathbf{W}(F,n,\delta)$ holds if and only if $\Sigma_F=\mathscr{D}_F$, and then the Hodge conjecture holds in every degree for every member of the family with $\mathrm{Hg}(A_s)=G_F$.*

(ii) *Σ_F is stable under $G_F(\mathbb{Q})$, every $G_F(\mathbb{Q})$-orbit in $\mathscr{D}_F$ is dense, and Σ_F is dense.*

(iii) *Σ_F is the preimage under p of a countable union of Zariski closed subsets of $\mathscr{S}_F$; either $\Sigma_F=\mathscr{D}_F$, or Σ_F is meagre.*

(iv) *The following are equivalent: Σ_F is closed; Σ_F has an interior point; $\Sigma_F=\mathscr{D}_F$; the Hodge conjecture holds in every degree for A_s for all s outside a countable union of proper closed analytic subsets of $\mathscr{D}_F$.*

Proof. (i) By Proposition A.4(i) and (iv), every abelian variety of (F,n,δ)-Weil type has its Hodge structure at some point of $\mathscr{D}_F$, so $\mathbf{W}(F,n,\delta)$ is the algebraicity of W_F at every point, and by Proposition A.4(iii) it gives the conjecture at the members with full Hodge group.

(ii) An element $g\in G_F(\mathbb{Q})$ carries J to gJg^{-1} and induces, after clearing denominators, an isogeny $A_J\to A_{gJ}$; pull-back and push-forward along an isogeny carry algebraic classes to algebraic classes, and g fixes W_F pointwise by Proposition A.4(i). So Σ_F is stable.

For the density we show that the closure L of $G_F(\mathbb{Q})$ in $G_F(\mathbb{R})=\prod_\tau \mathrm{SU}(V_\tau,H_\tau)$ contains the identity component, which acts transitively on $\mathscr{D}_F$. First, the form H is isotropic over F_0. It is indefinite at every real place, its signature being (n,n). At every finite place v its trace form is a quadratic form in $4n\ge 8$ variables over the local field $F_{0,v}$, hence isotropic, and a hermitian form is isotropic as soon as its trace form is [77, Ch. 10, §1]. Since hermitian forms over the quadratic extension F/F_0 satisfy the Hasse principle [77, Ch. 10, §6], H is isotropic. Let $X=\{v\in V : H(v,v)=0\}$ be the cone of isotropic vectors, a quadric cone in the $4n$-dimensional F_0-space V with a nonzero F_0-point. Projection from that point shows that the projective quadric of its lines is F_0-rational, so it satisfies weak approximation at the archimedean places, F_0 being dense in $F_0\otimes\mathbb{R}=\prod_\tau\mathbb{R}$. Hence the F_0-points of X are dense in its points over $F_0\otimes\mathbb{R}$, which are the products $\prod_\tau X_\tau$ of the isotropic cones of the H_τ.

For an isotropic vector $v\in V$ let $U_v\subseteq\mathrm{SU}(V,H)$ be the unipotent radical of the stabiliser of the line Fv, a unipotent F_0-group. The exponential identifies $U_v(F_0)$ with an F_0-vector space and $U_v(F_0\otimes\mathbb{R})=\prod_\tau U_{v,\tau}(\mathbb{R})$ with the corresponding real space, so $U_v(F_0)$ is dense in $\prod_\tau U_{v,\tau}(\mathbb{R})$. Hence L contains $\prod_\tau U_{v,\tau}(\mathbb{R})$ for every rational isotropic v, and its Lie algebra $\mathfrak{l}$ contains $\mathfrak{u}_{v,\tau}$ for every τ and every rational isotropic v. The map $v\mapsto\mathfrak{u}_{v,\tau}$ is continuous on $X_\tau\setminus 0$ and rational v are dense in $\prod_\tau X_\tau$, so the closed subspace $\mathfrak{l}\cap\mathfrak{su}(V_\tau,H_\tau)$ contains $\mathfrak{u}_{v,\tau}$ for every isotropic $v\in V_\tau$. The span of these is stable under the adjoint action of $\mathrm{SU}(V_\tau,H_\tau)$, which permutes the isotropic lines, so it is a nonzero ideal of the simple Lie algebra $\mathfrak{su}(n,n)$, hence everything. Thus $\mathfrak{l}\supseteq\bigoplus_\tau\mathfrak{su}(V_\tau,H_\tau)$ and $L\supseteq G_F(\mathbb{R})^+$, so every $G_F(\mathbb{Q})$-orbit in $\mathscr{D}_F$ is dense. Since Σ_F is nonempty by Theorem A.6 and $G_F(\mathbb{Q})$-stable, it contains a dense orbit.

(iii) The quotient $\mathscr{S}_F$ is a quasi-projective variety by the theorem of Baily and Borel [9]; after passing to a neat subgroup Γ of finite index it is smooth and carries the polarised abelian scheme $\pi\colon\mathscr{A}\to\mathscr{S}_F$. By Proposition A.4(i) the group $\Gamma\subseteq G_F(\mathbb{Q})$ fixes W_F pointwise, so every $w\in W_F$ descends to a flat section of $R^{2n}\pi_*\mathbb{Q}$ over $\mathscr{S}_F$ that is a Hodge class at every point. Fix a $\mathbb{Q}$-basis $w_1,\dots,w_{2m}$ of W_F. By Lemma 4.2(i) the algebraic locus $\Sigma(w_j)\subseteq\mathscr{S}_F$ of each w_j is a countable union of Zariski closed subsets, and so is the finite intersection $\bigcap_j\Sigma(w_j)$, whose preimage is Σ_F; by Proposition A.4(ii) it equals each $\Sigma(w_j)$. The alternative is Lemma 4.2(ii), since p is a local homeomorphism and the preimage of a nowhere dense closed set is nowhere dense and closed.

(iv) If Σ_F is closed it is closed and dense by (ii), hence everything. If it has an interior point it is not meagre, by Baire's theorem, and (iii) gives $\Sigma_F = \mathscr{D}_F$. If $\Sigma_F = \mathscr{D}_F$, the Hodge conjecture holds for every member with $\mathrm{Hg}(A_s) = G_F$ by (i), and these are the members outside the set Z of Lemma A.5. Finally, if the Hodge conjecture holds outside a countable union Z' of proper closed analytic subsets, then Σ_F contains the complement of Z', since the Weil classes are Hodge classes at every point; this complement is not meagre, and (iii) gives $\Sigma_F = \mathscr{D}_F$. □

Corollary A.8 (The algebraic locus, for every CM field)**.** *For every CM field F of degree $2m$, every $n \geq 2$ and every discriminant δ, the polarised period domain $\mathscr{D}_F$ of the family of (F,n,δ)-Weil type is connected of dimension mn^2, its very general member has Hodge group $G_F = \mathrm{Res}_{F_0/\mathbb{Q}}\,\mathrm{SU}(V,H)$, and the algebraic locus Σ_F of the Weil classes*

(i) *contains a point s_0 at which A_{s_0} is isogenous to a product of abelian varieties with complex multiplication by F and the Weil classes are polynomials with rational coefficients in divisor classes;*

(ii) *is dense in $\mathscr{D}_F$;*

(iii) *is the preimage of a countable union of Zariski closed subsets of $\mathscr{S}_F$, and is either $\mathscr{D}_F$ or meagre.*

Proof. Proposition A.4(i), Lemma A.5, Theorem A.6 and Theorem A.7(ii), (iii). □

Proposition A.9 (The K-valued Weil class is multiplicative)**.** *Let $K = \mathbb{Q}(\sqrt{-d})$, write $\delta_K = \sqrt{-d} \in \mathbb{C}$, and let A be of Weil type for K with $H^1(A,\mathbb{Q}) = V$ and a K-basis $u_1,\dots,u_{2n}$ of V. Put $x_j = u_j$, $y_j = \sqrt{-d}\,u_j \in V$ and*

$$\omega(A) = d^{-n}\prod_{j=1}^{2n}\big(dx_j + \delta_K y_j\big) = \omega_1(A) + \delta_K\,\omega_2(A), \qquad \omega_1(A),\omega_2(A) \in \textstyle\bigwedge^{2n} V. \tag{A.2}$$

Then $\omega(A)$ spans $\bigwedge^{2n} V_-$, its conjugate spans $\bigwedge^{2n} V_+$, and $\omega_1(A),\omega_2(A)$ form a rational basis of $W(A)$. If A_1, A_2 are of Weil type for K, of dimensions $2n_1$, $2n_2$, then $A_1 \times A_2$ with the diagonal action and the product polarisation is of Weil type, and for the basis obtained by concatenation

$$\begin{aligned}\omega_1(A_1\times A_2) &= \omega_1(A_1)\,\omega_1(A_2) - d\,\omega_2(A_1)\,\omega_2(A_2),\\ \omega_2(A_1\times A_2) &= \omega_1(A_1)\,\omega_2(A_2) + \omega_2(A_1)\,\omega_1(A_2),\end{aligned} \tag{A.3}$$

that is, $\omega(A_1\times A_2) = \omega(A_1)\,\omega(A_2)$, the products being cup products after pull-back along the two projections.

Proof. The vector $dx_j + \delta_K y_j = du_j + \delta_K\sqrt{-d}\,u_j \in V_{\mathbb{C}}$ satisfies $\sqrt{-d}\cdot(dx_j+\delta_K y_j) = dy_j - d\,\delta_K x_j = -\delta_K(dx_j+\delta_K y_j)$, so it lies in the eigenspace on which $\sqrt{-d}$ acts by $-\delta_K$; call it V_-. These $2n$ vectors are the images of the K-basis under the projection $V \to V_-$, up to the factor $2d$, so they form a basis of V_- and their product spans $\bigwedge^{2n} V_-$. The conjugate spans $\bigwedge^{2n} V_+$, so $\omega_1 = \frac{1}{2}(\omega+\overline{\omega})$ and $\omega_2 = (2\delta_K)^{-1}(\omega-\overline{\omega})$ are rational, lie in $W(A)\otimes\mathbb{C}$ and span it. The product is of Weil type because $H^1(A_1\times A_2) = V_1 \oplus V_2$ as K-modules with polarisation $E_1 \oplus E_2$, so the hermitian form is $H_1 \oplus H_2$, of signature (n_1+n_2, n_1+n_2). In (A.2) for the concatenated basis the product splits as

$$\omega(A_1\times A_2) = \Big(d^{-n_1}\prod_{j\leq 2n_1}(dx_j+\delta_K y_j)\Big)\Big(d^{-n_2}\prod_{j>2n_1}(dx_j+\delta_K y_j)\Big) = \omega(A_1)\,\omega(A_2),$$

with no sign, the factors being taken in the order of the basis. Expanding $(\omega_1(A_1) + \delta_K\omega_2(A_1))(\omega_1(A_2)+\delta_K\omega_2(A_2))$ with $\delta_K^2 = -d$ and comparing rational parts and δ_K-parts, which is legitimate because the classes $\omega_j(A_i)$ are rational and $1,\delta_K$ are linearly independent over $\mathbb{Q}$, gives (A.3). □

A.3. **The polarised semiregularity criterion**

Setting A.10 (Polyvector fields, evaluation and semiregularity)**.** Let A be an abelian variety of dimension g. Write

$$HH^*(A) = \bigoplus_{p,q} H^p\big(A, \textstyle\bigwedge^q T_A\big) = \bigwedge^*\big(H^{0,1}(A)\oplus H^0(T_A)\big),$$

graded by $p+q$; by the Hochschild–Kostant–Rosenberg isomorphism this is the Hochschild cohomology of A, as a graded ring because the Todd class of A is trivial [22, 24]. Its part of degree one, $HH^1(A) = H^{0,1}(A) \oplus H^0(T_A)$, has dimension $2g$, and its part of degree two is

$$HT^2(A) = H^{0,2}(A) \oplus H^1(T_A) \oplus H^0(\textstyle\bigwedge^2 T_A), \qquad \dim HT^2(A) = \binom{2g}{2}.$$

The algebra $HH^*(A)$ acts on $H^*(A,\mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$: a class of $H^{0,1}$ acts by wedge product and a class of $H^0(T_A) = (H^{1,0})^\vee$ by contraction; the two families of operators involve disjoint generators, so they anticommute and the action is an action of the exterior algebra. We write $\xi \lrcorner \gamma$ for the action of ξ on γ, and $\mathrm{Ann}_{HH^k}(\gamma)$, $\mathrm{Ann}_{HT^2}(\gamma)$ for the classes ξ of the given degree with $\xi \lrcorner \gamma = 0$. On $H^1(T_A) = \mathrm{Hom}(H^{1,0}, H^{0,1})$ the action is the derivation that replaces one factor of type $(1,0)$ by its image, so $\mathrm{Ann}_{HT^2}(\gamma) \cap H^1(T_A)$ is the space of first order deformations of A along which γ stays of Hodge type to first order [37].

Let E be a nonzero perfect complex on A with Atiyah class $\mathrm{At}(E) \in \mathrm{Ext}^1(E, E \otimes \Omega^1_A)$, normalised so that $\mathrm{ch}(E) = \mathrm{tr}\exp \mathrm{At}(E)$. The evaluation map is

$$\mathrm{ev}_E \colon HT^2(A) \to \mathrm{Ext}^2(E,E), \qquad \mathrm{ev}_E(z,v,\pi) = z \cdot \mathrm{id}_E + v \lrcorner \mathrm{At}(E) + \tfrac{1}{2}\, \pi \lrcorner \mathrm{At}(E)^2,$$

the contraction with the exponential of the Atiyah class [44, Theorem A], [11], and the semiregularity map is

$$\sigma_E \colon \mathrm{Ext}^2(E,E) \longrightarrow \bigoplus_{q \geq 0} H^{q+2}(A, \Omega^q_A), \qquad \sigma_E(\zeta) = \sum_q \tfrac{1}{q!}\, \mathrm{tr}\big(\zeta \circ \mathrm{At}(E)^q\big).$$

They satisfy

$$\sigma_E\big(\mathrm{ev}_E(\xi)\big) = \xi \lrcorner \mathrm{ch}(E) \qquad (\xi \in HT^2(A)) \tag{A.4}$$

by [11, Proposition 6.2.1 and Corollary 6.3.2]. More generally ev_E extends to a homomorphism of graded rings $HH^*(A) \to \mathrm{Ext}^*(E,E)$ and σ_E to all degrees, with $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ [11, Section 6], [22, 24]; the form in degrees other than two is used only in Proposition A.47(iii). Buchweitz and Flenner use the opposite sign for the Atiyah class, which changes only the sign of the middle term of ev_E and does not affect any statement below.

Theorem A.11 (Buchweitz and Flenner [10]; Pridham [73])**.** *Let X be a smooth projective complex variety, E a perfect complex on X, and $X \hookrightarrow \mathscr{X} \to \mathrm{Spec}\, R$ a deformation of X over an Artin local $\mathbb{C}$-algebra along which $\mathrm{ch}(E)$ stays of Hodge type. If the semiregularity map*

$$\sigma_E \colon \mathrm{Ext}^2_X(E,E) \longrightarrow \bigoplus_{q \geq 0} H^{q+2}(X, \Omega^q_X), \qquad \sigma_E(\zeta) = \sum_q \tfrac{1}{q!}\, \mathrm{tr}\big(\zeta \circ \mathrm{At}(E)^q\big),$$

is injective, then E extends to a perfect complex on $\mathscr{X}$.

Lemma A.12 (Injective semiregularity deforms the complex)**.** *Let (S, s_0) be a germ of a complex manifold, $\mathscr{T} \to S$ a smooth proper family of complex tori with fibre T_0 over s_0, and $E^\bullet$ a bounded complex of holomorphic vector bundles on T_0 whose Chern character extends to a flat section of Hodge classes over S. If σ_E is injective, there is a neighbourhood U of s_0 such that for every $s \in U$ the fibre T_s carries a bounded complex of holomorphic vector bundles whose Chern character is the parallel transport of* $\mathrm{ch}(E)$.

Proof. By the theorem of Buchweitz and Flenner in Pridham's form (Theorem A.11, [10, 73]; the argument applies to compact complex manifolds), every obstruction to extending E along an Artinian deformation of T_0 inside S vanishes. The deformations of the complex structure inside S together with those of the holomorphic structures and differentials of $E^\bullet$ are governed by a differential graded Lie algebra of Dolbeault type with finite dimensional cohomology, and Kuranishi's method [51], in the form of [35, Sections 2–3], gives an analytic germ (M, m_0) with a morphism $\pi \colon (M, m_0) \to (S, s_0)$ and a complex of holomorphic vector bundles on the pulled back family of tori deforming $E^\bullet$, whose completion is a hull of the formal deformation functor. The vanishing of the obstructions says that π is formally smooth, hence smooth, so its image contains a neighbourhood U of s_0. Chern characters are locally constant in families, which gives the last statement. When $\mathscr{T} \to S$ is the analytic germ of a polarised abelian scheme and $\mathrm{Ext}^{<0}(E,E) = 0$, the same conclusion follows from Lieblich's algebraic stack of complexes [53], which is locally of finite presentation over the base with tangent

space $\mathrm{Ext}^1(E,E)$ and obstruction space $\mathrm{Ext}^2(E,E)$, and then U is the image of a smooth morphism, hence Zariski open. □

Theorem A.13 (The annihilator is the mixed part of a splitting)**.** *Let A be an abelian $2n$-fold of Weil type for K with $n \geq 2$, write $V_\pm^{1,0}$, $V_\pm^{0,1}$ for the Hodge components of $V_\pm$, each of dimension n, and $T_\pm = (V_\pm^{1,0})^\vee \subset H^0(T_A)$, and let $\omega = a\alpha_+ + b\alpha_- \in W(A)\otimes\mathbb{C}$ with $ab \neq 0$. Put*

$$P = \mathrm{Ann}_{HH^1}(\alpha_+) = V_+^{0,1} \oplus T_-, \qquad Q = \mathrm{Ann}_{HH^1}(\alpha_-) = V_-^{0,1} \oplus T_+.$$

(i) $\dim P = \dim Q = 2n$ *and* $HH^1(A) = P \oplus Q$*; in particular* $\mathrm{Ann}_{HH^1}(\omega) = 0$.

(ii) $\mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$*, of dimension* $4n^2$*; it is* $\big(V_+^{0,1}\otimes V_-^{0,1}\big) \oplus \mathrm{Hom}_K\big(H^{1,0}, H^{0,1}\big) \oplus \big(T_+ \otimes T_-\big)$*, and it equals* $\mathrm{Ann}_{HH^2}(\alpha_+) \cap \mathrm{Ann}_{HH^2}(\alpha_-)$.

(iii) *Contraction with α_- is injective on $\bigwedge^* P$, and contraction with α_+ is injective on $\bigwedge^* Q$; the two images meet in exactly one line, in degree $2n$.*

(iv) *For every $k \geq 1$,*

$$\mathrm{Ann}_{HH^k}(\omega) = \Big(\bigoplus_{\substack{i+j=k \\ i,j\geq 1}} \textstyle\bigwedge^i P \wedge \bigwedge^j Q \Big) \;\oplus\; \varepsilon_k,$$

where $\varepsilon_k = 0$ for $k \neq 2n$ and ε_{2n} is the line of (iii)*. So* $\dim \mathrm{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$ *for $k \geq 1$, while* $\mathrm{Ann}_{HH^0}(\omega) = 0$*, and the annihilator is the ideal generated by its degree-two part together with that one line.*

Proof. Choose bases $x_1,\dots,x_n$ of $V_+^{1,0}$ and $y_1,\dots,y_n$ of $V_-^{1,0}$, and bases $\bar{x}_i$ of $V_+^{0,1}$ and $\bar{y}_j$ of $V_-^{0,1}$ (the bar is notation only: complex conjugation carries $V_+^{1,0}$ to $V_-^{0,1}$), so that $\alpha_+ = x_1\cdots x_n\bar{x}_1\cdots\bar{x}_n$ and $\alpha_- = y_1\cdots y_n\bar{y}_1\cdots\bar{y}_n$, and write $\partial_{x_i}, \partial_{y_j}$ for the dual basis of $H^0(T_A) = T_+ \oplus T_-$.

(i) Of the $4n$ basis operators, $\bar{x}_i\wedge$ kills α_+ because $\bar{x}_i$ already occurs in it, and $\partial_{y_j}\lrcorner$ kills α_+ because no y occurs in it; so $P \supseteq V_+^{0,1} \oplus T_-$. Conversely, the classes $\bar{y}_j \wedge \alpha_+$ are linearly independent, as are the classes $\partial_{x_i}\lrcorner\alpha_+$, and the two families lie in different summands of H^*, so no nonzero combination of the operators $\bar{y}_j\wedge$ and $\partial_{x_i}\lrcorner$ kills α_+: $P = V_+^{0,1} \oplus T_-$, of dimension $2n$. The computation for Q is the same with the roles of the two eigenspaces exchanged, and $P \cap Q = 0$ with $\dim P + \dim Q = 4n$, so $HH^1 = P \oplus Q$. A class of degree one that kills ω kills α_+ and α_- separately, the two images lying in different summands and $ab \neq 0$, so it lies in $P \cap Q = 0$.

(iii) A monomial $\bar{x}_I\partial_{y_J} \in \bigwedge^* P$ sends α_- to $\pm\bar{x}_I\, y_{[n]\setminus J}\,\bar{y}_{[n]}$, and these are linearly independent for distinct pairs (I,J), which is the injectivity; the statement for Q is symmetric. For the intersection, the image of $\bigwedge^* P$ is spanned by monomials with no factor x and all n factors $\bar{y}$, and the image of $\bigwedge^* Q$, by the same computation, by monomials with no factor y and all n factors $\bar{x}$. Both images are spanned by monomials, so their intersection is spanned by the monomials common to both, and the only such monomial is $\bar{x}_{[n]}\bar{y}_{[n]}$. It is the image of $\pi_P = \bar{x}_{[n]}\partial_{y_{[n]}}$ and of $\pi_Q = \bar{y}_{[n]}\partial_{x_{[n]}}$, both of degree $2n$. So the intersection is the single line spanned by that class.

(ii) and (iv) Decompose $\bigwedge^k HH^1 = \bigoplus_{i+j=k} \bigwedge^i P \wedge \bigwedge^j Q$. A summand with $i,j \geq 1$ annihilates α_+, because it has a factor in P and the action is an action of the algebra, and annihilates α_- for the same reason with Q; so it lies in the annihilator of ω. The sum of these summands over all k is the ideal generated by $P \wedge Q$, since each of their monomials has, after reordering, a factor $p \wedge q$ with $p \in P$ and $q \in Q$. On the remaining two summands, $\xi_P \in \bigwedge^k P$ has $\xi_P\lrcorner\omega = b\,\xi_P\lrcorner\alpha_-$ and $\xi_Q \in \bigwedge^k Q$ has $\xi_Q\lrcorner\omega = a\,\xi_Q\lrcorner\alpha_+$, so a combination $\xi_P + \xi_Q$ annihilates ω if and only if $b\,\xi_P\lrcorner\alpha_- = -a\,\xi_Q\lrcorner\alpha_+$. By (iii) both contractions are injective and their images meet only in degree $2n$, in one line; so this happens only for $k = 2n$, and then only on the line spanned by $b^{-1}\pi_P - a^{-1}\pi_Q$, with π_P and π_Q scaled so that $\pi_P\lrcorner\alpha_- = \pi_Q\lrcorner\alpha_+$. In degree zero the annihilator is zero because $\omega \neq 0$. Taking $k = 2$ and $n \geq 2$, so that $2 \neq 2n$, gives (ii), the dimension being $\dim P \cdot \dim Q = 4n^2$; the general count is $\binom{4n}{k} - \binom{2n}{k} - \binom{2n}{k}$ plus the exceptional line. □

Setting A.14 (Polarised characters)**.** Let A be an abelian $2n$-fold of Weil type for the imaginary quadratic field K, $n \geq 2$, with polarisation class θ, in the notation of Definition A.2, and let

$\omega = u\alpha_+ + \bar{u}\,\alpha_-$, $u \neq 0$, be a nonzero real class of $W(A)\otimes\mathbb{R}$. For complex numbers $c_0,\dots,c_{2n}$ put

$$p(\theta) = \sum_{k=0}^{2n} c_k \theta^k, \qquad \gamma = \omega + p(\theta), \qquad \mu_k = k!\,c_k,$$

and let $\rho \in \{0,1,2,3\}$ be the rank of the $(2n-1)\times 3$ Hankel matrix $\mathsf{H} = [\mu_{j+i}]$, $0 \le j \le 2n-2$, $0 \le i \le 2$, the *Hankel rank* of γ. A class of polarised shape $N\omega + \sum c_k\theta^k$ with $N \neq 0$ is of this form after replacing ω by $N\omega$. Write $r(\gamma)$ for the rank of the contraction $HT^2(A) \to H^*(A,\mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$.

Choose a basis $e_0,\dots,e_{2n-1}$ of V_+ with $e_j \in V_+^{1,0}$ for $j < n$ and $e_j \in V_+^{0,1}$ for $j \ge n$. By (A.1) and since θ is nondegenerate of type $(1,1)$, there is a unique basis $f_0,\dots,f_{2n-1}$ of V_- with $\theta = \sum_j t_j$, $t_j = e_j \wedge f_j$, and then $f_j \in V_-^{0,1}$ for $j < n$ and $f_j \in V_-^{1,0}$ for $j \ge n$. Write $w(x)$ for the wedge product with a class x of type $(0,1)$ and $c(x)$ for the contraction with the tangent vector dual to a basis vector x of type $(1,0)$. For $a < n \le b$ put

$$z_{ab} = w(e_b)w(f_a), \qquad v^1_{ab} = w(e_b)c(e_a), \qquad v^2_{ab} = w(f_a)c(f_b), \qquad \pi_{ab} = c(e_a)c(f_b),$$

in $V_+^{0,1}\otimes V_-^{0,1} \subseteq H^{0,2}$, in $\mathrm{Hom}(V_+^{1,0},V_+^{0,1})$ and $\mathrm{Hom}(V_-^{1,0},V_-^{0,1}) \subseteq H^1(T_A)$, and in $T_+\otimes T_- \subseteq H^0(\bigwedge^2 T_A)$. These $4n^2$ classes form a basis of the space $P\wedge Q$ of Theorem A.13.

Theorem A.15 (The polarised criterion)**.** *In Setting A.14:*

(i) *For $n \ge 3$,*

$$\mathrm{Ann}_{HT^2}(\gamma) = \bigoplus_{a<n\le b} \left\{ x z_{ab} + y_1 v^1_{ab} + y_2 v^2_{ab} + w\,\pi_{ab} \;:\; (x,\, y_1+y_2,\, w) \in \ker \mathsf{H} \right\},$$

so that $\dim \mathrm{Ann}_{HT^2}(\gamma) = n^2(4-\rho)$ *and* $r(\gamma) = \binom{4n}{2} - n^2(4-\rho) = (4+\rho)n^2 - 2n$*, whatever $u \neq 0$ is. For $n = 2$ and $p = 0$ the same holds, with $\rho = 0$ and $r(\gamma) = 2n(2n-1)$.*

(ii) *The n^2 classes $v^1_{ab} - v^2_{ab}$ span the tangent space of the polarised family $\mathscr{D}_{n,n}$ at A, and they lie in* $\mathrm{Ann}_{HT^2}(\gamma)$*. For $n \ge 3$, if $c_k \neq 0$ for some $1 \le k \le 2n-1$, then* $\mathrm{Ann}_{HT^2}(\gamma) \cap H^1(T_A)$*, the first order Hodge locus of γ, is exactly this tangent space; and if $\rho = 3$ the whole of* $\mathrm{Ann}_{HT^2}(\gamma)$ *is this tangent space, with no component in $H^{0,2}$ or in $H^0(\bigwedge^2 T_A)$.*

(iii) *Let E be a perfect complex on A with* $\mathrm{ch}(E) = \gamma$*. Then* $\dim \mathrm{Ext}^2(E,E) \ge r(\gamma)$*, with no hypothesis on σ_E. If equality holds,* ev_E *is onto with kernel* $\mathrm{Ann}_{HT^2}(\gamma)$ *and σ_E is injective.*

(iv) *Let E be a perfect complex on A with* $\mathrm{ch}(E) = \gamma$ *and σ_E injective. Then E deforms with A over an open subset of $\mathscr{D}_{n,n}$, the Weil classes are algebraic on every member of the family $\mathscr{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member with Hodge group* $\mathrm{SU}(V,H)$*.*

Proof. (i) Let the torus $(\mathbb{C}^\times)^{2n}$ act on $V_\mathbb{C}$ by $e_j \mapsto \tau_j e_j$, $f_j \mapsto \tau_j^{-1} f_j$. It preserves the Hodge decomposition and θ, and scales $\alpha_\pm$ by $(\prod_j \tau_j)^{\pm 1}$; the action of $HH^*(A)$ is equivariant. The basis of $HH^1(A)$ formed by the operators $w(e_b)$, $w(f_a)$, $c(e_a)$, $c(f_b)$ consists of weight vectors of weights $\pm\varepsilon_j$, so a weight of $HT^2(A)$ has at most two nonzero coordinates, and for $n \ge 3$ two such weights never differ by a nonzero multiple of $\mathbf{1} = (1,\dots,1) \in \mathbb{Z}^{2n}$. Decompose $\xi \in HT^2(A)$ into weight components ξ_λ. Then $\xi_\lambda \lrcorner p(\theta)$, $\xi_\lambda \lrcorner \alpha_+$ and $\xi_\lambda \lrcorner \alpha_-$ have weights λ, $\lambda+\mathbf{1}$ and $\lambda-\mathbf{1}$, and all the weights so obtained from the different λ are distinct. Hence

$$\mathrm{Ann}_{HT^2}(\gamma) = \mathrm{Ann}_{HT^2}(p(\theta)) \cap \mathrm{Ann}_{HT^2}(\alpha_+) \cap \mathrm{Ann}_{HT^2}(\alpha_-),$$

and the last two meet in $P\wedge Q$ by Theorem A.13(ii) and its proof. For $n = 2$ and $p = 0$ the equality $\mathrm{Ann}_{HT^2}(\omega) = P\wedge Q$ is Theorem A.13(ii) itself. The space $P\wedge Q$ is the sum over $a < n \le b$ of the four-dimensional weight spaces of weight $\varepsilon_b - \varepsilon_a$ spanned by $z_{ab}, v^1_{ab}, v^2_{ab}, \pi_{ab}$, and $\mathrm{Ann}(p(\theta))$ is stable under the torus, so the computation splits pair by pair. Fix $a < n \le b$, drop the indices, and put $m = e_b \wedge f_a$ and $\theta' = \theta - t_a - t_b$. The t_j are even with $t_j^2 = 0$, so $\theta^k = \theta'^k + k\theta'^{k-1}(t_a+t_b) + k(k-1)\theta'^{k-2}t_a t_b$, and since $c(e_a)t_a = f_a$ and $c(f_b)t_b = -e_b$, direct evaluation gives

$$z \lrcorner \theta^k = m\,\theta'^k, \qquad v^1 \lrcorner \theta^k = k\,m\,\theta'^{k-1}, \qquad v^2 \lrcorner \theta^k = k\,m\,\theta'^{k-1}, \qquad \pi \lrcorner \theta^k = k(k-1)\,m\,\theta'^{k-2}.$$

So for $\xi = xz + y_1 v^1 + y_2 v^2 + w\pi$ and $\lambda = y_1 + y_2$,

$$\xi \lrcorner p(\theta) = m \wedge \sum_{j=0}^{2n-2} \frac{\theta'^j}{j!} \big(x\mu_j + \lambda \mu_{j+1} + w\mu_{j+2}\big).$$

The classes $m \wedge \theta'^j$, $0 \le j \le 2n-2$, are nonzero, because θ' is a nondegenerate form on the span of the $4n-4$ basis vectors other than e_a, f_a, e_b, f_b, and they have distinct degrees; so ξ kills $p(\theta)$ exactly when $(x, \lambda, w) \in \ker \mathsf{H}$. The map $(x, y_1, y_2, w) \mapsto (x, \lambda, w)$ is onto with kernel spanned by $v^1 - v^2$, which gives $1 + (3 - \rho)$ dimensions for each of the n^2 pairs.

(ii) A first order deformation of A in $H^1(T_A) = \mathrm{Hom}(H^{1,0}, H^{0,1})$ carries the action of K exactly when it is K-linear, that is, lies in $\mathrm{Hom}(V_+^{1,0}, V_+^{0,1}) \oplus \mathrm{Hom}(V_-^{1,0}, V_-^{0,1})$, with basis the v^1_{ab} and v^2_{ab}; and it keeps the polarisation exactly when it kills θ. By the formulas of (i) with $k = 1$, a combination $\sum y^1_{ab} v^1_{ab} + y^2_{ab} v^2_{ab}$ kills θ exactly when $y^1_{ab} + y^2_{ab} = 0$ for all pairs. So the tangent space of $\mathscr{D}_{n,n}$, which has dimension n^2 by Proposition A.4(i), is spanned by the classes $v^1_{ab} - v^2_{ab}$. These kill every θ^k by the same formulas, and they kill $\alpha_\pm$ because they lie in $P \wedge Q$; so they lie in $\mathrm{Ann}_{HT^2}(\gamma)$ for every $n \ge 2$. For $n \ge 3$, by (i) the intersection of $\mathrm{Ann}_{HT^2}(\gamma)$ with $H^1(T_A)$ consists of the combinations with $(0, y_1 + y_2, 0) \in \ker \mathsf{H}$ in each pair, that is, with $(y_1 + y_2)\mu_{j+1} = 0$ for $0 \le j \le 2n-2$; if some μ_k with $1 \le k \le 2n-1$ is nonzero this forces $y_1 + y_2 = 0$. If $\rho = 3$ then $\ker \mathsf{H} = 0$.

(iii) By (A.4), $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HT^2}(\gamma)$, so $\dim \mathrm{Ext}^2(E,E) \ge \mathrm{rank}\, \mathrm{ev}_E \ge \dim HT^2(A) - \dim \mathrm{Ann}_{HT^2}(\gamma) = r(\gamma)$. If equality holds, ev_E is onto and its kernel has the dimension of $\mathrm{Ann}_{HT^2}(\gamma)$, hence equals it. Every class of $\mathrm{Ext}^2(E,E)$ is then $\mathrm{ev}_E(\xi)$ for some ξ, and $\sigma_E(\mathrm{ev}_E(\xi)) = \xi \lrcorner \gamma$ vanishes only for $\xi \in \mathrm{Ann}_{HT^2}(\gamma) = \ker \mathrm{ev}_E$; so σ_E is injective.

(iv) Since $\mathrm{ch}(E) = \gamma$ is rational, ω is a nonzero rational Weil class and the c_k are rational. The classes θ and ω are flat sections of Hodge classes over $\mathscr{D}_{n,n}$ (Proposition A.4(i)), so γ is. Since A is projective, E is isomorphic to a bounded complex of vector bundles, and Lemma A.12 gives an open neighbourhood U of the point of A in $\mathscr{D}_{n,n}$ over which E deforms: every A_s, $s \in U$, carries a bounded complex of holomorphic vector bundles E_s with $\mathrm{ch}(E_s) = \gamma_s$. The abelian variety A_s is projective, so E_s is algebraic and its Chern character is the class of an algebraic cycle with rational coefficients. Its component of degree $2n$ is $\omega_s + c_n \theta^n$, so ω_s is algebraic, and then every class of $W(A_s)$ is algebraic by Proposition A.4(ii). So the algebraic locus Σ_K contains the open set U, and $\Sigma_K = \mathscr{D}_{n,n}$ by Theorem A.7(iv). The last statement is Theorem A.3. □

Remark A.16 (The range of n and the Hankel rank)**.** The formula $r(\gamma) = (4+\rho)n^2 - 2n$ is proved for $n \ge 3$, which is the range in which Theorem 6.1 is applied in the paper; the values used there are $r = 57$ for $n = 3$ and $r = 104$ for $n = 4$, both with $\rho = 3$. For $n = 2$ the two weights $\varepsilon_2 + \varepsilon_3$ and $-\varepsilon_0 - \varepsilon_1$ of HT^2 differ by $\mathbf{1}$, the three annihilators in the proof of (i) are coupled in one block, and $\dim \mathrm{Ann}_{HT^2}(\gamma)$ can exceed $4(4-\rho)$ by at most two; parts (ii) (first sentence), (iii) and (iv) hold for every $n \ge 2$ with $r(\gamma)$ defined as the rank of the contraction. The matrix H has three columns, so $\rho \le 3$; $\rho = 3$ for a general polynomial p, and $\rho \le 2$ exactly when $\mu_0, \dots, \mu_{2n}$ satisfy a linear recurrence $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$, $0 \le j \le 2n-2$, with $(x, \lambda, w) \ne 0$, that is, when p is one of $ae^{s\theta} + be^{t\theta}$, $(a + b\theta)e^{t\theta}$, $ae^{t\theta} + b\theta^{2n}$, $a\theta^{2n-1} + b\theta^{2n}$, according to the roots of $wT^2 + \lambda T + x$.

The criterion cannot be met by a complex whose Chern character is a multiple of a Weil class alone. This is proved on the larger family of Weil tori.

Setting A.17 (Weil tori)**.** Let A be an abelian $2n$-fold of Weil type for K, $V = H^1(A, \mathbb{Q})$ with its action of K, and $V \otimes \mathbb{C} = V_+ \oplus V_-$ the two eigenspaces, each of dimension $2n$. For complementary n-dimensional subspaces $X, Y \subset V_+$ the subspace $X \oplus \overline{Y}$ of $V \otimes \mathbb{C}$ is the space of holomorphic 1-forms of a complex structure on the real torus underlying A; call the resulting complex torus $T_{X,Y}$. On $T_{X,Y}$ the field K acts with signature (n,n), so $T_{X,Y}$ is a Weil torus in the sense of [91], and every complex structure on that real torus commuting with K and of signature (n,n) arises in this way. The pairs (X,Y) form a connected complex manifold S of dimension $2n^2$, containing the point of A; the polarised Weil family is the submanifold of dimension n^2 on which the polarisation of A is of type $(1,1)$, and the tangent space of S is the space $\mathrm{Hom}_K(H^{1,0}, H^{0,1})$ of K-linear first order deformations

in Theorem A.13(ii). Since α_+ spans $\bigwedge^{2n} V_+ = \bigwedge^n X \otimes \bigwedge^n Y$ and X is holomorphic while Y is anti-holomorphic, the Weil classes are of type (n,n) on every $T_{X,Y}$. We say that a property holds for a very general member of S if it holds outside a countable union of proper analytic subsets.

Lemma A.18 (The Hodge classes of a very general Weil torus)**.** *For a very general member T of S,* $\mathrm{Hdg}^k(T) = 0$ *for* $1 \le k \le 2n-1$, $k \neq n$, *and* $\mathrm{Hdg}^n(T)$ *is the rational Weil plane.*

Proof. There are countably many rational classes, and the Hodge locus of each is an analytic subset of the connected manifold S; so a rational class that is of Hodge type at a very general member is of Hodge type on all of S. Let $G = \mathrm{Res}_{K/\mathbb{Q}}\mathrm{GL}_{2n}$ be the group of K-linear automorphisms of V. Its real points act transitively on S, since $\mathrm{GL}(V_+)$ acts transitively on pairs of complementary subspaces. The Mumford–Tate group M of a very general member lies in G, since K acts by endomorphisms of every member, and it contains the Mumford–Tate group of every member, whose Hodge classes include those of the very general one. So the real Lie algebra of M contains the complex structure J_s of every member, which is the derivative of its Hodge cocharacter, and therefore contains the $\mathrm{Ad}\,G(\mathbb{R})$-span of one J_s. This span is an ideal of $\mathfrak{g}_{\mathbb{R}} = \mathfrak{gl}_{2n}(\mathbb{C})$ containing the non-central element J_s, which acts on V_+ by i on X and by $-i$ on Y; hence it contains $\mathfrak{sl}_{2n}(\mathbb{C})$, and M contains $G^{\mathrm{der}} = \mathrm{Res}_{K/\mathbb{Q}}\mathrm{SL}_{2n}$. Over $\mathbb{C}$ this is $\mathrm{SL}(V_+)\times\mathrm{SL}(V_-)$ acting on $\bigwedge^*(V_+\oplus V_-) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$, whose invariants are spanned by 1, α_+, α_- and $\alpha_+\alpha_-$. Hodge classes are invariant under M, which gives the statement. □

Lemma A.19 (A very general Weil torus has no subvarieties)**.** *A very general member T of S has no analytic subsets other than finite sets and T itself.*

Proof. Let $Z \subset T$ be an irreducible analytic subset of codimension c with $0 < c < 2n$. For every Kähler class κ the class $[Z]$ satisfies $\int_T [Z]\,\kappa^{2n-c} = \int_Z \kappa^{2n-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$. By Lemma A.18 this forces $c = n$ and $[Z]$ into the Weil plane. Take for κ_0 the class of a positive definite hermitian form that is block diagonal for the splitting $H^{1,0} = X \oplus \overline{Y}$. Its blocks lie in $X \otimes \overline{X}$ and $\overline{Y} \otimes Y$, both inside $V_+ \otimes V_-$, so every power of κ_0 has a factor from V_+ and a factor from V_- in each term, and $\alpha_\pm \wedge \kappa_0^n = 0$ because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So $\int_T [Z]\,\kappa_0^n = 0$, which is impossible. □

Proposition A.20 (Coherent sheaves on a very general Weil torus)**.** *Let $n \ge 2$ and let T be a very general member of S. Every coherent analytic sheaf F on T has* $\mathrm{ch}_k(F) = 0$ *in* $H^{2k}(T,\mathbb{Q})$ *for* $1 \le k \le 2n-1$. *For $n = 2$ this is Voisin's theorem that every coherent sheaf on a general Weil torus of dimension four has* $c_2(F) = 0$ [91, Theorem 10]*, whose proof we follow.*

Proof. Chern classes of coherent sheaves on T exist in rational Deligne cohomology, extend those of bundles and are additive [36], so they are Hodge classes. By Lemma A.19 the torsion of F, the locus where the torsion-free quotient is not locally free, and the cokernel of the map from that quotient to its reflexive hull are all supported on finite sets, and a sheaf supported on a finite set has Chern character in the top degree only. So it suffices to treat a reflexive sheaf G. Every coherent sheaf on T has $c_1 = 0$, since $\mathrm{Hdg}^1(T) = 0$ (Lemma A.18); so every torsion-free sheaf has slope zero for every Kähler class, and a torsion-free sheaf is stable for one Kähler class exactly when it has no saturated subsheaf of intermediate rank, which does not depend on the class. A Jordan–Hölder filtration of G therefore has stable torsion-free factors Q_j, stable for every Kähler class, and $\mathrm{ch}(G) = \sum_j \mathrm{ch}(Q_j)$. By the theorem of Bando and Siu [20] the stable reflexive sheaf Q_j^{**} carries an admissible Hermite–Einstein metric for every Kähler class κ and satisfies $\int_T \big(2rc_2 - (r-1)c_1^2\big)(Q_j^{**})\,\kappa^{2n-2} \ge 0$, with equality exactly when Q_j^{**} is a projectively flat vector bundle. Here $c_1 = 0$ and $c_2(Q_j^{**}) \in \mathrm{Hdg}^2(T)$. For $n \ge 3$ that group is zero; for $n = 2$ it is the Weil plane, and then $\int_T c_2(Q_j^{**})\,\kappa_0^2 = 0$ for the class κ_0 of the proof of Lemma A.19. In either case equality holds for some κ, so Q_j^{**} is a projectively flat bundle with $c_1 = 0$, hence flat, and its rational Chern classes of positive degree vanish. Since Q_j and Q_j^{**} differ only on a finite set, $\mathrm{ch}_k(Q_j) = 0$ for $1 \le k \le 2n-1$, and the same holds for G and for F. □

Theorem A.21 (The pure form of the criterion is never met)**.** *Let $n \ge 2$, let A be an abelian $2n$-fold of Weil type for K, ω a nonzero rational Weil class, and E a perfect complex on A with* $\mathrm{ch}(E) - N\omega \in H^0(A) \oplus H^{4n}(A)$, $N \neq 0$. *Then σ_E is not injective. In particular, if* $\mathrm{ch}(E) = N\omega$ *then* $\dim \mathrm{Ext}^2(E,E) > 2n(2n-1)$.

Proof. Suppose that σ_E is injective. Since A is projective, E is isomorphic to a bounded complex of holomorphic vector bundles. The class $\mathrm{ch}(E)$ is of Hodge type on every member of the family S of Setting A.17, so by Lemma A.12 every member T_s with s in a neighbourhood U of the point of A in S carries a bounded complex of holomorphic vector bundles $E_s^\bullet$ whose Chern character is the parallel transport of $\mathrm{ch}(E)$. Choose $s \in U$ very general. By Proposition A.20, $\mathrm{ch}_n(E_s) = \sum_i (-1)^i \mathrm{ch}_n(E_s^i) = 0$, while it is the transport of $N\omega$, which is not zero. This contradiction proves that σ_E is not injective. If $\mathrm{ch}(E) = N\omega$, then $\dim \mathrm{Ext}^2(E,E) \geq 2n(2n-1)$ and equality would force σ_E to be injective, by Theorem A.15(i) for $p = 0$ and (iii). □

A.4. **Convolutions of line bundles on E_0^6 and E_0^8**

This subsection treats the convolutions, the iterated cones of shifted line bundles, at the member E_0^{2n}, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, of the family of Weil type for $K = \mathbb{Q}(i)$. Their Chern character is the same signed sum of exponentials as for a direct sum, but their Ext groups are cut down by the maps between the pieces. Proposition A.24 writes a signed sum of $2 \cdot 4^{n-1}$ line bundles that is exactly a real Weil class; for the convolutions of those pieces with three multiples of the polarisation, Theorem A.32 gives $\dim \mathrm{Ext}^2 \geq 69 > 57$ at $n = 3$, and Theorem A.34 to Corollary A.45 give $\dim \mathrm{Ext}^2 \geq 640 > 104$ at $n = 4$ for the arrangements of shifts listed there. A perfect complex E with $\mathrm{ch}(E) = \gamma$ of polarised shape is said to *meet the polarised criterion* if $\dim \mathrm{Ext}^2(E,E) = r(\gamma)$, the equality case of Theorem A.15(iii).

Setting A.22 (Convolutions)**.** Let A be an abelian variety of dimension g with a flat Kähler metric, and let $L_1, \dots, L_m$ be line bundles on A with hermitian metrics of constant curvature, hermitian forms $H_1, \dots, H_m$ and pairwise distinct first Chern classes. For integers $d_1, \dots, d_m$ put $P_a = L_a[-d_a]$. A *convolution* of $P_1, \dots, P_m$ is an object E given by a one-sided twisted complex[13]

$$\Big(\bigoplus_a \mathscr{A}^{0,*}(L_a)[-d_a],\ \bar\partial + \delta \Big), \qquad \delta = (\delta_{ab})_{a<b}, \quad \delta_{ab} \in \mathscr{A}^{0,1+d_a-d_b}\big(L_a^{-1} \otimes L_b\big),$$

with $\bar\partial\delta + \delta \circ \delta = 0$, so that $\mathrm{ch}(E) = \sum_a (-1)^{d_a} e^{c_1(L_a)}$. Write $\mathscr{A}(a,b) = \mathscr{A}^{0,*}(L_a^{-1} \otimes L_b)$ and $\mathscr{H}(a,b)$ for its harmonic forms, so that $\mathscr{H}^q(a,b) \cong H^q(A, L_a^{-1} \otimes L_b)$, and write π, G and $h = \bar\partial^* G$ for the harmonic projection, the Green operator and the homotopy, with $1 - \pi = \bar\partial h + h\bar\partial$ and $\pi h = h\pi = h^2 = 0$. A form of degree q in $\mathscr{A}(a,b)$ is a morphism $P_a \to P_b$ of degree $w = q - d_a + d_b$; we call q its Ext degree and, when $w = 1$, call $q - 1$ its *excess*. When $H_b - H_a$ is nondegenerate, $\mathscr{H}(a,b)$ is concentrated in the single degree $n(H_b - H_a)$, the number of negative eigenvalues [16, Chapter 3].

Lemma A.23 (The harmonic model)**.** *In Setting A.22:*

(i) *E is isomorphic to a convolution of the same pieces with $h\delta = 0$, and then $\delta = \delta^H - h(\delta \circ \delta)$ with $\delta^H = \pi\delta$ harmonic. Substituting repeatedly, every δ_{ab} is a finite sum of planar binary trees whose leaves are components δ^H_{cd} with $c < d$, whose internal vertices compose, and whose internal edges carry $-h$.*

(ii) *$\mathrm{Ext}^*(E,E)$ is the cohomology of $\mathscr{H}_E = \bigoplus_{a,b} \mathscr{H}(a,b)$, graded by w, with the differential $d_E = \sum_{k \geq 0} \pi Q(-hQ)^k$, where $Q = [\delta, -]$.*

(iii) *For $x \in \mathscr{H}(a,b)$, the component of $d_E(x)$ in $\mathscr{H}(c,e)$ is a sum of terms $\pm\pi T$, one for each planar binary tree T with $k \geq 2$ leaves, internal vertices composing, $-h$ on the internal edges other than the root, whose leaves read in order are the components along a path $c = a_0 \to a_1 \to \cdots \to a_k = e$ in which one step is $x\colon a \to b$ and the others are components $\delta^H_{a_j a_{j+1}}$ with $a_j < a_{j+1}$. The* Ext *degree of such a term is the sum of the* Ext *degrees of its leaves minus $k - 2$.*

The terms with k leaves in (iii) are the Massey products of length k of the pieces;[14] (iii) is all that is used of them below.

[13] Every object obtained from $P_1, \dots, P_m$ by iterated cones in this order is of this form, because the Dolbeault complexes form a differential graded enhancement of the full subcategory of line bundles. All the objects of this subsection are given in this way.

[14] In the language of A_∞-categories they are the higher products m_k of the minimal model of the line bundles, evaluated on the Maurer–Cartan element δ^H of the twisted complex; for $k = 2$ they are cup products.

Proof. (i) Suppose $h\delta_{cd}=0$ for $d-c<r$. Conjugating by $1+g$, with g of degree 0 supported at distance r, replaces δ_{ab} for $b-a=r$ by $X_{ab}+\bar\partial g_{ab}$, where X_{ab} depends only on δ, and leaves the components at smaller distance unchanged. With $g_{ab}=-hX_{ab}$ the new component is $X_{ab}-\bar\partial hX_{ab}$, and $h\bar\partial h=h(1-\pi-h\bar\partial)=h$ gives $h(X_{ab}-\bar\partial hX_{ab})=0$. Induction on r gives $h\delta=0$, and then $\delta=\pi\delta+\bar\partial h\delta+h\bar\partial\delta=\delta^H-h(\delta\circ\delta)$ by the Maurer–Cartan equation. The component $(\delta\circ\delta)_{ab}$ involves only components at distance less than $b-a$, so the substitution ends. (ii) The endomorphism complex of E is $\bigoplus_{a,b}\mathscr{A}(a,b)$ with the differential $\bar\partial+Q$. Blockwise, the inclusion of harmonic forms, π and $-h$ form a strong deformation retraction of $(\bigoplus\mathscr{A}(a,b),\bar\partial)$ onto $(\mathscr{H}_E,0)$, and Q raises $b-a$, so hQ is nilpotent. The basic perturbation lemma gives a homotopy equivalence of $(\bigoplus\mathscr{A}(a,b),\bar\partial+Q)$ with $(\mathscr{H}_E,d_E)$, with the chain maps $\iota_E=\sum_k(-hQ)^k$ on $\mathscr{H}_E$ and $\pi_E=\pi\sum_k(-Qh)^k$. (iii) Each Q in d_E composes with one component of δ on the left or on the right; writing that component as in (i) gives the trees, the edges from (i) and those from d_E being exactly the internal edges other than the root. Composition adds degrees, h lowers them by one, and a binary tree with k leaves has $k-2$ internal edges. □

Proposition A.24 (A pure Weil class from line bundles)**.** *Let $E_0=\mathbb{C}/\mathbb{Z}[i]$, $A=E_0^{2n}$ with coordinates $z_1,\dots,z_{2n}$ and $n\ge 2$, let $K=\mathbb{Q}(i)$ act by i on $z_1,\dots,z_n$ and by $-i$ on $z_{n+1},\dots,z_{2n}$, and let θ be the product principal polarisation; then A is an abelian $2n$-fold of Weil type for $\mathbb{Q}(i)$, with complex multiplication. Write B_j for the surface with coordinates (z_j,z_{n+j}), so that $A=B_1\times\cdots\times B_n$. For $p\in\mathbb{Z}$ and $\zeta\in\mu_4^n$ let L_ζ be a line bundle whose hermitian form is $\left(\begin{smallmatrix}p&\zeta_j\\ \overline{\zeta}_j&p\end{smallmatrix}\right)$ on B_j and zero between different surfaces, and put $\alpha=\prod_j\frac{i}{2}dz_j\wedge d\bar z_{n+j}$.*

(i) *α and $\overline{\alpha}$ span the Weil classes of A, and*

$$\sum_{\prod\zeta=1}e^{c_1(L_\zeta)}-\sum_{\prod\zeta=-1}e^{c_1(L_\zeta)}=2\cdot4^{n-1}(\alpha+\overline{\alpha}).$$

(ii) *For distinct integers t_1,t_2,t_3 and signs σ_i, the character $\gamma=2\cdot4^{n-1}(\alpha+\overline{\alpha})+\sum_i\sigma_ie^{t_i\theta}$ has Hankel rank 3, so $r(\gamma)=7n^2-2n$ for $n\ge3$. A convolution with this character of the $2\cdot4^{n-1}$ bundles L_ζ, with $(-1)^d=\prod\zeta$, and of three line bundles M_i with $c_1(M_i)=t_i\theta$ has $2\cdot4^{n-1}+3$ pieces and $(2\cdot4^{n-1}+3)\,n(2n-1)$ diagonal classes of degree two: 525 against $r=57$ at $n=3$, and 3668 against 104 at $n=4$.*

Proof. The hermitian forms have imaginary parts integral on $\mathbb{Z}[i]^{2n}$, so the line bundles exist. On B_j write c_1 of the restriction as $p(a_j+b_j)+\zeta_jc_j+\overline{\zeta}_jc'_j$, where $a_j=\frac{i}{2}dz_j\wedge d\bar z_j$, $b_j=\frac{i}{2}dz_{n+j}\wedge d\bar z_{n+j}$, $c_j=\frac{i}{2}dz_j\wedge d\bar z_{n+j}$ and $c'_j=\frac{i}{2}dz_{n+j}\wedge d\bar z_j$. Since $c_j^2=c_j'^2=0$,

$$e^{c_1(L_\zeta)}=\prod_j f_j(\zeta_j),\qquad f_j(z)=e^{p(a_j+b_j)}\bigl(1+zc_j+\bar zc'_j+c_jc'_j\bigr).$$

For $u\in\mu_4$, $\frac12(u+\overline{u})$ is u when $u=\pm1$ and 0 when $u=\pm i$, so the left side of (i) is $\frac12\bigl(\prod_j\sum_z zf_j(z)+\prod_j\sum_z\bar zf_j(z)\bigr)$. Now $\sum_{z\in\mu_4}z=\sum z^2=0$ and $\sum|z|^2=4$, so $\sum_z zf_j(z)=4e^{p(a_j+b_j)}c'_j=4c'_j$, because $a_jc'_j$ and $b_jc'_j$ repeat a differential, and likewise $\sum_z\bar zf_j(z)=4c_j$. This gives $2\cdot4^{n-1}(\prod c'_j+\prod c_j)=2\cdot4^{n-1}(\overline{\alpha}+\alpha)$. The differentials dz_j, $j\le n$, and $d\bar z_{n+j}$ span the i-eigenspace of K on $H^1(A,\mathbb{C})$, and α spans its top exterior power, which is a Weil class, and $\overline{\alpha}$ the conjugate one. (ii) The moments $\mu_m=\sum_i\sigma_it_i^m$ make the Hankel matrix of Setting A.14 a product of Vandermonde matrices of rank 3, and Theorem A.15(i) gives r for $n\ge3$. The counts are immediate. □

The pieces L_ζ and $L_{\zeta'}$ have a degenerate difference as soon as ζ and ζ' share a coordinate. The rest of this subsection computes what the Massey products can do to the diagonal classes of such a convolution. The pieces are external products over the surfaces B_j, and the first tool is a bound on how far a product can lower the degree on each surface.

Lemma A.25 (Degree drop on a product)**.** *Let $A=B_1\times\cdots\times B_n$ with a product metric, and let every L_a be an external product with a product metric. Decompose the leaves of a term of Lemma A.23(iii)*

with k leaves into tensor products of harmonic forms on the factors. Then the output has degree $s_j - d_j$ on B_j, where s_j is the sum of the degrees of the leaves on B_j and $d_j \geq 0$, $\sum_j d_j = k-2$, and

$$d_j \leq \max(0, u_j - 2),$$

where u_j is the number of leaves whose component on B_j is not a multiple of the unit section of $\mathscr{O}_{B_j}$ (such a unit occurs only where source and target agree on B_j).

Proof. The Laplacian is $\Delta = \sum_j \Delta_j$, so $h = \sum_j h_j$ with $h_j = \bar{\partial}_j^* G = G\bar{\partial}_j^*$, which lowers the degree on B_j by one and keeps the others; expand every internal edge accordingly, and let d_j be the number of edges carrying h_j. Let V_j, U_j and W_j be the forms whose component on B_j is Δ_j-harmonic, a multiple of the unit, or in the image of $\bar{\partial}_j^*$. The operators G and h_i, $i \neq j$, preserve all three, h_j kills V_j and W_j, π kills W_j, and multiplication by an element of U_j preserves V_j, U_j and W_j. For an edge e carrying h_j let $N(e)$ be the set of leaves below e whose component on B_j is not a unit. If $|N(e)| \leq 1$ the value below e lies in V_j and the term vanishes. If e lies below the next such edge e' with $N(e) = N(e')$, everything composed in between lies in U_j, the value at e' lies in W_j, and the term vanishes; if $N(e)$ is the set of all u_j such leaves for the highest such edge, the value at the root lies in W_j and π kills it. So in a nonzero term the sets $N(e)$ are distinct, of size at least two, different from the whole set, and pairwise nested or disjoint. Together with the whole set they form a family of that kind, which has at most $u_j - 1$ members, since its members are the internal vertices of a forest with u_j leaves in which every internal vertex has at least two children. Hence $d_j \leq u_j - 2$. □

Proposition A.26 (Nothing enters the diagonal)**.** *Keep Proposition A.24 with $n \geq 3$, and let t_1, t_2, t_3 be distinct with $|t_i - p| \geq 2$. For every convolution E of the pieces L_ζ and M_i, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero, no element of degree one of $\mathscr{H}_E$ has a differential with a nonzero diagonal component. Consequently*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; \dim \ker\big(d_E|_{D^2}\big), \qquad D^2 = \bigoplus_a H^2(A, \mathscr{O}_A).$$

Proof. A term of Lemma A.23(iii) that reaches a diagonal class in $\mathscr{H}^2(c,c) = H^2(A, \mathscr{O}_A)$ has k leaves of degree one running around a closed path $c \to \cdots \to c$; the pieces before the step x increase from c and those after it increase to c, so the path passes through k distinct pieces; and the output has Ext degree $\sum_j q_j - (k-2) = 2$, where q_j are the Ext degrees of the leaves, so the total excess $\sum_j (q_j - 1)$ is 0. The step from L_ζ to $L_{\zeta'}$ has Ext degree 1 on each surface where $\zeta_j \neq \zeta'_j$, the difference there having eigenvalues $\pm|\zeta'_j - \zeta_j|$, and at least 0 elsewhere. If $\prod \zeta = \prod \zeta'$ the two differ on at least two surfaces; otherwise the step has even Ext degree, the shifts having opposite parities. Either way its excess is at least 1. The difference between L_ζ and M_i has eigenvalues $t_i - p \pm 1$ on every surface, so it is definite: the step has Ext degree 0 and excess -1 if it goes up the value ϕ, where $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$, and Ext degree $2n$ and excess $2n-1$ if it goes down; the same holds between the M_i. A maximal run of the cycle through the M_i starts and ends at the value p, or is the whole cycle, so it has at least one step down and at most three steps up; its excess is at least $2n - 1 - 3 \geq 2$. So every cycle has excess at least 1. Twisting by line bundles of degree zero leaves the forms unchanged and can only make some $\mathscr{H}(a,b)$ vanish. For the inequality, the diagonal components of cocycles of degree two vanish on coboundaries, so $\dim \mathrm{Ext}^2(E,E)$ is at least the dimension of their image, which contains the kernel of d_E on D^2. □

At $n = 2$ the bound on a run is 0 and a cycle through the three multiples can reach the diagonal; this is why the analysis below starts at $n = 3$.

Proposition A.27 (Cup products on the diagonal)**.** *Keep Proposition A.26 with $n = 3$, so that D^2 has dimension $35 \cdot 15 = 525$. The terms of d_E on D^2 with two leaves, the cup products, vanish on the 45 diagonal classes of the M_i, and on the 480 diagonal classes of the L_ζ they have a kernel of dimension at least 204. So their kernel on D^2 has dimension at least 249.*

Proof. For $y = (y_a) \in D^2$ the cup products in $\mathscr{H}(a,b)$ are $(y_a - y_b) \wedge \delta^H_{ab}$, since y has even degree. On a surface where the pieces differ, δ^H_{ab} has degree 1, the only degree of the cohomology there, and the product with a component of y of positive degree lands in a degree where the cohomology vanishes.

The pieces M_i differ from every other piece on every surface, which gives the first statement. Write y_a by its type $\tau \in \{0,1,2\}^3$, $|\tau| = 2$, the degrees on the three surfaces: three types $(2,0,0)$ of dimension one and three types $(1,1,0)$ of dimension four. The degree count on each surface and the parity of Ext degrees show that L_ζ and $L_{\zeta'}$ can carry a nonzero cup product on the type τ only if either ζ' differs from ζ by a sign in one coordinate outside the support of τ, or ζ' is ζ multiplied by (i,i) or $(-i,-i)$ in two coordinates and τ is 2 on the third. Let Γ_τ be the graph on the 32 pieces with these edges. If each component y^τ is constant on the connected components of Γ_τ, every cup product vanishes. For $\tau = (2,0,0)$ the coordinate ζ_1 is constant along the edges, and the eight pieces with a given ζ_1 are connected, so Γ_τ has 4 components; for $\tau = (1,1,0)$ the edges change the sign of ζ_3 only, and Γ_τ has 16 components. The kernel therefore contains a space of dimension $3\cdot 4\cdot 1 + 3\cdot 16\cdot 4 = 12 + 192 = 204$. □

Theorem A.28 (The diagonal on E_0^6)**.** *Let $n = 3$ and let E be a convolution of the 32 bundles L_ζ and of M_1, M_2, M_3 as in Proposition A.24, with $p+2 \le t_1 < t_2 < t_3$ and signs σ_i, so that* $\mathrm{ch}(E) = 32(\alpha + \overline{\alpha}) + \sum_i \sigma_i e^{t_i\theta}$ *and the polarised criterion asks for* $\dim \mathrm{Ext}^2(E,E) = 57$.

(i) *The Massey products of length at least three vanish on D^2, except those of length four along a path $M_2 \to M_3 \to L_\zeta \to M_1$ taken in this order in the convolution, which occur only if $\sigma_1 = \sigma_3 = -\sigma_2$ and take values in $\mathscr{H}^6(M_2, M_1) \cong H^6\big(A, \mathscr{O}_A(-(t_2-t_1)\theta)\big)$, of dimension $(t_2-t_1)^6$.*

(ii) $\dim \mathrm{Ext}^2(E,E) \ge 249 - \min\big(249, (t_2-t_1)^6\big)$, *and* $\dim \mathrm{Ext}^2(E,E) \ge 249$ *unless the order and the signs are as in (i). In particular E does not meet the polarised criterion when $t_2 - t_1 \le 2$, since $249 - 64 = 185 > 57$, nor when the signs or the order differ from (i).*

(iii) *If E meets the polarised criterion, then the length four products of (i) have rank at least $249 - 57 = 192$ on the kernel of the cup products, and $t_2 - t_1 \ge 3$.*

The same holds for $t_1 < t_2 < t_3 \le p-2$ with E replaced by its dual.

Proof. (i) By Lemma A.23(iii) a term with $k \ge 3$ leaves acting on $y \in H^2(\mathscr{O}_A)$ at a piece c runs along a chain $s = a_0 \to \cdots \to a_{k-1} = e$ of distinct pieces, increasing in the order of the convolution, whose $k-1$ steps are components of degree one, with y inserted at c. Its value lies in $\mathscr{H}(s,e)$, in Ext degree $2 + \sum_j q_j - (k-2) = 3 + \varepsilon$, where ε is the sum of the excesses of the steps; so $-3 \le \varepsilon \le 3$, and $\mathscr{H}(s,e)$ must be nonzero in degree $3+\varepsilon$. By the proof of Proposition A.26 a step between two L_ζ has excess at least 1, with equality only between pieces of opposite parities, and a step between pieces at different values of ϕ has excess -1 and Ext degree 0 going up, excess 5 and Ext degree 6 going down. Let U and D be the numbers of steps up and down and λ the total excess of the steps between two L_ζ, so that $\varepsilon = 5D - U + \lambda$. For two distinct L_ζ, $L_{\zeta'}$ differing in $m \ge 1$ coordinates, $\mathscr{H}(L_\zeta, L_{\zeta'})$ vanishes above degree $6-m$; for pieces at different values it sits in degree 0 if the target is higher and in degree 6 if it is lower.

Two observations remove chains. First, a chain of total excess -2 has no step down, and so consists of steps between L_ζ followed by at most three steps up. Second, let $a \to b \to c$ be two consecutive steps of Ext degree 0, sections x_{ab} and x_{bc} of line bundles. The Maurer–Cartan equation $\pi(\delta \circ \delta) = 0$ of the proof of Proposition A.30(i) has, in $\mathscr{H}^0(a,c)$, one term for each chain from a to c of total excess -2. If $a \to b \to c$ is the only such chain, the equation reads $x_{ab}x_{bc} = 0$, a product of holomorphic sections being holomorphic and so harmonic; since A is connected, x_{ab} or x_{bc} vanishes, and every term along a chain through both steps is zero.

Here every step up ends at one of M_1, M_2, M_3, so $U \le 3$, and $D \le 1$. If $D = 0$, the chain is $r \ge 0$ pieces L_ζ followed by multiples in increasing order. When it ends at an L_ζ, $k \ge 3$ gives $\lambda \ge 2$, so s and e differ in one coordinate and $\lambda = 2$; then the two steps change the parity twice, whereas s and e have opposite parities. When it ends at a multiple, e lies above s, so $\varepsilon = -3$, which forces $U = 3$, $\lambda = 0$ and the chain $L_\zeta \to M_1 \to M_2 \to M_3$; by the first observation $L_\zeta \to M_1 \to M_2$ is the only chain of excess -2 from L_ζ to M_2, and the second removes it. If $D = 1$, then $U \ge 2 + \lambda$, so (U,λ) is $(2,0)$, $(3,0)$ or $(3,1)$ and ε is 3, 2 or 3. For $\varepsilon = 3$ the piece s is a multiple and e lies below it. With $(3,1)$ the three steps up would end at three multiples other than s, which do not exist. With $(2,0)$ the chain has three steps, and a step down lands at an L_ζ, since a fourth multiple does not exist. It is $M_1 \to M_2 \to M_3 \to L_\zeta$, removed by the second observation at $M_1 \to M_2 \to M_3$; or $M_a \to M_b \to L_\zeta \to M_c$ with $t_a < t_b$ and $t_c < t_a$, that

is $M_2 \to M_3 \to L_\zeta \to M_1$; or $M_a \to L_\zeta \to M_b \to M_c$ with $t_b < t_c < t_a$, that is $M_3 \to L_\zeta \to M_1 \to M_2$, removed at $L_\zeta \to M_1 \to M_2$. For $\varepsilon = 2$, with $(U,\lambda) = (3,0)$, the pieces s and e are L_ζ differing in one coordinate; the last step is the step down, preceded by three steps up from s, so the chain is $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, removed at $L_\zeta \to M_1 \to M_2$.

Only $M_2 \to M_3 \to L_\zeta \to M_1$ remains. Its steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3} = d_{M_2} + 1$, $d_L = d_{M_3} - 5$ and $d_{M_1} = d_L + 1$; this gives the signs, and the value lies in $\mathscr{H}^{3+d_{M_2}-d_{M_1}}(M_2,M_1) = \mathscr{H}^6(M_2,M_1)$, whose dimension is $|\chi(\mathscr{O}_A((t_1-t_2)\theta))| = (t_2-t_1)^6$. (ii) By Proposition A.26 it suffices to bound the kernel of d_E on D^2. By Proposition A.27 the cup products have a kernel Y of dimension at least 249, and on Y the differential is the sum of the products of (i), of rank at most $(t_2-t_1)^6$. (iii) follows from (ii). For the dual, $E \mapsto E^\vee$ preserves Ext^2 and sends the configuration with p, t_i to the one with $-p, -t_i$, the set of the L_ζ being stable under $\zeta \mapsto -\zeta$. □

The placements of the t_i on both sides of p are not covered by Theorem A.28. There two families of fourfold products occur instead of one, and no convolution carries both.

Proposition A.29 (Placements on both sides of p)**.** *Keep Theorem A.28, but let $t_1 \le p-2$ and $p+2 \le t_2 < t_3$.*

(i) *The Massey products of length at least three vanish on D^2 unless $\sigma_1 = \sigma_2 = -\sigma_3$. In that case the only ones that act have length four and run along one of the paths*

$$\text{(a)}\ M_2 \to M_3 \to M_1 \to L_\zeta, \qquad \text{(b)}\ M_3 \to M_1 \to L_\zeta \to M_2,$$

taken in this order in the convolution, with $\prod \zeta = \sigma_3$. The products along (a) take values in $\mathscr{H}^6(M_2,L_\zeta)$, of dimension $\big((t_2-p)^2-1\big)^3$, and those along (b) in $\mathscr{H}^6(M_3,M_2)$, of dimension $(t_3-t_2)^6$. A convolution carries at most one of the two kinds, since (a) puts M_2 before M_3 and (b) puts M_3 before M_2.

(ii) *If E meets the polarised criterion, then $\sigma_1 = \sigma_2 = -\sigma_3$, and the products along (a) or those along (b) have rank at least 192 on the kernel of the cup products. For (b) this needs $t_3 - t_2 \ge 3$. The sixteen targets of (a) have dimension $16\big((t_2-p)^2-1\big)^3 \ge 432$, so for (a) the count gives no condition.*

For $t_1 < t_2 \le p-2$ and $t_3 \ge p+2$ the same holds for $E^\vee[1]$, and the condition on the signs becomes $\sigma_2 = \sigma_3 = -\sigma_1$.

Proof. (i) We use the notation and the two observations of the proof of Theorem A.28(i). Now $\phi(M_1) = t_1 < p < t_2 < t_3$; a step up ends at M_2, at M_3, or at an L_ζ coming from M_1, so again $U \le 3$ and $D \le 1$. If $D = 0$, the values do not decrease along the chain. When e lies above s, $\varepsilon = -3$ forces $U = 3$, $\lambda = 0$ and the chain $M_1 \to L_\zeta \to M_2 \to M_3$, removed at $L_\zeta \to M_2 \to M_3$, the only chain of excess -2 from L_ζ to M_3. When e and s have the same value, they are two L_ζ joined by steps between L_ζ, which is excluded as before. If $D = 1$, (U,λ) is $(2,0)$, $(3,0)$ or $(3,1)$. For $\varepsilon = 3$, e lies below s. With $(3,1)$ the steps up end at M_2, M_3 and an L_ζ following M_1; then s is an L_ζ, so $e = M_1$, which is followed by a step: impossible. With $(2,0)$ the chain has three steps and no two consecutive L_ζ. As M_1 is the lowest piece, it is never the end of a step up, and a step up into an L_ζ comes from M_1; so two consecutive steps up end at M_2 or M_3. The chain cannot start at M_1. If it starts at an L_ζ, it ends at M_1 with the step down, after two steps up: $L_\zeta \to M_2 \to M_3 \to M_1$, removed at $L_\zeta \to M_2 \to M_3$. If it starts at M_3, the first step goes down and the last two go up, ending at M_2; the middle piece is then an L_ζ entered from M_1, and the chain is $M_3 \to M_1 \to L_\zeta \to M_2$, path (b). If it starts at M_2, then $e \in \{L_\zeta, M_1\}$, so the last two steps are not both up; the first step goes up to M_3, the second down, and the third up into $e = L_\zeta$ from M_1: the chain is $M_2 \to M_3 \to M_1 \to L_\zeta$, path (a). For $\varepsilon = 2$, $(U,\lambda) = (3,0)$ and s, e are L_ζ differing in one coordinate. The last step enters e from M_1, since a step down into e would follow three steps up from s, one more than the values allow; M_1 is entered by the step down, after two steps up from s: $L_\zeta \to M_2 \to M_3 \to M_1 \to L_{\zeta'}$, removed at $L_\zeta \to M_2 \to M_3$. Along (a) the steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3} = d_{M_2} + 1$, $d_{M_1} = d_{M_3} - 5$ and $d_L = d_{M_1} + 1$. Along (b) they have Ext degrees 6, 0 and 0, which forces $d_{M_1} = d_{M_3} - 5$, $d_L = d_{M_1} + 1$ and $d_{M_2} = d_L + 1$. This gives the signs and puts the values in $\mathscr{H}^{3+d_{M_2}-d_L}(M_2,L_\zeta) = \mathscr{H}^6(M_2,L_\zeta)$ and in $\mathscr{H}^{3+d_{M_3}-d_{M_2}}(M_3,M_2) = \mathscr{H}^6(M_3,M_2)$.

On B_j the line bundle $L_\zeta \otimes M_2^{-1}$ has the hermitian form $\left(\begin{smallmatrix} p-t_2 & \zeta_j \\ \overline{\zeta}_j & p-t_2 \end{smallmatrix}\right)$, with eigenvalues $p-t_2 \pm 1 < 0$ and determinant $(t_2-p)^2-1$, which gives the first dimension. The second is that of Theorem A.28(i) with t_3, t_2 in place of t_2, t_1. (ii) Proposition A.26 and Proposition A.27 do not depend on the placement of the t_i, so as in Theorem A.28(ii) $\dim \mathrm{Ext}^2(E,E) \geq 249-\rho$, where ρ is the rank on the kernel of the cup products of the products of (i) that E carries, and $\rho = 0$ unless $\sigma_1 = \sigma_2 = -\sigma_3$. The criterion asks for 57, so $\rho \geq 192$, and $(t_3-t_2)^6 \geq 192$ exactly when $t_3 - t_2 \geq 3$. For the last statement, $E \mapsto E^\vee[1]$ preserves Ext^2, sends p and the t_i to $-p$ and the $-t_i$, reverses the order of the pieces and changes every sign. □

The Maurer–Cartan equation can be solved for one order of the pieces.

Proposition A.30 (An explicit convolution)**.** *Keep Theorem A.28 with $p=0$ and $t=(2,5,6)$, take the pieces to be external products $L_\zeta = N_{1,\zeta_1} \boxtimes N_{2,\zeta_2} \boxtimes N_{3,\zeta_3}$ and $M_i = M_{i,1} \boxtimes M_{i,2} \boxtimes M_{i,3}$ of line bundles on the B_j, and order them as M_2, M_3, the sixteen L_ζ with $\prod \zeta = 1$, M_1, the sixteen L_ζ with $\prod \zeta = -1$, with the shifts 0, 1, −4, −3, −5.*

(i) *The components that the degrees allow are $x_1 \in \mathscr{H}^0(M_2,M_3)$, $x_2^\zeta \in \mathscr{H}^6(M_3,L_\zeta)$, $x_3^\zeta \in \mathscr{H}^0(L_\zeta,M_1)$, components in $\mathscr{H}^2(L_\zeta,L_{\zeta'})$ for the $48+96$ pairs with $\prod\zeta = 1 = -\prod\zeta'$ that differ in one or two coordinates, and components in $\mathscr{H}^6(M_2,L_{\zeta'})$, $\prod\zeta' = -1$, which we take to be zero. For harmonic components the Maurer–Cartan equation is exactly*

$$x_1 x_2^\zeta = 0 \text{ in } H^6(A, M_2^{-1}\otimes L_\zeta) \text{ for every } \zeta, \qquad \sum_\zeta x_2^\zeta x_3^\zeta = 0 \text{ in } H^6(A, M_3^{-1}\otimes M_1),$$

and every solution is the harmonic part of a convolution E with $\mathrm{ch}(E) = 32(\alpha+\overline{\alpha}) + e^{5\theta} - e^{6\theta} - e^{2\theta}$.

(ii) *There are solutions with every $x_2^\zeta \neq 0$ and $x_1 x_2^\zeta = x_2^\zeta x_3^\zeta = 0$.*

(iii) *For every solution $\dim \mathrm{Ext}^2(E,E) \geq 2560$, so E does not meet the polarised criterion.*

Proof. (i) A component from P_a to P_b has Ext degree $1+d_a-d_b$, and the forms give the cohomology of each pair: $\mathscr{H}(M_2,M_3)$ and $\mathscr{H}(L_\zeta,M_1)$ sit in degree 0, $\mathscr{H}(M_3,L_\zeta)$, $\mathscr{H}(M_2,L_{\zeta'})$ and $\mathscr{H}(M_2,M_1)$ in degree 6, and $\mathscr{H}(L_\zeta,L_{\zeta'})$ in degree 2 exactly for the pairs listed; no other pair has cohomology in the degree a component needs. Put $\delta = \delta^H - h(\delta\circ\delta)$ recursively, as in Lemma A.23(i), and $\Phi = \bar\partial\delta + \delta\circ\delta$. Then $\Phi = \pi(\delta\circ\delta) + h\bar\partial(\delta\circ\delta)$ and $\bar\partial(\delta\circ\delta) = \Phi\circ\delta - \delta\circ\Phi$, so by induction on $b-a$ the twisted complex exists exactly when $\pi(\delta\circ\delta) = 0$. The components of $\pi(\delta\circ\delta)$ come from paths of two or more components; those that land in a nonzero group are $x_1x_2^\zeta$ in $\mathscr{H}^6(M_2,L_\zeta)$ and $\sum_\zeta x_2^\zeta x_3^\zeta$ in $\mathscr{H}^6(M_3,M_1)$, while the product along $M_2 \to M_3 \to L_\zeta \to M_1$ would lie in $\mathscr{H}^5(M_2,M_1) = 0$ and x_2^ζ composed with a component in $\mathscr{H}^2(L_\zeta,L_{\zeta'})$ in degree eight. The signs $(-1)^d$ are $+,-,+,-,-$, which gives the Chern character by Proposition A.24(i). (ii) Take x_1 and the x_3^ζ to be external products of nonzero sections on the B_j. On each B_j the products with their factors map $H^2(B_j, M_3^{-1}L_\zeta)$, of dimension 35, to spaces of dimensions 24 and 16, so they have kernels of dimensions at least 11 and 19. A sum of external products whose factor on one surface lies in the first kernel and whose factor on another lies in the second is a nonzero x_2^ζ with $x_1x_2^\zeta = x_2^\zeta x_3^\zeta = 0$, which solves (i).[15] (iii) The sixteen L_ζ with $\prod\zeta = 1$ have shift -4 and the others -5, so Lemma A.31 below applies to every pair of them that differ in all three coordinates: there are 112 such pairs, and their groups $\mathscr{H}^3$ have dimensions adding up to $16\cdot 160 = 2560$. □

The classes counted in (iii) are the ones no product can reach. Two pieces L_ζ, $L_{\zeta'}$ that differ in every coordinate have a difference that is indefinite on every surface, so $\mathscr{H}(L_\zeta,L_{\zeta'})$ is $\bigotimes_j H^1(B_j,\cdot)$, concentrated in degree three, of dimension $D(\zeta,\zeta') = \prod_j |\zeta_j - \zeta'_j|^2$; no component can join them, and if their shifts differ by one their classes have degree two.

[15] Such solutions are far from unique: the equations of (i) are linear in each of $x_1, x_2^\zeta, x_3^\zeta$ separately, and the components on the 144 pairs do not enter them at all.

Lemma A.31 (Pieces that differ everywhere)**.** *Keep Proposition A.26 with $n = 3$. Suppose that for some integer r either $p+2 \le t_1 < t_2 < t_3$ and M_1, M_2, M_3 have the shifts $r+1$, $r+4$, $r+5$, or $t_1 \le p-2 < p+2 \le t_2 < t_3$ and they have the shifts $r-1$, $r+3$, $r+4$ or $r-1$, $r+1$, $r+4$; these are the shifts forced by the paths of Theorem A.28(i) and of Proposition A.29(i), (a) and (b), with r the shift of the L_ζ on the path. Let L_ζ, $L_{\zeta'}$ differ in all three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Then d_E vanishes on $\mathscr{H}^3(L_\zeta, L_{\zeta'})$, and no element of degree one has a differential with a nonzero component there. Consequently*

$$\dim \mathrm{Ext}^2(E,E) \;\ge\; \dim \ker\bigl(d_E|_{D^2}\bigr) + \sum D(\zeta,\zeta'),$$

the sum running over the pairs of this kind.

Proof. Only the shifts are used, and the order of the pieces only removes components, so we ignore it. A component from P_a to P_b of Ext degree q has $d_b = d_a + 1 - q$. Between distinct L_ζ the degree q is at least 2: there is no $\mathscr{H}^0$, and $\mathscr{H}^1$ occurs only for pieces differing in one coordinate, which have opposite parities, while $q = 1$ would need equal shifts. So a component between two L_ζ lowers the shift by at least one, and by exactly one only in degree two. For a multiple M_i let μ_i and ν_i be the shifts of an L_ζ that a component from M_i reaches and from which a component reaches M_i: $\mu_i = d_{M_i} - 5$ and $\nu_i = d_{M_i} - 1$ when $t_i > p$, and $\mu_i = d_{M_i} + 1$, $\nu_i = d_{M_i} + 5$ when $t_i < p$. Between the multiples the degrees allow only $M_2 \to M_3$ in the first case, $M_2 \to M_3 \to M_1$ in the second, and $M_3 \to M_1$ in the third. Let λ_i be the largest μ_k and κ_i the least ν_k over the multiples M_k reachable from M_i, respectively reaching M_i, through these steps, M_i included. The values, in the order M_1, M_2, M_3 and relative to r, are

	μ			ν			λ			κ		
$r+1, r+4, r+5$	-4	-1	0	0	3	4	-4	0	0	0	3	3
$r-1, r+3, r+4$	0	-2	-1	4	2	3	0	0	0	2	2	2
$r-1, r+1, r+4$	0	-4	-1	4	0	3	0	-4	0	3	0	3

and in all three cases, for all i and k,

$$\lambda_i \le r \le \kappa_k, \qquad \kappa_i \ge \lambda_i + 2, \qquad \lambda_i \le \mu_i + 2, \qquad \nu_i \le \kappa_i + 2, \tag{A.5}$$

while $\mu_i \le \lambda_i$ and $\nu_i \ge \kappa_i$ by definition. A run $L \to M_k \to \cdots \to M_i \to L'$ through multiples goes from the shift $\nu_k \ge \kappa_k$ to $\mu_i \le \lambda_k \le \kappa_k - 2$, so every chain of components from one L to another lowers the shift, by at least two if it passes through a multiple and by exactly one only if it is a single component of degree two. In particular a chain from an L of shift x into M_i has $x \ge \kappa_i$, and a chain from M_i to an L of shift y has $y \le \lambda_i$.

By Lemma A.23(iii), a term of d_E on a class u of Ext degree q_u in $\mathscr{H}(a',b')$ runs along a chain $c \to \cdots \to a'$, the step u, and a chain $b' \to \cdots \to e$, and since the degrees of the components telescope, its value lies in $\mathscr{H}(c,e)$ in Ext degree

$$q_u + 1 + (d_c - d_{a'}) + (d_{b'} - d_e).$$

Write $a = L_\zeta$ and $b = L_{\zeta'}$, so that $d_a = d_b + 1$.

Nothing reaches $\mathscr{H}^3(a,b)$. Let u have degree one, so $q_u = 1 + d_{a'} - d_{b'}$, with chains from a to a' and from b' to b and $(a',b') \ne (a,b)$. If a' and b' are both pieces L, then $d_{a'} \le d_a$ and $d_{b'} \ge d_b$, not both with equality, so $q_u \le 1$; $q_u = 0$ is impossible, and $q_u = 1$ needs either $a' = b'$, and then a chain from a to b lowering the shift by one, a single component in $\mathscr{H}^2(a,b) = 0$, or a', b' differing in one coordinate, of opposite parities, against $d_{a'} = d_{b'}$. If $a' = M_i$ and b' is an L, then u lies in the degree of a component, so $d_{b'} = \mu_i$, while $d_{b'} \ge d_b = d_a - 1 \ge \kappa_i - 1 \ge \lambda_i + 1 > \mu_i$. If a' is an L and $b' = M_k$, then $d_{a'} = \nu_k$ and $d_a - d_b \ge \nu_k - \lambda_k \ge \kappa_k - \lambda_k \ge 2$. If $a' = M_i$ and $b' = M_k$, then $d_a - d_b \ge \kappa_i - \lambda_k$; for $i = k$ this is at least 2, and for $i \ne k$ the class u lies in the degree of a step $M_i \to M_k$, so $\lambda_k \le \lambda_i$ and again $d_a - d_b \ge 2$. Each case contradicts $d_a - d_b = 1$.

d_E vanishes on $\mathscr{H}^3(a,b)$. Here u is the class itself and a term lands in $\mathscr{H}(c,e)$, $(c,e) \ne (a,b)$, in degree $3 + d_c - d_e$, with chains from c to a and from b to e. If c and e are both pieces L, then $d_c - d_e \ge 2$; $c = e$ is impossible, and otherwise a class of degree at least 5 needs c, e differing in one coordinate, in degree 5, so $d_c - d_e = 2$ against their opposite parities. If $c = M_i$ and e is an L, the degree is that of a component when $d_e = \mu_i + 2$, but $d_e \le d_b < d_a \le \lambda_i \le \mu_i + 2$. If c is an L and $e = M_k$, it

is that of a component when $d_c = \nu_k - 2$, but $d_c \geq d_a > d_b \geq \kappa_k \geq \nu_k - 2$. If $c = M_i$ and $e = M_k$, then $d_a \leq \lambda_i \leq r \leq \kappa_k \leq d_b$, against $d_a > d_b$.

For the inequality: a coboundary has no component on D^2 (Proposition A.26) and none on these groups, so the cocycles of $\ker(d_E|_{D^2})$ and of these groups span a space that meets the coboundaries only in zero. Twisting the pieces by line bundles of degree zero does not change $D(\zeta,\zeta')$, the forms being nondegenerate on every surface, and can only remove components. □

Theorem A.32 (No convolution of these pieces meets the criterion)**.** *Keep Proposition A.26 with $n = 3$: let E be any convolution of the 32 bundles L_ζ and of M_1, M_2, M_3, with $t_1 < t_2 < t_3$ and $|t_i - p| \geq 2$, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero. Then $\dim \mathrm{Ext}^2(E,E) \geq 69$. In particular no such convolution meets the polarised criterion, which asks for 57.*

Proof. Replacing E by $E^\vee$ or $E^\vee[1]$, which preserves Ext^2, we may assume that the t_i are as in Theorem A.28 or in Proposition A.29. If no fourfold product of those results acts, $\dim \mathrm{Ext}^2(E,E) \geq 249$. Otherwise the path fixes the shifts of the M_i as in Lemma A.31, with r the shift of every L_ζ on a path; let P be the sixteen pieces of the parity of r, Q the other sixteen, and m the number of pieces on paths.

The fourfold products act on D^2 only through the classes at the M_i and at the m pieces on paths, so their rank is at most $45 + 15m$, and by Proposition A.27 $\dim \ker(d_E|_{D^2}) \geq 249 - 45 - 15m = 204 - 15m$. If $m \leq 9$ this is at least 69.

Let $m \geq 10$, so that at least ten pieces of P have the shift r. Every $W \in Q$ has 3, 6 and 7 partners in P differing from it in one, two and three coordinates, and for the seven $D = 16$ six times and $D = 64$ once.[16] If $d_W = r \pm 1$, at most six pieces of P lack the shift r, so one of the seven has it, and Lemma A.31 gives at least 16 classes. If $d_W \neq r \pm 1$, consider for each type τ of the form $(1,1,0)$ the partner $W^\tau \in P$ obtained by changing the sign of the coordinate where τ vanishes. By the proof of Proposition A.27, W^τ is the only piece with which W can carry a nonzero cup product on the type τ, and conversely; a cup product between them needs a component of degree two, hence shifts differing by one. If they do, the classes of type τ equal at W and W^τ and zero elsewhere have vanishing cup products, and W^τ, of shift $d_W \pm 1 \neq r$, lies on no path; if they do not, the classes of type τ at W alone have vanishing cup products. Either way these 4 classes lie in $\ker(d_E|_{D^2})$, 12 for each such W, and those of different W are independent, since $W \mapsto W^\tau$ is injective. Adding over Q, Lemma A.31 gives $\dim \mathrm{Ext}^2(E,E) \geq 16 \cdot 12 = 192$. □

At $n = 4$ there are 128 pieces L_ζ, and the classes that no product reaches are again those between pieces that differ in three coordinates and have adjacent shifts. The shifts along a chain are now controlled by counting alone.

Lemma A.33 (Shifts along chains at $n = 4$)**.** *Keep Proposition A.24 with $n = 4$, take the L_ζ to be external products $N_{1,\zeta_1} \boxtimes \cdots \boxtimes N_{4,\zeta_4}$ of line bundles on the B_j, let t_1, t_2, t_3 be distinct with $|t_i - p| \geq 2$, and let E be a convolution of the L_ζ and of M_1, M_2, M_3 with shifts of the given parities. Call a component of δ^H, or an element of degree one of $\mathscr{H}_E$ in one group $\mathscr{H}(a,b)$, a step from P_a to P_b; a step of* Ext *degree q has $d_b = d_a + 1 - q$. Put $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$.*

(i) *A step between distinct L_ζ has* Ext *degree at least two, so it lowers the shift by at least one, and by exactly one only in* Ext *degree two. A step in $\mathscr{H}^1(a,a) = H^1(A, \mathscr{O}_A)$ keeps the shift.*

(ii) *A step between an L_ζ and a multiple, or between two multiples, has* Ext *degree 0 if it goes up the value ϕ and 8 if it goes down, so it raises the shift by one or lowers it by seven.*

(iii) *A sequence of steps from an L_ζ to an $L_{\zeta'}$ through multiples, distinct before one marked step and distinct after it, where the marked step may lie in $\mathscr{H}^1(a,a)$ at a multiple, lowers the shift by at least four. Consequently a sequence of steps from one L_ζ to another, at most one of them not a component, lowers the shift by at least one, and by exactly one only if it consists of steps between pieces L_ζ, one of* Ext *degree two and the others in $\mathscr{H}^1(a,a)$.*

[16] The ratios ζ'_j/ζ_j of two pieces of opposite parities have product -1; when all three differ from 1 they are $(-1,-1,-1)$, with $D = 4^3$, or a permutation of $(i,-i,-1)$, with $D = 2 \cdot 2 \cdot 4$.

Proof. (i) For $\zeta \neq \zeta'$ the group $\mathscr{H}(L_\zeta, L_{\zeta'})$ is the tensor product over the surfaces of H^1 of a line bundle with a nondegenerate form, on the $k \geq 1$ surfaces where ζ and ζ' differ, and of $H^*(B_j, \mathscr{O}_{B_j})$ on the others, the factors of L_ζ and $L_{\zeta'}$ on those being equal. So its degrees are at least k, and degree one needs $k = 1$. Then the ratio $\zeta'_j/\zeta_j \neq 1$ in the one coordinate is the ratio of $\prod \zeta'$ and $\prod \zeta$, which is -1, and the shifts have opposite parities, while a step of Ext degree one keeps the shift. (ii) The forms of $M_i \otimes L_\zeta^{-1}$ have the eigenvalues $t_i - p \pm 1$ on every surface, which have the sign of $t_i - p$; the form of $M_k \otimes M_i^{-1}$ is $(t_k - t_i)\,\mathrm{Id}$. A definite form has cohomology in degree 0 if it is positive and $2n = 8$ if it is negative [16, Chapter 3].[17] (iii) Let U and D be the numbers of steps up and down outside $\mathscr{H}^1(a,a)$; by (ii) the shift changes by $U - 7D$. The values start and end at p and pass through the values of multiples, so $D \geq 1$. If $D = 1$, the values increase strictly before the step down and strictly after it, so those before it lie above p and those after it below p, each multiple occurs once, and $U \leq 3$: the drop $7 - U$ is at least four.[18] If $D \geq 2$, at most six multiples occur, counted with repetition, so there are at most seven steps outside $\mathscr{H}^1(a,a)$, $U \leq 7 - D$, and the drop $7D - U \geq 8D - 7$ is at least nine. For the consequence, split the sequence at the pieces L_ζ it visits: each part through multiples lowers the shift by at least four, and by (i) each other part lowers it by at least one, unless it lies in $\mathscr{H}^1(a,a)$. □

Theorem A.34 (Two shifts on E_0^8)**.** *Keep Lemma A.33, and let L_ζ and $L_{\zeta'}$ differ in exactly three coordinates and have the shifts $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Then $\mathscr{H}(L_\zeta, L_{\zeta'})$ has its classes of degree two in $\mathscr{H}^3(L_\zeta, L_{\zeta'})$, of dimension $D(\zeta,\zeta') = \prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$, which is 16 or 64, and:*

(i) *for every convolution, no element of degree one has a differential with a nonzero component in $\mathscr{H}^3(L_\zeta, L_{\zeta'})$;*

(ii) *if the L_ζ occupy two consecutive shifts, those with $\prod \zeta = \varepsilon$ the shift s and the others $s - 1$, for a sign ε and an integer s, then d_E vanishes on these groups, and for every order of the pieces, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 64 \cdot 640 = 40960.$$

In particular E does not meet the polarised criterion, which asks for $\dim \mathrm{Ext}^2(E,E) = 104$.

Proof. Write $W = L_\zeta$ and $V = L_{\zeta'}$. By the proof of Lemma A.33(i), $\mathscr{H}(W,V)$ has degrees 3, 4 and 5, the last two from the surface where ζ and ζ' agree, and its classes of degree $w = 2$ have Ext degree $2 + d_W - d_V = 3$; the dimension of $\mathscr{H}^3(W,V)$ is the product over the three other surfaces of the determinants $|\zeta_j - \zeta'_j|^2$, as in Lemma A.31.

(i) As in the proof of Lemma A.31, a term of the differential of an element u of degree one in $\mathscr{H}(a',b')$ with a component in $\mathscr{H}(W,V)$ runs along a chain of components $W \to \cdots \to a'$, the step u, and a chain $b' \to \cdots \to V$, with $(a',b') \neq (W,V)$. This is a sequence of steps from W to V, one of them u, lowering the shift by one. By Lemma A.33(iii) it consists of steps between pieces L_ζ, one of Ext degree two and at most one, u, in $\mathscr{H}^1(a,a)$. If u is not in $\mathscr{H}^1(a,a)$, it is the step of degree two and both chains are empty, against $(a',b') \neq (W,V)$. If u is in $\mathscr{H}^1(a,a)$, the rest is a single component $W \to V$ of Ext degree two, in $\mathscr{H}^2(W,V) = 0$.

(ii) A term of d_E on $x \in \mathscr{H}^3(W,V)$ runs along chains of components $c \to \cdots \to W$ and $V \to \cdots \to e$, not both empty, and lands in $\mathscr{H}(c,e)$ in Ext degree $3 + 1 + (d_c - d_W) + (d_V - d_e) = 3 + d_c - d_e$. No piece $L_{\zeta''}$ other than W lies on the first chain: by Lemma A.33(iii) it would have the shift at least $d_W + 1 = s + 1$. Likewise only multiples follow V on the second chain, and each chain has at most three pieces besides W or V. Let U and D be the numbers of steps up and down along both chains, so that $U + D \leq 6$; by Lemma A.33(ii), $d_W = d_c + U_1 - 7D_1$ and $d_e = d_V + U_2 - 7D_2$, and the Ext degree

[17] This is Mumford's index theorem: a line bundle on an abelian variety whose hermitian form is nondegenerate with i negative eigenvalues has cohomology only in degree i. On $B_j = E_0 \times E_0$ the form of $M_i \otimes L_\zeta^{-1}$ is $\left(\begin{smallmatrix} t_i - p & -\zeta_j \\ -\bar{\zeta}_j & t_i - p \end{smallmatrix}\right)$, with eigenvalues $t_i - p \pm 1$, and $|t_i - p| \geq 2$ makes both of the sign of $t_i - p$; over the four surfaces the index is 0 or 8. The hypothesis $|t_i - p| \geq 2$ is what keeps the steps between a piece and a multiple out of the middle degrees.

[18] The bound is attained: with p below the three values t_i, the run $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, up three times and down once, changes the shift by $3 - 7 = -4$. With p in the middle, $t_1 < p < t_2 < t_3$, the run up through M_2 and M_3, down to M_1 and up to $L_{\zeta'}$ changes it by $3 - 7 = -4$ as well.

is

$$3+d_c-d_e=4-U+7D.$$

If $c=e$, this degree is 3, so $U=7D+1$, and $D\geq 1$ because the values return to $\phi(c)\neq p$; then $U\geq 8$, which is impossible. If $c\neq e$, the group $\mathscr{H}(c,e)$ sits in degree 0 if $\phi(e)>\phi(c)$ and in degree 8 if $\phi(e)<\phi(c)$, by Lemma A.33(ii). Degree 0 needs $U=4+7D\geq 4$ with $D=0$, so the values increase strictly, the multiples are distinct and $U\leq 3$. Degree 8 needs $7D=U+4$, so $D=1$ and $U=3$; the values increase strictly before the step down, from $\phi(c)$, and after it, up to $\phi(e)<\phi(c)$, so the five values along the path, with W and V counted once, are distinct, and four of them are values of multiples, of which there are three. So no term reaches a nonzero group, and d_E vanishes on $\mathscr{H}^3(W,V)$.

By (i) and (ii) the classes of these groups are cocycles of degree two, and no nonzero combination of them is a coboundary, since every coboundary has zero components there. Each piece of the shift s has 28 partners of the other parity differing from it in exactly three coordinates,[19] all of the shift $s-1$, and the 28 groups have dimensions adding up to 640; there are 64 pieces of the shift s. □

The hypothesis of (ii) is the arrangement of the explicit convolution of Proposition A.30 at $n=3$, where the sixteen pieces of each parity share a shift. The pieces must not be twisted by line bundles of degree zero here, unlike in Theorem A.32: a twist that is nontrivial on the surface where ζ and ζ' agree removes $\mathscr{H}(L_\zeta,L_{\zeta'})$.

With a third shift, a component of Ext degree two can follow or precede a class of $\mathscr{H}^3(L_\zeta,L_{\zeta'})$, so the differential no longer vanishes on these classes. When the three shifts are consecutive, such terms reach only the groups between the highest and the lowest shift, which are too small to remove the classes.

Theorem A.35 (Three consecutive shifts on E_0^8)**.** *Keep Lemma A.33, and suppose that the shifts of the L_ζ lie in $\{s,s-1,s-2\}$ for an integer s. By their parities, for a sign ε, the pieces with $\prod\zeta=\varepsilon$ have the shift $s-1$ and the others the shift s or $s-2$; let T and B be the sets of pieces of the shifts s and $s-2$. Then, for every order of the pieces, every choice of the shifts of the M_i and every placement of the t_i:*

(i) *every two pieces L_ζ, $L_{\zeta'}$ that differ in exactly three coordinates have shifts differing by one, and the groups $\mathscr{H}^3(L_\zeta,L_{\zeta'})$ of Theorem A.34, taken from the piece of the larger shift, have dimensions adding up to* 40960*;*

(ii) *d_E maps the sum of these groups into $\bigoplus_{c\in T,e\in B}\mathscr{H}^5(c,e)$, and on them it consists of cup products with a single component of* Ext *degree two;*

(iii)
$$\dim\mathrm{Ext}^2(E,E)\ \geq\ 40960-\sum_{c\in T,e\in B}\dim\mathscr{H}^5(c,e)\ \geq\ 40960-18432=22528.$$

In particular E does not meet the polarised criterion. When T or B is empty this is Theorem A.34(ii).

Proof. (i) The ratios ζ'_j/ζ_j in the three coordinates have product -1, so the two pieces have opposite parities: one has the shift $s-1$ and the other s or $s-2$. Each of the 64 pieces of the shift $s-1$ has 28 such partners, whose groups have dimensions adding up to 640, as in the proof of Theorem A.34; this gives $64\cdot 640=40960$.

(ii) Let $x\in\mathscr{H}^3(a,b)$ with $d_a=d_b+1$. As in the proof of Theorem A.34(ii), a term of $d_E(x)$ runs along chains of components $c\to\cdots\to a$ and $b\to\cdots\to e$, not both empty, and lands in $\mathscr{H}(c,e)$ in Ext degree $3+d_c-d_e$. The shifts of the pieces L_ζ differ by at most two, so by Lemma A.33(iii) no part of a chain between two pieces L_ζ passes through a multiple. Hence a chain $c\to\cdots\to a$ that starts at a multiple M_i is a run $M_i=m_0\to\cdots\to m_r\to L_1$ of distinct multiples, $r\leq 2$, followed by pieces L_ζ only; let U_1 and D_1 be the numbers of its steps up and down the value ϕ before L_1, so that $d_{L_1}=d_{M_i}+U_1-7D_1$ by Lemma A.33(ii) and $U_1+D_1=r+1\leq 3$. Likewise a chain $b\to\cdots\to e$ ending at a multiple M_k ends with a run from a piece L_2 with U_2 steps up and D_2 down, $U_2+D_2\leq 3$

[19] The ratios ζ'_j/ζ_j in the three coordinates have product -1, so they are $(-1,-1,-1)$, with $D=64$, or a permutation of $(i,-i,-1)$, with $D=16$; with four places for the coordinate where they agree this gives $4\cdot 7=28$ partners and $4\cdot(64+6\cdot 16)=640$ classes.

and $d_{M_k} = d_{L_2} + U_2 - 7D_2$. Put $L_1 = c$ if c is a piece L_ζ and $L_2 = e$ if e is one, and $\delta = d_{L_1} - d_{L_2}$; then $1 = d_a - d_b \le \delta \le 2$.

If c and e are both pieces L_ζ, each nonempty chain lowers the shift by at least one, so $3 + d_c - d_e \ge 5$, while $\delta \le 2$ gives at most 5. So the degree is 5, $c \in T$, $e \in B$, and the two chains together consist of one component, of Ext degree two by Lemma A.33(i): the term is a cup product of x with that component.

If $c = M_i$ and e is a piece L_ζ, the degree is $3 + \delta - U_1 + 7D_1$, and $\mathscr{H}(c,e)$ sits in degree 0 or 8 by Lemma A.33(ii). Degree 0 needs $U_1 \ge 3 + \delta \ge 4$, and degree 8 needs $7D_1 - U_1 = 5 - \delta \in \{3,4\}$, so $D_1 \ge 1$ and $U_1 \ge 3$; both contradict $U_1 + D_1 \le 3$. The case of a piece c and a multiple e is the same with the second run.

If $c = M_i$ and $e = M_k$, the degree is $3 + \delta - U + 7D$ with $U = U_1 + U_2 \le 6$ and $D = D_1 + D_2$. For $i = k$ it is 3, so $U - 7D = \delta \ge 1$; $D = 0$ is impossible, since the values would increase from t_i to p along the first run and from p to t_i along the second, and $D \ge 1$ gives $U \ge 8$. For $i \ne k$, $\mathscr{H}(c,e)$ sits in degree 0 if $t_k > t_i$ and 8 if $t_k < t_i$. Degree 0 needs $U = 3 + \delta + 7D$, so $D = 0$ and $U \ge 4$; but then the multiples of the first run have values below p and those of the second values above p, so they are distinct and $U \le 3$. Degree 8 needs $7D - U = 5 - \delta$, so $D = 1$ and $U \ge 3$: the runs have at least four steps, so some multiple lies on both. One run increases strictly. If it is the second, its values lie in $(p, t_k]$, while those of the first lie at or above $t_i > t_k$ before its step down and below p after it. If it is the first, its values lie in $[t_i, p)$, while those of the second lie above p before its step down and at or below $t_k < t_i$ after it. Either way no multiple lies on both runs, a contradiction.

(iii) By Theorem A.34(i), which holds for every convolution, no coboundary has a component in these groups, so the kernel of d_E on their sum injects into $\mathrm{Ext}^2(E,E)$, and by (ii) its dimension is at least 40960 minus the dimension of the targets. For pieces c, e of the same parity, $\mathscr{H}^5(c,e)$ is nonzero only when they differ in two or three coordinates, and each piece has such partners with groups $\mathscr{H}^5$ of dimensions adding up to 960.[20] So the sum in (iii) is at most $\min(|T|,|B|) \cdot 960 \le 32 \cdot 960 = 30720$, which already gives $\dim \mathrm{Ext}^2(E,E) \ge 10240$. The dimension of $\mathscr{H}^5(c,e)$ depends only on the ratio e/c, so the sum is the weight of the cut (T,B) of a weighted Cayley graph on the group of ratios of two pieces of one parity, of order 64 and degree 960. For the vector $x = \pm 1$ of the cut, the weight is $\frac{1}{4}\big(64 \cdot 960 - x^{\mathsf{T}} A x\big) \le 16\,(960 - \lambda)$, with A the weighted adjacency matrix and λ its least eigenvalue. Its eigenvalues are the character sums $\sum_g \dim \mathscr{H}^5(1,g)\,\chi(g)$. A character is $\chi(g) = \prod_j g_j^{k_j}$ with $k \in (\mathbb{Z}/4)^4$, up to adding $(1,1,1,1)$; put $\omega_j = i^{k_j}$, $W = \sum_j \omega_j$, $\rho = \sum_j \omega_j^2$ and $\Omega = \prod_j \omega_j$. By the list in the footnote, the ratios that move the pair of coordinates $\{j,l\}$ contribute $64\,\omega_j^2\omega_l^2 + 32\,\Re(\omega_j \overline{\omega}_l)$, and those that fix only the coordinate o contribute $32\,\Re\big(\Omega\overline{\omega}_o(W - \omega_o)\big)$. Summing,

$$\chi\big(\textstyle\sum_g \dim \mathscr{H}^5(1,g)\,g\big) = 32\rho^2 + 16|W|^2 - 192 + 32\,\Re(\Omega)\big(|W|^2 - 4\big).$$

If $\Re\Omega = 0$, the sum of the k_j is odd, so an odd number of them are odd, $|\rho| = 2$, and the value is at least -64. If $\Re\Omega = 1$, an even number are odd: with none or four, $|\rho| = 4$ and the value is at least 192; with two, $\rho = 0$, and $W = 0$ would force the k_j to be $0,1,2,3$, whose sum is not divisible by 4, so $|W|^2 \ge 4$ and the value is at least -128. If $\Re\Omega = -1$, the value is $32\rho^2 - 16|W|^2 - 64$, and $|W|^2 \le 8 + 2\rho^2$: when all k_j have the same parity, $|W| \le 4 = |\rho|$, and when two are odd, $\rho = 0$ and W is a sum of two of ± 1 and two of $\pm i$, so $|W|^2 \le 8$. Hence $\lambda \ge -192$,[21] and the sum is at most $16 \cdot 1152 = 18432$. □

The groups of Theorem A.34 join two pieces whose shifts differ by one. All groups between pieces whose shifts differ by three, and thirteen kinds between pieces whose shifts differ by two, are also left alone by the differential, and with them the argument covers every arrangement of the shifts within five consecutive values. For pieces a and b call $d_a - d_b$ the *spread* of $\mathscr{H}(a,b)$; its classes of degree two have Ext degree $2 + d_a - d_b$. For $\zeta, \zeta' \in \mu_4^4$ call $\zeta'/\zeta = (\zeta'_j/\zeta_j)_j$ their *ratio*; it *moves* the coordinates

[20] The ratios $e_j/c_j \ne 1$ have product 1. In two coordinates they are $(-1,-1)$, with $D = 16$, or $(i,-i)$, $(-i,i)$, with $D = 4$; then $\mathscr{H}^5 = \mathscr{H}^1 \otimes \mathscr{H}^1 \otimes \mathscr{H}^3(B_j \times B_k, \mathscr{O})$ has dimension $4D$, and the six pairs of coordinates give $6 \cdot 64 + 12 \cdot 16 = 576$. In three coordinates they are a permutation of $(i,i,-1)$ or of $(-i,-i,-1)$, with $D = 16$, and $\mathscr{H}^5 = (\mathscr{H}^1)^{\otimes 3} \otimes H^2(B_l, \mathscr{O})$ has dimension 16; the four triples give $4 \cdot 6 \cdot 16 = 384$. In four coordinates $\mathscr{H}(c,e)$ sits in degree 4.

[21] The eight values are 960, 384, 192, 96, -32, -64, -128 and -192. The bound 18432 is attained by the cut $\zeta_3\zeta_4 \in \{1,i\}$ against $\zeta_3\zeta_4 \in \{-1,-i\}$, so 22528 is the best bound that this count gives.

where it differs from 1, and it changes the parity, $\prod\zeta' = -\prod\zeta$, exactly when the product of its entries is -1. As before, $D(\zeta,\zeta') = \prod_{\zeta_j\neq\zeta'_j}|\zeta_j-\zeta'_j|^2$.

Lemma A.36 (Groups of spread two and three)**.** *Keep Lemma A.33, and let E be any convolution of the L_ζ and the M_i, in any order, with any shifts of the given parities and any placement of the t_i.*

(i) *For a class x of degree two in $\mathscr{H}(L_\zeta,L_{\zeta'})$ of positive spread, every term of $d_E(x)$ has all its leaves between pieces $L_{\zeta''}$.*

(ii) *Let $\prod\zeta = \prod\zeta'$, let ζ'/ζ move all four coordinates, one of them by -1, and let $d_{L_\zeta} = d_{L_{\zeta'}}+2$. The classes of degree two of $\mathscr{H}(L_\zeta,L_{\zeta'})$ form $\mathscr{H}^4(L_\zeta,L_{\zeta'})$, of dimension $D(\zeta,\zeta')\in\{64,256\}$; no element of degree one has a differential with a nonzero component in it, and d_E vanishes on it.*

(iii) *Let $\prod\zeta = -\prod\zeta'$ and $d_{L_\zeta} = d_{L_{\zeta'}}+3$. The classes of degree two of $\mathscr{H}(L_\zeta,L_{\zeta'})$ lie in $\mathscr{H}^5(L_\zeta,L_{\zeta'})$, and d_E vanishes on them.*

(iv) *Let ζ and ζ' differ in exactly three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}}+1$. Every term of d_E on $\mathscr{H}^3(L_\zeta,L_{\zeta'})$ runs along chains of components $c\to\cdots\to L_\zeta$ and $L_{\zeta'}\to\cdots\to e$ between pieces $L_{\zeta''}$ that together lower the shift by one or two.*

Proof. Write $a = L_\zeta$, $b = L_{\zeta'}$ and $\delta = d_a - d_b$. As in the proof of Theorem A.34(ii), a term of d_E on a class of degree two in $\mathscr{H}(a,b)$ runs along chains of components $c\to\cdots\to a$ and $b\to\cdots\to e$, not both empty, and lands in $\mathscr{H}(c,e)$ in Ext degree $3+d_c-d_e$, which is 3 if $c=e$.

(i) Let L_1 be the first piece $L_{\zeta''}$ of the first chain and L_2 the last of the second, and put $X = (d_{L_1}-d_a)+(d_b-d_{L_2})$. By Lemma A.33(iii), $X\geq 0$, and $X\geq 4$ if a multiple lies between L_1 and a or between b and L_2. Before L_1 the first chain is a run of distinct multiples, with U_1 steps up and D_1 steps down the value ϕ and $U_1+D_1\leq 3$; after L_2 the second chain is a run with U_2 and D_2 steps, $U_2+D_2\leq 3$. By Lemma A.33(ii) the term has Ext degree $3+\delta+X-U+7D$, where $U = U_1+U_2$ and $D = D_1+D_2$. If c and e are pieces, this is $3+\delta+X$; a multiple on a chain makes it at least 8, but $\mathscr{H}^8(c,e)\neq 0$ only for $c=e$, where the degree is 3. If $c = M_i$ and e is a piece, $\mathscr{H}(c,e)$ sits in degree 0 if $t_i<p$ and in degree 8 if $t_i>p$. Degree 0 needs $U_1\geq 3+\delta+X\geq 4$. Degree 8 needs a step down on the run from t_i to p, so $D_1\geq 1$, and then $U_1 = 7D_1-5+\delta+X\geq 3$. Both contradict $U_1+D_1\leq 3$, and a piece c with a multiple e is the same. Let $c = M_i$ and $e = M_k$. For $i=k$ the degree is 3, so $U-7D = \delta+X\geq 1$; $D\geq 1$ would give $U\geq 8$, and $D=0$ would make the values increase from t_i to p and then from p to t_i. For $i\neq k$, $\mathscr{H}(c,e)$ sits in degree 0 if $t_k>t_i$ and in degree 8 if $t_k<t_i$. Degree 0 needs $U = 3+\delta+X+7D$, so $D=0$ and $U\geq 4$; but then the first run lies below p and the second above, so the multiples are distinct and $U\leq 3$. Degree 8 needs $D\geq 1$, since with $D=0$ the values would satisfy $t_i<p<t_k$; then $7D-U = 5-\delta-X\leq 4$ gives $D=1$ and $U\geq 3$, so the runs have at least four steps and share a multiple, which the last paragraph of the proof of Theorem A.35(ii) excludes.

(ii) By the proof of Lemma A.33(i), $\mathscr{H}(a,b)$ is the tensor product of four groups H^1 and sits in degree 4, of dimension $D(\zeta,\zeta')$; the ratio is $(-1,-1,-1,-1)$, with $D=256$, or a permutation of $(i,-i,-1,-1)$, with $D=64$.[22] A coboundary with a component in $\mathscr{H}(a,b)$ is a sum of terms of Lemma A.23(iii) along a path $a\to\cdots\to b$ of morphisms of degree one, one of them an element u and the others components, that lowers the shift by two. By Lemma A.33(iii) no multiple lies on it, and by Lemma A.33(i) each step between distinct pieces lowers the shift by at least one. So the path consists of a component of Ext degree three from a to b, which lies in $\mathscr{H}^3(a,b)=0$, and a class of $\mathscr{H}^1(a,a)$ or $\mathscr{H}^1(b,b)$; or of two steps $a\to c\to b$ of Ext degree two, possibly with a class of $H^1(A,\mathscr{O}_A)$ at one of the three pieces. The output has degree one on every surface. Without the class of H^1 the two steps have degrees adding up to one on every surface, so they move disjoint pairs of coordinates and have degree zero elsewhere; a step of Ext degree two changes the parity, so its two moved entries have product -1 and are (i,i) or $(-i,-i)$, and ζ'/ζ has no entry -1. With the class of H^1, Lemma A.25 gives a drop of one on a surface where all three leaves have positive degree, and the output has degree at least two there.

[22] A ratio that moves every coordinate has entries in $\{i,-1,-i\}$ and product 1. With a -1 among them it is $(-1,-1,-1,-1)$ or has exactly two entries -1 and the entries i and $-i$, twelve ratios; together thirteen, and $256+12\cdot 64 = 1024$. The other eight ratios have all four entries in $\{i,-i\}$ and $D=16$; (ii) makes no statement about them.

Now let a term of d_E on $\mathscr{H}^4(a,b)$ carry the components $y_1,\dots,y_m$, $m \geq 1$, between pieces by (i), and land in $\mathscr{H}(c,e)$; let K be the set of coordinates fixed by the ratio of c and e, and $X \geq m$ the drop of the chains. The output has Ext degree $5+X$. Let n_j be the number of the y_i of positive degree on B_j; the class x has degree one there, so Lemma A.25 bounds the drop on B_j by $\max(0,n_j-1)$. For $j \notin K$ the output has degree 1 on B_j, so the degrees of the y_i there add up to at most $\max(0,n_j-1)$, which forces $n_j = 0$. For $j \in K$ the y_i undo the entry $\zeta'_j/\zeta_j \neq 1$, so $n_j \geq 1$, and the output has degree at least $1+n_j-(n_j-1) = 2$ there, hence exactly 2. So $5+X = (4-|K|)+2|K|$, that is $X = |K|-1$. The components change the parity X times modulo two, and together their ratio is ζ/ζ' on K and 1 elsewhere, so

$$\prod_{j\in K} \zeta'_j/\zeta_j = (-1)^{|K|-1}.$$

For the ratio $(-1,-1,-1,-1)$ the left side is $(-1)^{|K|}$. For a permutation of $(i,-i,-1,-1)$ the left side is real only if K contains both or neither of the coordinates moved by $\pm i$, and then it is again $(-1)^{|K|}$. So no term is nonzero.

(iii) The classes of degree two have Ext degree 5. By (i) a term of d_E on them lands in Ext degree $6+X$, with $X \geq 1$ the drop of the chains between pieces. For $X \geq 2$ this is at least 8, which needs $c = e$ and degree 3. For $X = 1$ the pieces c, e have shifts differing by four, hence the same parity, so their ratio moves at least two coordinates and $\mathscr{H}(c,e)$ has no degree above 6.

(iv) By (i) a term lands in Ext degree $4+X$, with $X \geq 1$ the drop of the chains between pieces. For $X \geq 4$ the degree is at least 8, which needs $c = e$. For $X = 3$ the degree is 7, which needs a ratio moving one coordinate, hence opposite parities, while $d_c - d_e = 4$ is even. □

The groups of (ii) are reached by no coboundary and left by no term in every convolution, so they survive in $\mathrm{Ext}^2(E,E)$ as soon as the shifts give them the spread two. That happens along every edge of a Cayley graph that crosses between two shift values of one parity, and a cut of that graph is never light.

Lemma A.37 (Cuts of a Cayley graph)**.** *Let G be a finite abelian group, $S \subset G \setminus \{0\}$ a generating set and $w\colon S \to \mathbb{R}_{>0}$ a weight, extended by 0 to G. For every subset $T \subset G$ with $T \neq \emptyset, G$,*

$$\sum_{c\in T,\ e\notin T} w(e-c) \ \geq\ \min_{H} |H| \cdot w(S \setminus H),$$

the minimum over the proper subgroups H of G. For the group G_0 of the 64 ratios of pieces of one parity, with S the thirteen ratios of Lemma A.36(ii) and $w = D$, the minimum is 1024, attained only at $H = 0$, so every such cut weighs at least 1024.

Proof. The left side is $\sum_{s\in S} w(s)\,|(T+s)\setminus T|$. Let H be the stabiliser of T, a proper subgroup since $T \neq \emptyset, G$. Then T and $T+s$ are unions of cosets of H, and for $s \notin H$ the set $(T+s)\setminus T$ is nonempty, since $|T+s| = |T|$, so it has at least $|H|$ elements. This gives the inequality. In G_0, written additively as the quadruples in $(\mathbb{Z}/4)^4$ of sum 0, the ratios of type $(i,-i,-1,-1)$ generate, since the differences of two of them give every $e_j - e_k$. The weights are 256 once and 64 otherwise, and $|S \cap H| \leq |H|-1$. So $|H|\,w(S\setminus H)$ is 1024 for $H = 0$, at least $|H|\,(1024-256-64(|H|-2))$, which is 1536, 2560 and 3072, for $|H| = 2,4,8$, and at least $64|H| \geq 1024$ for $|H| \geq 16$, because some generator lies outside H.[23] □

Theorem A.38 (One parity at two or three values)**.** *Keep Lemma A.33, and suppose that the shifts of the pieces L_ζ with $\prod \zeta = \varepsilon$ take exactly two values s and $s-2$, or exactly three values $s+2$, s and $s-2$, for a sign ε and an integer s. Then, for every order of the pieces, all shifts of the other pieces, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E,E) \ \geq\ 1024.$$

In particular E does not meet the polarised criterion.

[23] The orders of the subgroups of $G_0 \cong (\mathbb{Z}/4)^3$ are powers of two. A cut at one vertex weighs exactly 1024, so the bound is sharp.

Proof. Let T be the set of pieces of parity ε at the shift s; it is neither empty nor all of them. Two pieces of parity ε, one in T and the other not, have shifts differing by exactly two. When their ratio is one of the thirteen of Lemma A.36(ii), the group $\mathscr{H}^4$ from the piece of the larger shift to the other is reached by no coboundary and killed by no term, so its classes inject into $\operatorname{Ext}^2(E,E)$, and those of different pairs lie in different summands $\mathscr{H}(a,b)$. Their dimensions add up to the weight of the cut $(T,\complement T)$ of the Cayley graph of Lemma A.37, since the pieces of parity ε form a coset of G_0; so they add up to at least 1024. □

Theorem A.39 (Shifts three apart)**.** *Keep Lemma A.33, and suppose that the pieces L_ζ with $\prod\zeta=\varepsilon$ all have the shift s and the others have shifts in $\{s-3,s+3\}$. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim\operatorname{Ext}^2(E,E)\ \geq\ 64\cdot 1072-128\cdot 8=67584.$$

Proof. Every piece W of parity $-\varepsilon$ and every piece V of parity ε have shifts differing by three. By Lemma A.36(iii) the classes of degree two of the group from the higher to the lower of the two lie in $\mathscr{H}^5$, and d_E vanishes on them. For each W these groups have dimensions adding up to 1072.[24] A coboundary with a component in such a group runs along a path of morphisms of degree one that lowers the shift by three; no multiple lies on it, by Lemma A.33(iii), and a step between distinct pieces lowers the shift by at least one, while no piece has a shift strictly between those of W and V. So the path is the component between W and V together with a class of $H^1(A,\mathscr{O}_A)$ at one of them, and the components of all coboundaries in these groups come from $\bigoplus_a\mathscr{H}^1(a,a)$ over the 128 pieces, of dimension $128\cdot 8=1024$. The $64\cdot 1072=68608$ classes are cocycles, and at most 1024 dimensions of them are coboundaries. □

When one parity sits at a single shift, the groups $\mathscr{H}^3$ of Theorem A.34 survive next to it unless pieces of the other parity crowd it from both sides.

Proposition A.40 (A gap of four)**.** *Keep Lemma A.33, let the pieces L_ζ with $\prod\zeta=\varepsilon$ all have the shift y, and let T be the set of pieces of parity $-\varepsilon$ at the shift $y+1$. If T is not empty and no piece has the shift $y-1$ or $y+3$, then for every order, every choice of the shifts of the M_i and every placement of the t_i, the groups $\mathscr{H}^3(W,V)$ with $W\in T$ and V differing from W in exactly three coordinates are reached by no coboundary and killed by no term, and*

$$\dim\operatorname{Ext}^2(E,E)\ \geq\ 640\,|T|\ \geq\ 640.$$

The same holds with $y+1$, $y-1$, $y+3$ replaced by $y-1$, $y+1$, $y-3$ and the groups $\mathscr{H}^3(V,W)$.

Proof. By Theorem A.34(i), which holds for every convolution, no coboundary reaches these groups. By Lemma A.36(iv) a term of d_E on $\mathscr{H}^3(W,V)$ runs along chains of components between pieces into W and out of V that lower the shift by one or two in all; the first would pass through a piece at the shift $y+2$ or $y+3$, the second through one at $y-1$ or $y-2$. The shifts $y\pm 2$ have parity ε, whose pieces are all at y, and $y-1$, $y+3$ are excluded. Each $W\in T$ has 28 partners V, all at the shift y, with groups of dimensions adding up to 640. The second statement has the same proof, with the chains into V and out of W in place of those into W and out of V. □

Corollary A.41 (Five consecutive shifts)**.** *Keep Lemma A.33. If the shifts of the 128 pieces L_ζ lie in five consecutive integers, then, for every order, every choice of the shifts of the M_i and every placement of the t_i, $\dim\operatorname{Ext}^2(E,E)\geq 640$; no such convolution meets the polarised criterion.*

Proof. Let the values be $s,\dots,s-4$, with one parity at values in $\{s,s-2,s-4\}$ and the other at values in $\{s-1,s-3\}$. If the second parity takes both its values, or the first takes two values two apart or all three, Theorem A.38 applies. Otherwise the second parity sits at one value y and the first at one value or at $\{s,s-4\}$. One value each, at an odd distance at most three: Theorem A.34 or Theorem A.39.

[24] Let the ratio move k coordinates. For $k=1$ it is -1 in one coordinate, $D=4$, and $\mathscr{H}^5$ is H^1 tensored with the part of degree four of $H^*(\mathscr{O})$ on the other three surfaces, of dimension $\binom{6}{4}=15$: four partners with 60 each. For $k=2$ it is (i,i) or $(-i,-i)$, $D=4$, and the part of degree three on the other two surfaces has dimension $\binom{4}{3}=4$: twelve partners with 16 each. For $k=3$ the part of degree two on the last surface has dimension one, and the 28 partners give 640 as in Theorem A.34. For $k=4$ the group sits in degree 4. So $240+192+640=1072$.

The first at $\{s, s-4\}$: for $y = s-1$ the pieces at $y+1 = s$ are not empty and $y-1$, $y+3$ are empty, and for $y = s-3$ the pieces at $y-1 = s-4$ are not empty and $y+1$, $y-3$ are empty, so Proposition A.40 applies. The least of the bounds is 640. □

Within six consecutive values two arrangements are left, up to exchanging the parities: each parity at one shift, five apart, and the parities at $\{s, s-4\}$ and $\{s-1, s-5\}$. The first is settled by the diagonal, whose classes at a shift with no other piece nearby can be reached only through all three multiples, and the second by the groups of Theorem A.34 once more.

Lemma A.42 (The diagonal at a lonely shift)**.** *Keep Lemma A.33, and let the pieces L_ζ with $\prod \zeta = \varepsilon$ all have the shift s and the others the shift $s-g$, for a sign ε and an odd integer $g \geq 5$. Let $y \in \mathscr{H}^2(a,a)$ be a diagonal class at a piece a, and let a term of $d_E(y)$ in Lemma A.23(iii) be nonzero. Then a is the only piece on its path, the path passes through M_1, M_2 and M_3, and the shifts of a, M_1, M_2, M_3 are pairwise incongruent modulo* 4.

Proof. Write the path as $c \to \cdots \to a \to \cdots \to e$, with the chain of components into a and the chain out of a not both empty, so that $c \neq e$, since the components of a convolution go up the order of the pieces. Let L_1 and L_2 be the first and the last piece on it and $X = d_{L_1} - d_{L_2}$; before L_1 and after L_2 the path is a run of distinct multiples, with U steps up and D steps down the value ϕ in all. As in the proof of Lemma A.36(i), now with $\delta = 0$, the term has Ext degree $3 + X - U + 7D$ by Lemma A.33(ii). A sequence of components from one piece to another lowers the shift by Lemma A.33(iii), and the pieces have the shifts s and $s-g$; so $X \in \{0, g\}$, and $X = 0$ exactly when a is the only piece on the path.

If c and e are pieces, then $U = D = 0$ and $X = g$, and $\mathscr{H}^{3+g}(c,e) = 0$ since $3 + g \geq 8$ and $c \neq e$. If c is a multiple and e a piece, then $\mathscr{H}(c,e)$ sits in degree 0 or 8. Degree 0 needs $U = 3 + X + 7D$, so $X = D = 0$ and $U = 3$. Degree 8 needs $7D - U = 5 - X$: for $X = 0$ this gives $D = 1$ and $U = 2$; for $X = g$ it gives $D = 0$ and $U = g - 5$, which is at least one because the run before L_1 is not empty and at most three, so $g = 7$; but then the path lowers the shift by seven from L_1 to e, by a single component, of Ext degree eight between distinct pieces, or by a run through multiples, and neither is possible.[25] A piece c with a multiple e is the same with the runs exchanged. If c and e are multiples, both runs are nonempty and together pass through at most three multiples, so $U + D \leq 3$ and $U + D \geq 2$. Degree 0 needs $U = 3 + X + 7D$, so $X = D = 0$ and $U = 3$. Degree 8 needs $7D - U = 5 - X$; $D = 0$ gives $U = X - 5 \geq 2$, so $X = g = 7$, which is excluded as before; $D = 1$ gives $U = 2 + X \leq 2$, so $X = 0$ and $U = 2$; $D \geq 2$ gives $U \geq 9$.

So in every case $X = 0$ and $U + D = 3$: the path consists of a and the three multiples, with three steps. A step up raises the shift by one and a step down lowers it by seven, so every step adds one modulo 4, and the four nodes of the path have the shifts $d_c, d_c + 1, d_c + 2, d_c + 3$ modulo 4 in the order of the path.[26] □

Proposition A.43 (Single shifts)**.** *Keep Lemma A.33, and suppose that the pieces L_ζ with $\prod \zeta = \varepsilon$ all have the shift s and the others all have the shift $s-g$, for a sign ε and an odd integer g. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 64 \cdot 28 = 1792.$$

Proof. Exchanging the parities, we may take $g > 0$. For $g = 1$ and $g = 3$ this is Theorem A.34 and Theorem A.39. Let $g \geq 5$. If a term of d_E were nonzero on a diagonal class at a piece of shift s and another on a diagonal class at a piece of shift $s-g$, then by Lemma A.42 the shifts of the three multiples would be pairwise incongruent modulo 4 and miss both the residue of s and that of $s-g$, which differ, since g is odd. So at one of the two shifts d_E vanishes on the 28 diagonal classes of each

[25] Between distinct pieces $\mathscr{H}^8 = 0$. A run from a piece through $m \leq 3$ distinct multiples to a piece has $m+1$ steps, at least one up and one down, and lowers the shift by $7D' - U'$ with $U' + D' = m + 1$: by 6, 5, 4 for $D' = 1$, by 13, 12 for $D' = 2$ and by 20 for $D' = 3$, never by seven.

[26] With the values $t_1 < t_2 < t_3$ and p, the path with $U = 3$ visits the four nodes in the order of their values; a path with $U = 2$ and $D = 1$ ending in degree eight visits the largest j values in increasing order and then the others, $j = 1, 2, 3$. So each placement has exactly four such paths, and for each the shifts of the multiples are determined by d_a.

of the 64 pieces there, and by Proposition A.26, which holds for $n = 4$, these 1792 classes inject into $\mathrm{Ext}^2(E,E)$.[27] □

Proposition A.44 (Interleaved shifts)**.** *Keep Lemma A.33, and suppose that the shifts of the pieces L_ζ with $\prod\zeta = \varepsilon$ take exactly the two values s and $s-4$, and those of the others exactly the two values $s-1$ and $s-5$. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i, $\dim\mathrm{Ext}^2(E,E) \geq 640$.*

Proof. Let A and A' be the pieces of parity ε at the shifts s and $s-4$, and B and B' those of parity $-\varepsilon$ at $s-1$ and $s-5$; all four are nonempty. Let W and V differ in exactly three coordinates, with $d_W = d_V + 1$, that is $W \in A$ and $V \in B$, or $W \in A'$ and $V \in B'$. By Theorem A.34(i) no coboundary reaches $\mathscr{H}^3(W,V)$, and by Lemma A.36(iv) a term of d_E on it passes through a piece at the shift $d_W + 1$ or $d_W + 2$ or at $d_V - 1$ or $d_V - 2$. These are $s+1$, $s+2$, $s-2$, $s-3$ for $W \in A$ and $s-3$, $s-2$, $s-6$, $s-7$ for $W \in A'$, and no piece has these shifts. So the groups inject into $\mathrm{Ext}^2(E,E)$.

Let Γ be the group of the ratios of two pieces, of order 128, which acts simply transitively on the pieces, and $S_3 \subset \Gamma$ the 28 ratios that move exactly three coordinates and change the parity, with the weight D, which adds up to 640 (Theorem A.34). With $X = A \cup B'$ the pairs above are exactly the edges of the Cayley graph of (Γ, S_3) between X and its complement $A' \cup B$; so the dimensions add up to the weight of that cut, and X is neither empty nor everything. Written additively, as the quadruples in $(\mathbb{Z}/4)^4$ of even sum, the differences of two ratios of type $(i,-i,-1,1)$ give every $e_j - e_k$, which generate the quadruples of sum 0 modulo 4, and the elements of S_3 have sum 2 modulo 4; so S_3 generates Γ. In Lemma A.37, $H = 0$ gives 640; for $|H| = 2,4,8,16$ at most $|H|-1$ elements of S_3 lie in H, at most four of them of weight 64, so $|H|\,w(S_3 \setminus H)$ is at least $2\cdot 576$, $4\cdot 448$, $8\cdot 336$ and $16\cdot 208$; for $|H| = 32$ some $r \in S_3$ lies outside H, and so does $-r$, which gives weight at least 32, or 64 if $r = -r$; and for $|H| = 64$ the weight outside H is at least 16. So every such cut weighs at least 640.[28] □

Corollary A.45 (Six consecutive shifts)**.** *Keep Lemma A.33. If the shifts of the 128 pieces L_ζ lie in six consecutive integers, then, for every order, every choice of the shifts of the M_i and every placement of the t_i, $\dim\mathrm{Ext}^2(E,E) \geq 640$; no such convolution meets the polarised criterion.*

Proof. By Corollary A.41 we may assume that the shifts take both values s and $s-5$ of the window $s,\dots,s-5$; let the parity at s have its values in $\{s, s-2, s-4\}$ and the other in $\{s-1, s-3, s-5\}$. If one of them takes two values two apart, or three, Theorem A.38 applies. Otherwise the first takes $\{s\}$ or $\{s, s-4\}$ and the second $\{s-5\}$ or $\{s-1, s-5\}$. For $\{s\}$ and $\{s-5\}$ this is Proposition A.43, and for $\{s, s-4\}$ and $\{s-1, s-5\}$ it is Proposition A.44. For $\{s\}$ and $\{s-1, s-5\}$, Proposition A.40 applies in its second form with $y = s$: the shift $s-1$ is taken and $s+1$, $s-3$ are not. For $\{s, s-4\}$ and $\{s-5\}$ it applies in its first form with $y = s-5$: the shift $s-4$ is taken and $s-6$, $s-2$ are not. The least of the bounds is 640. □

A.5. **The count for sextic CM fields**

Setting A.46 (The flat secant space of a sextic CM field)**.** Let $F = F_0(\sqrt{-q})$ be a sextic CM field with totally real cubic subfield F_0, real places τ_1, τ_2, τ_3 and $q \in F_0$ totally positive, and let X be a principally polarised abelian sixfold with real multiplication by the ring of integers of F_0, so that $H^1(X,\mathbb{C}) = U_1 \oplus U_2 \oplus U_3$ with U_j the eigenspace of τ_j, of dimension four and Hodge type $(2,2)$. Let $\theta_j \in H^{1,1}(X)$ be the component of the polarisation on U_j, so that $\theta = \sum_j \theta_j$, $\theta_j^3 = 0$ and $\int_X \theta_1^2\theta_2^2\theta_3^2 = 8$. For $t \in F_0$ and $\varepsilon \in \{\pm 1\}^3$ put $s_j = \sqrt{-1}\,\sqrt{\tau_j(q)}$ and $e_\varepsilon = \exp\bigl(\sum_j (\tau_j(t) + \varepsilon_j s_j)\theta_j\bigr)$, and let $S(t,q)$ be the span of the eight e_ε. The Galois group permutes the e_ε, so the span is a rational subspace of the algebra generated by the θ_j; since $\theta_j^3 = 0$, $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$ with $B_j = 1 - \frac{1}{2}\tau_j(q)\theta_j^2$, so the eight

[27] The proof does not look at the classes between the two parities. For $g = 5$ these are $64\cdot 16 = 1024$ classes of degree two in groups $\mathscr{H}^7$, on which d_E vanishes by Lemma A.36(i); but the coboundaries of the loops in $\mathscr{H}^1(a,a)$, and, when two multiples have suitable shifts, the products along runs $W \to M_i \to M_j \to V$, reach them, and counting alone gives nothing there.

[28] The subgroups of $\Gamma \cong (\mathbb{Z}/4)^3 \times \mathbb{Z}/2$ have orders that are powers of two. A cut at a single vertex weighs exactly 640, so the bound is sharp.

classes are independent and $\dim S(t,q)=8$; and $S(t,q)=e^{\sum_j \tau_j(t)\theta_j}S(0,q)$. A real class $v=\sum_\varepsilon w_\varepsilon e_\varepsilon$ has $w_{-\varepsilon}=\overline{w}_\varepsilon$. Write N_w for the number of nonzero w_ε; $A_{jj'}$ for the number of sign pairs (σ,σ') such that $w_\varepsilon\neq 0$ for some ε with $\varepsilon_j=\sigma$ and $\varepsilon_{j'}=\sigma'$; ρ_j for the rank of the 2×4 matrix (w_ε) whose rows are indexed by ε_j and whose columns by the two other signs; and, for $\{j,j',j''\}=\{1,2,3\}$ and $\sigma'=\pm 1$, $M_{\sigma'}^{(j;j')}$ for the 2×2 matrix $(w_\varepsilon)_{\varepsilon_j,\varepsilon_{j''}}$ with $\varepsilon_{j'}=\sigma'$ fixed. Let $r^k(v)$ be the rank of the contraction $HT^k(X)\to H^*(X,\mathbb{C})$, $\xi\mapsto \xi\lrcorner v$, where $HT^k(X)$ is the part of degree k of $HH^*(X)$ in Setting A.10, and write $\chi(v,v')=\int_X v^\vee v'$, where $v^\vee$ is v with the sign of its components of degree 2 mod 4 changed. For $c_0,c_3\in\mathbb{Q}$ and $f,g\in F_0$ let

$$v(c_0,f,g,c_3)=c_0B_1B_2B_3+\sum_j \tau_j(f)\,\theta_jB_kB_l+\sum_j \tau_j(g)\,B_j\theta_k\theta_l+c_3\,\theta_1\theta_2\theta_3,$$

where $\{j,k,l\}=\{1,2,3\}$ in each sum.

Proposition A.47 (The count in degree six)**.** *In Setting A.46, let $v\in S(t,q)$ be a nonzero real class.*

(i) $r^1(v)=12$,

$$r^2(v)=4\,(A_{12}+A_{13}+A_{23})+\rho_1+\rho_2+\rho_3,\qquad r^3(v)=8N_w+2\sum_{j\neq j'}\big(\mathrm{rk}\,M_+^{(j;j')}+\mathrm{rk}\,M_-^{(j;j')}\big).$$

So $r^2(v)\le 54$ and $r^3(v)\le 112$, with equality when every w_ε is nonzero and every $M_{\sigma'}^{(j;j')}$ is invertible, which holds outside a proper Zariski closed subset of $S(t,q)$. When the support of w is a pair $\{\varepsilon_0,-\varepsilon_0\}$, so that $N_w=2$, one has $(r^2,r^3)=(30,40)$; this is the case of a class in the secant plane of an imaginary quadratic subfield of F.

(ii) $\chi(v,v)=\int_X v^\vee v=-64\,\mathrm{N}_{F_0/\mathbb{Q}}(q)\sum_\varepsilon|w_\varepsilon|^2<0$. *For a CM field of degree $2m$, with X of dimension $2m$ and 2^m exponentials, the same computation gives $(-4)^m\mathrm{N}(q)\sum_\varepsilon|w_\varepsilon|^2$: the Euler form on the secant space is definite, of sign $(-1)^m$.*

(iii) *Let $\mathscr{F}$ be a perfect complex on X with $\mathrm{ch}(\mathscr{F})=v$ and put $e_k=\dim\mathrm{Ext}^k(\mathscr{F},\mathscr{F})$. Then $e_k\ge r^k(v)$ for every k. If $e_0=1$, $e_1=12$ and $e_2=r^2(v)$, the least values, then $e_3=2r^2(v)-22-\chi(v,v)$, and this is compatible with $e_3\ge r^3(v)$ exactly when*

$$\chi(v,v)\le-\big(r^3(v)-2r^2(v)+22\big),$$

the condition of the count*: $\chi(v,v)\le -26$ when $(r^2,r^3)=(54,112)$, and $\chi(v,v)\le -2$ when $N_w=2$.*

Proof. (i) Choose a basis x_{j1},x_{j2} of $U_j^{1,0}$ and y_{j1},y_{j2} of $U_j^{0,1}$ with $\theta_j=x_{j1}y_{j1}+x_{j2}y_{j2}$, and let $\omega_j=y_{j1}y_{j2}$ span $\bigwedge^2U_j^{0,1}$. Then $H^*(X,\mathbb{C})$ and $HT^*(X)=\bigwedge^*(H^{0,1}\oplus T)$ are the tensor products over j of the corresponding algebras of the three factors, $HT^k(X)=\bigoplus_{a_1+a_2+a_3=k}HT_1^{a_1}\otimes HT_2^{a_2}\otimes HT_3^{a_3}$, and contraction is compatible with the factorisation; $v=\sum_\varepsilon w_\varepsilon\bigotimes_j e^{\lambda_j(\varepsilon_j)\theta_j}$ with $\lambda_j(\pm1)=\tau_j(t)\pm s_j$, two distinct numbers. The blocks of $HT^k(X)$ shift the three Hodge bidegrees of $H^*(X)$ by different amounts, so their images are independent and $r^k(v)$ is the sum of their ranks. Inside a block the bookkeeping rests on two facts about one factor: the two classes $e^{\lambda_j(\pm1)\theta_j}$ are linearly independent, and so are the subspaces $U_j^{0,1}e^{\lambda_j(\pm1)\theta_j}$ of $H^*(U_j)$, since $u\,e^{\lambda\theta_j}=u+\lambda\,u\theta_j$ for $u\in U_j^{0,1}$; whereas $\omega_je^{\lambda\theta_j}=\omega_j$ for every λ, so that the summands of a block through HT_j^2 share the factor ω_j and are combined, not juxtaposed. On one factor, contraction into $e^{\lambda\theta_j}$ is as follows. For $\xi\in HT_j^1$, $\xi\lrcorner e^{\lambda\theta_j}=y^\lambda(\xi)\,e^{\lambda\theta_j}$ with $y^\lambda(\xi)=\xi^{0,1}+\lambda\,\xi_T\lrcorner\theta_j\in U_j^{0,1}$, because contraction with a tangent vector is a derivation; y^λ is onto, its kernel K_j^λ has dimension two, and $HT_j^1=K_j^{\lambda_j(+1)}\oplus K_j^{\lambda_j(-1)}$ because $\lambda_j(+1)\neq\lambda_j(-1)$ and $\xi_T\lrcorner\theta_j$ determines ξ_T. For $\xi\in HT_j^2$, since $\omega_j\theta_j=0$, one finds $\omega_j\lrcorner e^{\lambda\theta_j}=\omega_j$, $(y_{ja}\otimes\partial_{x_{jb}})\lrcorner e^{\lambda\theta_j}=\pm\lambda\,\omega_j$ for $a\neq b$ and 0 for $a=b$, and $(\partial_{x_{j1}}\wedge\partial_{x_{j2}})\lrcorner e^{\lambda\theta_j}=\pm\lambda^2\omega_j$: the image of HT_j^2 is the line $\mathbb{C}\omega_j$, $\xi\lrcorner e^{\lambda\theta_j}=c(\xi,\lambda)\,\omega_j$, and the pairs $\big(c(\xi,\lambda_j(+1)),c(\xi,\lambda_j(-1))\big)$ fill $\mathbb{C}^2$. For $k\ge3$, HT_j^k kills $e^{\lambda\theta_j}$, because every element of it contains ω_j or $\partial_{x_{j1}}\wedge\partial_{x_{j2}}$ together with a further factor, and the further factor kills ω_j.

Now count. The block HT_j^1 acts on v by $\xi \mapsto \sum_\varepsilon w_\varepsilon y^{\lambda_j(\varepsilon_j)}(\xi)\bigotimes_{j'} e^{\lambda_{j'}(\varepsilon_{j'})\theta_{j'}}$; on the summand $K_j^{\lambda_j(-\sigma)}$ only the ε with $\varepsilon_j=\sigma$ contribute, and they map it isomorphically onto $U_j^{0,1}$ tensored with a nonzero class as soon as one such w_ε is nonzero. Both signs occur in the support of a nonzero real w, since $w_{-\varepsilon}=\overline{w}_\varepsilon$, so the block has rank 4 and $r^1(v)=12$. The block $HT_j^1\otimes HT_{j'}^1$ is the sum of the four summands $K_j^{\lambda_j(-\sigma)}\otimes K_{j'}^{\lambda_{j'}(-\sigma')}$, each of dimension four; the summand indexed by (σ,σ') maps isomorphically onto $U_j^{0,1}\otimes U_{j'}^{0,1}$ tensored with the class $\sum_{\sigma''} w_\varepsilon e^{\lambda_{j''}(\sigma'')\theta_{j''}}$, which is nonzero exactly when the pair (σ,σ') occurs in the support, and to zero otherwise; so the block has rank $4A_{jj'}$. The block HT_j^2 maps ξ to $\sum_\sigma c(\xi,\lambda_j(\sigma))z_\sigma$ with $z_\sigma=\omega_j\otimes\sum_\mu w_{\sigma\mu}e_\mu$, where μ runs over the four sign patterns of the other two places and the e_μ are independent; as $(c(\xi,\lambda_j(+1)),c(\xi,\lambda_j(-1)))$ fills $\mathbb{C}^2$, the rank is the dimension of the span of z_+,z_-, which is ρ_j. This gives $r^2(v)$. In degree three the block $HT_1^1\otimes HT_2^1\otimes HT_3^1$ is the sum of eight summands $\bigotimes_j K_j^{\lambda_j(-\varepsilon_j)}$, of dimension eight, on each of which exactly one term of v acts, isomorphically when $w_\varepsilon\neq 0$ and by zero otherwise: rank $8N_w$. The block $HT_j^2\otimes HT_{j'}^1$ factors through $\mathbb{C}^2\otimes HT_{j'}^1$ by $c\otimes\mathrm{id}$, which is onto, and on $\mathbb{C}^2\otimes K_{j'}^{\lambda_{j'}(-\sigma')}$ the map is $M_{\sigma'}^{(j;j')}\otimes y^{\lambda_{j'}(\sigma')}$ followed by tensoring with independent classes, of rank $2\,\mathrm{rk}\,M_{\sigma'}^{(j;j')}$; the two values of σ' give independent images. The blocks HT_j^3 contribute nothing. This gives $r^3(v)$. The maximal values are $48+6=54$ and $64+6\cdot 8=112$; the conditions for them are the nonvanishing of the w_ε and of the determinants of the $M_{\sigma'}^{(j;j')}$, which are polynomials in the coordinates of v, nonzero on $S(t,q)$. If the support of w is a pair $\{\varepsilon_0,-\varepsilon_0\}$, then $A_{jj'}=2$, $\rho_j=2$, $N_w=2$ and every $M_{\sigma'}^{(j;j')}$ has one nonzero entry, so $(r^2,r^3)=(24+6,16+24)$.

(ii) Since $\theta_j^3=0$, $\theta^{2m}=\frac{(2m)!}{2^m}\theta_1^2\cdots\theta_m^2$, and $\int_X\theta^{2m}/(2m)!=1$ gives $\int_X\theta_1^2\cdots\theta_m^2=2^m$ and $\int_X\exp(\sum_j c_j\theta_j)=\prod_j c_j^2$. The dual of $e^{\lambda\theta_j}$ is $e^{-\lambda\theta_j}$, so $\int_X e_\varepsilon^\vee e_{\varepsilon'}=\prod_j\big(\lambda_j(\varepsilon_j')-\lambda_j(\varepsilon_j)\big)^2$, which is 0 unless $\varepsilon'=-\varepsilon$, where it is $\prod_j(2s_j)^2=(-4)^m\mathrm{N}(q)$. Hence $\chi(v,v)=(-4)^m\mathrm{N}(q)\sum_\varepsilon w_\varepsilon w_{-\varepsilon}=(-4)^m\mathrm{N}(q)\sum_\varepsilon|w_\varepsilon|^2$.

(iii) The evaluation map and the semiregularity map of Setting A.10 in degree k satisfy $\sigma_{\mathscr{F}}\circ\mathrm{ev}_{\mathscr{F}}=(\cdot)\lrcorner\mathrm{ch}(\mathscr{F})$, so the kernel of $\mathrm{ev}_{\mathscr{F}}\colon HT^k(X)\to\mathrm{Ext}^k(\mathscr{F},\mathscr{F})$ lies in the annihilator of v and $e_k\geq r^k(v)$. Serre duality on the sixfold gives $e_k=e_{6-k}$, so $\chi(\mathscr{F},\mathscr{F})=2(e_0-e_1+e_2)-e_3=2r^2(v)-22-e_3$ for the least values, and by Riemann–Roch on an abelian variety $\chi(\mathscr{F},\mathscr{F})=\int_X v^\vee v=\chi(v,v)$; this determines e_3, and $e_3\geq r^3(v)$ is the stated inequality. With the maximal ranks it reads $\chi\leq-(112-108+22)=-26$, and with $N_w=2$ it reads $\chi\leq-(40-60+22)=-2$. □

Proposition A.48 (Integral flat characters in degree six)**.** *Keep the notation of Proposition A.47.*

(i) *For every perfect complex $\mathscr{F}$ on a complex torus of dimension six, $\chi(\mathscr{F},\mathscr{F})$ is even. So the condition of Proposition A.47(iii) holds for every $\mathscr{F}$ whose character has $N_w=2$.*

(ii) *For every nonzero real $v\in S(t,q)$, $r^3(v)-2r^2(v)\leq 4$, with equality only if $(r^2(v),r^3(v))=(54,112)$. So for every shape the condition of Proposition A.47(iii) is implied by $\chi(F_i,F_i)\leq -26$.*

(iii) *The rational points of $S(0,q)$ are the classes $v(c_0,f,g,c_3)$, and*

$$\chi(v,v)=-8\big(\mathrm{N}(q)\,c_0^2+\mathrm{Tr}\big(\mathrm{N}(q)\,q^{-1}f^2\big)+\mathrm{Tr}(q\,g^2)+c_3^2\big).$$

(iv) *For $q=1$, the classes $\mathrm{Re}\,e^{\sqrt{-1}\,\theta}=v(1,0,-1,0)$ and $\mathrm{Im}\,e^{\sqrt{-1}\,\theta}=v(0,1,0,-1)$ are integral points of $S(0,1)$ with $\chi=-32$ and $N_w=2$, so they satisfy the condition of Proposition A.47(iii).*

Proof. (i) The Chern character $v=\mathrm{ch}(\mathscr{F})$ of a perfect complex on a complex torus is integral, since by the Künneth theorem the topological K-theory of a real torus is generated by products of classes pulled back from its circle factors, whose Chern characters are integral; and $\chi(\mathscr{F},\mathscr{F})=\int_X v^\vee v$ by Riemann–Roch. Write $v=\sum_k v_{2k}$, so that $\int_X v^\vee v=\sum_k(-1)^k\int_X v_{2k}v_{12-2k}$. The terms k and $6-k$ are equal, since $(-1)^k=(-1)^{6-k}$ and classes of even degree commute, so they contribute an even number when $k\neq 3$. For $k=3$ write $v_6=\sum_I a_I e_I$ in the basis of $\bigwedge^6 H^1(X,\mathbb{Z})$ given by the products e_I of six elements of a basis of $H^1(X,\mathbb{Z})$; since $e_Ie_I=0$ and $e_Ie_J=e_Je_I$, $v_6^2=2\sum_{I<J}a_Ia_Je_Ie_J$ is even. For

$N_w = 2$ the condition of Proposition A.47(iii) is $\chi(\mathscr{F},\mathscr{F}) \le -2$, and $\chi(\mathscr{F},\mathscr{F})$ is a negative even integer by Proposition A.47(ii).

(ii) For j and $\sigma = \pm 1$ let $W_{j,\sigma}$ be the 2×2 slice $(w_\varepsilon)_{\varepsilon_j=\sigma}$, with rows and columns indexed by the signs at the two other places. The matrices $M_\sigma^{(j';j)}$ are $W_{j,\sigma}$ and its transpose, so $\sum_{j\neq j'}\big(\operatorname{rk} M_+^{(j;j')} + \operatorname{rk} M_-^{(j;j')}\big) = 2\sum_{j,\sigma}\operatorname{rk} W_{j,\sigma}$. Since $w_{-\varepsilon} = \overline{w}_\varepsilon$, $W_{j,-}$ is the complex conjugate of $W_{j,+}$ with its rows and its columns reversed, so the two slices have the same rank k_j. Let S be the support of w and $N = N_w = |S|$. By Proposition A.47(i),

$$r^3(v) - 2r^2(v) = 8N - 8\,(A_{12}+A_{13}+A_{23}) + \sum_j \big(8k_j - 2\rho_j\big). \tag{A.6}$$

The set S is stable under $\varepsilon \mapsto -\varepsilon$, which exchanges the two slices along each place; so N is even, each slice has $N/2$ nonzero entries, $k_j \ge 1$, and $A_{jj'} \in \{2,4\}$, the projection of S to the signs at j and j' being stable under negation. Also $1 \le \rho_j \le 2$ and $k_j \le 2$, and a slice whose nonzero entries lie in one row or one column has rank one. If $\rho_j = 1$, then $w_\varepsilon = a_{\varepsilon_j} W(\varepsilon')$, with ε' the signs at the two other places, so every slice along a place $j' \neq j$ is $a \otimes$ (a row of W) and $k_{j'} \le 1$. Now take the four values of N in turn.

If $N = 8$, every $A_{jj'}$ is 4 and (A.6) is $-32 + \sum_j (8k_j - 2\rho_j)$. When every ρ_j is 2 this is at most $-32 + 3\cdot 12 = 4$, with equality exactly when every k_j is 2, and then $r^2 = 48+6 = 54$ and $r^3 = 64+48 = 112$. When some ρ_j is 1, the two other k are at most 1 and (A.6) is at most $-32+14+6+6 = -6$.

If $N = 6$, S misses one pair $\{\pm\varepsilon_0\}$. As ε_0 and $-\varepsilon_0$ differ at every place, they never have the same signs at two places, so every sign pair at two places occurs in S, every $A_{jj'}$ is 4, and (A.6) is at most $48 - 96 + 3\cdot 14 = -6$.

If $N = 4$, $S = \{\pm\varepsilon, \pm\varepsilon'\}$, and after replacing ε' by $-\varepsilon'$ if necessary, ε and ε' differ at exactly one place j. For the two other places k and l one has $A_{kl} = 2$ and $A_{jk} = A_{jl} = 4$. The two nonzero entries of a slice along k are $w_{\pm\varepsilon}$ and $w_{\pm\varepsilon'}$ with the same sign, whose indices agree at l; they lie in one row or one column, so $k_k = 1$, and likewise $k_l = 1$. Hence (A.6) is at most $32 - 80 + 8\,(2+1+1) - 6 = -22$.

If $N = 2$, $S = \{\pm\varepsilon_0\}$, every $A_{jj'}$ is 2, every k_j is 1, and every ρ_j is 2, the two slices along j having their nonzero entries at different positions; so (A.6) is $16 - 48 + 24 - 12 = -20$, the value $40 - 60$ of Proposition A.47(i).

So $r^3 - 2r^2 \le 4$, with equality only at $(54, 112)$, and the threshold $r^3 - 2r^2 + 22$ of Proposition A.47(iii) is at most 26.

(iii) Since $\theta_j^3 = 0$ and $s_j^2 = -\tau_j(q)$, $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$, so the eight exponentials and the eight products $\prod_j \beta_j$ with $\beta_j \in \{B_j, \theta_j\}$ span the same complex space, the change of basis being the tensor cube of an invertible 2×2 matrix. The Galois group permutes the τ_j, the θ_j and the B_j compatibly, so $\sum_a C_a \prod_j \beta_j$, indexed by the subsets a of $\{1,2,3\}$ with $\beta_j = \theta_j$ for $j \in a$, is rational exactly when $C_{\pi(a)} = \sigma(C_a)$ for every σ inducing π; the stabiliser of $\{j\}$, and of its complement, is the stabiliser of τ_j, which gives the form $v(c_0, f, g, c_3)$. The involution $v \mapsto v^\vee$ and the integral are multiplicative over the three factors, with $\int_X \theta_1^2\theta_2^2\theta_3^2 = 8$; on one factor $B_j^\vee B_j = 1 - \tau_j(q)\theta_j^2$, $\theta_j^\vee\theta_j = -\theta_j^2$ and the mixed products are odd in θ_j. So $\chi(v,v) = -8\sum_a C_a^2 \prod_{j\notin a}\tau_j(q)$, which is the formula.

(iv) For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = \prod_j e^{\sqrt{-1}\,\theta_j} = e^{\sqrt{-1}\,\theta}$. The polarisation is principal, so each $\theta^k/k!$ is integral, and so are the real and imaginary parts $\sum_k (-1)^k\theta^{2k}/(2k)!$ and $\sum_k (-1)^k\theta^{2k+1}/(2k+1)!$. They are $\frac{1}{2}(e_{(1,1,1)} + e_{(-1,-1,-1)})$ and $\frac{1}{2\sqrt{-1}}(e_{(1,1,1)} - e_{(-1,-1,-1)})$, so $N_w = 2$ and $\chi = -64\,(\frac{1}{4} + \frac{1}{4}) = -32$ by Proposition A.47(ii); expanding $\prod_j (B_j + \sqrt{-1}\,\theta_j)$ gives the coordinates. □

A.6. **The rules of the closure graph**

The statements (L), (M), (V) and (F3) are those of Section 3.3. Recall André's definition [6]: a class $\xi \in H^{2p}(X,\mathbb{Q})$ on a smooth projective variety X is *motivated* if there are a smooth projective variety Y and algebraic classes α, β on $X\times Y$ with $\xi = \mathrm{pr}_{X*}(\alpha \cup *\beta)$, where $*$ is the Lefschetz involution of $X\times Y$ for a product polarisation. An algebraic class is motivated (take Y a point and $\beta = \pm h^{\dim X}$).

Proposition A.49 (The conjecture for the varieties that are not abelian is the conjecture)**.** *Let A be a complex abelian variety of dimension $g \ge 1$ and let $X = A \times \mathbb{P}^1$. Then X is a smooth complex projective*

variety that is not an abelian variety, and if every Hodge class of degree $2p$ on X is algebraic, then so is every Hodge class of degree $2p$ on A. Consequently the statement

> *the Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety*

implies the Hodge conjecture for every abelian variety, and it is equivalent to **HC**.

Proof. X is smooth and projective of dimension $g+1$. By the Künneth formula $h^{1,0}(X) = h^{1,0}(A) + h^{1,0}(\mathbb{P}^1) = g$, while an abelian variety of dimension $g+1$ has $h^{1,0} = g+1$; so X is not an abelian variety. (Equally, X contains the rational curves $\{a\}\times\mathbb{P}^1$, and every morphism from $\mathbb{P}^1$ to a complex torus is constant, since it lifts to the universal cover $\mathbb{C}^{g+1}$ [16, Chapter 1].)

Let $\mathrm{pr}\colon X \to A$ be the projection and $D = A\times\{0\}$, a smooth divisor that pr maps isomorphically onto A, so that $\mathrm{pr}_*[D] = 1$ in $H^0(A,\mathbb{Q})$. Let α be a Hodge class of degree $2p$ on A. Pull-back along a morphism is a morphism of Hodge structures, so $\mathrm{pr}^*\alpha$ is a Hodge class of degree $2p$ on X, and by hypothesis $\mathrm{pr}^*\alpha = [Z]$ for an algebraic cycle Z on X with rational coefficients. The projection formula gives

$$\mathrm{pr}_*\big([Z]\cup[D]\big) = \mathrm{pr}_*\big(\mathrm{pr}^*\alpha\cup[D]\big) = \alpha\cup\mathrm{pr}_*[D] = \alpha.$$

The cycle class map is compatible with intersection products and with proper push-forward [33, Chapter 19], so the left side is the class of the cycle $\mathrm{pr}_*(Z\cdot D)$ and α is algebraic. A point, the abelian variety of dimension zero, satisfies the conjecture trivially.

So the displayed statement gives the conjecture for every abelian variety, and with it the conjecture for every smooth complex projective variety. The converse is immediate: the displayed statement is a special case of **HC**. □

Theorem A.50 (The rules with premises (L), (M), (V))**.**

(i) (L) *and* (M) *together imply* **HC**.

(ii) (L) *implies* (V).

(iii) (V) *implies* **AB**.

(iv) (L) *implies* **AB**.

Proof. (i) Let ξ be a Hodge class on X. By (M), $\xi = \mathrm{pr}_{X*}(\alpha\cup *\beta)$ with α,β algebraic on $X\times Y$. By (L) the Lefschetz standard conjecture holds for $X\times Y$, so the involution $*$ is induced by an algebraic correspondence [48, Proposition 1.4.4 and Theorem 2.9]; hence $*\beta$, the product $\alpha\cup *\beta$ and its push-forward ξ are algebraic. This is André's observation that motivated classes are algebraic as soon as the Lefschetz involutions are [6, §2.1].

(ii) Let $f\colon \mathscr{X}\to S$ be a smooth projective morphism over a smooth connected base, ξ a flat section of $R^{2p}f_*\mathbb{Q}$ that is a Hodge class at every point, and $s\in S$ with ξ_s algebraic. Then ξ_s is motivated, so ξ_t is motivated for every $t\in S$ by André's deformation theorem [6, Théorème 0.5], and a motivated class is algebraic under (L) by the proof of (i).

(iii) A polarised abelian scheme over a smooth connected base is a smooth projective family, so (V) gives statement (iii) of Theorem 4.5, which is equivalent to **AB** by that theorem. Its proof uses Theorem A.6, André's theorem on Hodge classes of CM type [5] and Proposition 4.4, and no statement of this subsection.

(iv) By (ii) and (iii). □

A.7. **Mumford fourfolds**

The notation of this subsection refines Setting 3.6; the polarisation class of X is θ, and $\mathrm{Hdg}^k(Y)$ denotes the space of Hodge classes in $H^{2k}(Y,\mathbb{Q})$.

Proposition A.51 (The Hodge ring is not generated by divisor and Weil classes)**.** *Let X be a complex abelian fourfold whose Hodge group is a $\mathbb{Q}$-form of an almost direct product of three copies of SL_2, acting on $V = H^1(X,\mathbb{Q})$ through the tensor product $V_1\otimes V_2\otimes V_3$ of the three standard representations, with $\mathrm{End}^0(X) = \mathbb{Q}$; such fourfolds exist and form families of dimension one, by Mumford* [63]*, and by* [66, 2.5(i)] *they are exactly the fourfolds with $\mathrm{End}^0(X) = \mathbb{Q}$ and Hodge group smaller than Sp_8. Then:*

(i) *the Hodge ring of X is generated by the polarisation:* $\dim_{\mathbb{Q}}\mathrm{Hdg}^1(X) = \dim_{\mathbb{Q}}\mathrm{Hdg}^2(X) = 1$*;*

(ii) $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X\times X)=3$ *and* $\dim_{\mathbb{Q}} \mathrm{Hdg}^2(X\times X)=8$*, while the products of divisor classes span a subspace of dimension* 6 *in* $H^4(X\times X,\mathbb{Q})$*;*

(iii) *the two remaining classes lie in* $H^4(X\times X,\mathbb{Q})$ *and are Weil classes of no CM field acting on* $X\times X$*, on a quotient of it, or on an abelian subvariety of it.*

In particular the statement that every Hodge class on an abelian variety lies in the subring generated by divisor classes and Weil classes of quotients is false in dimension eight, though true through dimension five [66].

Proof. Hodge classes are the invariants of the Hodge group, so $\mathrm{Hdg}^k(Y)\otimes\mathbb{C}$ is the space of $\mathrm{Hg}(Y)_{\mathbb{C}}$-invariants in $\bigwedge^{2k}H^1(Y,\mathbb{C})$, and $\mathrm{Hg}(X\times X)=\mathrm{Hg}(X)$ acting diagonally on $V\oplus V$. Over $\mathbb{C}$ the group is SL_2^3 and the weights of V under its maximal torus are the eight sign vectors $(\pm1,\pm1,\pm1)$. Write $V=V_1\otimes W$ with $W=V_2\otimes V_3$, on which $\mathrm{SL}_2^2=\mathrm{Spin}(4)$ acts orthogonally. The Cauchy formula $\bigwedge^k(V_1\otimes W)=\bigoplus_{|\lambda|=k}S_\lambda V_1\otimes S_{\lambda'}W$ [31, §6.1], with $S^2V_1=S_{(3,1)}V_1=V(2)$, $\bigwedge^2V_1=S_{(2,2)}V_1=\mathbf{1}$ and $S^4V_1=V(4)$, gives $\bigwedge^2V=V(2)\otimes\bigwedge^2W\oplus S^2W$ and $\bigwedge^4V=V(4)\otimes\bigwedge^4W\oplus V(2)\otimes S_{(2,1,1)}W\oplus S_{(2,2)}W$. Under $\mathrm{Spin}(4)$, $\bigwedge^2W=V(2,0)\oplus V(0,2)$, $S^2W=V(2,2)\oplus\mathbf{1}$, $\bigwedge^4W=\mathbf{1}$, $S_{(2,1,1)}W\cong\mathfrak{sl}(W)=\bigwedge^2W\oplus S^2_0W$, and $S_{(2,2)}W=V(4,0)\oplus V(0,4)\oplus V(2,2)\oplus\mathbf{1}$, the decomposition of the algebraic curvature tensors in dimension four into the self-dual and anti-self-dual Weyl parts, the traceless Ricci part and the scalar part. Hence

$$\bigwedge^2V=V(2,2,0)\oplus V(2,0,2)\oplus V(0,2,2)\oplus\mathbf{1},$$
$$\bigwedge^4V=V(2,2,2)\oplus\textstyle\bigoplus_{\mathrm{cyc}}V(4,0,0)\oplus\bigoplus_{\mathrm{cyc}}V(2,2,0)\oplus\mathbf{1},$$

each with one trivial summand, which is (i). For (ii), $\bigwedge^4(V\oplus V)=\bigoplus_a\bigwedge^aV\otimes\bigwedge^{4-a}V$ and the invariants in $\bigwedge^aV\otimes\bigwedge^{4-a}V$ are $\mathrm{Hom}_{\mathrm{SL}_2^3}(\bigwedge^aV,\bigwedge^{4-a}V)$, since every summand is self-dual; the counts are $1,1,4,1,1$, total 8, the 4 coming from the four distinct irreducible summands of $\bigwedge^2V$. Likewise the invariants in $\bigwedge^2(V\oplus V)$ number $1+1+1=3$. The divisor classes of $X\times X$ are the three symplectic forms ψ_{11}, ψ_{12}, ψ_{22} on $V\oplus V$ built from the polarisation ψ of V. Their six pairwise products are linearly independent: they lie in the summands $\bigwedge^aV\otimes\bigwedge^{4-a}V$ with $a=4,3,2,2,1,0$, each is nonzero, and in the summand $a=2$ the classes $\psi_{11}\psi_{22}$ and ψ_{12}^2 correspond, as maps $\bigwedge^2V\to\bigwedge^2V$, to a multiple of the projection onto $\mathbb{C}\psi$ and to a multiple of the identity. The Sp_8-invariants of $\bigwedge^4(V\oplus V)$ number $1+1+2+1+1=6$ by the same count, since $\bigwedge^1V=V$, $\bigwedge^2V=\bigwedge^2_0V\oplus\mathbf{1}$ and $\bigwedge^3V=\bigwedge^3_0V\oplus V$ under Sp_8; so the six products are all the classes a Hodge group equal to Sp_8 would allow. For (iii), the Weil classes of an abelian variety Y with respect to a CM field $F\subseteq\mathrm{End}^0(Y)$ are, by definition, the classes in $\bigwedge^{2k}_FH^1(Y,\mathbb{Q})\subseteq H^{2k}(Y,\mathbb{Q})$ with $2k=\dim_FH^1(Y,\mathbb{Q})$ (Setting A.1; here F denotes a CM field). A CM field acting on $X\times X$ lies in $\mathrm{End}^0(X\times X)=M_2(\mathbb{Q})$, so it is imaginary quadratic, $2k=8$, and its Weil classes lie in H^8, while the two classes of (ii) have degree four. Every proper abelian subvariety and every proper quotient of $X\times X$ is isogenous to X, because X is simple, and $\mathrm{End}^0(X)=\mathbb{Q}$ contains no CM field, so X carries no Weil classes at all. This is the statement of [66, 0.1(iii)] that the exceptional classes of $X\times X$ are not of Weil type. The last sentence is (ii) and (iii) together with the theorem of [66] for dimension at most five. □

Setting A.52 (Mumford type)**.** Let X be as in Proposition A.51, with $V=H^1(X,\mathbb{Q})$, polarisation ψ and polarisation class $\theta\in H^2(X,\mathbb{Q})$, and let $G=\mathrm{Hg}(X)$. Over $\bar{\mathbb{Q}}$ the Hodge cocharacter is nontrivial on exactly one of the three factors of G: on V_i it has weights $\pm a_i/2$ with $a_i\geq0$, and V has only the two weights $\pm\frac12$, so exactly one a_i is nonzero. Since $\mathrm{End}^0(X)=\mathbb{Q}$, it follows that G is $\mathbb{Q}$-simple: the Lie algebra of a $\mathbb{Q}$-factor not containing that geometric factor would consist of Hodge classes in $\mathrm{End}(V)$, that is, of endomorphisms of X. So G is the image in $\mathrm{GL}(V)$ of $\mathrm{Res}_{F/\mathbb{Q}}D^1$, where F is a totally real cubic field, D is a quaternion algebra over F split at one real place σ_3 and definite at the other two, σ_1 and σ_2, with $\mathrm{Cor}_{F/\mathbb{Q}}D\cong M_8(\mathbb{Q})$, and D^1 is the group of elements of reduced norm one [63, 34]; the kernel is the central subgroup of triples of signs with product one. Over $\bar{\mathbb{Q}}$, $V=V_1\otimes V_2\otimes V_3$, where V_i is the standard representation of $D\otimes_{F,\sigma_i}\bar{\mathbb{Q}}=M_2(\bar{\mathbb{Q}})$; the Galois group permutes the three factors as it permutes the three embeddings, and $H^{1,0}(X)=V_1\otimes V_2\otimes V_3^{1,0}$. Put

$$T=\mathrm{Lie}\,G=D^0,\qquad q_\lambda(x,y)=\mathrm{Tr}_{F/\mathbb{Q}}\big(\lambda\,\mathrm{trd}(xy)\big)\quad(\lambda\in F^\times),$$

where D^0 is the space of quaternions of reduced trace zero, regarded as a $\mathbb{Q}$-vector space of dimension nine, and trd is the reduced trace. Over $\bar{\mathbb{Q}}$, $T = T_1 \oplus T_2 \oplus T_3$ with $T_i = \mathfrak{sl}(V_i)$, and $a \in F$ acts on T_i by $\sigma_i(a)$. Write m_a for multiplication by a. The space $T \subset \mathrm{End}(V)$ is a sub-Hodge structure of weight zero.

Proposition A.53 (The two classes are a real multiplication)**.** *In Setting A.52:*

(i) $H^2(X,\mathbb{Q}) = \mathbb{Q}\theta \oplus U$, *where* U *is the primitive part, of dimension* 27, *and over* $\bar{\mathbb{Q}}$

$$U = U_{12} \oplus U_{13} \oplus U_{23}, \qquad U_{ij} = S^2V_i \otimes S^2V_j \otimes \textstyle\bigwedge^2 V_k, \quad \{i,j,k\} = \{1,2,3\},$$

each of dimension nine. The summand U_{12} *is of type* $(1,1)$, *and* U_{13} *and* U_{23} *have Hodge numbers* $(3,3,3)$.

(ii) *The algebra of Hodge endomorphisms of* $H^2(X,\mathbb{Q})$ *is* $\mathbb{Q} \times F$, *acting by scalars on* $\mathbb{Q}\theta$, *with* $a \in F$ *acting on* U_{ij} *by* $\sigma_k(a)$. *Through Poincaré duality and the Lefschetz isomorphism* $\theta^2 \colon H^2 \to H^6$ *this algebra is the space of Hodge classes in the Künneth component* $H^2(X) \otimes H^2(X)$ *of* $H^4(X \times X, \mathbb{Q})$; *the subalgebra* $\mathbb{Q} \times \mathbb{Q}$ *is spanned by the products of divisor classes that lie there, and the two classes of Proposition A.51(ii) may be taken to be the elements of* F *of trace zero.*

(iii) $T(-1)$ *is a Hodge structure of K3 type with Hodge numbers* $(1,7,1)$, *and its Hodge endomorphisms are the* m_a, $a \in F$. *The* G*-invariant symmetric forms on* T *are the* q_λ, *and* q_λ *polarises* $T(-1)$ *if and only if* λ *is totally positive.*

Proof. (i) is the decomposition of $\bigwedge^2(V_1 \otimes V_2 \otimes V_3)$ in the proof of Proposition A.51, where U_{ij} appears as the summand of highest weight 2 on the i-th and j-th factors. The Hodge cocharacter acts trivially on V_1 and V_2 and splits $V_3 = V_3^{1,0} \oplus V_3^{0,1}$, so $\bigwedge^2 V_3$ has type $(1,1)$ and S^2V_3 has one dimension of each of the types $(2,0)$, $(1,1)$, $(0,2)$.

(ii) A Hodge endomorphism commutes with the Mumford–Tate group $G \cdot \mathbb{G}_m$, and $\mathbb{G}_m$ acts on H^2 by scalars, so the Hodge endomorphisms are the G-endomorphisms. Over $\bar{\mathbb{Q}}$ the four summands $\mathbb{Q}\theta$, U_{12}, U_{13}, U_{23} are irreducible and pairwise non-isomorphic, so the G-endomorphisms are the combinations of the four projections. The Galois group fixes the projection onto $\mathbb{Q}\theta$ and permutes the other three as it permutes the indices k, that is, as it permutes the embeddings of F. For $a \in F$ an endomorphism acting on U_{ij} by $\sigma_k(a)$ is therefore Galois invariant, hence rational, and with the scalars on $\mathbb{Q}\theta$ these exhaust the rational points, which form $\mathbb{Q} \times F$. Poincaré duality identifies $\mathrm{End}H^2(X,\mathbb{Q})$ with $H^6(X) \otimes H^2(X)$, and $\theta^2 \otimes 1$ carries $H^2 \otimes H^2$ onto it; both are given by algebraic classes, and on an abelian variety so is the inverse of the Lefschetz isomorphism [48, §2]. The products of divisor classes in $H^2 \otimes H^2$ are $\psi_{11}\psi_{22}$ and ψ_{12}^2 in the notation of the proof of Proposition A.51. They are invariant under the whole of $\mathrm{Sp}(V,\psi)$, whose endomorphisms of $H^2 = \mathbb{Q}\theta \oplus U$ are spanned by the identity and the projection onto $\mathbb{Q}\theta$, and they are linearly independent, being a multiple of the projection and a multiple of the identity (proof of Proposition A.51); so they span $\mathbb{Q} \times \mathbb{Q}$, and any complement of $\mathbb{Q} \times \mathbb{Q}$ in $\mathbb{Q} \times F$ may serve as the two remaining classes.

(iii) The adjoint action of the Hodge cocharacter is trivial on T_1 and T_2 and has weights -1, 0, 1 on T_3, so T has Hodge numbers $(1,7,1)$ in the types $(-1,1)$, $(0,0)$, $(1,-1)$, and $T(-1)$ is of K3 type. Its Hodge endomorphisms are again the G-endomorphisms, and the argument of (ii), applied to the three pairwise non-isomorphic summands T_i, gives F acting by multiplication. On each T_i the invariant symmetric forms are the multiples of $\mathrm{tr}(xy)$, and $\mathrm{tr}(x_iy_i) = \sigma_i(\mathrm{trd}(xy))$ for $x,y \in D^0$, so the Galois-invariant ones are the q_λ. At σ_1 and σ_2 the algebra $D \otimes \mathbb{R}$ is Hamilton's quaternions, T_i is of type $(1,1)$, and $\mathrm{trd}(x^2) = -2\,\mathrm{nrd}(x)$ is negative on nonzero pure quaternions. At σ_3 the form $\mathrm{tr}(xy)$ on $\mathfrak{sl}_2(\mathbb{R})$ is negative on the line of type $(1,1)$, the Lie algebra of the rotations fixed by the Hodge cocharacter, and positive on its orthogonal complement, the real points of the types $(2,0)$ and $(0,2)$. The Hodge–Riemann relations for a structure of K3 type ask for exactly these signs after multiplication by $\sigma_i(\lambda)$, so q_λ polarises $T(-1)$ precisely when every $\sigma_i(\lambda)$ is positive. □

Lemma A.54 (The product map)**.** *For* $x,y \in T$ *let* $\mu(x \otimes y) \in H^2(X,\mathbb{Q})$ *be the class that corresponds under* ψ *to the alternating form* $(v,w) \mapsto \psi(xv,yw) + \psi(yv,xw)$ *on* V. *Then* μ *is a* G*-equivariant map* $T \otimes T \to H^2(X,\mathbb{Q})$, *and:*

(i) *for* $i \neq j$ *it restricts to an isomorphism* $T_i \otimes T_j \to U_{ij}$, *and it carries* $T_i \otimes T_i$ *onto* $\mathbb{Q}\theta$;

(ii) *for $e,\lambda\in F^\times$ put $\mu_e=\mu\circ(m_e\otimes m_e)$ and let $\mu_e^\dagger$ be its adjoint for the form $q_\lambda\otimes q_\lambda$ on $T\otimes T$ and the Lefschetz pairing $B(\alpha,\beta)=\int_X\alpha\beta\,\theta^2$ on $H^2(X,\mathbb{Q})$. Then $\mu_e\mu_e^\dagger$ preserves $\mathbb{Q}\theta$ and acts on U as the element*

$$\nu\,\mathrm{N}_{F/\mathbb{Q}}(g)\,g^{-1}\in F,\qquad g=e^2/\lambda,$$

where ν is a nonzero rational number independent of e and λ.

Proof. Equivariance is clear, since μ is built from ψ and the action of $\mathrm{End}(V)$ on V. An element $x\in T_i$ acts on V through the i-th factor and lies in the symplectic Lie algebra of $\psi=\varepsilon_1\otimes\varepsilon_2\otimes\varepsilon_3$, so $\psi(xv,yw)=-\psi(v,xyw)$ and $\mu(x\otimes y)$ is the form $-\psi(v,(xy+yx)w)$. If $x\in T_i$ and $y\in T_j$ with $i\neq j$, then $xy=yx$ acts as $x\otimes y$ on $V_i\otimes V_j$, and the form is -2 times the product of $\varepsilon_i(\cdot,x\cdot)$, $\varepsilon_j(\cdot,y\cdot)$ and ε_k; the first two are symmetric because x and y lie in $\mathfrak{sp}(\varepsilon)=\mathfrak{sl}_2$, and $x\mapsto\varepsilon(\cdot,x\cdot)$ is an isomorphism $\mathfrak{sl}_2\to S^2V^\vee$, so μ maps $T_i\otimes T_j$ isomorphically onto U_{ij}. If $i=j$, then $xy+yx=\mathrm{tr}(xy)$ on V_i, because x and y are trace-free 2×2 matrices, and $\mu(x\otimes y)=-\mathrm{tr}(xy)\,\theta$.

For (ii), m_e is self-adjoint for q_λ, and the adjoint of a map out of $(T\otimes T,q_\lambda\otimes q_\lambda)$ is $(m_\lambda\otimes m_\lambda)^{-1}$ times its adjoint for $q_1\otimes q_1$, so $\mu_e\mu_e^\dagger=\mu\,(m_g\otimes m_g)\,\mu^{\dagger_1}$ with $\mu^{\dagger_1}$ the adjoint for $q_1\otimes q_1$. This map is G-equivariant, so it preserves $\mathbb{Q}\theta$ and each U_{ij}, and $\mu^{\dagger_1}$ carries U_{ij} into $(T_i\otimes T_j)\oplus(T_j\otimes T_i)$, on which $m_g\otimes m_g$ is multiplication by $\sigma_i(g)\sigma_j(g)=\mathrm{N}(g)/\sigma_k(g)$. Hence $\mu_e\mu_e^\dagger$ is $\mathrm{N}(g)\sigma_k(g)^{-1}$ times $\mu\mu^{\dagger_1}$ on U_{ij}. The summands of $H^2(X,\bar{\mathbb{Q}})$ are pairwise non-isomorphic and self-dual, so B is nondegenerate on U_{ij}, and by Schur's lemma the pull-back of B along the isomorphism $\mu_0=\mu|_{T_i\otimes T_j}$ is $c_k\,q_1\otimes q_1$ with $c_k\neq0$; then $\mu_0\mu_0^{\dagger_1}=c_k$ on U_{ij}. As μ is symmetric in its two arguments and the exchange of the factors is an isometry of $q_1\otimes q_1$, $\mu^{\dagger_1}u=\mu_0^{\dagger_1}u+s\,\mu_0^{\dagger_1}u$ with s that exchange, and $\mu\mu^{\dagger_1}=2c_k$ on U_{ij}. On the split model $V=\bar{\mathbb{Q}}^2\otimes\bar{\mathbb{Q}}^2\otimes\bar{\mathbb{Q}}^2$, $\psi=\varepsilon\otimes\varepsilon\otimes\varepsilon$, the permutations of the three factors preserve ψ, hence θ, B and μ, and also $q_1=\sum_i\mathrm{tr}(x_iy_i)$, while they permute the U_{ij}; so $c_1=c_2=c_3$, and $\mu\mu^{\dagger_1}$ is one and the same nonzero number ν on the three summands U_{ij}, rational because $\mu\mu^{\dagger_1}$ is defined over $\mathbb{Q}$. By Proposition A.53(ii) an endomorphism of U acting on U_{ij} by $\nu\mathrm{N}(g)/\sigma_k(g)$ is the element $\nu\mathrm{N}(g)g^{-1}$ of F. □

Theorem A.55 (One K3 surface for the two classes)**.** *In Setting A.52, let $\lambda\in F$ be totally positive.*

(i) *There is a projective K3 surface S_λ of Picard number thirteen and an isomorphism of Hodge structures $T(S_\lambda)_\mathbb{Q}\cong T(-1)$ carrying the intersection form to q_λ. Its Kuga–Satake variety is isogenous to X^{32}.*

Proof. (i) The quadratic space (T,q_λ) has dimension nine and signature $(2,7)$ by Proposition A.53(iii), and H^2 of a K3 surface with its intersection form has dimension 22 and signature $(3,19)$. Over $\mathbb{Q}_p$ a nondegenerate quadratic space embeds isometrically in every nondegenerate space of dimension at least three larger, and over $\mathbb{R}$ the signatures allow it, so by the Hasse–Minkowski theorem for representations of forms by forms [69, §§63, 66] there is an isometric embedding of (T,q_λ) into $\Lambda_\mathbb{Q}$, where Λ is the K3 lattice. The line $\ell=T(-1)^{2,0}\subset T_\mathbb{C}\subset\Lambda_\mathbb{C}$ satisfies $(\ell,\ell)=0$ and $(\sigma,\bar\sigma)>0$ for $0\neq\sigma\in\ell$, so by the surjectivity of the period map there is a K3 surface S_λ with a marking $H^2(S_\lambda,\mathbb{Z})\cong\Lambda$ that carries $H^{2,0}(S_\lambda)$ to ℓ [45, Chs. 6–7]. A rational class is of type $(1,1)$ exactly when it is orthogonal to ℓ, and its component in T is then a rational class of type $(1,1)$ in $T(-1)$, that is a G-invariant vector of $T=\mathrm{Lie}\,G$; there is none, since G is semisimple. So $\mathrm{NS}(S_\lambda)_\mathbb{Q}=T^\perp$, of dimension 13 and signature $(1,12)$, and $T(S_\lambda)_\mathbb{Q}=T$ with the form q_λ. A class of positive square in $\mathrm{NS}(S_\lambda)$ makes S_λ projective [45, Ch. 8].

For the Kuga–Satake variety, note first that over $\bar{\mathbb{Q}}$ the space T is the orthogonal sum of the three ternary spaces T_i, so $C(T)$ is the graded tensor product of the $C(T_i)$. Scaling a ternary form does not change its even Clifford algebra, so D^1 is the spin group of $(D^0,\lambda\,\mathrm{trd})$ for every λ, and $\mathrm{Res}_{F/\mathbb{Q}}D^1$ maps to $\mathrm{Spin}(T,q_\lambda)$ by multiplication in the Clifford algebra, with the same kernel as its map to $\mathrm{GL}(V)$, so that G is a subgroup of $\mathrm{Spin}(T,q_\lambda)$; as a representation of the i-th factor, $C(T_i)$ is four copies of V_i. Hence $C^+(T)$ is $V^{\oplus32}$ as a representation of G over $\bar{\mathbb{Q}}$, and over $\mathbb{Q}$ as well, because V is absolutely irreducible. The Hodge structures on $C^+(T)=H^1(\mathrm{KS}(S_\lambda),\mathbb{Q})$ and on $V^{\oplus32}=H^1(X^{32},\mathbb{Q})$ are both given by lifts of weight one of the same homomorphism $\mathbb{S}\to\mathrm{MT}(T)_\mathbb{R}$ through $G\cdot\mathbb{G}_m$, and two such

lifts coincide.[29] So $H^1(\mathrm{KS}(S_\lambda),\mathbb{Q}) \cong H^1(X^{32},\mathbb{Q})$ as Hodge structures, and $\mathrm{KS}(S_\lambda)$ is isogenous to X^{32}. For the form used by Galluzzi this is her theorem [34]; van Geemen proves the corresponding splitting for every K3 surface with real multiplication [86]. □

Remark A.56. In Theorem A.55 the field F is the totally real cubic field of the Mumford fourfold, an arbitrary one admitting a quaternion algebra as in Setting A.52; it need not be Galois over $\mathbb{Q}$, so the Galois group of its normal closure may be S_3. By Proposition A.53(iii) the field of Hodge endomorphisms of $T(S_\lambda)_\mathbb{Q}$ is F, so S_λ has real multiplication by F with $\dim_F T(S_\lambda)_\mathbb{Q} = 3$. The Kuga–Satake variety has dimension $2^{9-2} = 2^7 = 128 = \dim X^{32}$.

Theorem A.57 (Every power of a Mumford fourfold)**.** *In Setting A.52, suppose that the two exceptional classes of Proposition A.51(ii) are algebraic on $X \times X$. Then every Hodge class on X^n is algebraic, for every $n \geq 1$.*

Proof. Over $\bar{\mathbb{Q}}$ keep $V = V_1 \otimes V_2 \otimes V_3$ and $\psi = \varepsilon_1 \otimes \varepsilon_2 \otimes \varepsilon_3$, as in the proof of Lemma A.54. For $m \geq 1$ regard $V^{\otimes m} = H^1(X)^{\otimes m}$ as the Künneth component of $H^m(X^m,\mathbb{Q})$ of multidegree $(1,\dots,1)$, whose projector is algebraic [48, §2]. Call an endomorphism of $V^{\otimes m}$ algebraic if it is the action on $V^{\otimes m}$ of an algebraic cycle on $X^m \times X^m$ that preserves $V^{\otimes m}$. Composing with the Künneth projector, the algebraic endomorphisms form a $\mathbb{Q}$-algebra, and they carry algebraic classes in $V^{\otimes m}$ to algebraic classes.

Step 1: reduction to multilinear invariants. Let $n \geq 1$ and $p \geq 0$. The map $V^{\otimes 2p} \otimes (\mathbb{Q}^n)^{\otimes 2p} \to \bigwedge^{2p}(V \otimes \mathbb{Q}^n) = H^{2p}(X^n,\mathbb{Q})$ sending $(v_1 \otimes \cdots \otimes v_{2p}) \otimes (a_1 \otimes \cdots \otimes a_{2p})$ to $(v_1 \otimes a_1) \wedge \cdots \wedge (v_{2p} \otimes a_{2p})$ is surjective and G-equivariant for the trivial action on $\mathbb{Q}^n$, so it is surjective on G-invariants, G being reductive. For $a_1,\dots,a_{2p} \in \mathbb{Z}^n$ the image of $t \otimes (a_1 \otimes \cdots \otimes a_{2p})$ is f^*t, where $f\colon X^n \to X^{2p}$ is the homomorphism with rows $a_1,\dots,a_{2p}$, and such tensors span $(\mathbb{Q}^n)^{\otimes 2p}$. So $\mathrm{Hdg}^p(X^n)$ is spanned by the classes f^*t with f a homomorphism and $t \in (V^{\otimes 2p})^G$, and it suffices to show that every element of $(V^{\otimes 2p})^G$ is algebraic.

Step 2: degree four. The space $(V^{\otimes 4})^G$ has dimension eight, the product of the dimensions of the three spaces of SL_2-invariants in $V_k^{\otimes 4}$. Let $\mathscr{P}_0 \subset (V^{\otimes 4})^G$ be the span of the multilinear components of the classes $f^*\xi$, for $\xi \in \mathrm{Hdg}^2(X \times X)$ and $f\colon X^4 \to X^2$ a homomorphism, and of the products of two divisor classes of X^4. By Proposition A.51(ii) $\mathrm{Hdg}^2(X \times X)$ is spanned by the six products of divisor classes and the two exceptional classes, so under the hypothesis every element of $\mathscr{P}_0$ is algebraic. For $t \in V^{\otimes 4}$, read as a correspondence from the first two factors of X^4 to the last two, let $\hat{t}$ be the endomorphism $v \mapsto t_*\big((\theta^3 \times \theta^3) \cup v\big)$ of $V \otimes V$. The pairing $(v,w) \mapsto \int_X v \cup w \cup \theta^3$ on V is nondegenerate by hard Lefschetz and G-invariant, so $t \mapsto \hat{t}$ is an isomorphism $(V^{\otimes 4})^G \to \mathrm{End}_G(V \otimes V)$, and $\hat{t}$ is algebraic when t is. We show that the $\hat{t}$ and the composites $\hat{u}\hat{t}$, for $t,u \in \mathscr{P}_0$, span $\mathrm{End}_G(V \otimes V)$. Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V \otimes V) = \bigotimes_k \mathrm{End}_{\mathrm{SL}_2}(V_k \otimes V_k)$ is the algebra of functions on the eight sign patterns $\varepsilon \in \{\pm 1\}^3$, where ε_k is the eigenvalue of the exchange s_k of the two copies of V_k; the patterns with $\varepsilon_1\varepsilon_2\varepsilon_3 = -1$, the odd ones, make up $\bigwedge^2 V$.

(a) Let $f = (x_1 + x_2, x_3 + x_4)\colon X^4 \to X^2$. The multilinear component of $f^*\xi$ depends only on the Künneth component of ξ in $H^2(X) \otimes H^2(X)$, and it is that component read as a tensor $t \in \bigwedge^2 V \otimes \bigwedge^2 V$, so $\hat{t}$ vanishes on S^2V and takes values in $\bigwedge^2 V$. These components fill $(H^2 \otimes H^2)^G = \mathrm{End}_G(H^2)$, of dimension 4 by Proposition A.53(ii); so the $\hat{t}$ span the functions supported on the four odd patterns.

(b) The map $g = (x_1 + x_3, x_2 + x_4)$ is f composed with the exchange of the factors 2 and 3 of X^4, which permutes the factors of $V^{\otimes 4}$ up to sign. On $V_k^{\otimes 4}$ the SL_2-invariants are spanned by $\varepsilon_{13}\varepsilon_{24}$, $\varepsilon_{14}\varepsilon_{23}$ and $\varepsilon_{12}\varepsilon_{34} = \varepsilon_{13}\varepsilon_{24} - \varepsilon_{14}\varepsilon_{23}$, where ε_{ab} is ε_k on the factors a and b; up to a common sign their hats are 1, s_k and $1 - s_k$. The exchange of the factors 2 and 3 swaps the first and the third and changes the sign of the second, so on functions of ε_k it carries $\frac{1}{2}(1 - s_k)$ to $\frac{1}{2}$ and $\frac{1}{2}(1 + s_k)$ to $\frac{1}{2} - s_k$. Applied to the indicator of the odd pattern $(1,1,-1)$ it gives $\frac{1}{2}(\frac{1}{2} - s_1)(\frac{1}{2} - s_2)$. The identity is $\hat{t}$ for the multilinear component of a product of two divisor classes, the pull-backs of the class ψ on $X \times X$ along the projections of X^4 to the factors $(1,3)$ and $(2,4)$, so $h = (1 - 2s_1)(1 - 2s_2) - 1$ lies in the span of the $\hat{t}$, $t \in \mathscr{P}_0 \otimes \bar{\mathbb{Q}}$. As $(\varepsilon_1,\varepsilon_2)$ runs through $(1,1)$, $(1,-1)$, $(-1,1)$, $(-1,-1)$, h takes the values $0,-4,-4,8$, so $h(h+4)/96$ is the indicator of $\varepsilon_1 = \varepsilon_2 = -1$; subtracting the indicator of $(-1,-1,-1)$, from (a),

[29] They differ by a homomorphism from the anisotropic torus $\mathbb{S}/\mathbb{G}_m$ to the centre of $G \cdot \mathbb{G}_m$, whose identity component is $\mathbb{G}_m$, and such a homomorphism is trivial.

leaves that of $(-1,-1,1)$. The same argument with the factors V_k permuted gives the other two even patterns with two minus signs, and the indicator of $(1,1,1)$ is 1 minus the other seven.

So the $\hat{t}$ and $\hat{u}\hat{t}$ span $\mathrm{End}_G(V\otimes V)\otimes\bar{\mathbb{Q}}$. Since composition is bilinear and $\mathscr{P}_0\otimes\bar{\mathbb{Q}}$ is spanned by the same pull-backs and products of a basis of Hodge classes, the $\mathbb{Q}$-algebra of algebraic G-endomorphisms of $V\otimes V$ has dimension 8: every G-endomorphism of $V\otimes V$ is algebraic.

Step 3: the partial swaps. Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V\otimes V)=\bigotimes_k \mathrm{End}_{\mathrm{SL}_2}(V_k\otimes V_k)$, and $\mathrm{End}_{\mathrm{SL}_2}(V_k\otimes V_k)$ is spanned by 1 and the exchange s_k of the two copies of V_k. In a basis of V_k the matrix units E_{ab} and E_{ba} are dual for the trace form of $\mathfrak{gl}(V_k)$, and $\sum_{a,b}E_{ab}\otimes E_{ba}=s_k$; removing the centre, spanned by the identity 1 with $\mathrm{tr}(1\cdot 1)=2$, gives $s_k=\mathrm{Cas}_k+\frac{1}{2}$, with $\mathrm{Cas}_k=\sum_c x_c\otimes x^c$ for dual bases (x_c), (x^c) of $\mathfrak{sl}(V_k)$ under the trace form. So the s_k come from the three summands T_k of T. The Galois group permutes the s_k as it permutes the embeddings σ_k, so for $a\in F$ the endomorphism $r_a=\sum_k\sigma_k(a)s_k$ is rational, hence algebraic by Step 2. For $m\geq 2$ and $1\leq i<m$ let $r_a^{(i)}$ and $s_k^{(i)}$ act as r_a and s_k on the factors i and $i+1$ of $V^{\otimes m}$ and as the identity on the others; $r_a^{(i)}$ is algebraic, being induced by the product of a cycle inducing r_a with the diagonals of the other factors. By Schur–Weyl duality $\mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$ is the image of $\bar{\mathbb{Q}}[S_m]$, generated by the adjacent transpositions, and the commutant of SL_2^3 in $\bigotimes_k V_k^{\otimes m}$ is the tensor product of the three commutants; so $\mathrm{End}_G(V^{\otimes m})\otimes\bar{\mathbb{Q}}$ is generated by the $s_k^{(i)}$. For a basis a_1,a_2,a_3 of F the matrix $(\sigma_k(a_j))$ is invertible, the square of its determinant being the discriminant of F, so each $s_k^{(i)}$ is a $\bar{\mathbb{Q}}$-linear combination of the $r_{a_j}^{(i)}$. The $\mathbb{Q}$-algebra generated by the $r_a^{(i)}$ therefore lies in $\mathrm{End}_G(V^{\otimes m})$ and spans it over $\bar{\mathbb{Q}}$, so it is $\mathrm{End}_G(V^{\otimes m})$: every G-endomorphism of $V^{\otimes m}$ is algebraic. Its dimension is the cube of $\dim\mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$, the sum of the squares of the multiplicities in $V_k^{\otimes m}$: $8=2^3$ for $m=2$ and $125=5^3$ for $m=3$, where $V_k^{\otimes 3}=S^3V_k\oplus V_k^{\oplus 2}$.

Step 4: all multilinear invariants. By the first fundamental theorem for SL_2 the invariants in $V_k^{\otimes 2m}$ are spanned by the images of $\varepsilon_k^{\otimes m}$ under the permutations of the $2m$ factors. So $(V^{\otimes 2m})^G\otimes\bar{\mathbb{Q}}$ is spanned by the images of $\psi^{\otimes m}=\bigotimes_k\varepsilon_k^{\otimes m}$ under the operators that permute the factors of the three $V_k^{\otimes 2m}$ separately, which lie in $\mathrm{End}_G(V^{\otimes 2m})\otimes\bar{\mathbb{Q}}$, and the map $\mathrm{End}_G(V^{\otimes 2m})\to(V^{\otimes 2m})^G$, $e\mapsto e(\psi^{\otimes m})$, is surjective. The class ψ in $V\otimes V$ is that of the divisor $m^*\theta-p_1^*\theta-p_2^*\theta$ on $X\times X$ (proof of Theorem A.55(ii)), up to a nonzero factor, and $\psi^{\otimes m}$ is, up to sign, the product of its pull-backs along the projections of X^{2m} to the pairs of factors $(2j-1,2j)$; so it is algebraic, and by Step 3 so is every element of $(V^{\otimes 2m})^G$. Step 1 concludes. □

Proposition A.58 (The Hodge locus of an exceptional class)**.** *In Setting A.52, let $\mathfrak{H}$ be the period domain of polarised Hodge structures of weight one on (V,ψ), a Siegel upper half space of dimension ten, let $h_X\in\mathfrak{H}$ be the point of X, and let $e=m_a$, $a\in F\setminus\mathbb{Q}$, be an exceptional class, regarded by Proposition A.53(ii) as an endomorphism of $\bigwedge^2 V$. The connected component through h_X of the locus in $\mathfrak{H}$ where e is a Hodge class is the orbit $G(\mathbb{R})^+\cdot h_X$, of dimension one, and its image in the moduli space of polarised abelian fourfolds is a compact curve. In particular no fibre of a degenerating family of polarised abelian fourfolds at which e is still of Hodge type is singular.*

Proof. Let $M\subseteq\mathrm{Sp}(V,\psi)$ be the stabiliser of e and $\mathfrak{m}$ its Lie algebra. Since e is G-invariant, $\mathfrak{m}$ is a G-stable subspace of $\mathfrak{sp}(V)=S^2V$ containing $\mathfrak{g}=\mathrm{Lie}\,G$. Over $\bar{\mathbb{Q}}$,

$$S^2(V_1\otimes V_2\otimes V_3)=\bigoplus_i S^2V_i\otimes\bigwedge^2V_j\otimes\bigwedge^2V_k\ \oplus\ S^2V_1\otimes S^2V_2\otimes S^2V_3,$$

the sum of $\mathfrak{g}=T_1\oplus T_2\oplus T_3$ and of an irreducible representation of dimension 27. So $\mathfrak{m}$ is $\mathfrak{g}$ or $\mathfrak{sp}(V)$. In the second case e would commute with $\mathrm{Sp}(V,\psi)$ on $\bigwedge^2V=\mathbb{Q}\theta\oplus U$, where U is irreducible for $\mathrm{Sp}(V,\psi)$, and would act on U by a rational scalar, against $a\notin\mathbb{Q}$. Hence $\mathfrak{m}=\mathfrak{g}$ and the identity component of M is G. The class e is a Hodge class at $h\in\mathfrak{H}$ exactly when the homomorphism $h\colon\mathrm{U}(1)\to\mathrm{Sp}(V,\psi)(\mathbb{R})$ takes values in $M(\mathbb{R})$, hence in $G(\mathbb{R})$. Homomorphisms from the compact group $\mathrm{U}(1)$ to $G(\mathbb{R})$ are locally rigid up to conjugation, so the connected component of this locus through h_X is the orbit $G(\mathbb{R})^+\cdot h_X$. Since $G(\mathbb{R})$ is $\mathrm{SL}_2(\mathbb{R})\times\mathrm{SU}(2)\times\mathrm{SU}(2)$ modulo a finite central subgroup and the stabiliser of h_X contains the compact factors and a circle of the first factor, the orbit is a copy of the upper half plane. Its image in the moduli space is the quotient by an arithmetic subgroup of $G(\mathbb{Q})$. The algebra D is a division algebra, being definite at two real places, so G is anisotropic

over $\mathbb{Q}$ and the quotient is compact [72, Theorem 4.12]; this is the compactness of Mumford's curves [63]. A singular fibre of a degenerating family gives a point of the boundary of the moduli space in the closure of the locus, and a compact curve has no such point. □

A.8. **The very general K3 surface with real multiplication**

Setting A.59 (A family with real multiplication)**.** Let E be a totally real field of degree $r \geq 2$ and T a finite-dimensional $\mathbb{Q}$-vector space with an action of E and a nondegenerate symmetric form q for which E acts by self-adjoint maps, so that $q = \mathrm{Tr}_{E/\mathbb{Q}} q_E$ for a unique E-bilinear form q_E. Put $m = \dim_E T \geq 3$, and suppose that q_E has signature $(2, m-2)$ at one real place σ_1 of E and is negative definite at the others. Let $T_\sigma = T \otimes_{E,\sigma} \mathbb{R}$, $T_1 = T_{\sigma_1}$ and

$$\mathscr{D}_E = \{\, [\omega] \in \mathbb{P}(T_1 \otimes_{\mathbb{R}} \mathbb{C}) :\ q(\omega,\omega) = 0,\ q(\omega,\bar\omega) > 0 \,\}^+,$$

one of its two components, a bounded symmetric domain of dimension $m-2$. For $\omega \in \mathscr{D}_E$ let T_ω be the Hodge structure of K3 type on T with $T_\omega^{2,0} = \mathbb{C}\omega$, $T_\omega^{0,2} = \mathbb{C}\bar\omega$ and $T_\omega^{1,1}$ their orthogonal complement; it is polarised by $-q$. Each $e \in E$ acts on $T_1 \otimes \mathbb{C}$ by the scalar $\sigma_1(e)$ and is self-adjoint, so it is an endomorphism of the Hodge structure T_ω. Let $G = \mathrm{Res}_{E/\mathbb{Q}} \mathrm{SO}(T, q_E)$, so that $G(\mathbb{R}) = \mathrm{SO}(T_1) \times \prod_{\sigma \neq \sigma_1} \mathrm{SO}(T_\sigma)$, and write $G(\mathbb{R})^+$ for the identity component, which acts transitively on $\mathscr{D}_E$ through its first factor. When (T, q) is embedded isometrically in $\Lambda_{\mathbb{Q}}$ for the K3 lattice Λ, the surjectivity of the period map gives for every ω a K3 surface S_ω with $H^{2,0}(S_\omega) = \mathbb{C}\omega$, and then $T(S_\omega)_{\mathbb{Q}} \subseteq T$ is the orthogonal complement in T of the rational classes of type $(1,1)$ of T_ω.

Proposition A.60 (The very general member)**.** *In Setting A.59 there is a countable union Z of proper closed analytic subsets of $\mathscr{D}_E$ such that for every $\omega \notin Z$ the Hodge structure T_ω has no nonzero rational class of type $(1,1)$ and $\mathrm{End}_{\mathrm{Hdg}}(T_\omega) = E$. In the geometric situation, $T(S_\omega)_{\mathbb{Q}} = T$ and the field of Hodge endomorphisms of $T(S_\omega)_{\mathbb{Q}}$ is E for $\omega \notin Z$. The proof uses the hypothesis on m only through $m \geq 3$.*

Proof. The Hodge structure T_ω is given by $u_\omega \colon \mathrm{U}(1) \to \mathrm{SO}(T_{\mathbb{R}})$, acting by z^2 on $\mathbb{C}\omega$, by z^{-2} on $\mathbb{C}\bar\omega$ and trivially on their orthogonal complement; it commutes with E, lies in $G(\mathbb{R})^+$, and $u_{g\omega} = g u_\omega g^{-1}$ for $g \in G(\mathbb{R})^+$.

(a) *No nonzero $t \in T$ is orthogonal to every $\omega \in \mathscr{D}_E$.* The lines $\mathbb{C}\omega$, $\omega \in \mathscr{D}_E$, span $T_1 \otimes \mathbb{C}$, since $\mathscr{D}_E$ is a nonempty open subset of the nondegenerate quadric $q(\omega,\omega) = 0$ in $\mathbb{P}(T_1 \otimes \mathbb{C})$, so such a t would have zero image in T_1; but $T \to T_1$ is injective, since T is a free E-module and σ_1 is injective.

(b) *An endomorphism f of T that is an endomorphism of Hodge structures at every $\omega \in \mathscr{D}_E$ is multiplication by an element of E.* Such an f commutes with every u_ω, hence with the elements X_ω of the Lie algebra of $G(\mathbb{R})$ that span the Lie algebras of the groups $u_\omega(\mathrm{U}(1))$. They lie in $\mathfrak{so}(T_1) \oplus 0$, which acts on $T' = \bigoplus_{\sigma \neq \sigma_1} T_\sigma$ by zero, and their span is stable under $G(\mathbb{R})^+$, since $u_{g\omega} = g u_\omega g^{-1}$, so it is an ideal of $\mathfrak{so}(T_1)$. That ideal has nonzero projection to every simple ideal, as the eigenvalues $2i, -2i, 0, \ldots, 0$ of X_ω on $T_1 \otimes \mathbb{C}$ show also for $m = 4$, where $\mathfrak{so}(T_1) \otimes \mathbb{C} \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ and $T_1 \otimes \mathbb{C}$ is the tensor product of the two standard representations; so it is all of $\mathfrak{so}(T_1)$, and f commutes with $\mathfrak{so}(T_1)$. For $m \geq 3$ the representation of $\mathfrak{so}(T_1)$ on $T_1 \otimes \mathbb{C}$ is irreducible and nontrivial, so f preserves $T_1 = \mathfrak{so}(T_1) \cdot T_{\mathbb{R}}$ and T' and acts on $T_1 \otimes \mathbb{C} = T \otimes_{E,\sigma_1} \mathbb{C}$ by a scalar λ. An automorphism τ of $\mathbb{C}$, acting on $T \otimes_{\mathbb{Q}} \mathbb{C}$ through the second factor, commutes with f and maps $T \otimes_{E,\sigma_1} \mathbb{C}$ onto $T \otimes_{E,\tau\sigma_1} \mathbb{C}$, so f acts on the latter by $\tau(\lambda)$; and every embedding of E is of the form $\tau\sigma_1$. So $f \otimes 1$ is multiplication by an element e of $E \otimes_{\mathbb{Q}} \mathbb{C}$ on $T \otimes_{\mathbb{Q}} \mathbb{C} = T \otimes_E (E \otimes_{\mathbb{Q}} \mathbb{C})$, and e is fixed by $\mathrm{Aut}(\mathbb{C})$ because f is defined over $\mathbb{Q}$; hence $e \in E$.

For $t \in T$ the set of ω with $t \perp \omega$ is a closed analytic subset of $\mathscr{D}_E$, and for $f \in \mathrm{End}_{\mathbb{Q}}(T)$ so is the set where f is an endomorphism of Hodge structures, namely where $f(\omega) \in \mathbb{C}\omega$ and $f^*(\omega) \in \mathbb{C}\omega$, f^* the adjoint. By (a) and (b) these sets are proper for $t \neq 0$ and $f \notin E$. There are countably many of them; let Z be their union. A rational class of T is of type $(1,1)$ for T_ω exactly when it is orthogonal to ω, so outside Z there is none, and $\mathrm{End}_{\mathrm{Hdg}}(T_\omega) = E$. In the geometric situation this gives $T(S_\omega)_{\mathbb{Q}} = T$. □

TABLE 8. The Ext groups between two pieces at $A_0 = E_0^6$, from Lemma 6.8; $m = b-a$, $\chi_Z = 8$ for the threefolds Z_ζ.

from	to	degrees k	$\dim \mathrm{Ext}^k$
$\mathscr{O}_Z L^a$	$\mathscr{O}_Z L^a$	$0,\dots,6$	$\binom{6}{k}$: $1,6,15,20,15,6,1$
$\mathscr{O}_Z L^a$	$\mathscr{O}_Z L^b$, $m>0$	$0,\dots,3$	$\binom{3}{k}\,8m^3$
$\mathscr{O}_Z L^a$	$\mathscr{O}_Z L^b$, $m<0$	$3,\dots,6$	$\binom{3}{k-3}\,8\lvert m\rvert^3$
$\mathscr{O}_Z L^a$	$\mathscr{O}_{Z'} L^b$	3	$(Z\cdot Z') \in \{8,16,32,64\}$
$\mathscr{O}_Z L^a$	L^b, $m>0$	3	$8m^3$
$\mathscr{O}_Z L^a$	L^b, $m<0$	6	$8\lvert m\rvert^3$
$\mathscr{O}_Z L^a$	L^a	$3,\dots,6$	$\binom{3}{k-3}$
L^a	$\mathscr{O}_Z L^b$, $m>0$	0	$8m^3$
L^a	$\mathscr{O}_Z L^b$, $m<0$	3	$8\lvert m\rvert^3$
L^a	$\mathscr{O}_Z L^a$	$0,\dots,3$	$\binom{3}{k}$
L^a	L^b, $m>0$	0	m^6
L^a	L^b, $m<0$	6	$\lvert m\rvert^6$
L^a	L^a	$0,\dots,6$	$\binom{6}{k}$

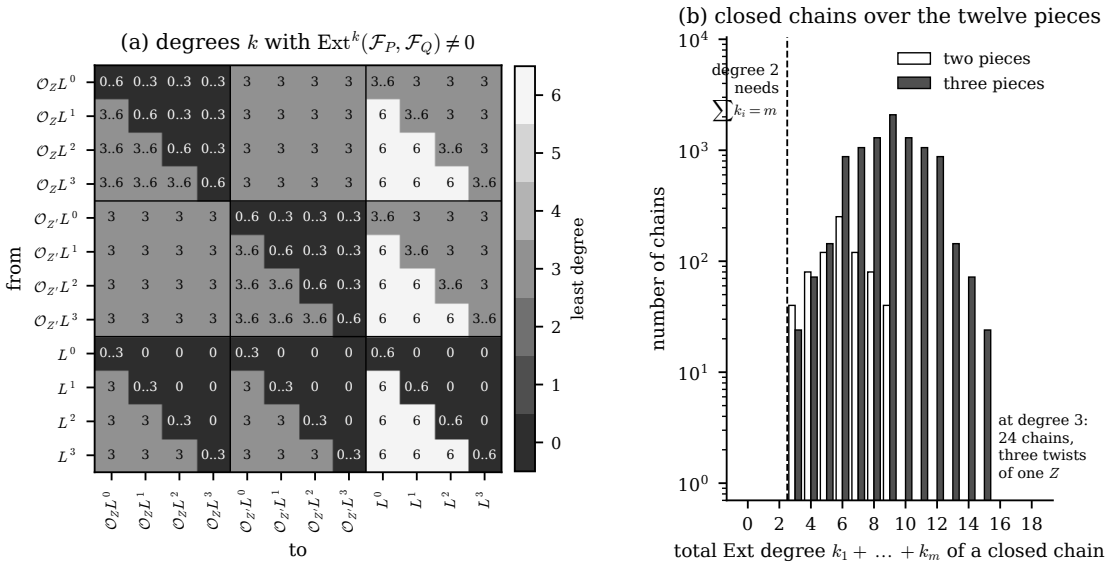


FIGURE 12. The degrees of the Ext groups between the twelve pieces $\mathscr{O}_Z L^k$, $\mathscr{O}_{Z'} L^k$, L^k, $k=0,\dots,3$, of Table 8. Left: the degrees in which $\mathrm{Ext}^k(\mathscr{F}_P,\mathscr{F}_Q)$ is nonzero, shaded by the least one; degree zero occurs only from a lower twist to a higher one and from L^a to $\mathscr{O}_Z L^a$. Right: the number of closed chains of two and of three distinct pieces by total degree, on a logarithmic scale; no two-piece chain has total degree below three, and the three-piece chains of total degree three are those of Proposition 6.10.

APPENDIX B. THE POINT OF COMPLEX MULTIPLICATION IN COORDINATES

This appendix works out, in coordinates, the objects of Lemma 6.8 and Proposition 6.10 on $A_0 = E_0^6$: the Hodge ring, the Weil plane, an explicit family of abelian threefolds whose classes combine to the Weil class, their intersection numbers, and the dimensions of every Ext group between the pieces built from them.

B.1. **The forms and the Hodge ring**

Write $z_1,\dots,z_6$ for the coordinates of $A_0 = E_0^6$, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, group the factors into the three surfaces $B_j = E_0^2$ with coordinates (z_j, z_{3+j}), $j=1,2,3$, and let $K=\mathbb{Q}(i)$ act by i on z_1,z_2,z_3 and by $-i$ on

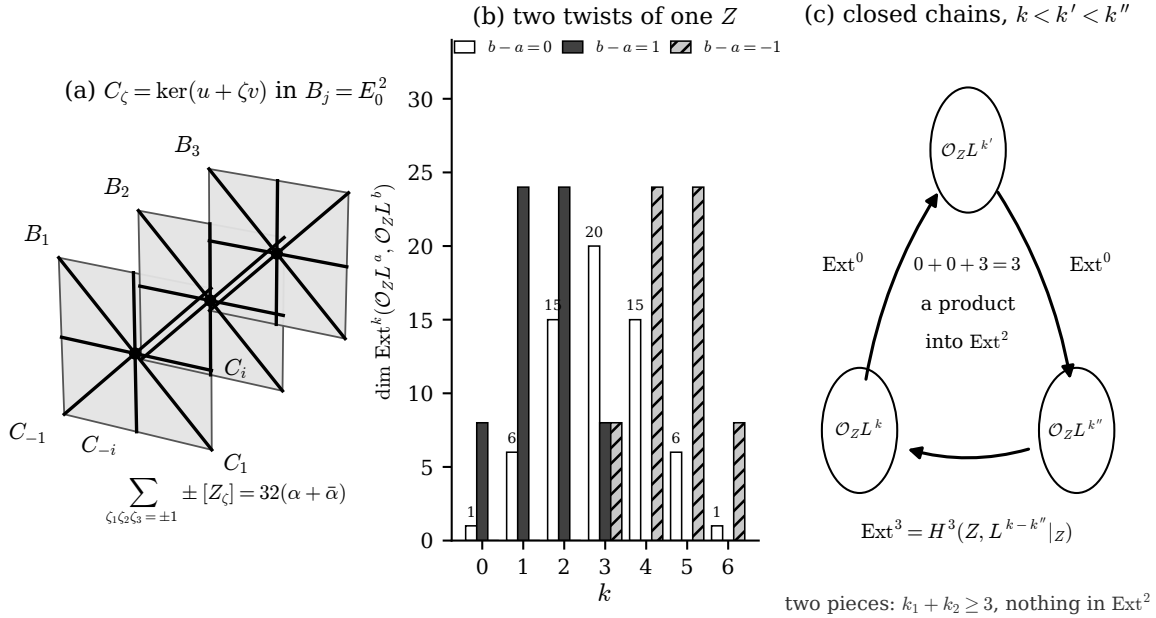


FIGURE 13. The point E_0^6 of complex multiplication. Left: the three Weil surfaces $B_j = E_0^2$ and the four elliptic curves C_ζ in each, kernels of $u+\zeta v$; a threefold Z_ζ picks one curve in each surface, and the 32 threefolds with $\zeta_1\zeta_2\zeta_3 = \pm 1$, signed by that product, add up to $32(\alpha+\bar\alpha)$. Middle: the dimensions of $\mathrm{Ext}^k(\mathscr{O}_Z\otimes L^a, \mathscr{O}_Z\otimes L^b)$ by degree for $b-a=0$ (the $\binom{6}{k}$), $b-a=1$ (degrees 0 to 3) and $b-a=-1$ (degrees 3 to 6). Right: the closed chains of Proposition 6.10; every two-step chain has total degree at least three, and the shortest chains reaching degree two run through three twists of one threefold with degrees $0,0,3$.

z_4, z_5, z_6. Put

$$\begin{aligned} a_j &= \tfrac{i}{2}\, dz_j \wedge d\bar z_j, \qquad b_j = \tfrac{i}{2}\, dz_{3+j} \wedge d\bar z_{3+j}, \\ c_j &= \tfrac{i}{2}\, dz_j \wedge d\bar z_{3+j}, \quad \bar c_j = \tfrac{i}{2}\, dz_{3+j} \wedge d\bar z_j. \end{aligned} \tag{B.1}$$

In the real coordinates $z_j = x_j + iy_j$ one has $a_j = dx_j \wedge dy_j$, the volume form of the j-th factor, of integral one over $\mathbb{C}/\mathbb{Z}[i]$,[30] and $c_j = \frac{1}{2}(dx_j \wedge dy_{3+j} - dy_j \wedge dx_{3+j}) + \frac{i}{2}(dx_j \wedge dx_{3+j} + dy_j \wedge dy_{3+j})$, so that the real and imaginary parts of c_j are rational classes and $\bar c_j$ is the complex conjugate of c_j. All four forms have type $(1,1)$. The polarisation is

$$\theta = \sum_{j=1}^{3} (a_j + b_j), \qquad \theta^6 = 6!\, a_1 b_1 a_2 b_2 a_3 b_3, \qquad \int_{A_0} \frac{\theta^6}{6!} = 1, \tag{B.2}$$

the principal polarisation of the product; the hermitian forms $\sum_{j\leq 3} a'_j z_j \bar w_j - \sum_{j>3} a'_j z_j \bar w_j$ of Lemma 6.8 with all $a'_j = 1$ give θ itself.

The Hodge group of A_0 is the circle acting by z on $H^{1,0}$ and by $\bar z$ on $H^{0,1}$, so the Hodge classes of degree $2k$ are exactly the rational classes of type (k,k), of dimension $\binom{6}{k}^2$:

$$\dim \mathrm{Hdg}^{2k}(A_0) = 1, 36, 225, 400, 225, 36, 1 \quad (k = 0, \dots, 6). \tag{B.3}$$

In degree two these are the 36 hermitian forms on $\mathbb{Z}[i]^6$, the Picard number of E_0^6. The K-eigenspaces of $H^1(A_0, \mathbb{C})$ are spanned by $dz_1, dz_2, dz_3, d\bar z_4, d\bar z_5, d\bar z_6$ (eigenvalue i) and by their conjugates, so the Weil plane $W = \bigwedge^6_K H^1(A_0, K) \subseteq H^6(A_0, \mathbb{Q})$ is spanned over $\mathbb{C}$ by

$$\alpha = c_1 c_2 c_3 \propto dz_1 dz_2 dz_3\, d\bar z_4 d\bar z_5 d\bar z_6 \qquad \text{and} \qquad \bar\alpha = \bar c_1 \bar c_2 \bar c_3, \tag{B.4}$$

[30] The fundamental domain of $\mathbb{Z}[i]$ is the unit square, so $\int_{E_0} dx \wedge dy = 1$; this normalisation makes every form in (B.1) integral on the lattice $\mathbb{Z}[i]^6$, and all intersection numbers below are integers computed as integrals of top forms.

and over $\mathbb{Q}$ by $\alpha+\bar\alpha$ and $i(\alpha-\bar\alpha)$. The relation $W\theta=0$ is visible term by term: c_ja_j contains dz_j twice and c_jb_j contains $d\bar z_{3+j}$ twice, so $\alpha(a_k+b_k)=0$ for each k, since α already contains dz_k and $d\bar z_{3+k}$.

B.2. The elliptic curves C_ζ and the threefolds Z_ζ

For $\zeta\in\mu_4=\{1,i,-1,-i\}$ let $C_\zeta\subseteq B_j$ be the kernel of $(u,v)\mapsto u+\zeta v$, an elliptic curve isomorphic to E_0, and for $\zeta=(\zeta_1,\zeta_2,\zeta_3)\in\mu_4^3$ let $Z_\zeta=C_{\zeta_1}\times C_{\zeta_2}\times C_{\zeta_3}\subseteq A_0$, an abelian threefold.

Lemma B.1. (i) *$[C_\zeta]=a_j+b_j+\bar\zeta\,c_j+\zeta\,\bar c_j$ in $H^2(B_j)$, and $[Z_\zeta]=\prod_j[C_{\zeta_j}]$.*

(ii) *$(C_\zeta\cdot C_{\zeta'})=|\zeta-\zeta'|^2\in\{0,2,4\}$ on B_j; $Z_\zeta\cap Z_{\zeta'}$ is finite exactly when $\zeta_j\neq\zeta_j'$ for all j, and then $|Z_\zeta\cap Z_{\zeta'}|=\prod_j|\zeta_j-\zeta_j'|^2\in\{8,16,32,64\}$.*

(iii) *$\theta|_{Z_\zeta}$ has degree two on each factor, $(\theta|_{Z_\zeta})^3=48$ and $\chi_{Z_\zeta}=8$; in particular $[Z_\zeta]\theta\neq0$.*

(iv) *Writing $S(\omega)=\sum_{\zeta_1\zeta_2\zeta_3=\omega}[Z_\zeta]$ for $\omega\in\mu_4$,*

$$\sum_{\omega\in\mu_4}\omega^mS(\omega)=\prod_{j=1}^3T_{m,j},\qquad T_{m,j}=\sum_{\zeta\in\mu_4}\zeta^m[C_\zeta],$$
$$T_{m,j}=4(a_j+b_j),\ 4c_j,\ 0,\ 4\bar c_j\quad\text{for } m\equiv0,1,2,3\pmod 4, \tag{B.5}$$

and therefore

$$S(1)-S(-1)=\sum_{\zeta_1\zeta_2\zeta_3=1}[Z_\zeta]-\sum_{\zeta_1\zeta_2\zeta_3=-1}[Z_\zeta]=32(\alpha+\bar\alpha). \tag{B.6}$$

Proof. (i) C_ζ is the fibre over 0 of the surjective homomorphism $f_\zeta\colon B_j\to E_0$, $w=u+\zeta v$, so $[C_\zeta]=f_\zeta^*[0]$ with $[0]=\frac{i}{2}dw\wedge d\bar w$, and

$$\begin{aligned}dw\wedge d\bar w&=(du+\zeta\,dv)\wedge(d\bar u+\bar\zeta\,d\bar v)\\&=du\wedge d\bar u+dv\wedge d\bar v+\bar\zeta\,du\wedge d\bar v+\zeta\,dv\wedge d\bar u,\end{aligned}$$

which is the formula. A product of subvarieties of the factors has the product class.

(ii) On B_j the only nonzero products of two of the four forms are a_jb_j and $c_j\bar c_j=\bar c_jc_j=-a_jb_j$ (reorder $du\,d\bar v\,dv\,d\bar u$ to $-du\,d\bar u\,dv\,d\bar v$), and $\int_{B_j}a_jb_j=1$. Hence $[C_\zeta][C_{\zeta'}]=2-\bar\zeta\zeta'-\zeta\bar\zeta'=|\zeta-\zeta'|^2$. Directly: a point of $C_\zeta\cap C_{\zeta'}$ has $(\zeta-\zeta')v=0$, so v runs over the kernel of $\zeta-\zeta'$ on E_0, of order $|\zeta-\zeta'|^2$, and $u=-\zeta v$ is then determined. The statement for Z_ζ follows factor by factor.

(iii) $\int_{C_\zeta}(a_j+b_j)=\int_{B_j}(a_j+b_j)[C_\zeta]=2$, since a_jc_j, $a_j\bar c_j$, b_jc_j, $b_j\bar c_j$ all repeat a differential. So $\theta|_{Z_\zeta}=2(p_1+p_2+p_3)$ with p_j the point class of the j-th factor, $(\theta|_{Z_\zeta})^3=3!\cdot8=48$, and $\chi_{Z_\zeta}=48/6=8$.

(iv) The left side of (B.5) is $\sum_\zeta(\zeta_1\zeta_2\zeta_3)^m\prod_j[C_{\zeta_j}]$, which factors as the product of the three sums $T_{m,j}$; these are read off from (i), using $\sum_{\zeta\in\mu_4}\zeta^m=4$ or 0 according as $4\mid m$ or not. Then $S(1)-S(-1)=\frac12\sum_\omega(\omega+\omega^3)S(\omega)=\frac12\big(\prod_jT_{1,j}+\prod_jT_{3,j}\big)=\frac12(64\alpha+64\bar\alpha)$. □

Equation (B.6) is the analogue, with abelian threefolds in place of line bundles, of the signed sum of the 32 line bundles on E_0^6 in Proposition A.24: the 32 threefolds Z_ζ with $\zeta_1\zeta_2\zeta_3=\pm1$, taken with the sign $\zeta_1\zeta_2\zeta_3$, have classes adding up to 32 times the real Weil class. Together with three powers of L this is a Chern character of polarised shape and Hankel rank three at A_0, and by Lemma 6.8(i) it is one of the shapes a convolution must realise. Two of these threefolds with $\zeta_j=\zeta_j'$ for one or two indices j meet along a curve or a surface, so Lemma 6.8(ii) does not apply to them and Lemma 6.9 gives their groups; Proposition 6.11 shows that no convolution of these thirty-five pieces meets the criterion. Figure 13 draws the curves, the dimensions of the Ext groups and the chains.

B.3. The dimensions in Lemma 6.8

For the threefolds $Z=Z_\zeta$ the numbers of Lemma 6.8 are explicit. Table 8 lists, for every pair of piece types, the degrees k in which Ext^k is nonzero and its dimension; here $m=b-a$ is the difference of the twists, $\chi_Z=8$, $Z'=Z_{\zeta'}$ with $\zeta_j\neq\zeta_j'$ for all j, and $(Z\cdot Z')=\prod_j|\zeta_j-\zeta_j'|^2$.

Each row of the table restates the corresponding item of Lemma 6.8, with $\chi(L^m|_Z) = m^3\chi_Z = 8m^3$ from (iii) above and $\chi(L^m) = m^6\theta^6/720 = m^6$ from (B.2). Two features of the table drive Proposition 6.10. Every group between two pieces of different twist is concentrated in degree 0, in degree 3 or 6, or in the range 0 to 3 for two twists of one threefold; and a group in degree at most two between distinct pieces occurs only from a lower twist to a higher one, or from L^a to $\mathcal{O}_Z L^a$. The bookkeeping of the proposition is then the following count. A closed chain of m pieces whose edges have Ext degrees $k_1,\dots,k_m$ carries an m-fold product into $\mathrm{Ext}^{\sum k_i - m + 2}$ of the first piece. For $m = 2$ the table gives, for each unordered pair of distinct piece types, the two degrees and their sum:

$$\begin{array}{lll} (\mathcal{O}_Z L^a, \mathcal{O}_{Z'} L^b) & 3+3 & = 6, \\ (\mathcal{O}_Z L^a, \mathcal{O}_Z L^b),\ b > a & (0 \text{ to } 3) + (3 \text{ to } 6) & \geq 3, \\ (\mathcal{O}_Z L^a, L^b),\ b > a & 3+3 & = 6, \\ (\mathcal{O}_Z L^a, L^b),\ b < a & 6+0 & = 6, \\ (\mathcal{O}_Z L^a, L^a) & (3 \text{ to } 6) + (0 \text{ to } 3) & \geq 3, \\ (L^a, L^b) & 0+6 & = 6, \end{array} \tag{B.7}$$

so no two-step product reaches degree two. For $m = 3$ the sum must be 3 with two edges of degree zero, and the table leaves exactly the ascending chains $\mathcal{O}_Z L^k \to \mathcal{O}_Z L^{k'} \to \mathcal{O}_Z L^{k''}$, $k < k' < k''$, closed by the degree-three edge $H^3(Z, L^{k-k''}|_Z)$, of dimension $8(k''-k)^3$. The program `julia/cm_point_ext.jl` enumerates the chains of two and three pieces over four twists and two threefolds and confirms both statements, together with the identities of Lemma B.1. Figure 12 shows the table as the matrix of the twelve pieces, and the distribution of the total degree over every closed chain of two and of three distinct pieces: the two-piece chains begin at total degree three, and the twenty-four ordered three-piece chains of total degree three are the eight ascending triples of twists of one threefold, each read from its three starting pieces.

Appendix C. Machine checks

Every finite quantity used in the proofs is recomputed by the Julia programs of the archive [8], and the combinatorial and arithmetic statements among them are also proved in Lean 4. No statement proved in the text rests on these checks; the enumerations of Proposition 6.13, Remarks 6.28 and 6.43 and the Picard number of Remark 5.10 are their output and are stated as such, and the rest guard against slips in the hand computations.

Lean. The file `lean/HodgeClosure.lean` (Lean 4.24.0, no Mathlib) contains sixty-three theorems. Section 1 encodes the eight statements and the rule set of Table 2 as data, computes closures by iteration, and proves by evaluation that the closures of $\{\mathrm{F2},\mathrm{F3}'\}$, $\{\mathrm{F3}\}$, $\{\mathrm{L},\mathrm{M}\}$, $\{\mathrm{L},\mathrm{F3}'\}$ and $\{\mathrm{V},\mathrm{F3}'\}$ contain **HC**, that the closures of $\{\mathrm{F2}\}$, $\{\mathrm{F3}'\}$, $\{\mathrm{L}\}$, $\{\mathrm{M}\}$, $\{\mathrm{V}\}$, $\{\mathbf{AB}\}$, $\{\mathrm{L},\mathrm{V}\}$ and $\{\mathrm{M},\mathrm{F3}'\}$ do not, and that the list of minimal sufficient sets, computed from the rule set alone, has exactly six members, those of Proposition 3.9. Section 2 proves the dimension counts of Section 6: $3n < n^2$ for $n \geq 4$, $m^2 + m'^2 < (m+m')^2$ for $m, m' \geq 1$ with $2mm'$ mixed directions, the values $r = 57$ and 104 of the lower bound, and the Hodge class count 26 of Theorem 6.47(i) as a sum of Künneth contributions. Section 3 proves the polynomial identities behind the Fricke families of Section 5.4: $c_4^3 - c_6^2 = 1728\,t\,(t^2+13t+49)^2$ at level 7 and $1728\,t\,(t^2+22t+125)^3$ at level 5, by exact arithmetic on coefficient lists, together with the root products 7^2 and 5^3, the Euler numbers 12 and 24, the Shioda–Tate counts giving Picard number at least 16, the transcendental rank 6, the dimensions $(n+1)/2$ of the dihedral Hecke algebras, and the polynomial identity behind the sign of Lemma 5.8: with A', B' the Vélu coefficients of the quotient of (5.1) by C_t, computed from the kernel polynomial by Newton's identities on coefficient lists, and $\tilde{A}(t) = t^4 A(N^k/t)$, $\tilde{B}(t) = t^6 B(N^k/t)$, the identity $N^{k-1}B'\tilde{A} + A'\tilde{B} = 0$ holds at both levels, which is $\lambda(t)^2 = -t^2/N^{k-1}$. Section 4 proves the Dickson identities $p_7(t+t^{-1}) = t^7 + t^{-7}$ and $p_9(t+t^{-1}) = t^9 + t^{-9}$ behind the families of Theorem 5.14, that 2 has order three in $(\mathbb{Z}/7)^\times/\pm 1$ and in $(\mathbb{Z}/9)^\times/\pm 1$, so that 2 is inert in both cubic fields, and the sign pattern of the conjugates of θ at the three real places. Section 5 proves the finite statements of Appendix B: the binomial identity $\sum_q \binom{3}{q}\binom{3}{k-q} = \binom{6}{k}$, the intersection numbers $(C_\zeta \cdot C_{\zeta'}) = |\zeta - \zeta'|^2$ and $\theta \cdot C_\zeta = 2$ in the basis of the $(1,1)$-forms of B_j with the pairing $a_j b_j = 1$, $c_j \bar{c}_j = -1$, the twisted sums $T_{m,j}$, the identity $S(1) - S(-1) = 32(\alpha + \bar{\alpha})$ as

an equality of the 64 coefficients of the triple products over $\mathbb{Z}[i]$, the counts 48, 8, 75 and 18 with the Euler characteristics, and the two chain statements of Proposition 6.10 over the twelve pieces of Appendix B.3, each by `decide`. Section 6 encodes the pieces of Table 5 and the Ext dimensions of Lemmas 6.8 and 6.9, and proves that the isolated blocks of Lemma 6.12 sum to 7213, 2605, 2605, 0, 0 and 5325 at the six configurations named in the table; the reachability relation along blocks of total degree one is supplied as a certificate, which the kernel checks to be reflexive, to contain every such block and to be closed under one more step, before the sum is evaluated by `decide +kernel`. From the same certified relation it proves that the configurations (a) and (c) of the table have 32 and 22 threefold pieces outside every directed cycle of the graph Γ of Section 6.6; and, by evaluating the Ext dimensions on every pair, that no edge of Γ enters a piece at shift 1 from the thirty-two pieces or from the pieces $\mathscr{O}_{Z_\zeta} \otimes L^k[t]$, $\zeta \in \mu_4^3$, $1 \le k \le 3$, and $L^l[t]$, $1 \le l \le 4$, with $-3 \le t \le 2$, that $L^1[3]$, $L^4[3]$ and $\mathscr{O}_{Z_{(1,1,1)}} \otimes L^1[3]$ do enter, and that $L^1[-2], \dots, L^6[3]$ put each of the thirty-two pieces on a cycle of length six. Section 7 proves the arithmetic of the newer sections: the four values $r = 57$, 48, 39, 30 of $(4+\rho)n^2 - 2n$ at $n = 3$ (Remark 6.18), the two Gram blocks $\left(\begin{smallmatrix} 0 & 2d \\ 2d & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} -1 & 0 \\ 0 & -d \end{smallmatrix}\right)$ of T_d and its determinant $16d^5$ for the squarefree $d \le 30$ (Lemma 5.22), the numerical conditions of Nikulin's criterion and the order $16d$ of the discriminant group of L_d, the discriminants 49, 81 and 148 of the cubic fields of Theorems 5.14 and 6.51, and the factorials in the coefficients of θ_h^k used in Remark 6.18. Every theorem is checked by `#print axioms`: fifty-three use no axiom, the two inequalities proved with the arithmetic tactic use `propext` and `Quot.sound`, and the six isolated-block counts and the two chain-free counts of Section 6 use `propext`; `native_decide` is not used. The file compiles in about two minutes on one core.

Julia. The directory `julia/` holds twenty-seven programs for Julia 1.11.7 that use only the standard modules `LinearAlgebra`, `Random` and `Printf`. Each prints one line `PASS` or `FAIL` per check and a final count, and exits with a nonzero code if a check fails; `julia/run_all.jl` runs all of them. Together they make 665 checks, all of which pass; the outputs are deterministic, with fixed seeds where generic data are drawn. Arithmetic is exact (integers, `Rational{BigInt}`, Gaussian rationals, $\mathbb{Q}(\sqrt{-3})$, residues modulo primes with rational reconstruction) except in three places: the ranks of `product_object.jl` and `convolution_checks.jl`, computed from singular values with relative tolerance 10^{-7}, which agree with the closed formulas, and the roots of $j(t) = j(\tau)$ in `fricke_families.jl`, computed with 256-bit floating point. Table 9 lists the programs.

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AUTHOR CONTRIBUTIONS

Deep Bhattacharjee, who leads the project, posed the problem in the form treated here, carried out the research and wrote the paper. The proofs are by Deep Bhattacharjee: the reduction and the dichotomy, the theorems on real multiplication and on the Fricke families, the transcendental rank six construction, the sixfold link and its behaviour at the points of complex multiplication, the constructions and exclusions at the points E_0^6 and E_0^8, and the appendices. Deep Bhattacharjee made all the computations, the Lean 4 file and the Julia checks of Appendix C and the figures, and brought the work to its present state. Ushashi Bhattacharya read the manuscript and verified its arguments and references. Both authors approved the final version.

TABLE 9. The Julia programs of the archive, the results they check and their number of checks.

program	what it checks	checks
`quadratic_forms.jl`	Hilbert symbols, Hasse invariants and the Hasse–Minkowski classification, shared by the lattice programs	10
`closure_graph.jl`	closures of the 256 subsets of the eight statements, the six minimal sets (Prop. 3.9), the rules with the literature adjoined and the blow-up bookkeeping of Prop. 7.1	18
`sqrt6_chain.jl`	Thm. 5.5: the 165 forms over $\mathbb{Q}(\sqrt{6})$ (98 embed, 67 do not), the complements, both orders of the chain, Varesco's $\sqrt{2}$, $\sqrt{3}$ as controls	25
`fricke_families.jl`	fibre types, Euler numbers and $\rho \geq 16$ of the Fricke families; the Fricke involution on j; the twist identity; the Hecke algebras for $n = 5,7,9,11$	41
`fricke_sign.jl`	Lemma 5.8: kernel polynomials, Vélu's quotient, $\lambda(t)^2$, the twist constants, $\Lambda^2 = \mp N$	32
`fricke_picard.jl`	Rem. 5.10: point counts over $\mathbb{F}_{\ell^i}$, the characteristic polynomial of Frobenius on T and its factorisation over $\mathbb{Q}(\sqrt{-N})$	12
`fricke_lattice.jl`	Prop. 5.19(i): $T \cong U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -N\rangle$ and the rational isometries of root lattices	21
`zeta_invariants.jl`	Thm. 5.14: the trace forms of $(-1, \pm\theta)_F$ against the transcendental lattices for $n = 7,9$, three contrasting algebras, the Dickson identity	19
`kummer_type.jl`	Prop. 5.17 against a direct search; the counts $1084/1296$ and $1304/4320$	15
`rank_six.jl`	Lemma 5.22 and Thm. 5.23 for the 305 squarefree $d \leq 500$; the embeddings in V_{F}	35
`product_object.jl`	Thm. 6.3 in the exterior-algebra model at $(m,m') = (2,1),(2,2)$: Hodge loci, cross terms, Hankel rank, $r = 57$, 104	32
`convolution_checks.jl`	Prop. 6.5: contraction ranks, the annihilators under the orthogonality hypotheses, the block trace	26
`annihilator_exact.jl`	Thm. 6.6 over $\mathbb{Q}(\sqrt{-3})$: formula (6.4) and the annihilator of dimension $m^2 + m'^2$	34
`riemann_roch_pieces.jl`	the polynomial (6.5) and the thirty-one supports of Prop. 6.7	8
`cm_point_ext.jl`	Lemmas 6.8, B.1 and Prop. 6.10: Koszul cohomology, the twisted groups, the closed chains of two and three pieces	14
`cm_point_counts.jl`	Table 5 and Lemma 6.12: the Chern characters, C^1, C^2, C^3 and the isolated blocks	13
`cm_point_chains.jl`	Prop. 6.14, Cor. 6.15, Rem. 6.16: the graph Γ and its directed cycles	12
`line_bundle_chains.jl`	Thm. 6.25, Rem. 6.28: the arrangements of $n \leq 8$ line bundles and their counts	6
`line_bundle_fine.jl`	rule E3* (Lemma 6.27), conditions C1–C4 in two implementations, the 165 and 269 states, the nine-bundle arrangements	17
`chern_h.jl`	Rem. 6.18: the weight-zero Chern character against θ_h for $h \in \{1,2,3\}^6$, the ranks 57, 48, 39, 30	9
`threefold_supports.jl`	Section 6.8: Thm. 6.32, Lemmas 6.31, 6.37, 6.35 and Thm. 6.36 in exact arithmetic	16
`cover_boundary.jl`	Prop. 6.42: smoothness of B_λ, the divisor of $y^3 + 1/\lambda$, $\eta \neq 0$, the rank nine	10
`sextic_lattice.jl`	Rem. 6.43: the integral points of $S(0,q)$ over the twelve cubic fields and their least values of $-\chi$	24
`sixfold_link.jl`	Thm. 6.47: ψ_c, the K^1-invariants, the count 26, the weights of ξ; the shared isogeny factor at $L = K$	55
`mumford_cm.jl`	Thms. 6.50, 6.51: invariants by Künneth type, Pohlmann's criterion on $E^a \times A^b$ and on X_t	51
`mumford_disc.jl`	Thm. 6.50(iii): $\det H = b\, N_{F/\mathbb{Q}}(a)\, d_F$ and discriminant one, for four cubic fields	24
`mumford_residual.jl`	Rem. 6.53: the residual case $L = K$, the Hodge torus, the lattices and dimensions, the fields of degree twelve, discriminant-one polarisations	86
total		665

OPEN GLUON AND ANAXIA

Two tools built by D. B. stand behind the way this project was organised. Open Gluon is a workspace D. B. assembled and maintains, which keeps pinned copies of the mathematical libraries, the source record of the project and the checks in one reproducible place. Anaxia, the engine of the eBLLM project, is the construction engine of D. B.: it stores each construction as a route with its

input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact rational arithmetic. D. B. wrote both and developed them from the constructions of earlier papers. Three of their constructions were used here. The first is the encoding of the stopping points of the paper as typed targets, from which Anaxia built the candidate configurations at E_0^6 of Table 5, the quotient chains of Proposition 5.25 and the first enumerations of the chains of line bundles behind Remark 6.28. The second is the reading of supports from a localised obstruction, which led to the statements of Section 6.8. The third is a Frobenius test on elliptic K3 surfaces, which first showed that a Fricke self-map is a complex and not a real multiplication, the observation behind Lemma 5.8. The engine proposes routes and tests them; every statement of this paper is proved in the text.

DECLARATIONS

Data availability. No data were generated. The paper, the Lean 4 file and the Julia checks of Appendix C are archived at doi:10.5281/zenodo.23439577 [8]. Nothing in the paper rests on them: every statement proved in the text has its proof there, and the statements obtained by enumeration are marked as such.

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Ethics. The work raises no ethical question; it involves no human participants, no animals and no personal data.

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